数学物理方程

3 Bessel函数

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3.0 为什么要有Bessel函数

物质世界 三维空间系统在时域内的单向演化→详实描述变量时空演化→必须拓展至高维

PDE向高维拓展的新问题

$$u = u(x,t)$$

向高维拓展

1 时间维度不变→空间维度增多→变量总数增多

$$u = u(x,y,t) = u(\rho,\theta,t) = u(\rho,\theta,h) = u(x,y,z)$$

2 变量空间分布的表达→强烈依赖于空间维度的组织形式→即坐标系的选取

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \Delta u = \frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho}\frac{\partial u}{\partial \rho} + \frac{1}{\rho^2}\frac{\partial^2 u}{\partial \theta^2}$$

$$\frac{\partial^2 u}{\partial t^2} - a^2 \Delta u = 0$$

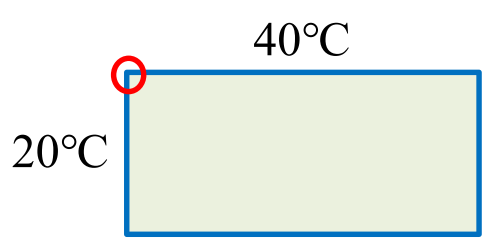
平面直角系

$$\frac{\partial^2 u}{\partial t^2} - a^2\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) = 0$$

极坐标系

$$\frac{\partial^2 u}{\partial t^2} - a^2\left(\frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho}\frac{\partial u}{\partial \rho} + \frac{1}{\rho^2}\frac{\partial^2 u}{\partial \theta^2}\right) = 0$$

3 边界条件可能出现奇点



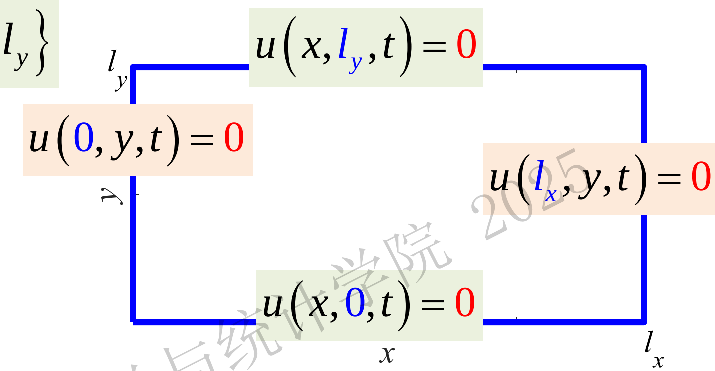
二维导热

分离变量法能否直接使用？

3.0 为什么要有Bessel函数

平面直角系上的波动方程

$$\Omega=\{(x,y)\,|\,0<x<l_x,0<y<l_y\}$$



例1

$$\begin{cases} u_{tt}-a^2(u_{xx}+u_{yy})=0, & (x,y)\in\Omega,\ t>0 \\ u(0,y,t)=u(l_x,y,t)=0, & 0\le y\le l_y \\ u(x,0,t)=u(x,l_y,t)=0, & 0\le x\le l_x \\ u(x,y,0)=\varphi(x,y),u_t(x,y,0)=\psi(x,y), & (x,y)\in\overline{\Omega} \end{cases}$$

波动/二维/齐次	
1齐	1齐
1齐	1齐

解 第1步 时间/空间分离

设 $u(x,y,t)=T(t)w(x,y)$ 代入原PDE

分离变量有 $T''(t)w(x,y)-a^2T(t)\Delta w(x,y)=0$

$$\frac{T''(t)}{a^2T(t)}=\frac{\Delta w(x,y)}{w(x,y)}=-\mu$$

时空分离常数

定义运算

$$A=-\frac{d^2}{dx^2} \qquad -\Delta=-\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right)$$

初值问题

线性二阶常系数齐次ODE

$$T''(t)+\mu a^2T(t)=0$$

二维边值问题

$$\begin{cases} -\Delta w(x,y)=\mu w(x,y), & (x,y)\in\Omega \\ w|_{\partial\Omega}=0 \end{cases}$$

运算 (元素$_i$) = $\lambda\cdot$元素

边界条件

$$\left.\begin{aligned} u(0,y,t)&=T(t)w(0,y)=0\\ u(l_x,y,t)&=T(t)w(l_x,y)=0\\ u(x,0,t)&=T(t)w(x,0)=0\\ u(x,l_y,t)&=T(t)w(x,l_y)=0 \end{aligned}\right\}$$

原PDE齐次边界由边值问题实现

$T(t)$ 不恒为0

二维边值问题=算子$-\Delta$的特征值问题

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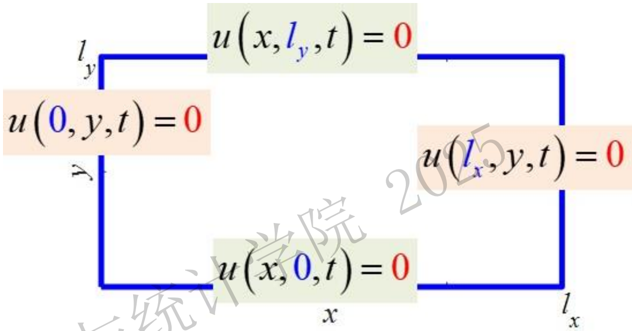

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1齐	1齐
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第2步 空间各维度分离

$u(x,y,t)=T(t)w(x,y)$

$$\begin{cases} \Delta w(x,y)+\mu w(x,y)=0, & (x,y)\in\Omega \\ w|_{\partial\Omega}=0 \end{cases}$$

设 $w(x,y)=X(x)Y(y)$ 代入二维边值问题, 按直角系展开Laplace算子 Δ

$$X''(x)Y(y)+X(x)Y''(y)+\mu X(x)Y(y)=0$$

$$-\frac{Y''(y)+\mu Y(y)}{Y(y)}=\frac{X''(x)}{X(x)}=-\lambda$$

空间分离常数

y方向边值问题 $=\lambda_y$

$$\begin{cases} Y''(y)+(\mu-\lambda_x)Y(y)=0 \\ Y(0)=Y(l_y)=0 \end{cases}$$

特征值问题 $A_1=-\dfrac{d^2}{dx^2}$

x方向边值问题

$$\begin{cases} X''(x)+\lambda_x X(x)=0 \\ X(0)=X(l_x)=0 \end{cases}$$

$\lambda_y=\mu-\lambda_x$ → $\mu=\lambda_x+\lambda_y$

直角系:时空分离常数=各维特征值之和= A_2=-Δ的特征值

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波动/二维/齐次	
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算子 $A=-\frac{d^2}{dx^2}$ 不同齐次边界条件下的特征值/特征函数

齐次边界类型	特征值问题	特征值/特征函数
[1,1]	$\begin{cases}X''(x)+\lambda X(x)=0\\X(0)=X(l)=0\end{cases}$	$\lambda_n=\left(\frac{n\pi}{l}\right)^2, X_n(x)=\sin\left(\frac{n\pi}{l}x\right),(n=1,2,3\cdots)$
[1,2]	$\begin{cases}X''(x)+\lambda X(x)=0\\X(0)=X'(l)=0\end{cases}$	$\lambda_n=\left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x)=\sin\left[\frac{(2n+1)\pi}{2l}x\right],(n=0,1,2,3\cdots)$
[2,1]	$\begin{cases}X''(x)+\lambda X(x)=0\\X'(0)=X(l)=0\end{cases}$	$\lambda_n=\left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x)=\cos\left[\frac{(2n+1)\pi}{2l}x\right],(n=0,1,2,3\cdots)$
[2,2]	$\begin{cases}X''(x)+\lambda X(x)=0\\X'(0)=X'(l)=0\end{cases}$	$\lambda_n=\left(\frac{n\pi}{l}\right)^2, X_n(x)=\cos\left(\frac{n\pi}{l}x\right),(n=0,1,2,3\cdots)$
周期	$\begin{cases}\Phi''(\theta)+\lambda\Phi(\theta)=0\\\Phi(0)=\Phi(2\pi),\\\Phi'(0)=\Phi'(2\pi)\end{cases}$	$\lambda_0=0,\ \Phi_0(\theta)=1,\ (n=0)$ $\lambda_n=n^2,\ \Phi_n(\theta)=C_1\cos n\theta+C_2\sin n\theta,\ (n=1,2,3\cdots)$

第3步 空间维度上的特征值

$$w(x,y)=X(x)Y(y)$$

$$\begin{cases}X''(x)+\lambda X(x)=0\\X(0)=X(l_x)=0\end{cases}\qquad\begin{cases}Y''(y)+(\mu-\lambda)Y(y)=0\\Y(0)=Y(l_y)=0\end{cases}$$

$$\underbrace{\lambda_{nx}=\left(\frac{n_x\pi}{l_x}\right)^2,\ X_{nx}(x)=\sin\left(\frac{n_x\pi}{l_x}x\right),\ (n_x=1,2,3\cdots)\qquad \mu-\lambda_{nx}=\left(\frac{n_y\pi}{l_y}\right)^2,\ Y_{ny}(y)=\sin\left(\frac{n_y\pi}{l_y}y\right),\ (n_y=1,2,3\cdots)}$$

$$\begin{cases}\Delta w(x,y)+\mu w(x,y)=0, & (x,y)\in\Omega\\ w|_{\partial\Omega}=0\end{cases}$$

二维边值问题特征值/特征函数

二维特征值函数与各维函数

$$-\Delta w(x,y)=-\Delta\left[X(x)Y(y)\right]$$
$$=-X''(x)Y(y)-X(x)Y''(y)$$
$$=\lambda_{nx}X(x)Y(y)+(\mu-\lambda_{nx})X(x)Y(y)$$
$$=\mu X(x)Y(y)$$

两组序号

$$\mu_{nx,ny}=\lambda_{nx}+\lambda_{ny}=\left(\frac{n_x\pi}{l_x}\right)^2+\left(\frac{n_y\pi}{l_y}\right)^2,\ (n_x,n_y=1,2,3\cdots)$$

$$W_{nx,ny}(x,y)=\sin\left(\frac{n_x\pi}{l_x}x\right)\sin\left(\frac{n_y\pi}{l_y}y\right)$$

直角系

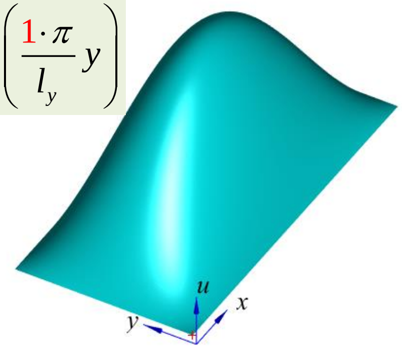
二维边值问题特征值=各维度特征值之和
二维边值问题特征函数=各维特征函数之积

3.0 为什么要有Bessel函数

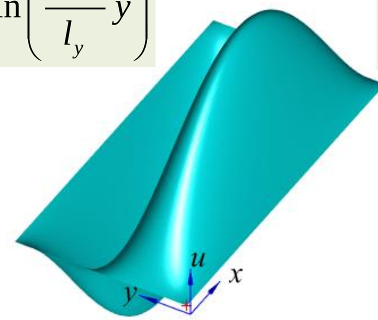
$$\left(\frac{n_x\pi}{l_x}\right)^2+\left(\frac{n_y\pi}{l_y}\right)^2,\ \sin\left(\frac{n_x\pi}{l_x}x\right)\sin\left(\frac{n_y\pi}{l_y}y\right),\ \left(n_x,n_y=1,2,3\cdots\right)$$

二维特征函数系

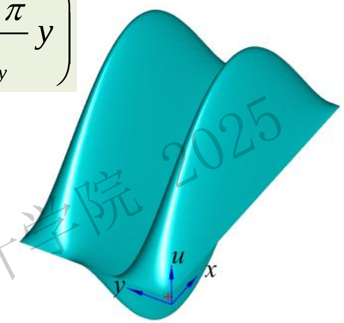
$$\sin\left(\frac{1\cdot\pi}{l_x}x\right)\sin\left(\frac{1\cdot\pi}{l_y}y\right)$$



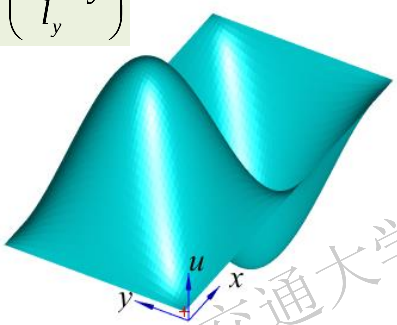
$$\sin\left(\frac{1\cdot\pi}{l_x}x\right)\sin\left(\frac{2\cdot\pi}{l_y}y\right)$$



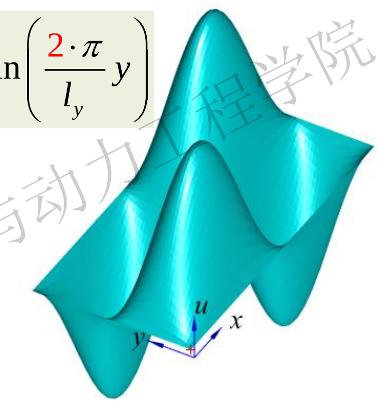
$$\sin\left(\frac{1\cdot\pi}{l_x}x\right)\sin\left(\frac{3\cdot\pi}{l_y}y\right)$$



$$\sin\left(\frac{2\cdot\pi}{l_x}x\right)\sin\left(\frac{1\cdot\pi}{l_y}y\right)$$



$$\sin\left(\frac{2\cdot\pi}{l_x}x\right)\sin\left(\frac{2\cdot\pi}{l_y}y\right)$$



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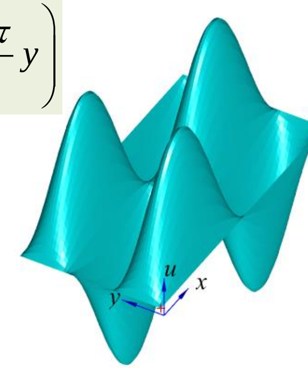
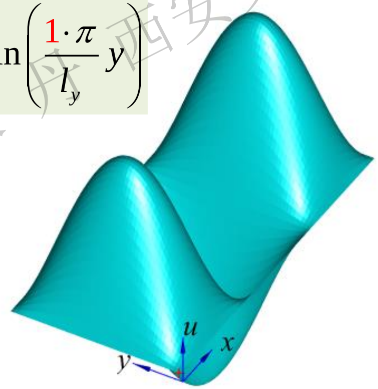
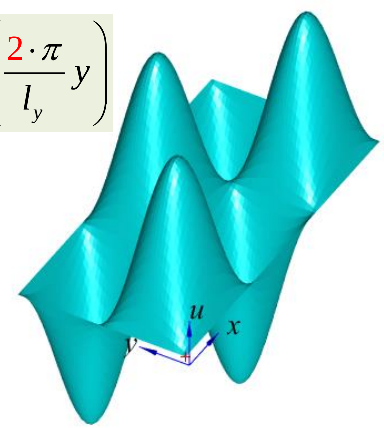


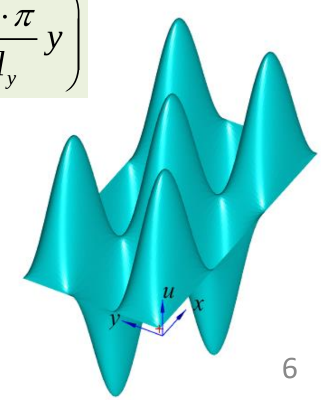
$$\sin\left(\frac{3\cdot\pi}{l_x}x\right)\sin\left(\frac{1\cdot\pi}{l_y}y\right)$$



$$\sin\left(\frac{3\cdot\pi}{l_x}x\right)\sin\left(\frac{2\cdot\pi}{l_y}y\right)$$



$$\sin\left(\frac{3\cdot\pi}{l_x}x\right)\sin\left(\frac{3\cdot\pi}{l_y}y\right)$$



3.0 为什么要有Bessel函数

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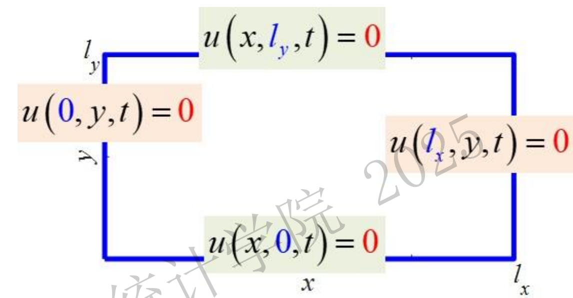

第4步 求解初值ODE $T''(t)+\mu_{nx,ny}a^2T(t)=0$ 时空分离常数

$$\mu_{nx,ny}=\left(\frac{n_y\pi}{l_y}\right)^2+\left(\frac{n_y\pi}{l_y}\right)^2,\ \left(n_y,n_y=1,2,3\cdots\right)$$

$$W_{nx,ny}(x,y)=\sin\left(\frac{n_x\pi}{l_x}x\right)\sin\left(\frac{n_y\pi}{l_y}y\right)$$

通解为 $T_{nx,ny}(t)=C_{1n}\cos\left(a\sqrt{\mu_{nx,ny}}t\right)+C_{2n}\sin\left(a\sqrt{\mu_{nx,ny}}t\right)$

由叠加原理, 原定解问题的形式解为

$$u(x,y,t)=\sum_{ny=1}^{\infty}\sum_{nx=1}^{\infty}\left[T_{nx,ny}(t)\cdot W_{nx,ny}(x,y)\right]$$

序号二重累加

$$=\sum_{ny=1}^{\infty}\sum_{nx=1}^{\infty}\left[C_{1n}\cos\left(a\sqrt{\mu_{nx,ny}}t\right)+C_{2n}\sin\left(a\sqrt{\mu_{nx,ny}}t\right)\right]\cdot\sin\left(\frac{n_x\pi}{l_x}x\right)\sin\left(\frac{n_y\pi}{l_y}y\right)$$

对时间偏导为

$$u_t(x,y,t)=\sum_{ny=1}^{\infty}\sum_{nx=1}^{\infty}\left[-a\sqrt{\mu_{nx,ny}}C_{1n}\sin\left(a\sqrt{\mu_{nx,ny}}t\right)+a\sqrt{\mu_{nx,ny}}C_{2n}\cos\left(a\sqrt{\mu_{nx,ny}}t\right)\right]\cdot\sin\left(\frac{n_x\pi}{l_x}x\right)\sin\left(\frac{n_y\pi}{l_y}y\right)$$

利用形式解实现初始条件→确定系数 C_{1n},C_{2n}

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$$W_{nx,ny}(x,y)=\sin\left(\frac{n_x\pi}{l_x}x\right)\sin\left(\frac{n_y\pi}{l_y}y\right)$$

$$u(x,y,t)=\sum_{ny=1}^{\infty}\sum_{nx=1}^{\infty}\begin{bmatrix} C_{1n}\cos\left(a\sqrt{\mu_{nx,ny}}t\right) \\ +C_{2n}\sin\left(a\sqrt{\mu_{nx,ny}}t\right)\end{bmatrix}\cdot\sin\left(\frac{n_x\pi}{l_x}x\right)\sin\left(\frac{n_y\pi}{l_y}y\right)$$

第4步 解初值ODE

$$u(x,y,0)=\sum_{ny=1}^{\infty}\sum_{nx=1}^{\infty}C_{1n}\cdot\sin\left(\frac{n_x\pi}{l_x}x\right)\sin\left(\frac{n_y\pi}{l_y}y\right)=\varphi(x,y)$$

$$u_t(x,y,t)=\sum_{ny=1}^{\infty}\sum_{nx=1}^{\infty}\begin{bmatrix} -a\sqrt{\mu_{nx,ny}}C_{1n}\sin\left(a\sqrt{\mu_{nx,ny}}t\right) \\ +a\sqrt{\mu_{nx,ny}}C_{2n}\cos\left(a\sqrt{\mu_{nx,ny}}t\right)\end{bmatrix}\cdot\sin\left(\frac{n_x\pi}{l_x}x\right)\sin\left(\frac{n_y\pi}{l_y}y\right)$$

$$u_t(x,y,0)=\sum_{ny=1}^{\infty}\sum_{nx=1}^{\infty}\left[a\sqrt{\mu_{nx,ny}}C_{2n}\sin\left(\frac{n_x\pi}{l_x}x\right)\sin\left(\frac{n_y\pi}{l_y}y\right)\right]=\psi(x,y)$$

$$C_{1n}=\frac{4}{l_xl_y}\int_0^{l_y}\int_0^{l_x}\varphi(x,y)\sin\left(\frac{n_x\pi}{l_x}x\right)\sin\left(\frac{n_y\pi}{l_y}y\right)dxdy$$

二元函数Fourier正交分解

$$C_{2n}=\frac{4}{l_xl_ya\sqrt{\mu_{nx,ny}}}\int_0^{l_y}\int_0^{l_x}\psi(x,y)\sin\left(\frac{n_x\pi}{l_x}x\right)\sin\left(\frac{n_y\pi}{l_y}y\right)dxdy$$

原定解问题的解

$$u(x,y,t)=\sum_{ny=1}^{\infty}\sum_{nx=1}^{\infty}\left[C_{1n}\cos\left(a\sqrt{\mu_{nx,ny}}t\right)+C_{2n}\sin\left(a\sqrt{\mu_{nx,ny}}t\right)\right]\cdot\sin\left(\frac{n_x\pi}{l_x}x\right)\sin\left(\frac{n_y\pi}{l_y}y\right)$$

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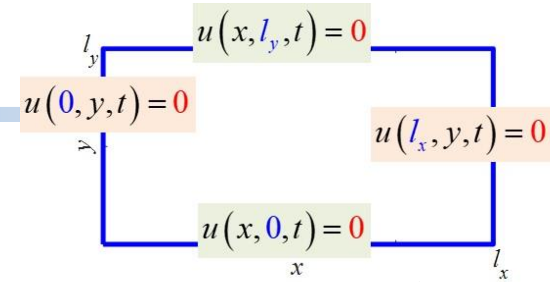

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$$W_{nx,ny}(x,y)=\sin\left(\frac{n_x\pi}{l_x}x\right)\sin\left(\frac{n_y\pi}{l_y}y\right)$$

讨论 初始位移 $u(x,y,0)=\varphi(x,y)=Ax(x-l_x)y(y-l_y)$

初始速度 $u_t(x,y,0)=\psi(x,y)=0$

$$C_{1n}=\frac{4}{l_xl_y}\int_0^{l_y}\int_0^{l_x}\varphi(x,y)\sin\left(\frac{n_x\pi}{l_x}x\right)\sin\left(\frac{n_y\pi}{l_y}y\right)dxdy$$

$$=\frac{4A}{l_xl_y}\int_0^{l_x}x(x-l_x)\sin\left(\frac{n_x\pi}{l_x}x\right)dx\cdot\int_0^{l_y}y(y-l_y)\cdot\sin\left(\frac{n_y\pi}{l_y}y\right)dy$$

由Fourier展开得 $C_{2n}=0$

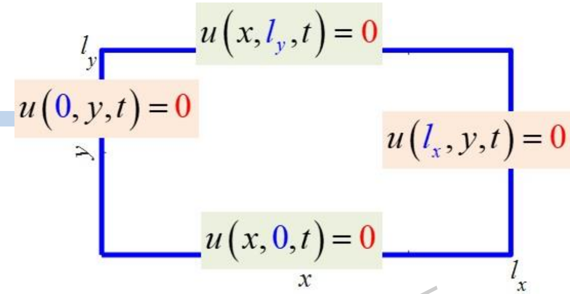
1 $$\int_0^{l_x}x(x-l_x)\sin\left(\frac{n_x\pi}{l_x}x\right)dx=\left[-\frac{l_x}{n_x\pi}x^2\cos\frac{n_x\pi}{l_x}x+\frac{l_x^2}{n_x\pi}x\cos\frac{n_x\pi}{l_x}x+\frac{2l_x^3}{(n_x\pi)^3}\cos\frac{n_x\pi}{l_x}x\right]_0^{l_x}=\frac{2l_x^3}{n_x^3\pi^3}\left[(-1)^{n_x}-1\right]$$

2 $$\int_0^{l_y}y(y-l_y)\sin\left(\frac{n_y\pi}{l_y}y\right)dy=\left[-\frac{l_y}{n_y\pi}y^2\cos\frac{n_y\pi}{l_y}y+\frac{l_y^2}{n_y\pi}y\cos\frac{n_y\pi}{l_y}y+\frac{2l_y^3}{(n_y\pi)^3}\cos\frac{n_y\pi}{l_y}y\right]_0^{l_y}=\frac{2l_y^3}{n_y^3\pi^3}\left[(-1)^{n_y}-1\right]$$

得展开系数

$$C_{1n}=\frac{16Al_x^2l_y^2}{\pi^6}\cdot\frac{\left[(-1)^{n_x}-1\right]\left[(-1)^{n_y}-1\right]}{n_x^3n_y^3}$$

3.0 为什么要有Bessel函数



平面直角系上的波动方程

$$\Omega=\{(x,y)|0<x<l_x,0<y<l_y\}$$

例1

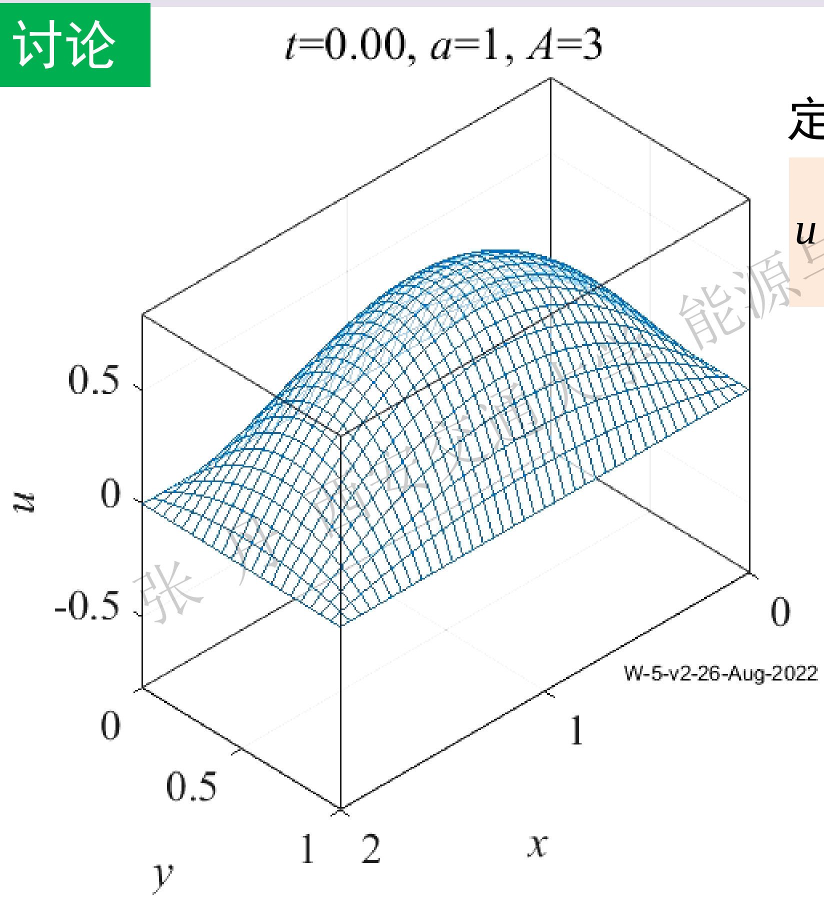
$$\begin{cases} u_{tt}-a^2(u_{xx}+u_{yy})=0, & (x,y)\in\Omega,\ t>0 \\ u(0,y,t)=u(l_x,y,t)=0, & 0\le y\le l_y \\ u(x,0,t)=u(x,l_y,t)=0, & 0\le x\le l_x \\ u(x,y,0)=\varphi(x,y),u_t(x,y,0)=\psi(x,y), & (x,y)\in\bar{\Omega} \end{cases}$$

波动/二维/齐次	
1齐	1齐
1齐	1齐

$$\mu_{nx,ny}=\left(\frac{n_x\pi}{l_x}\right)^2+\left(\frac{n_y\pi}{l_y}\right)^2,(n_y,n_y=1,2,3\cdots)$$

$$W_{nx,ny}(x,y)=\sin\left(\frac{n_x\pi}{l_x}x\right)\sin\left(\frac{n_y\pi}{l_y}y\right)$$

讨论



定解问题的解 $\varphi(x,y)=Ax(x-l_x)y(y-l_y),\ \psi(x,y)=0$

$$u(x,y,t)=\sum_{ny=1}^{\infty}\sum_{nx=1}^{\infty}C_{1n}\cos\left(a\sqrt{\mu_{nx,ny}}t\right)\cdot\sin\left(\frac{n_x\pi}{l_x}x\right)\sin\left(\frac{n_y\pi}{l_y}y\right)$$

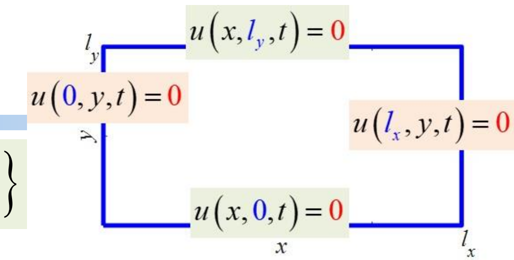
$$C_{1n}=\frac{16Al_x^2l_y^2}{\pi^6}\cdot\frac{\left[(-1)^{n_x}-1\right]\left[(-1)^{n_y}-1\right]}{n_x^3n_y^3}$$

分离变量法圆满解决了定义在平面直角系的二维波动/导热方程定解问题

3.0 为什么要有Bessel函数

平面直角系上的波动方程

$$\Omega=\left\{(x,y)\middle|0<x<l_x,0<y<l_y\right\}$$



例1

$$\begin{cases} u_{tt}-a^2\left(u_{xx}+u_{yy}\right)=0,\ (x,y)\in\Omega,\ t>0 \\ u(0,y,t)=u(l_x,y,t)=0,\ 0\le y\le l_y \\ u(x,0,t)=u(x,l_y,t)=0,\ 0\le x\le l_x \\ u(x,y,0)=\varphi(x,y),u_t(x,y,0)=\psi(x,y),\ (x,y)\in\bar{\Omega} \end{cases}$$

波动/二维/齐次	
1齐	1齐
1齐	1齐

$$\mu_{nx,ny}=\left(\frac{n_x\pi}{l_x}\right)^2+\left(\frac{n_y\pi}{l_y}\right)^2,\left(n_y,n_y=1,2,3\cdots\right)$$

$$W_{nx,ny}(x,y)=\sin\left(\frac{n_x\pi}{l_x}x\right)\sin\left(\frac{n_y\pi}{l_y}y\right)$$

小结

$$u(x,y,t)=T(t)w(x,y) \rightarrow \frac{T''(t)}{a^2T(t)}=\frac{\Delta w(x,y)}{w(x,y)}=-\mu$$

时空分离常数

$$T''(t)w(x,y)-a^2T(t)\Delta w(x,y)=0$$

初值问题 $T''(t)+\mu a^2T(t)=0$

二维边值问题

$$\begin{cases}\Delta w(x,y)+\mu w(x,y)=0,\ (x,y)\in\Omega \\ w|_{\partial\Omega}=0\end{cases}$$

$$w(x,y)=X(x)Y(y) \rightarrow \frac{X''(x)}{X(x)}=-\frac{Y''(y)+\mu Y(y)}{Y(y)}=-\lambda$$

空间分离常数

$$X''(x)Y(y)+X(x)Y''(y)+\mu X(x)Y(y)=0$$

1. 两级分离→分离变量法可用
2. 时空分离常数 μ；空间分离常数 λ
3. 二维特征值由2个维度联合确定
4. 二次分离后的2个边值问题/特征值问题都是常系数ODE

x方向边值问题

$$\begin{cases}X''(x)+\lambda X(x)=0 \\ X(0)=X(l_x)=0\end{cases}$$

特征值问题

y方向边值问题

$$\begin{cases}Y''(y)+(\mu-\lambda)Y(y)=0 \\ Y(0)=Y(l_y)=0\end{cases}$$

二维空间用直角系描述→分离变量法求解没有障碍

3.0 为什么要有Bessel函数 →表达径向Bessel方程的解

极坐标上的导热方程 $\dfrac{\partial u}{\partial t}-a^2\Delta u=0$ $u(\rho,\theta,t)=T(t)w(\rho,\theta)$

$$T'(t)w(\rho,\theta)-a^2T(t)\Delta w(\rho,\theta)=0$$

$$\frac{T'(t)}{a^2T(t)}=\frac{\Delta w(\rho,\theta)}{w(\rho,\theta)}=-\mu$$

时空分离常数

❷ 尚未确定

初值问题

$$T'(t)+\mu a^2T(t)=0$$

二维边值问题/特征值问题 $A_2=-\Delta$

$$\begin{cases}-\Delta w(\rho,\theta)=\mu w(\rho,\theta),\ (\rho,\theta)\in\Omega\\ w|_{\partial\Omega}=0\end{cases}$$

$$\Delta w=\frac{\partial^2 w}{\partial\rho^2}+\frac{1}{\rho}\frac{\partial w}{\partial\rho}+\frac{1}{\rho^2}\frac{\partial^2 w}{\partial\theta^2}$$

$$w(\rho,\theta)=R(\rho)\Phi(\theta)$$

$$R''(\rho)\Phi(\theta)+\frac{1}{\rho}R'(\rho)\Phi(\theta)+\frac{1}{\rho^2}R(\rho)\Phi''(\theta)=-\mu R(\rho)\Phi(\theta)$$

空间分离常数

$$\frac{\Phi''(\theta)}{\Phi(\theta)}=-\frac{\mu R(\rho)+R''(\rho)+\frac{1}{\rho}R'(\rho)}{\frac{1}{\rho^2}R(\rho)}=-\lambda$$

周向 径向

$$\lambda_n=n^2,$$
$$\Phi_n(\theta)=\{1,\cos n\theta,\sin n\theta\}$$
$$(n=0,1,2,3\cdots)$$

$$\begin{cases}\Phi''(\theta)+\lambda\Phi(\theta)=0\\ \Phi(\theta)=\Phi(\theta+2\pi)\end{cases}\qquad\begin{cases}|R(0)|<+\infty,\ R(\rho_0)=0\\ \rho^2R''(\rho)+\rho R'(\rho)+(\mu\rho^2-\lambda)R(\rho)=0\end{cases}$$

相当于原方程是Poisson方程

Euler方程 $\mu=0$

$$\rho^2R''(\rho)+\rho R'(\rho)-n^2R(\rho)=0$$

Bessel方程 $\mu>0$ **1824**

$$\rho^2R''(\rho)+\rho R'(\rho)+(\mu\rho^2-n^2)R(\rho)=0$$

线性二阶变系数齐次ODE ❶ 如何求解

F.W.Bessel
1784-1846
German Math.
Astronomer

3.0 为什么要有Bessel函数

求解Bessel方程是求解极坐标下波动/导热方程/柱系下Poisson方程的技术瓶颈

二维波动/导热方程求解框架

$$\frac{\partial^2 u}{\partial t^2}-a^2\Delta u=0 \qquad \frac{\partial u}{\partial t}-a^2\Delta u=0$$

$$\frac{T''(t)}{a^2T(t)}=\frac{\Delta w}{w}=-\mu \qquad \frac{T'(t)}{a^2T(t)}=\frac{\Delta w}{w}=-\mu$$

时空分离常数

初值问题

$$T''(t)+\mu a^2T(t)=0$$
$$T'(t)+\mu a^2T(t)=0$$

二维边值问题/特征值问题

$$\begin{cases}\Delta w+\mu w=0, & (\rho,\theta)\in\Omega\\ w|_{\partial\Omega}=0\end{cases}$$

平面直角系

$$\begin{cases}\Delta w(x,y)+\mu w(x,y)=0, & (x,y)\in\Omega\\ w|_{\partial\Omega}=0\end{cases}$$

$$w(x,y)=X(x)Y(y) \qquad \Delta w=\frac{\partial^2 w}{\partial x^2}+\frac{\partial^2 w}{\partial y^2}$$

$$\frac{X''(x)}{X(x)}=-\frac{Y''(y)+\mu Y(y)}{Y(y)}=-\lambda$$

空间分离常数

x方向

$$\begin{cases}X''(x)+\lambda X(x)=0\\ X(0)=X(l_x)=0\end{cases}$$

y方向

$$\begin{cases}Y''(y)+(\mu-\lambda)Y(y)=0\\ Y(0)=Y(l_y)=0\end{cases}$$

极坐标系

$$\begin{cases}\Delta w(\rho,\theta)+\mu w(\rho,\theta)=0, & (\rho,\theta)\in\Omega\\ w|_{\partial\Omega}=0\end{cases}$$

$$w(\rho,\theta)=R(\rho)\Phi(\theta) \qquad \Delta w=\frac{\partial^2 w}{\partial\rho^2}+\frac{1}{\rho}\frac{\partial w}{\partial\rho}+\frac{1}{\rho^2}\frac{\partial^2 w}{\partial\theta^2}$$

$$-\frac{\mu R(\rho)+R''(\rho)+\frac{1}{\rho}R'(\rho)}{\frac{1}{\rho^2}R(\rho)}=\frac{\Phi''(\theta)}{\Phi(\theta)}=-\lambda$$

空间分离常数

径向

$$\begin{cases}|R(0)|<+\infty,\ R(\rho_0)=0\\ \boxed{\rho^2R''(\rho)+\rho R'(\rho)+(\mu\rho^2-\lambda)R(\rho)=0}\end{cases}$$

周向

$$\begin{cases}\Phi''(\theta)+\lambda\Phi(\theta)=0\\ \Phi(\theta)=\Phi(\theta+2\pi)\end{cases}$$

本章重点解决Bessel方程

1. 数学建模和基本原理介绍
2. 分离变量法
3. Bessel函数
   - 3.1 二阶线性常微分方程的幂级数解法
     - 常系数二阶线性ODE的基解组
     - 变系数二阶线性ODE的幂级数解法
   - 3.2 Bessel函数
   - 3.3 多个自变量分离变量法举例
4. 积分变换法
5. Green函数法
6. 特征线法
7. Legendre多项式

线性ODE的解法

$$x^{<n>}+P_1x^{<n-1>}+P_2x^{<n-2>}+\cdots+P_{n-1}x^{<1>}+P_nx^{<0>}=F$$

线性常微分方程解法

- 一阶
  - 齐次 $x'+P(t)x=0$ 分离变量法
  - 非齐次 $x'+P(t)x=f(t)$ 常数变易法
- 二阶(高阶)
  - 常系数
    - 齐次
      - 特征方程法 $\lambda^2+p\lambda+q=0$
        - 互异实根 $\{e^{\lambda_1 t},e^{\lambda_2 t}\}$
        - 相同实根 $\{te^{\lambda t},e^{\lambda t}\}$
        - 共轭复根 $\{e^{\alpha t}\cos\beta t,e^{\alpha t}\sin\beta t\}$
      - 1 Liouville公式 基解1→基解2 $x_2(t)=x_1(t)\int\frac{1}{x_1^2(t)}e^{-\int p_1(t)dt}dt$
      - 摄动法 2
    - 非齐次
      - 通解=通+特
      - 求非齐特解
        - 待定函数法 $x''(t)+a_1x'(t)+a_0x(t)=\varphi(t)e^{\mu t}\sin\nu t$ $s=\mu\pm\nu i$
          $\bar{x}(t)=t^ke^{\mu t}\left[\Phi_1(t)\cos\nu t+\Phi_2(t)\sin\nu t\right]$
        - 观察法 $\ddot{x}+x=\sin 2t,\ \bar{x}=C\sin 2t$
  - 变系数
    - Euler方程 $t^n\frac{d^nx}{dt^n}+a_1t^{n-1}\frac{d^{n-1}x}{dt^{n-1}}+\cdots+a_{n-1}t\frac{dx}{dt}+a_nx=F(t)$, $t=e^s$ } 变量代换法
    - Bernoulli方程 $\frac{dy}{dx}+P(x)\cdot y=Q(x)\cdot y^{\alpha}$, $(\alpha\neq 0,1)$, $u=y^{-\alpha+1}$ } 变量代换法
    - Bessel方程 $x^2\frac{d^2y}{dx^2}+x\frac{dy}{dx}+(x^2-r^2)y=0$ } 幂级数法 3
    - Legendre方程 $(1-x^2)\frac{d^2y}{dx^2}-2x\frac{dy}{dx}+n(n+1)y=0$ } 幂级数法 3
- 高阶(可降阶)
  - $y^{(n)}=f(x)$ 无其他导数项, 直接积分法
  - $y''=f(x,y')$ 不显含y，令 $p=y',y''=\frac{dp}{dx}=f(x,p)$ } 变量代换法
  - $y''=f(y,y')$ 不显含x，令$p=y',y''=\frac{dp}{dx}=\frac{dp}{dy}\frac{dy}{dx}=p\frac{dp}{dy}$ } 变量代换法

3.1 二阶线性ODE幂级数解法

常系数二阶线性ODE的基解组

例1 用特征方程法求解下列齐次ODE

(1) $y''-3y'+2y=0$

解 二阶线性常系数ODE

特征方程 $\lambda^2-3\lambda+2=0$

特征根 $\lambda_1=1,\ \lambda_2=2$

互异实根,基解组 $\{e^x,\ e^{2x}\}$

通解 $y=C_1e^x+C_2e^{2x}$

(2) $y''+4y'+13y=0$

解 二阶线性常系数ODE

特征方程 $\lambda^2+4\lambda+13=0$

特征根 $\lambda_{1,2}=-2\pm 3i$

共轭复根,基解组 $\{e^{-2x}\cos 3x,\ e^{-2x}\sin 3x\}$

通解 $y=C_1e^{-2x}\cos 3x+C_2e^{-2x}\sin 3x$

$$x''(t)+px'(t)+qx(t)=0$$

$$\lambda^2+p\lambda+q=0$$

- 互异实根 $\{e^{\lambda_1 t},e^{\lambda_2 t}\}$
- 相同实根 $\{te^{\lambda t},e^{\lambda t}\}$
- 共轭复根 $\{e^{\alpha t}\cos\beta t,e^{\alpha t}\sin\beta t\}$

(3) $y''+4y'+4y=0$

解法1 特征方程法 二阶线性常系数ODE

特征方程 $\lambda^2+4\lambda+4=0$

特征根 $\lambda_{1,2}=-2$

相同实根,基解组 $\{e^{-2x},xe^{-2x}\}$

通解 $y=C_1e^{-2x}+C_2xe^{-2x}$

二阶齐次ODE求解→寻找2个线性无关基解

$$C_1f_1(x)+C_2f_2(x)=0$$

线性无关: 仅有C_1, C_2同时为0才成立

3.1 二阶线性ODE幂级数解法

常系数二阶线性ODE的基解组

(3) $y''+4y'+4y=0$

解法2 摄动法 设原方程常系数有微小变化

摄动方程 $y''+(4+\varepsilon)y'+(4+2\varepsilon)y=0$

特征方程 $\lambda^2+(4+\varepsilon)\lambda+(4+2\varepsilon)=0$

特征根 $\lambda_1=-2,\ \lambda_2=-2-\varepsilon$ 基解组$\{e^{-2x},e^{-2x-\varepsilon x}\}$

基解的线性组合仍然是解

$$y_\varepsilon=\frac{e^{-2x-\varepsilon x}-e^{-2x}}{-\varepsilon}=e^{-2x}\frac{e^{-\varepsilon x}-1}{-\varepsilon}$$

当 $\varepsilon\to0$时, 摄动方程趋近于原方程

$$y=\lim_{\varepsilon\to0}y_\varepsilon=\lim_{\varepsilon\to0}e^{-2x}\frac{e^{-\varepsilon x}-1}{-\varepsilon}$$

L'Hospital 法则
对ε求导

$$=\lim_{\varepsilon\to0}e^{-2x}\frac{-xe^{-\varepsilon x}}{-1}=xe^{-2x}$$

第1类Bessel函数→摄动法→第2类Bessel函数

摄动方程中为什么取 $4y\to(4+2\varepsilon)y$

摄动方程当 $\varepsilon\to0$时, 不仅两个方程要形似, 还需保持原方程性态→二阶ODE方程性态→取决于特征方程判别式

原方程 $y''+4y'+4y=0$

特征方程 $s^2+4s+4=0$

判别式 $\Delta=4^2-4\cdot4=0$ 相同实根

摄动方程 $y''+(4+\varepsilon)y'+(4+\alpha\varepsilon)y=0$

特征方程 $s^2+(4+\varepsilon)s+(4+\alpha\varepsilon)=0$

判别式
$$\Delta_\varepsilon=(4+\varepsilon)^2-4\cdot(4+\alpha\varepsilon)$$
$$=4^2+8\varepsilon+\varepsilon^2-4^2-4\alpha\varepsilon$$
$$=8\varepsilon+\varepsilon^2-4\alpha\varepsilon$$

欲使 $\lim_{\varepsilon\to0}\Delta_\varepsilon=\lim_{\varepsilon\to0}(8\varepsilon+\varepsilon^2-4\alpha\varepsilon)=0=\Delta$

需 $\alpha=2$

3.1 二阶线性ODE幂级数解法

$$x''(t)+p(t)x'(t)+q(t)x(t)=0$$
$$x_2(t)=x_1(t)\int\frac{1}{x_1^2(t)}e^{-\int p(t)dt}dt$$

Liouville公式

二阶常/变系数齐次ODE

$$x''(t)+p(t)x'(t)+q(t)x(t)=0$$

二阶系数为1

求解的关键是确定2个基解 $\{x_1(t),x_2(t)\}$

如何根据 基解1→基解2 ?

假设已知1个基解 $x_1(t)$

另一个基解 $x_2(t)$与 $x_1(t)$ 线性无关

$$\frac{x_2(t)}{x_1(t)}=\varphi(t)\neq C \qquad x_2(t)=\varphi(t)x_1(t)$$

各阶导数 $x_2'(t)=\varphi'(t)x_1(t)+\varphi(t)x_1'(t)$

$$\begin{aligned}x_2''(t)&=\varphi''(t)x_1(t)+\varphi'(t)x_1'(t)\\&\quad+\varphi'(t)x_1'(t)+\varphi(t)x_1''(t)\\&=\varphi''(t)x_1(t)+2\varphi'(t)x_1'(t)+\varphi(t)x_1''(t)\end{aligned}$$

代入原ODE

$$x_2''(t)+p(t)x_2'(t)+q(t)x_2(t)=0$$

$$\begin{aligned}&\varphi''(t)x_1(t)+2\varphi'(t)x_1'(t)+\varphi(t)x_1''(t)\\&+p(t)\left[\varphi'(t)x_1(t)+\varphi(t)x_1'(t)\right]\\&+q(t)\varphi(t)x_1(t)=0\end{aligned}$$

$$\begin{aligned}&\varphi(t)\underbrace{\left[x_1''(t)+p(t)x_1'(t)+q(t)x_1(t)\right]}_{=0}\\&+\varphi''(t)x_1(t)+\varphi'(t)\left[2x_1'(t)+p(t)x_1(t)\right]=0\end{aligned}$$

J.Liouville
1809-1882
French Math.

$$\varphi''(t)x_1(t)+\varphi'(t)\left[2x_1'(t)+p(t)x_1(t)\right]=0$$

分离变量法 $\dfrac{\varphi''(t)}{\varphi'(t)}=-2\dfrac{x_1'(t)}{x_1(t)}-p(t)$

$$\varphi'(t)=\frac{1}{x_1^2(t)}\int e^{-\int p(t)dt} \;\Rightarrow\; \varphi(t)=\int\frac{1}{x_1^2(t)}\int e^{-\int p(t)dt}dt$$

另一线性无关基解 $x_2(t)=x_1(t)\displaystyle\int\frac{1}{x_1^2(t)}e^{-\int p(t)dt}dt$

3.1 二阶线性ODE幂级数解法

$$x''(t)+p(t)x'(t)+q(t)x(t)=0$$
$$x_2(t)=x_1(t)\int\frac{1}{x_1^2(t)}e^{-\int p(t)dt}dt$$

常系数二阶线性ODE的基解组

(3) $y''+4y'+4y=0$

解法3 Liouville公式

二阶变系数ODE $y''(x)+p(x)y'(x)+q(x)y(x)=0$

本例已知一个基解为 $y_1(x)=e^{-2x}$

$$y_2(x)=y_1(x)\int\frac{1}{y_1^2(x)}e^{-\int p(x)dx}dx$$
$$=e^{-2x}\int\frac{1}{\left(e^{-2x}\right)^2}e^{-\int 4dx}dx$$
$$=e^{-2x}\int e^{4x}e^{-4x+C}dx$$
$$=e^{-2x}\int e^{C}dx$$
$$=e^{-2x}\left(e^{C}x+C_2\right)$$
$$=e^{C}xe^{-2x}+C_2e^{-2x}$$
$$=C_1xe^{-2x}+C_2e^{-2x}$$

$p(x)=4$

J.Liouville
1809-1882
French Math.

于是原方程另1个线性无关的基解为

$$y_2(x)=xe^{-2x}$$

3.1 二阶线性ODE幂级数解法

$$x''(t)+p(t)x'(t)+q(t)x(t)=0$$
$$x_2(t)=x_1(t)\int\frac{1}{x_1^2(t)}e^{-\int p(t)dt}dt$$

常系数二阶线性ODE的基解组

例2 用Liouville公式解Euler方程

$$t^2\frac{d^2x}{dt^2}+t\frac{dx}{dt}+a_nx=0\quad(a_n<0)$$

J.Liouville
1809-1882
French Math.

L.Euler
1707-1783
Swiss Math.
Physical Scientist

解 原方程变型为 $\dfrac{d^2x}{dt^2}+\dfrac{1}{t}\dfrac{dx}{dt}+\dfrac{a_n}{t^2}x=0$ 　基解 $\left\{t^{\sqrt{-a_n}},\ t^{-\sqrt{-a_n}}\right\}\quad(a_n<0)$

$$x_2(t)=x_1(t)\int\frac{1}{x_1^2(t)}e^{-\int p(t)dt}dt$$
$$=t^{\sqrt{-a_n}}\int\frac{1}{t^{2\sqrt{-a_n}}}e^{-\int\frac{1}{t}dt}dt=t^{\sqrt{-a_n}}\int\frac{1}{t^{2\sqrt{-a_n}}}e^{-\ln t+C_1}dt$$
$$=t^{\sqrt{-a_n}}\int\frac{1}{t^{2\sqrt{-a_n}}}\frac{C_1}{t}dt=t^{\sqrt{-a_n}}C_1\int t^{-(1+2\sqrt{-a_n})}dt$$
$$=t^{\sqrt{-a_n}}\left[C_1\frac{t^{-(1+2\sqrt{-a_n})+1}}{-(1+2\sqrt{-a_n})+1}+C_2\right]=t^{\sqrt{-a_n}}C_1\frac{1}{-2\sqrt{-a_n}}t^{-2\sqrt{-a_n}}+C_2t^{\sqrt{-a_n}}$$
$$=C_1t^{-\sqrt{-a_n}}+C_2t^{\sqrt{-a_n}}$$

吸收至C_1中

于是原方程另1个线性无关的基解为 $x_2(t)=t^{-\sqrt{-a_n}}$

1. 数学建模和基本原理介绍
2. 分离变量法
3. Bessel函数
   - 3.1 二阶线性常微分方程的幂级数解法
     - 常系数二阶线性ODE的基解组
     - 变系数二阶线性ODE的幂级数解法
   - 3.2 Bessel函数
   - 3.3 多个自变量分离变量法举例
4. 积分变换法
5. Green函数法
6. 特征线法
7. Legendre多项式

3.1 二阶线性ODE幂级数解法

$$y''(x)+p(x)y'(x)+q(x)y(x)=0$$

变系数二阶线性ODE的幂级数解法

1. 基解组仍为2个线性无关的解→基解一般求解困难
2. 广义待定函数法→方程两边用幂级数展开→比较/确定系数→得级数形式解
3. 关键: 幂级数的选择/收敛性

幂级数解存在定理1

对二阶ODE $y''(x)+p(x)y'(x)+q(x)y(x)=0$，若 $p(x)$, $q(x)$ 在的邻域 $\{x\in R|\ |x-x_0|<\delta\}$ 内解析, 即可展开为Talyor级数, 则原方程在该邻域内存在幂级数解 $y(x)=\sum_{k=0}^{\infty}a_k(x-x_0)^k$ 其中系数 a_k 可由待定系数法求出

Taylor级数

$$f(x)=f(x_0)+f'(x_0)(x-x_0)+\frac{f''(x_0)}{2!}(x-x_0)^2+\frac{f^{<3>}(x_0)}{3!}(x-x_0)^3+\cdots+\frac{f^{<n>}(x_0)}{n!}(x-x_0)^n+\cdots$$

推论 若 $p(x)$, $q(x)$ 在 $x=0$ 可展开, 收敛半径无穷大→幂级数解可定义为

$$y(x)=\sum_{k=0}^{\infty}a_kx^k$$

序号/幂次一致
k从0开始

3.1 二阶线性ODE幂级数解法

$$y''(x)+p(x)y'(x)+q(x)y(x)=0$$

例1 幂级数法同样适于常系数 $y'(x)-ay(x)=0$ 分离变量法 $y(x)=Ce^{ax}=y(0)e^{ax}$

初始条件

解 幂级数法 变系数 $p(x)=-a,\ q(x)=0$ 在实数域解析→有解

幂级数展开 多项式求导 对应幂次系数相等 得到递推式

$$y(x)=\sum_{k=0}^{\infty}b_kx^k$$

$y(x)=$
$+b_kx^k$
$+b_{k-1}x^{k-1}$
$+b_{k-2}x^{k-2}$
$+\dots$
$+b_3x^3$
$+b_2x^2$
$+b_1x^1$
$+b_0x^0$

$b_0=x(0)$是初始条件/是已知的

$y'(x)=$
$+k\cdot b_kx^{k-1}$
$+(k-1)\cdot b_{k-1}x^{k-2}$
$+(k-2)\cdot b_{k-2}x^{k-3}$
$+\dots$
$+3\cdot b_3x^2$
$+2\cdot b_2x^1$
$+1\cdot b_1x^0$
$+0$

$ay(x)=y'(x)$

$a\cdot b_{k-1}x^{k-1}=k\cdot b_kx^{k-1}$
$a\cdot b_{k-2}x^{k-2}=(k-1)\cdot b_{k-1}x^{k-2}$
$a\cdot b_{k-3}x^{k-3}=(k-2)\cdot b_{k-2}x^{k-3}$
$\dots$
$a\cdot b_2x^2=3\cdot b_3x^2$
$a\cdot b_1x^1=2\cdot b_2x^1$
$a\cdot b_0x^0=1\cdot b_1x^0$

$$b_k=\frac{a}{k}\cdot b_{k-1}$$
$$b_{k-1}=\frac{a}{(k-1)}\cdot b_{k-2}$$
$$b_{k-2}=\frac{a}{(k-2)}b_{k-3}$$
$$\dots$$
$$b_3=\frac{a}{3}\cdot b_2$$
$$b_2=\frac{a}{2}\cdot b_1$$
$$b_1=\frac{a}{1}\cdot b_0$$

逐级回代

$$b_k=\frac{a}{k}\cdot\frac{a}{(k-1)}\cdot\frac{a}{(k-2)}\cdots\frac{a}{3}\cdot\frac{a}{2}\cdot\frac{a}{1}\cdot b_0=\frac{a^k}{k!}\cdot b_0$$

$$y(x)=y(0)e^{ax}=y(0)\sum_{k=0}^{\infty}\frac{(ax)^k}{k!}=b_0\sum_{k=0}^{\infty}\frac{a^k}{k!}\cdot x^k=\sum_{k=0}^{\infty}b_kx^k$$

$$e^x=\sum_{k=0}^{\infty}\frac{x^k}{k!}=1+x+\frac{x^2}{2!}+\cdots+\frac{x^k}{k!}+\cdots$$

$$y(0)=b_0$$

3.1 二阶线性ODE幂级数解法

$$y''(x)+p(x)y'(x)+q(x)y(x)=0$$

例1 幂级数法同样适于常系数 $x'-ax=0$

分离变量法 $x(t)=Ce^{at}=x(0)e^{at}$ 初始条件

幂级数法 $x(t)=\sum_{k=0}^{\infty} b_k t^k = x(0)\sum_{k=0}^{\infty}\frac{a^k}{k!}\cdot t^k$

一阶齐次ODE幂级数解法的意义

线性高阶非齐次ODE $y^{(n)}+a_{n-1}y^{(n-1)}+\cdots+a_1\dot{y}+a_0y=u$

变量代换

$$\begin{cases} x_1=y \\ x_2=\dot{y} \\ \vdots \\ x_n=y^{(n-1)} \end{cases}$$

改造为线性ODE组

$$\begin{cases} \dot{x}_1=x_2 \\ \dot{x}_2=x_3 \\ \vdots \\ \dot{x}_{n-1}=x_n \\ \dot{x}_n=-a_0x_1-\cdots-a_{n-1}x_n+u \end{cases}$$

写成矩阵形式

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_{n-1} \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \boldsymbol{A} & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} u$$

高阶ODE求解统一化/程序化

齐次部分 $\boldsymbol{x}=\sum_{k=0}^{\infty}\boldsymbol{b}_k t^k$

$$\boldsymbol{x}(t)=\sum_{k=0}^{\infty}\frac{\boldsymbol{A}^k}{k!}\boldsymbol{x}(0)t^k=\boldsymbol{x}(0)\sum_{k=0}^{\infty}\frac{\boldsymbol{A}^k}{k!}t^k=e^{\boldsymbol{A}t}\boldsymbol{x}(0)$$

$y=x_1$

$$x(t)=\sum_{k=0}^{\infty} b_k t^k = x(0)\sum_{k=0}^{\infty}\frac{a^k}{k!}\cdot t^k$$

$$e^x=\sum_{k=0}^{\infty}\frac{x^k}{k!}=1+x+\frac{x^2}{2!}+\cdots+\frac{x^k}{k!}+\cdots$$

$y''(x)+p(x)y'(x)+q(x)y(x)=0$

变系数二阶线性ODE的幂级数解法

例2 用幂级数法解变系数ODE $y''+xy'+y=0$

解 变系数 $p(x)=x,\ q(x)=1$ 在实数域解析

存在幂级数解

$$y(x)=\sum_{k=0}^{\infty}a_kx^k=a_0+a_1x+a_2x^2+\cdots+a_kx^k+\cdots$$

$$y'(x)=a_1+2a_2x^1+\cdots+a_kkx^{k-1}+\cdots=\sum_{k=1}^{\infty}a_kkx^{k-1}$$

起点已变

$$y''(x)=2a_2+\cdots+a_kk(k-1)x^{k-2}+\cdots=\sum_{k=2}^{\infty}a_kk(k-1)x^{k-2}$$

起点已变

代入原方程

$$\sum_{k=2}^{\infty}a_kk(k-1)x^{k-2}+x\sum_{k=1}^{\infty}a_kkx^{k-1}+\sum_{k=0}^{\infty}a_kx^k=0$$

$$\sum_{k=0}^{\infty}a_{k+2}(k+2)(k+2-1)x^k+\sum_{k=0}^{\infty}a_kkx^k+\sum_{k=0}^{\infty}a_kx^k=0$$

变量代换, 从0累加

首项为0, 累加直接改自0起

序号/幂次同时调整一致, 同次幂系数相等

$$a_{k+2}(k+2)(k+1)+a_k(k+1)=0,\quad (k\ge 0)$$

得到递推式 $a_{k+2}=-\dfrac{1}{k+2}a_k,\quad (k\ge 0)$

系数间隔传递, 分奇/偶讨论

$$a_{2k}=\frac{(-1)^k}{(2k)!!}a_0,\quad a_{2k+1}=\frac{(-1)^k}{(2k+1)!!}a_1,\quad (k\ge 0)$$

半阶乘

$$(2k)!!=2k\cdot(2k-2)\cdot(2k-4)\cdots 2$$

$$(2k+1)!!=(2k+1)\cdot(2k-1)\cdot(2k-3)\cdots 1$$

半阶乘 双叹号不是阶乘两次

$(2k)!!$ 不大于 $(2k)$ 的偶数乘积

$(2k+1)!!$ 不大于 $(2k+1)$ 的奇数乘积

例如 $(6)!!=2\cdot4\cdot6=48$

$(7)!!=1\cdot3\cdot5\cdot7=105$

$(2k+1)!!(2k)!!=(2k+1)!$

原方程的解

$$y(x)=a_0\sum_{k=0}^{\infty}\frac{(-1)^k}{(2k)!!}x^{2k}+a_1\sum_{k=0}^{\infty}\frac{(-1)^k}{(2k+1)!!}x^{2k+1}$$

本例幂级数方程序号/幂次可同时调整一致

$y''(x)+p(x)y'(x)+q(x)y(x)=0$

变系数二阶线性ODE的幂级数解法

例3 用幂级数法解ODE $y''+x^2y'=0$

解 易知 a_0 是原方程的解，找另1个线性无关解

因 $p(x)=x^2$ 在实数域解析→存在幂级数解

$$y(x)=\sum_{n=0}^{\infty}a_nx^n,\ (a_0\neq 0)$$

$$y'(x)=\sum_{n=1}^{\infty}na_nx^{n-1},\quad y''(x)=\sum_{n=2}^{\infty}n(n-1)a_nx^{n-2}$$

$$\sum_{n=2}^{\infty}n(n-1)a_nx^{n-2}+x^2\sum_{n=1}^{\infty}na_nx^{n-1}=0$$

$$\sum_{n=2}^{\infty}n(n-1)a_nx^{n-2}+\sum_{n=1}^{\infty}na_nx^{n+1}=0$$

幂次/起点难以调整一致

列表

$n=1$		a_1x^2
$n=2$	$2\cdot1\cdot a_2x^0$	$2a_2x^3$
$n=3$	$3\cdot2\cdot a_3x^1$	$3a_3x^4$
$n=4$	$4\cdot3\cdot a_4x^2$	$4a_4x^5$
$n=5$	$5\cdot4\cdot a_5x^3$	$5a_5x^6$
$n=6$	$6\cdot5\cdot a_6x^4$	
$n=7$		

同次幂系数为0

x^0	$2\cdot1\cdot a_2=0$
x^1	$3\cdot2\cdot a_3=0$
x^2	$4\cdot3\cdot a_4+a_1=0$
x^3	$5\cdot4\cdot a_5+2a_2=0$
x^4	$6\cdot5\cdot a_6+3a_3=0$
x^5	$7\cdot6\cdot a_7+4a_4=0$

解得

$$a_2=0$$
$$a_3=0$$
$$a_4=-\frac{1}{4\cdot3}a_1=\frac{(-1)^1}{3^1\cdot(3\cdot1+1)\cdot1!}a_1$$
$$a_5=-\frac{2}{5\cdot4}a_2=0$$
$$a_6=-\frac{3}{6\cdot5}a_3=0$$
$$a_7=(-1)^2\frac{4}{7\cdot6}\frac{1}{4\cdot3}a_1=(-1)^2\frac{1}{3^2\cdot(3\cdot2+1)\cdot2!}a_1$$
$$\cdots$$

只有能递推至a_1的系数不为0

系数隔3传递 $\cdots\to a_7\to a_4\to a_1$

用连续下标 k 表示

$$a_{3k+1}=(-1)^k\frac{1}{3^k\cdot(3\cdot k+1)\cdot k!}a_1$$

原ODE解为

a_1是 x^1 的系数

$$y(x)=a_0+a_1\left[x+\sum_{k=1}^{\infty}\frac{(-1)^k}{3^k\cdot(3k+1)\cdot k!}x^{3k+1}\right]$$

1. 本例幂级数方程序号/幂次难以同时调整一致→列表计算系数
2. 注意验证常数项是否为解

$$y''(x)+p(x)y'(x)+q(x)y(x)=0$$

变系数二阶线性ODE的幂级数解法

$$x''(t)+p(t)x'(t)+q(t)x(t)=0$$

$$x_2(t)=x_1(t)\int\frac{1}{x_1^2(t)}e^{-\int p(t)dt}dt$$

例3 解ODE $y''+x^2y'=0$

J.Liouville
1809-1882
French Math.

解法2 Liouville公式 易知 a_0 是原方程的解，找另1个线性无关解

$$x_2(t)=x_1(t)\int\frac{1}{x_1^2(t)}e^{-\int p(t)dt}dt=a_0\int\frac{1}{a_0^2}e^{-\int x^2dx}dx$$

$$=C_1\int e^{-\int x^2dx}dx$$

$$=C_1\int e^{-\frac{x^3}{3}}dx$$

$$e^x=\sum_{k=0}^{\infty}\frac{x^k}{k!}=1+x+\frac{x^2}{2!}+\cdots+\frac{x^k}{k!}+\cdots$$

$$=C_1\int\left[\sum_{k=0}^{\infty}\frac{1}{k!}\left(-\frac{x^3}{3}\right)^k\right]dx$$

累加/积分交换次序

$$=C_1\sum_{k=0}^{\infty}\int\frac{(-1)^k}{3^k k!}x^{3k}dx$$

$$=C_2+C_1\sum_{k=0}^{\infty}\frac{(-1)^k x^{3k+1}}{3^k(3k+1)k!}$$

另1个基解

$$y(x)=a_0+a_1\left[x+\sum_{k=1}^{\infty}\frac{(-1)^k}{3^k\cdot(3k+1)\cdot k!}x^{3k+1}\right]$$

与解法1相同

3.1 二阶线性ODE幂级数解法

$y''(x)+p(x)y'(x)+q(x)y(x)=0$

变系数二阶线性ODE的幂级数解法 小结

幂级数法基本步骤

1. 判定解存在→给出幂级数解的基本形式→序号不一定从0开始
2. 幂级数解求导/展开$p(x), q(x)$→代入原方程→根据需要调整序号起点/幂次
3. 按照 x 同次幂合并同类项→得出幂级数解系数的递推公式
4. 系数向 a_0, a_1化简→ 得基解组/原方程通解→ a_0, a_1是幂级数前两项系数/通解常数
5. 注意验证常数项 a_0 是否为解

幂级数解存在定理2 拓展 求解Bessel方程的理论支撑

对二阶ODE $x''(t)+p(x)x'(t)+q(x)x(t)=0$, 若 $(x-x_0)p(x),\ (x-x_0)^2q(x)$ 在 x_0 的邻域$\{x\in R|\ |x-x_0|<\delta\}$ 内解析, 即 x_0 最多为 $p(x), q(x)$ 的一阶和二阶极点, 则原方程在该去心邻域 $\{x\in R|\ 0<|x-x_0|<\delta\}$ 内存在幂级数解

$$y(x)=(x-x_0)^{\rho}\sum_{k=0}^{\infty}a_k(x-x_0)^k=\sum_{k=0}^{\infty}a_k(x-x_0)^{k+\rho}$$

相当于调整了序号起点

其中常数 ρ, $a_k\ (a_0\neq 0)$ 可由待定系数法求出

$$p(x)=\frac{A(x)}{x-x_0}$$

一阶极点

$$q(x)=\frac{B(x)}{(x-x_0)^2}$$

二阶极点

$$y(x)=\sum_{k=0}^{\infty}a_kx^k$$

序号/幂次可独立调整
序号不一定从0开始

$$y''(x)+p(x)y'(x)+q(x)y(x)=0$$

变系数二阶线性ODE的幂级数解法

例4 用幂级数法解Euler方程 $x^2y''+xy'-\alpha^2y=0,\quad (\alpha>0)$

解 方程变型 $y''+\frac{1}{x}y'-\frac{\alpha^2}{x^2}y=0$

因存在 $x\cdot p(x)=x\cdot\frac{1}{x}=1,\quad -x^2\cdot q(x)=x^2\cdot\frac{\alpha^2}{x^2}=-1$ 在实数域内解析，原方程有幂级数解存在

设解为 $y(x)=x^\rho\sum_{n=0}^{\infty}a_nx^n=\sum_{n=0}^{\infty}a_nx^{n+\rho},\ (a_0\neq 0)$ → 各阶导数

$$y'(x)=\sum_{n=0}^{\infty}(n+\rho)a_nx^{n+\rho-1}$$

$$y''(x)=\sum_{n=0}^{\infty}(n+\rho)(n+\rho-1)a_nx^{n+\rho-2}$$

代入原方程 $y''+\frac{1}{x}y'-\frac{\alpha^2}{x^2}y=0$

$$\sum_{n=0}^{\infty}(n+\rho)(n+\rho-1)a_nx^{n+\rho-2}+\frac{1}{x}\sum_{n=0}^{\infty}(n+\rho)a_nx^{n+\rho-1}-\frac{\alpha^2}{x^2}\sum_{n=0}^{\infty}a_nx^{n+\rho}=0$$

$$\sum_{n=0}^{\infty}(n+\rho)(n+\rho-1)a_nx^{n+\rho-2}+\sum_{n=0}^{\infty}(n+\rho)a_nx^{n+\rho-2}-\alpha^2\sum_{n=0}^{\infty}a_nx^{n+\rho-2}=0$$

$$(n+\rho)(n+\rho-1)+(n+\rho)-\alpha^2=0$$

$$(n+\rho)^2=\alpha^2$$

$$n+\rho=\pm\alpha$$

←

$$y(x)=\sum_{n=0}^{\infty}a_nx^{n+\rho}=\sum_{n=0}^{\infty}a_{n1}x^{+\alpha}+\sum_{n=0}^{\infty}a_{n2}x^{-\alpha}$$

$$=C_1x^{+\alpha}+C_2x^{-\alpha}$$

3.1 二阶线性ODE幂级数解法

$$y''(x)+p(x)y'(x)+q(x)y(x)=0$$

变系数二阶线性ODE的幂级数解法

例5 用幂级数法求方程 $y''-2xy'-4y=0$ 满足初始条件 $y(0)=0, y'(0)=1$ 的解

解 因 $p(x)=-2x,\quad q(x)=-4$ 在实数域内解析, 原方程有幂级数解存在

设解为 $y(x)=\sum_{n=0}^{\infty}a_nx^n=a_0+a_1x^1+a_2x^2+\cdots+a_nx^n+\cdots$

由初始条件 $y(0)=a_0=0$ $\quad a_1=1$

$$y'(0)=a_1+a_2\cdot 0^2+\cdots+a_n\cdot 0^n+\cdots=1$$

各阶导数

$$y'(x)=\sum_{n=1}^{\infty}na_nx^{n-1}$$

$$y''(x)=\sum_{n=2}^{\infty}n(n-1)a_nx^{n-2}$$

代入原方程

$$\sum_{n=2}^{\infty}n(n-1)a_nx^{n-2}-2x\sum_{n=1}^{\infty}na_nx^{n-1}-4\sum_{n=0}^{\infty}a_nx^n=0$$

$$\sum_{n=2}^{\infty}n(n-1)a_nx^{n-2}-2\sum_{n=1}^{\infty}na_nx^n-4\sum_{n=0}^{\infty}a_nx^n=0$$

变量代换 调整序号

$$\sum_{n=0}^{\infty}(n+2)(n+2-1)a_{n+2}x^n-\sum_{n=0}^{\infty}2na_nx^n-\sum_{n=0}^{\infty}4a_nx^n=0$$

$$(n+2)(n+1)a_{n+2}-2(n+2)a_n=0$$

$$a_{n+2}=\frac{2}{n+1}a_n$$

偶数项系均为0 ← 仅需考虑奇数项 ← 换成连续下标 k

$$a_{2k+1}=\frac{2}{2k}\cdot a_{2k-1}=\frac{2}{2k}\cdot\frac{2}{2(k-1)}a_{2k-3}=\frac{2}{2k}\cdot\frac{2}{2(k-1)}\cdots\frac{2}{2}a_1=\frac{1}{k!}1$$

原方程的解

$$y(x)=\sum_{k=0}^{\infty}\frac{a_1}{k!}x^{2k+1}=1\cdot x^1+\frac{x^3}{2!}+\frac{x^5}{3!}+\cdots+\frac{x^{2k+1}}{k!}+\cdots=x\left[1+\frac{1}{2!}\cdot x^2+\frac{1}{3!}\cdot\left(x^2\right)^2+\cdots+\frac{1}{k!}\cdot\left(x^2\right)^k+\cdots\right]=xe^{x^2}$$

1. 数学建模和基本原理介绍
2. 分离变量法
3. Bessel函数
    - 3.1 二阶线性常微分方程的幂级数解法
    - 3.2 Bessel函数
        - Gamma函数
        - Bessel方程和Bessel函数
        - Bessel函数的性质
        - Bessel方程的特征值问题
        - 圆域上Laplace算子的特征值问题
    - 3.3 多个自变量分离变量法举例
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$$\Gamma(\alpha)=\int_0^{\infty}x^{\alpha-1}e^{-x}dx$$

Gamma函数

当 $\alpha>0$ 时, 广义积分 $\int_0^{\infty}x^{\alpha-1}e^{-x}dx$ 收敛

→可定义以 α 为自变量的函数→Γ函数

→难以用初等函数表示→超越函数

自变量是 α

$$\Gamma(\alpha)=\int_0^{\infty}x^{\alpha-1}e^{-x}dx=\int_0^{\infty}\frac{x^{\alpha}}{xe^{x}}dx,\quad(\alpha>0)$$

如何研究超越函数

1. 紧抓定义式→可否利用递推拓展定义域?
2. 特殊点是否有简单表达式?
3. 是否有递推式?
4. 和差积商初等计算规则
5. 微/积分计算规则

Γ函数的基本性质

(1) 几个基本值 $\Gamma(1)=1,\ \Gamma\left(\frac{1}{2}\right)=\sqrt{\pi}$

证 1

$$\Gamma(1)=\int_0^{\infty}x^{1-1}e^{-x}dx=\int_0^{\infty}e^{-x}dx=-e^{-x}\Big|_0^{\infty}=1$$

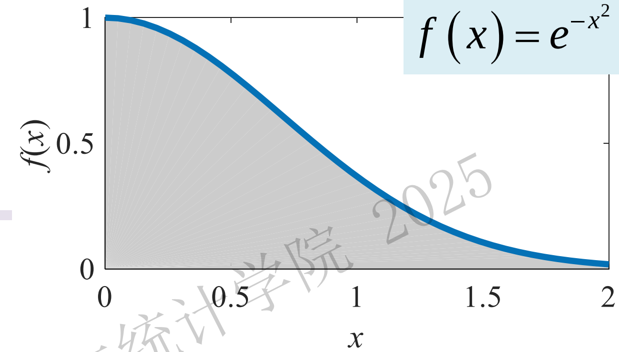
得证

3.2 Bessel函数

$$\Gamma(\alpha)=\int_0^{\infty}x^{\alpha-1}e^{-x}dx$$

Gamma函数

(1) 几个基本值 $\Gamma(1)=1$ $\Gamma\left(\frac{1}{2}\right)=\sqrt{\pi}$



证 2 先证明预备等式 $I=\int_0^{\infty}e^{-x^2}dx=\frac{\sqrt{\pi}}{2}$

$I=\int_0^{\infty}e^{-x^2}dx$ 不易积分→先积分其平方

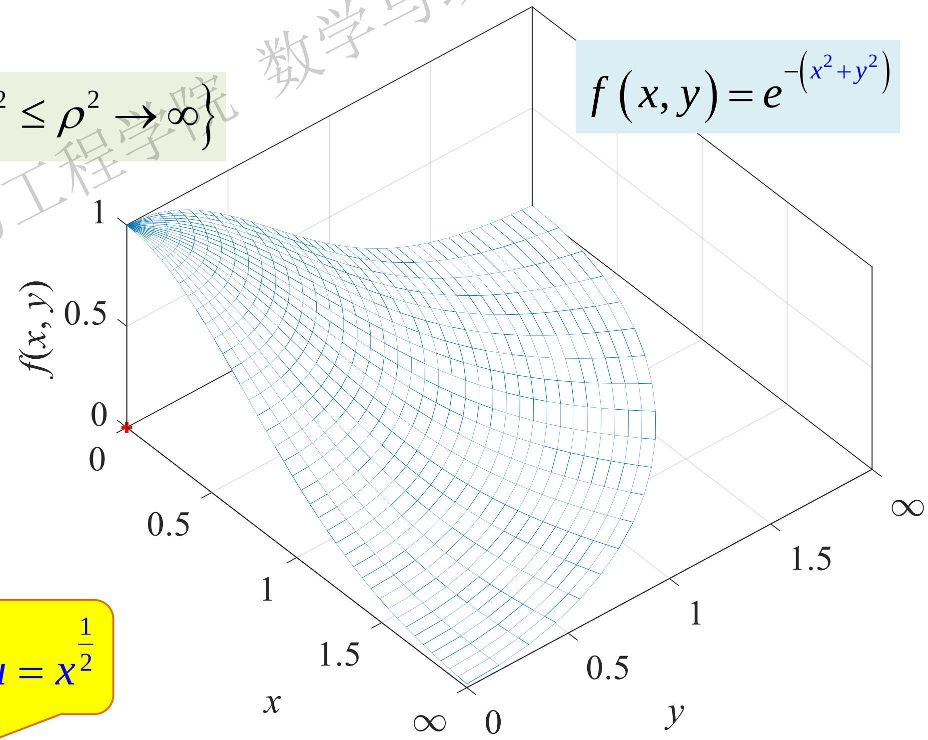
定义域为半径∞扇域 $\left\{(x,y)\middle|\,x\ge 0,y\ge 0,x^2+y^2\le\rho^2\to\infty\right\}$

$$I^2=\int_0^{\infty}e^{-x^2}dx\cdot\int_0^{\infty}e^{-x^2}dx=\int_0^{\infty}e^{-x^2}dx\cdot\int_0^{\infty}e^{-y^2}dy$$

$$=\int_0^{\infty}\int_0^{\infty}e^{-\left(x^2+y^2\right)}dxdy=\lim_{\rho\to\infty}\int_0^{\frac{\pi}{2}}d\theta\int_0^{\infty}e^{-\rho^2}\rho d\rho$$

扇域

$$=\frac{\pi}{4}\lim_{\rho\to\infty}\left(-e^{-\rho^2}\right)\Big|_0^{\rho}=\frac{\pi}{4}$$



即 $I^2=\frac{\pi}{4}$ → $I=\frac{\sqrt{\pi}}{2}$

变量代换 $u=x^{\frac{1}{2}}$

$$\Gamma\left(\frac{1}{2}\right)=\int_0^{\infty}x^{\frac{1}{2}-1}e^{-x}dx=\int_0^{\infty}x^{-\frac{1}{2}}e^{-x}dx=2\int_0^{\infty}e^{-x}d\left(x^{\frac{1}{2}}\right)=2\int_0^{\infty}e^{-u^2}du=2\cdot I=2\cdot\frac{\sqrt{\pi}}{2}=\sqrt{\pi}$$

$$\Gamma(\alpha)=\int_0^{\infty} x^{\alpha-1}e^{-x}dx$$

Gamma函数

(2) 递推公式 $\Gamma(\alpha+1)=\alpha\cdot\Gamma(\alpha),\quad(\alpha>0)$

分部积分

证 $$\Gamma(\alpha+1)=\int_0^{\infty} x^{\alpha+1-1}e^{-x}dx=\int_0^{\infty} x^{\alpha}e^{-x}dx=\int_0^{\infty} x^{\alpha}d\left(-e^{-x}\right)$$

$$=\underbrace{x^{\alpha}\left(-e^{-x}\right)\Big|_0^{\infty}}_{=0}-\alpha\int_0^{\infty} x^{\alpha-1}\left(-e^{-x}\right)dx=0+\alpha\int_0^{\infty} x^{\alpha-1}e^{-x}dx=\alpha\cdot\Gamma(\alpha)$$

推论 若α为正整数n $\quad \Gamma(n+1)=n!,\quad \Gamma\left(n+\frac{1}{2}\right)=\frac{(2n-1)!!}{2^n}\sqrt{\pi}$

证 $$\Gamma(n+1)=n\cdot\Gamma(n)$$
$$=n\cdot(n-1)\cdot\Gamma(n-1)$$
$$=n\cdot(n-1)\cdot(n-2)\cdot\Gamma(n-2)$$
$$=\cdots$$
$$=n\cdot(n-1)\cdot(n-2)\cdots\cdots 1\cdot\underbrace{\Gamma(1)}_{=1}$$
$$=n!$$

推论 $\Gamma(2)=\Gamma(1+1)=1\cdot\Gamma(1)=1$

证 $$\Gamma\left(n+\frac{1}{2}\right)=\Gamma\left(n-\frac{1}{2}+1\right)=\left(n-\frac{1}{2}\right)\Gamma\left(n-\frac{1}{2}\right)$$
$$=\frac{2n-1}{2}\Gamma\left(n-\frac{1}{2}\right)=\frac{2n-1}{2}\cdot\frac{2n-3}{2}\Gamma\left(n-\frac{3}{2}\right)$$
$$=\frac{2n-1}{2}\cdot\frac{2n-3}{2}\cdot\frac{2n-5}{2}\Gamma\left(n-\frac{5}{2}\right)$$

奇数

$$=\frac{2n-1}{2}\cdot\frac{2n-3}{2}\cdots\cdots\frac{2n-(2n-1)}{2}\Gamma\left(\frac{1}{2}\right)$$

半阶乘

$$=\frac{(2n-1)!!}{2^n}\Gamma\left(\frac{1}{2}\right)=\frac{(2n-1)!!}{2^n}\sqrt{\pi}$$

1. Γ函数是阶乘由整数→实数域内的推广
2. 两个推论为定积分的求解拓展了思路

$\Gamma(\alpha)=\int_0^{\infty} x^{\alpha-1}e^{-x}dx$

Gamma函数 原始定义 当 $\alpha>0$ 时, 广义积分 $\int_0^{\infty} x^{\alpha-1}e^{-x}dx$ 收敛→定义为 $\Gamma(\alpha)$ 函数

(3) 负实数域延拓

❶ 当 $\alpha>0$ 时 $\Gamma(\alpha)$ 可按原广义积分定义计算

$$\Gamma(\alpha)=\int_0^{\infty} x^{\alpha-1}e^{-x}dx$$

$$\Gamma(\alpha+1)=\alpha\cdot\Gamma(\alpha) \quad \Rightarrow \quad \Gamma(\alpha)=\frac{\Gamma(\alpha+1)}{\alpha}$$

❷ 当 $-1<\alpha<0$ 时 $0<\alpha+1<1$, $\Gamma(\alpha+1)$ 可按原广义积分定义计算→递推式变型计算 $\Gamma(\alpha)$

例

$\Gamma\left(\frac{1}{2}\right)=\sqrt{\pi}$

$$\Gamma(-0.5)=\frac{\Gamma(-0.5+1)}{-0.5}=\frac{\Gamma(0.5)}{-0.5}=-2\sqrt{\pi}$$

❸ 当 $-2<\alpha<-1$ 时, $-1<\alpha+1<0$, $\Gamma(\alpha+1)$ 已按上述递推定义→再次利用递推式计算 $\Gamma(\alpha)$

例

$$\Gamma(-1.5)=\frac{\Gamma(-1.5+1)}{-1.5}=\frac{\Gamma(-0.5)}{-1.5}=\frac{4\sqrt{\pi}}{3}$$

❹ 如此类推→可将定义域推广至除0/负整数外的所有实数域

❺ 规定0/负整数处 $\Gamma(\alpha)=\infty \quad (\alpha=0,Z^-)$

不同定义域下计算汇总

$$\Gamma(\alpha)=\begin{cases}\int_0^{\infty} x^{\alpha-1}e^{-x}dx=\int_0^{\infty}\frac{x^{\alpha}}{xe^{x}}dx, & (\alpha>0)\\ \frac{\Gamma(\alpha+1)}{\alpha}, & (\alpha<0,\alpha\neq Z^-)\\ \infty, & (\alpha=0,Z^-)\end{cases}$$

$$\Gamma(\alpha)=\int_0^{\infty}x^{\alpha-1}e^{-x}dx$$

Gamma函数　图线

$$\Gamma(\alpha)=\begin{cases}\int_0^{\infty}x^{\alpha-1}e^{-x}dx=\int_0^{\infty}\dfrac{x^{\alpha}}{xe^{x}}dx, & (\alpha>0)\\ \dfrac{\Gamma(\alpha+1)}{\alpha}, & (\alpha<0,\alpha\neq Z^-)\\ \infty, & (\alpha=0,Z^-)\end{cases}$$

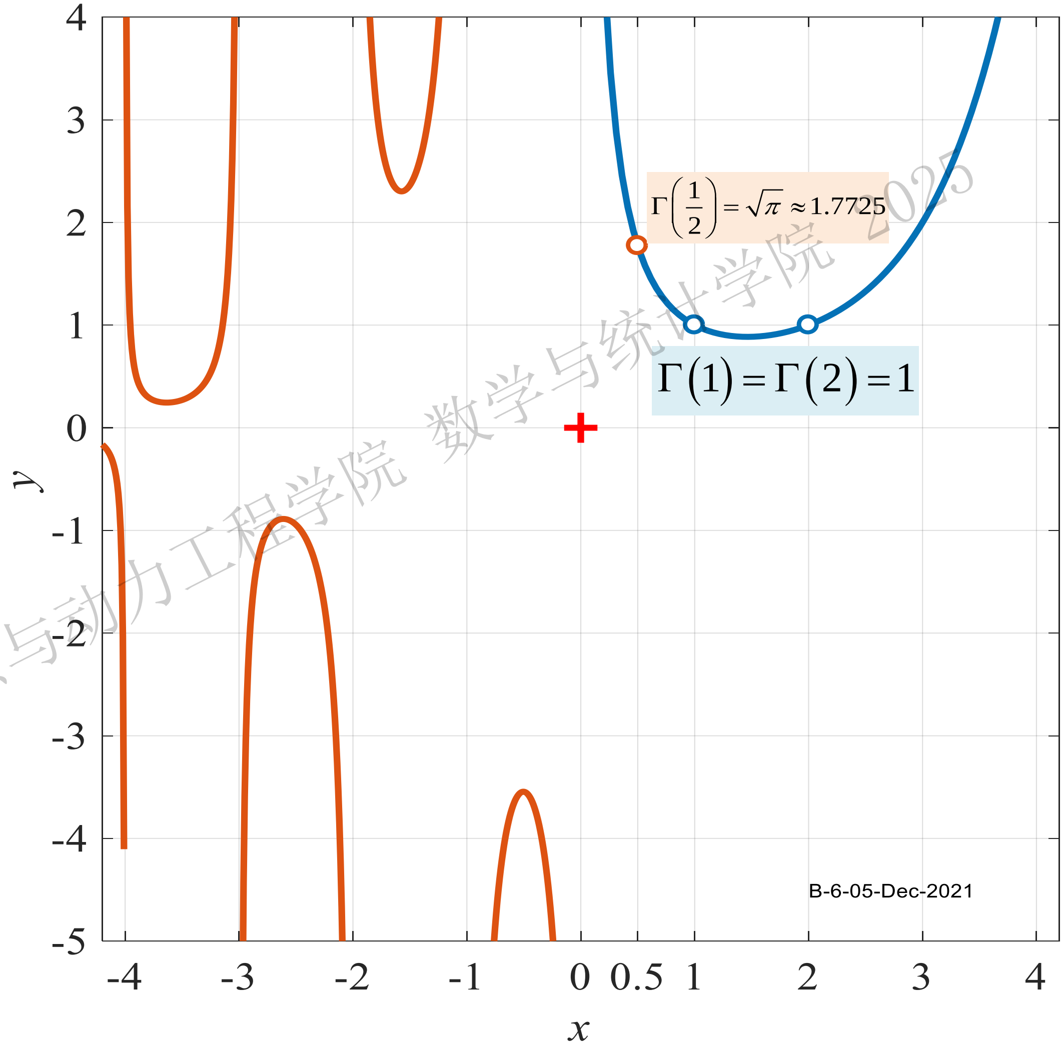
推论 ❶ $\Gamma(\alpha)\neq 0$

❷ $\Gamma(\alpha+1)=\alpha\cdot\Gamma(\alpha)$　　$\dfrac{1}{\alpha}=\dfrac{\Gamma(\alpha)}{\Gamma(\alpha+1)}$

超越函数之比为实数

$$\underbrace{2=\frac{1}{1/2}=\frac{\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{1}{2}+1\right)}=\frac{\sqrt{\pi}}{\sqrt{\pi}/2}=2}$$

Bessel函数化简发挥重要作用



Gamma函数 小结

Γ函数常用计算

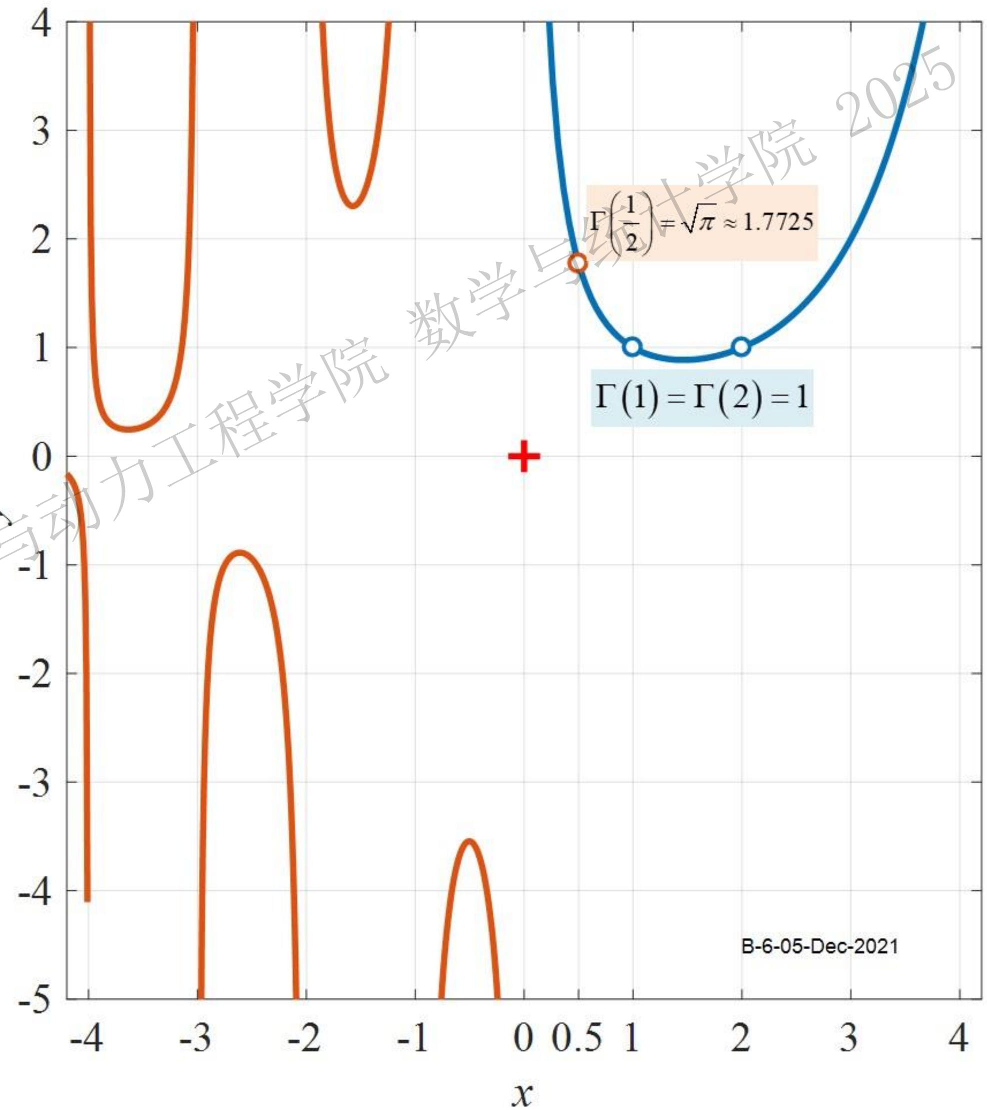
$$\Gamma(\alpha)=\begin{cases}\int_0^{\infty} x^{\alpha-1}e^{-x}dx, & (\alpha>0)\\ \dfrac{\Gamma(\alpha+1)}{\alpha}, & (\alpha<0,\alpha\neq Z^-)\\ \infty, & (\alpha=0,Z^-)\end{cases}$$

$$\Gamma(\alpha)\neq 0,\ \Gamma(2)=\Gamma(1)=1,\ \Gamma\left(\frac{1}{2}\right)=\sqrt{\pi}$$

$$\Gamma(\alpha+1)=\alpha\cdot\Gamma(\alpha),\ \frac{1}{\alpha}=\frac{\Gamma(\alpha)}{\Gamma(\alpha+1)}$$

$$\Gamma(n+1)=n!$$

$$\Gamma\left(n+\frac{1}{2}\right)=\frac{(2n-1)!!}{2^n}\sqrt{\pi}$$



$$\Gamma(\alpha)=\int_0^{\infty}x^{\alpha-1}e^{-x}dx$$

Gamma函数

$$\Gamma(\alpha)=\begin{cases}\int_0^{\infty}x^{\alpha-1}e^{-x}dx, & (\alpha>0)\\ \dfrac{\Gamma(\alpha+1)}{\alpha}, & (\alpha<0,\alpha\neq Z^-)\\ \infty, & (\alpha=0,Z^-)\end{cases}$$

$$\Gamma(\alpha+1)=\alpha\cdot\Gamma(\alpha)$$

$$\Gamma(n+1)=n!,\ \Gamma(2)=\Gamma(1)=1$$

$$\Gamma\left(n+\frac{1}{2}\right)=\frac{(2n-1)!!}{2^n}\sqrt{\pi},\ \Gamma\left(\frac{1}{2}\right)=\sqrt{\pi}$$

例1 计算下列Gamma函数值

$\Gamma(0)$ $\quad$ $\Gamma(0)=+\infty$

$\Gamma(1)$ $\quad$ $\Gamma(1)=1$

$\Gamma(2)$ $\quad$ $\Gamma(2)=1$

$\Gamma\left(\frac{1}{2}\right)$ $\quad$ $\Gamma\left(\frac{1}{2}\right)=\sqrt{\pi}$

$\Gamma\left(\frac{3}{2}\right)$ $\quad$ $\Gamma\left(\frac{3}{2}\right)=\Gamma\left(1+\frac{1}{2}\right)=\frac{(2-1)!!}{2^1}\sqrt{\pi}=\frac{\sqrt{\pi}}{2}$

$\Gamma\left(\frac{5}{2}\right)$ $\quad$ $\Gamma\left(\frac{5}{2}\right)=\Gamma\left(2+\frac{1}{2}\right)=\frac{(2\cdot2-1)!!}{2^2}\sqrt{\pi}=\frac{(1\cdot3)}{2^2}\sqrt{\pi}=\frac{3\sqrt{\pi}}{4}$

$$\Gamma\left(\frac{5}{2}\right)=\Gamma\left(\frac{3}{2}+1\right)=\frac{3}{2}\Gamma\left(\frac{3}{2}\right)=\frac{3}{2}\Gamma\left(\frac{1}{2}+1\right)=\frac{3}{2}\cdot\frac{1}{2}\Gamma\left(\frac{1}{2}\right)=\frac{3\sqrt{\pi}}{4}$$

$\Gamma\left(-\frac{1}{2}\right)$ $\quad$ $\Gamma\left(-\frac{1}{2}\right)=\dfrac{\Gamma\left(-\frac{1}{2}+1\right)}{-\frac{1}{2}}=-2\Gamma\left(\frac{1}{2}\right)=-2\sqrt{\pi}$

$$\Gamma(\alpha)=\int_0^{\infty} x^{\alpha-1}e^{-x}dx$$

Gamma函数

应用之一→为广义积分的计算另辟蹊径

例2 计算积分 $\int_0^{\infty} x^6 e^{-x^2}dx$

解 变量代换 $t=x^2, x=\sqrt{t}$

$$\int_0^{\infty} x^6 e^{-x^2}dx$$

$$=\int_0^{\infty}\left(x^2\right)^3 e^{-x^2}dx$$

调整积分变量

$$=\int_0^{\infty} t^3 e^{-t}d\left(\sqrt{t}\right)=\frac{1}{2}\int_0^{\infty} t^3 e^{-t}\cdot t^{-\frac{1}{2}}dt$$

Γ函数定义

$$=\frac{1}{2}\int_0^{\infty} t^{\frac{5}{2}}e^{-t}dt=\frac{1}{2}\int_0^{\infty} t^{\frac{7}{2}-1}e^{-t}dt$$

$$=\frac{1}{2}\Gamma\left(\frac{7}{2}\right)=\frac{1}{2}\Gamma\left(3+\frac{1}{2}\right)$$

$$\Gamma\left(n+\frac{1}{2}\right)=\frac{(2n-1)!!}{2^n}\sqrt{\pi}$$

$$=\frac{1}{2}\cdot\frac{(2\cdot 3-1)!!}{2^3}\sqrt{\pi}=\frac{1}{2}\cdot\frac{(1\cdot 3\cdot 5)}{2^3}\sqrt{\pi}$$

$$=\frac{15}{16}\sqrt{\pi}$$

技巧 通过变量代换向Γ函数定义转化

例3 计算积分 $\int_{-\infty}^{\infty} x^4 e^{-\frac{x^2}{2}}dx$

解 变量代换 $t=\frac{x^2}{2},\ x^2=2t, x=\sqrt{2t}$

$$\int_{-\infty}^{\infty} x^4 e^{-\frac{x^2}{2}}dx=2\int_0^{\infty}\left(x^2\right)^2 e^{-\frac{x^2}{2}}dx$$

偶函数

调整积分变量

$$=2\int_0^{\infty}(2t)^2 e^{-t}d\left(\sqrt{2t}\right)=2\int_0^{\infty}4t^2 e^{-t}\sqrt{2}d\left(t^{\frac{1}{2}}\right)$$

$$=2\int_0^{\infty}4t^2 e^{-t}\cdot\sqrt{2}\frac{1}{2}(t)^{-\frac{1}{2}}dt$$

Γ函数定义

$$=4\sqrt{2}\int_0^{\infty} t^{\frac{3}{2}}e^{-t}dt=4\sqrt{2}\int_0^{\infty} t^{\frac{5}{2}-1}e^{-t}dt$$

$$=4\sqrt{2}\cdot\Gamma\left(\frac{5}{2}\right)=4\sqrt{2}\Gamma\left(2+\frac{1}{2}\right)$$

$$=4\sqrt{2}\cdot\frac{(2\cdot 2-1)!!}{2^2}\sqrt{\pi}$$

$$=4\sqrt{2}\cdot\frac{(1\cdot 3)}{2^2}\sqrt{\pi}=3\sqrt{2\pi}$$

$$\Gamma(\alpha)=\int_0^{\infty} x^{\alpha-1}e^{-x}dx$$

Gamma函数

例3 下列表达式错误的是

$$\int_0^{+\infty} x^{\frac{5}{2}}e^{-x^2}dx=\frac{3}{8}\Gamma\left(\frac{3}{4}\right) \rightarrow x^2=t, x=t^{\frac{1}{2}} \rightarrow$$

$$\int_0^{+\infty} x^{\frac{5}{2}}e^{-x^2}dx=\int_0^{+\infty} t^{\frac{1}{2}\cdot\frac{5}{2}}e^{-t}d\left(t^{\frac{1}{2}}\right)$$

$$=\frac{1}{2}\int_0^{+\infty} t^{\frac{1}{2}\cdot\frac{5}{2}}e^{-t}t^{-\frac{1}{2}}dt$$

$$=\frac{1}{2}\int_0^{+\infty} t^{\frac{3}{4}}e^{-t}dt$$

$$=\frac{1}{2}\int_0^{+\infty} t^{\frac{7}{4}-1}e^{-t}dt$$

$$=\frac{1}{2}\Gamma\left(\frac{7}{4}\right)=\frac{1}{2}\Gamma\left(1+\frac{3}{4}\right)$$

$$=\frac{3}{8}\Gamma\left(\frac{3}{4}\right)$$

$$\Gamma(-2)=\infty$$

$$\Gamma\left(-\frac{1}{2}\right)=-\sqrt{\pi}$$

$$\Gamma\left(-\frac{1}{2}\right)=\frac{\Gamma\left(-\frac{1}{2}+1\right)}{-\frac{1}{2}}=-2\Gamma\left(\frac{1}{2}\right)=-2\sqrt{\pi}$$

$$\Gamma\left(\frac{3}{2}\right)=\frac{\sqrt{\pi}}{2}$$

快速计算手册

边界条件齐次化常用辅助函数 $u(x,t)=v(x,t)+w(x,t)$

边界类型	边界条件	辅助函数
[1,1]	$u(0,t)=g_1(t),\ u(l,t)=g_2(t)$	$w(x,t)=\dfrac{g_2(t)-g_1(t)}{l}x+g_1(t)$
[2,1]	$u_x(0,t)=g_1(t),\ u(l,t)=g_2(t)$	$w(x,t)=g_1(t)(x-l)+g_2(t)$
[1,2]	$u(0,t)=g_1(t),\ u_x(l,t)=g_2(t)$	$w(x,t)=g_2(t)x+g_1(t)$
[2,2]	$u_x(0,t)=g_1(t),\ u_x(l,t)=g_2(t)$	$w(x,t)=\dfrac{g_2(t)-g_1(t)}{2l}x^2+g_1(t)x$

算子 $A=-\dfrac{d^2}{dx^2}$ 不同齐次边界条件下的特征值/特征函数

齐次边界类型	特征值问题	特征值/特征函数
[1,1]	$\begin{cases}X''(x)+\lambda X(x)=0\\X(0)=X(l)=0\end{cases}$	$\lambda_n=\left(\dfrac{n\pi}{l}\right)^2,\ X_n(x)=\sin\left(\dfrac{n\pi}{l}x\right),(n=1,2,3\cdots)$
[1,2]	$\begin{cases}X''(x)+\lambda X(x)=0\\X(0)=X'(l)=0\end{cases}$	$\lambda_n=\left[\dfrac{(2n+1)\pi}{2l}\right]^2,X_n(x)=\sin\left[\dfrac{(2n+1)\pi}{2l}x\right],(n=0,1,2,3\cdots)$
[2,1]	$\begin{cases}X''(x)+\lambda X(x)=0\\X'(0)=X(l)=0\end{cases}$	$\lambda_n=\left[\dfrac{(2n+1)\pi}{2l}\right]^2,X_n(x)=\cos\left[\dfrac{(2n+1)\pi}{2l}x\right],\ (n=0,1,2,3\cdots)$
[2,2]	$\begin{cases}X''(x)+\lambda X(x)=0\\X'(0)=X'(l)=0\end{cases}$	$\lambda_n=\left(\dfrac{n\pi}{l}\right)^2,\ X_n(x)=\cos\left(\dfrac{n\pi}{l}x\right),(n=0,1,2,3\cdots)$
周期	$\begin{cases}\Phi''(\theta)+\lambda\Phi(\theta)=0\\\Phi(0)=\Phi(2\pi),\\\Phi'(0)=\Phi'(2\pi)\end{cases}$	$\lambda_0=0,\ \Phi_0(\theta)=1,\ (n=0)$ $\lambda_n=n^2,\ \Phi_n(\theta)=C_1\cos n\theta+C_2\sin n\theta,\ (n=1,2,3\cdots)$

线性ODE的解法 $x^{<n>}+P_1x^{<n-1>}+P_2x^{<n-2>}+\cdots+P_{n-1}x^{<1>}+P_nx^{<0>}=F$

线性常微分方程解法
- 一阶
  - 齐次 $x'+P(t)x=0$ 分离变量法
  - 非齐次 $x'+P(t)x=f(t)$ 常数变易法
- 二阶(高阶)
  - 常系数
    - 齐次
      - 特征方程法 $\lambda^2+p\lambda+q=0$
        - 互异实根 $\{e^{\lambda_1 t},e^{\lambda_2 t}\}$
        - 相同实根 $\{te^{\lambda t},e^{\lambda t}\}$
        - 共轭复根 $\{e^{\alpha t}\cos\beta t,e^{\alpha t}\sin\beta t\}$
      - 基解1→基解2
        - Liouville公式 $x_2(t)=x_1(t)\int\dfrac{1}{x_1^2(t)}e^{-\int p_1(t)dt}dt$
        - 摄动法
    - 非齐次
      - 通解=通+特
      - 求非齐特解
        - 待定函数法 $x''(t)+a_1x'(t)+a_0x(t)=\varphi(t)e^{\mu t}\sin\nu t$，$s=\mu\pm\nu i$，$\bar{x}(t)=t^ke^{\mu t}\left[\Phi_1(t)\cos\nu t+\Phi_2(t)\sin\nu t\right]$
        - 观察法 $\ddot{x}+x=\sin 2t,\ \bar{x}=C\sin 2t$
  - 变系数
    - Euler方程 $t^n\dfrac{d^nx}{dt^n}+a_1t^{n-1}\dfrac{d^{n-1}x}{dt^{n-1}}+\cdots+a_{n-1}t\dfrac{dx}{dt}+a_nx=F(t),\ t=e^s$；Bernoulli方程 $\dfrac{dy}{dx}+P(x)\cdot y=Q(x)\cdot y^\alpha,\ (\alpha\neq 0,1),\ u=y^{-\alpha+1}$ } 变量代换法
    - Bessel方程 $x^2\dfrac{d^2y}{dx^2}+x\dfrac{dy}{dx}+(x^2-r^2)y=0$；Legendre方程 $(1-x^2)\dfrac{d^2y}{dx^2}-2x\dfrac{dy}{dx}+n(n+1)y=0$ } 幂级数法
- 高阶(可降阶)
  - $y^{(n)}=f(x)$ 无其他导数项，直接积分法
  - $y''=f(x,y')$ 不显含y，令 $p=y',y''=\dfrac{dp}{dx}=f(x,p)$；$y''=f(y,y')$ 不显含x，令$p=y',y''=\dfrac{dp}{dx}=\dfrac{dp}{dy}\dfrac{dy}{dx}=p\dfrac{dp}{dy}$ } 变量代换法

2

Γ函数常用计算

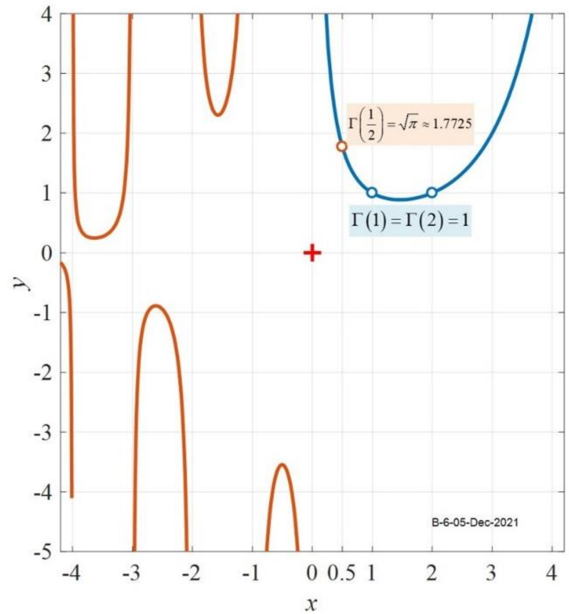
$$\Gamma(\alpha)=\begin{cases}\int_0^\infty x^{\alpha-1}e^{-x}dx, & (\alpha>0)\\ \dfrac{\Gamma(\alpha+1)}{\alpha}, & (\alpha<0,\alpha\neq Z^-)\\ \infty, & (\alpha=0,Z^-)\end{cases}$$

$\Gamma(\alpha)\neq 0$

$\Gamma(\alpha+1)=\alpha\cdot\Gamma(\alpha),\ \dfrac{1}{\alpha}=\dfrac{\Gamma(\alpha)}{\Gamma(\alpha+1)}$

$\Gamma(n+1)=n!,\ \Gamma(2)=\Gamma(1)=1$

$\Gamma\left(n+\dfrac{1}{2}\right)=\dfrac{(2n-1)!!}{2^n}\sqrt{\pi},\ \Gamma\left(\dfrac{1}{2}\right)=\sqrt{\pi}$



1. 数学建模和基本原理介绍
2. 分离变量法
3. Bessel函数
   - 3.1 二阶线性常微分方程的幂级数解法
   - 3.2 Bessel函数
     - Gamma函数
     - Bessel方程和Bessel函数
     - Bessel函数的性质
     - Bessel方程的特征值问题
     - 圆域上Laplace算子的特征值问题
   - 3.3 多个自变量分离变量法举例
4. 积分变换法
5. Green函数法
6. 特征线法
7. Legendre多项式

3.2 Bessel函数

→表达径向Bessel方程的解

极坐标上的波动方程

$$\frac{\partial^2 u}{\partial t^2} - a^2\Delta u = 0 \qquad u(\rho,\theta,t) = T(t)w(\rho,\theta)$$

$$T''(t)w(\rho,\theta) - a^2 T(t)\Delta w(\rho,\theta) = 0$$

$$\frac{T''(t)}{a^2T(t)} = \frac{\Delta w(\rho,\theta)}{w(\rho,\theta)} = -\mu$$

时空分离常数

❷ 尚未确定

初值问题

$$T''(t) + \mu a^2 T(t) = 0$$

二维边值问题/特征值问题 $A_2 = -\Delta$

$$\begin{cases} -\Delta w(\rho,\theta) = \mu w(\rho,\theta), & (\rho,\theta)\in\Omega \\ w|_{\partial\Omega} = 0 \end{cases}$$

$$\Delta w = \frac{\partial^2 w}{\partial \rho^2} + \frac{1}{\rho}\frac{\partial w}{\partial \rho} + \frac{1}{\rho^2}\frac{\partial^2 w}{\partial \theta^2}$$

$$w(\rho,\theta) = R(\rho)\Phi(\theta)$$

$$R''(\rho)\Phi(\theta) + \frac{1}{\rho}R'(\rho)\Phi(\theta) + \frac{1}{\rho^2}R(\rho)\Phi''(\theta) = -\mu R(\rho)\Phi(\theta)$$

$$\frac{\Phi''(\theta)}{\Phi(\theta)} = -\frac{\mu R(\rho) + R''(\rho) + \frac{1}{\rho}R'(\rho)}{\frac{1}{\rho^2}R(\rho)} = -\lambda$$

空间分离常数

$\lambda_0 = 0,\ \Phi_0(\theta) = 1,$

$(n=0)$

$\lambda_n = n^2, \Phi_n(\theta) = \{\cos n\theta, \sin n\theta\}$

$(n = 1,2,3\cdots)$

周向

$$\begin{cases} \Phi''(\theta) + \lambda\Phi(\theta) = 0 \\ \Phi(\theta) = \Phi(\theta + 2\pi) \end{cases}$$

径向

$$\begin{cases} |R(0)| < +\infty,\ R(\rho_0) = 0 \\ \rho^2 R''(\rho) + \rho R'(\rho) + (\mu\rho^2 - \lambda)R(\rho) = 0 \end{cases}$$

相当于原方程是Poisson方程

Euler方程 $\mu = 0$

$$\rho^2 R''(\rho) + \rho R'(\rho) - n^2 R(\rho) = 0$$

Bessel方程 $\mu > 0$ **1824**

$$\rho^2 R''(\rho) + \rho R'(\rho) + (\mu\rho^2 - n^2)R(\rho) = 0$$

线性二阶变系数齐次ODE ❶ 如何求解

F.W.Bessel
1784-1846
German Math.
Astronomer

3.2 Bessel函数

Bessel方程的标准形式 预处理

F.W.Bessel
1784-1846
German Math.
Astronomer

极坐标径向ODE

$$\begin{cases} |R(0)| < +\infty,\ R(\rho_0) = 0 \\ \rho^2 R''(\rho) + \rho R'(\rho) + (\mu\rho^2 - n^2) R(\rho) = 0 \end{cases}$$

时空分离常数

空间分离常数 $\lambda = n^2$

变量代换

$$x = \sqrt{\mu}\rho,\ \rho = \frac{x}{\sqrt{\mu}},$$

$$y(x) = R(\rho) = R\left(\frac{x}{\sqrt{\mu}}\right)$$

各阶导数

$$y'(x) = R'(\rho)\frac{d\rho}{dx} = \frac{1}{\sqrt{\mu}} R'(\rho)$$

$$y''(x) = \frac{1}{\sqrt{\mu}} R''(\rho)\frac{d\rho}{dx} = \frac{1}{\mu} R''(\rho)$$

$$R'(\rho) = \sqrt{\mu}\, y'(x)$$

$$R''(\rho) = \mu y''(x)$$

$$\frac{x^2}{\mu}\mu y''(x) + \frac{x}{\sqrt{\mu}}\sqrt{\mu}\, y'(x) + \left(\mu\frac{x^2}{\mu} - n^2\right) y(x) = 0$$

$$x^2 y''(x) + x y'(x) + (x^2 - n^2) y(x) = 0$$

定义Bessel方程阶数 $r^2 = n^2$

整数→实数
默认 $r \geqslant 0$

Bessel方程标准形式 $$x^2 y''(x) + x y'(x) + (x^2 - r^2) y(x) = 0$$

1. 阶数是周向特征值=非负整数→全面研究Bessel方程→阶数拓展→实数域
2. Bessel方程是线性二阶变系数齐次ODE→阶数 r 是唯一参数→决定了解的形式
3. 观察法→显然 $y(x)=0$ 是Bessel方程的解→关心更有意义的非零解

3.2 Bessel函数

$$x''(t)+p(x)x'(t)+q(x)x(t)=0$$

Bessel方程/Bessel函数

Bessel方程→幂级数求解→Bessel函数

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$$x^2y''+xy'+\left(x^2-r^2\right)y=0$$

阶数 $r\geq 0$

求解思路:
判定幂级数解存在→幂级数展开→2个形式解→判定线性相关性→确定基解

1 解的存在性

方程变型 $y''+\frac{1}{x}y'+\left(1-\frac{r^2}{x^2}\right)y=0$ $p(x)=\frac{1}{x}$, $q(x)=1-\frac{r^2}{x^2}$

而 $x\cdot p(x)=x\cdot\frac{1}{x}=1$, $x^2\cdot q(x)=x^2\cdot\left(1-\frac{r^2}{x^2}\right)=x^2-r^2$ 在实域内解析→解存在

Bessel方程存在幂级数解

$$y(x)=x^{\rho}\sum_{n=0}^{\infty}a_nx^n=\sum_{n=0}^{\infty}a_nx^{n+\rho},\quad (a_0\neq 0)$$

探寻非零解

1个参数 r →2组未知系数 ρ, a_n

幂级数解存在定理2　　求解Bessel方程的理论支撑

对二阶ODE $x''(t)+p(x)x'(t)+q(x)x(t)=0$, 若 $(x-x_0)p(x)$, $(x-x_0)^2q(x)$ 在 x_0 的邻域 $\{x\in R|\ |x-x_0|<\delta\}$ 内解析, 即 x_0 最多为 $p(x)$, $q(x)$ 的一阶和二阶极点, 则原方程在该去心邻域 $\{x\in R|\ 0<|x-x_0|<\delta\}$ 内存在幂级数解

$$y(x)=(x-x_0)^{\rho}\sum_{k=0}^{\infty}a_k(x-x_0)^k$$

其中常数 ρ, a_k $(a_0\neq 0)$ 可由待定系数法求出

3.2 Bessel函数

$$x^2y''+xy'+\left(x^2-r^2\right)y=0$$

Bessel方程/Bessel函数

2 寻找递推式

幂级数解 $y(x)=\sum_{n=0}^{\infty}a_nx^{n+\rho},\ (a_0\neq 0)$

各阶导数 $y'(x)=\sum_{n=0}^{\infty}(n+\rho)a_nx^{n+\rho-1},\ y''(x)=\sum_{n=0}^{\infty}(n+\rho)(n+\rho-1)a_nx^{n+\rho-2}$

代入原方程

$$x^2\sum_{n=0}^{\infty}(n+\rho)(n+\rho-1)a_nx^{n+\rho-2}+x\sum_{n=0}^{\infty}(n+\rho)a_nx^{n+\rho-1}+\left(x^2-r^2\right)\sum_{n=0}^{\infty}a_nx^{n+\rho}=0$$

$$\sum_{n=0}^{\infty}(n+\rho)(n+\rho-1)a_nx^{n+\rho}+\sum_{n=0}^{\infty}(n+\rho)a_nx^{n+\rho}-r^2\sum_{n=0}^{\infty}a_nx^{n+\rho}+\sum_{n=0}^{\infty}a_nx^{n+\rho+2}=0$$

约去x^ρ

$$\sum_{n=0}^{\infty}\left[(n+\rho)^2-r^2\right]a_nx^{n+\rho}+\sum_{n=0}^{\infty}a_nx^{n+\rho+2}=0$$

$$\sum_{n=0}^{\infty}\left[(n+\rho)^2-r^2\right]a_nx^{n}+\sum_{n=0}^{\infty}a_nx^{n+2}=0$$

展开前两项

$$\left(\rho^2-r^2\right)a_0+\left[(1+\rho)^2-r^2\right]a_1x+\sum_{n=2}^{\infty}\left[(n+\rho)^2-r^2\right]a_nx^{n}+\sum_{n=2}^{\infty}a_{n-2}x^{n}=0$$

调整起点→比较同幂次系数

比较系数→原始递推关系

$$\begin{cases}\left(\rho^2-r^2\right)a_0=0\\ \left[(1+\rho)^2-r^2\right]a_1=0\\ \left[(n+\rho)^2-r^2\right]a_n+a_{n-2}=0,\ (n\geq 2)\end{cases}$$

1. Bessel方程的阶数r决定→幂级数解中 x 的幂次 ρ
2. 后续根据幂次 ρ的取值/系数序号奇偶→化简递推

3.2 Bessel函数

$$x^2y''+xy'+\left(x^2-r^2\right)y=0$$

Bessel方程/Bessel函数

$$y(x)=\sum_{n=0}^{\infty}a_n x^{n+\rho},\ (a_0\neq 0)$$

2 递推式简化

$$y(x)=\sum_{n=0}^{\infty}a_n x^{n+\rho},\ (a_0\neq 0)$$

$$\begin{cases}\left(\rho^2-r^2\right)a_0=0 & ❶\\ \left[(1+\rho)^2-r^2\right]a_1=0 & ❷\\ \left[(n+\rho)^2-r^2\right]a_n+a_{n-2}=0,\ (n\geq 2) & ❸\end{cases}$$

❶ $\rho=\pm r,\quad (r\geq 0)$

❷ $\left[(1+\rho)^2-r^2\right]a_1=0$

$$\left(1+2\rho+\rho^2-r^2\right)a_1=0$$

$$(1+2\rho)a_1=0$$

❸ $\left[(n+\rho)^2-r^2\right]a_n+a_{n-2}=0,\ (n\geq 2)$

$$\left(n^2+2n\rho+\rho^2-r^2\right)a_n+a_{n-2}=0$$

$$\left(n^2+2n\rho\right)a_n+a_{n-2}=0$$

$$n(n+2\rho)a_n+a_{n-2}=0$$

递推式的简化

$$\begin{cases}\rho=\pm r & ❶\\ (1+2\rho)a_1=0 & ❷\\ n(n+2\rho)a_n+a_{n-2}=0,\ (n\geq 2) & ❸\end{cases}$$

1.已经确定 ρ

2.无论 r 是否为整数; 无论 ρ=+r 还是 ρ=-r ; 递推式均成立

3.但 ρ 的正负/取值→递推式❷ ❸ 的运用→需对ρ分类讨论

$$\rho=\begin{cases}r>0 & \\ -r<0 & \begin{cases}r=N^+\\ r\neq N^+\end{cases},\quad (r\geq 0)\\ r=0 & \end{cases}$$

3.2 Bessel函数

$$\rho=\begin{cases} \boxed{r>0} \\ -r<0 \begin{cases} r=N^+ \\ r\neq N^+ \end{cases}, \quad (r\geq 0) \\ r=0 \end{cases}$$

$$x^2y''+xy'+\left(x^2-r^2\right)y=0$$

$$y(x)=\sum_{n=0}^{\infty}a_n x^{n+\rho},\ (a_0\neq 0)$$

Bessel方程/Bessel函数

3 阶数 r 分类讨论

1 当 $\rho=r>0$ 时

❶ $\rho=\pm r$ → 此时只可能 $\rho=+r$

❷ $(1+2\rho)a_1=0$ → $\rho=r>0$ → $(1+2\rho)\neq 0$ → 只可能 $a_1=0$

❸ $n(n+2\rho)a_n+a_{n-2}=0,\ (n\geq 2)$ → $n(n+2\rho)\neq 0,\ (n\geq 2)$

↓ 可除至分母

$$a_n=-\frac{a_{n-2}}{n(n+2\rho)}=-\frac{a_{n-2}}{n(n+2r)},\ (n\geq 2)$$

说明系数跨位传递→
需对序号分奇偶讨论

1.1 当下标为奇数时→最终均可递推至 a_1→奇数项系数均为0

n 不连续→改用连续的 k 表示 $a_{2k-1}=a_{2k-3}=\cdots=a_1=0,\ (k\geq 1)$

3.2 Bessel函数

Bessel方程/Bessel函数

3 阶数 r 分类讨论

$$\rho=\begin{cases}\boxed{r>0} \\ -r<0\ \begin{cases} r=N^{+} \\ r\neq N^{+}\end{cases}, \quad (r\ge 0) \\ r=0\end{cases}$$

$$x^2y''+xy'+\left(x^2-r^2\right)y=0$$

$$y(x)=\sum_{n=0}^{\infty}a_n x^{n+\rho},\ (a_0\neq 0)$$

$$\begin{cases} ① & \rho=\pm r \\ ② & (1+2\rho)a_1=0 \\ ③ & n(n+2\rho)a_n+a_{n-2}=0,\ (n\ge 2)\end{cases}$$

1 当 $\rho=r>0$ 时

1.2 下标为偶数时 $a_0\neq 0$

$$a_n=-\frac{a_{n-2}}{n(n+2\rho)}=-\frac{a_{n-2}}{n(n+2r)},\ (n\ge 2)$$

$$a_2=-\frac{a_0}{2(2+2r)}=-\frac{a_0}{2^2\cdot 1!\cdot(1+r)}$$

$$a_4=-\frac{a_2}{4(4+2r)}=-\frac{a_2}{2^3(2+r)}=(-1)^2\frac{1}{2^3(2+r)}\cdot\frac{a_0}{2^2\cdot 1!\cdot(1+r)}=(-1)^2\frac{a_0}{2^4\cdot 2!\cdot(2+r)(1+r)}$$

$$a_6=-\frac{a_4}{6(6+2r)}=-\frac{a_4}{2\cdot 3\cdot 2\cdot(3+r)}=(-1)^3\frac{1}{2^2\cdot 3\cdot(3+r)}\cdot\frac{a_0}{2^4\cdot 2!\cdot(2+r)(1+r)}$$

$$=(-1)^3\frac{a_0}{2^6\cdot 3!\cdot(3+r)(2+r)(1+r)}$$

$$a_{2k}\to a_{2k-2}\to a_{2k-4}\to\cdots\to a_4\to a_2\to a_0$$

$(n\ge 2)$

$$a_{2k}=(-1)^k\frac{a_0}{2^{2k}\cdot k!\cdot\underbrace{(k+r)(k-1+r)\cdots(1+r)}_{\text{如何化简连乘}}},\ (k\ge 1)$$

偶数项系数不为0→均可递推至a_0 →
但系数表达式冗长连乘→如何化简?

3.2 Bessel函数

$$\rho=\begin{cases}\boxed{r>0}\\ -r<0\quad\begin{cases}r=N^+\\ r\neq N^+\end{cases},\quad (r\geq 0)\\ r=0\end{cases}$$

$$x^2y''+xy'+\left(x^2-r^2\right)y=0$$

Bessel方程/Bessel函数

3 阶数 r 分类讨论

$$y(x)=\sum_{n=0}^{\infty}a_nx^{n+\rho},\quad (a_0\neq 0)$$

1 当 $\rho=r>0$ 时

① $\rho=\pm r$

② $(1+2\rho)a_1=0$

③ $n(n+2\rho)a_n+a_{n-2}=0,\ (n\geq 2)$

1.2 下标为偶数　分母连乘的化简

$$a_{2k}=(-1)^k\frac{a_0}{2^{2k}\cdot k!\cdot(k+r)(k-1+r)\cdots(1+r)},\quad (k\geq 1)$$

$$=(-1)^k\frac{1}{2^{2k}\cdot k!}\cdot\frac{1}{k+r}\cdot\frac{1}{k-1+r}\cdots\frac{1}{3+r}\cdot\frac{1}{2+r}\cdot\frac{1}{1+r}a_0$$

$$\frac{1}{\alpha}=\frac{\Gamma(\alpha)}{\Gamma(\alpha+1)}\leftarrow\Gamma(\alpha+1)=\alpha\cdot\Gamma(\alpha)$$

$$=(-1)^k\frac{1}{2^{2k}\cdot k!}\cdot\frac{\Gamma(k+r)}{\Gamma(k+r+1)}\cdot\frac{\Gamma(k-1+r)}{\Gamma(k+r)}\cdots\frac{\Gamma(3+r)}{\Gamma(4+r)}\cdot\frac{\Gamma(2+r)}{\Gamma(3+r)}\cdot\frac{\Gamma(1+r)}{\Gamma(2+r)}a_0$$

$$=(-1)^k\frac{1}{2^{2k}\cdot k!}\cdot\frac{\Gamma(1+r)}{\Gamma(k+r+1)}a_0$$

$$a_0=\frac{1}{2^r\cdot\Gamma(1+r)}$$

系数 a_0 为任意常数不妨定义为

Bessel方程给定
→阶数r为定值

$$=(-1)^k\frac{\Gamma(1+r)}{2^{2k}\cdot k!\cdot\Gamma(k+r+1)}\frac{1}{2^r\cdot\Gamma(1+r)}$$

$$=\frac{(-1)^k}{2^r\cdot 2^{2k}\cdot k!\cdot\Gamma(k+r+1)}$$

a_0消失了

偶数项系数表达式 →

$$a_{2k}=\frac{(-1)^k}{2^r\cdot 2^{2k}\cdot k!\cdot\Gamma(k+r+1)}$$

3.2 Bessel函数

$$x^2y''+xy'+\left(x^2-r^2\right)y=0$$

Bessel方程/Bessel函数

3 阶数 r 分类讨论

$$\rho=\begin{cases}\boxed{r>0}\\ -r<0\ \begin{cases}r=N^+\\ r\neq N^+\end{cases},\quad (r\geq 0)\\ r=0\end{cases}$$

$$y(x)=\sum_{n=0}^{\infty}a_nx^{n+\rho},\ \left(a_0\neq 0\right)$$

① $\rho=\pm r$

② $(1+2\rho)a_1=0$

③ $n(n+2\rho)a_n+a_{n-2}=0,\ (n\geq 2)$

1 当 $\rho=r>0$ 时 小结

奇数项 $a_{2k-1}=\cdots=a_1=0,\ (k\geq 1)$ $a_{2k}=\dfrac{(-1)^k}{2^r\cdot 2^{2k}\cdot k!\,\Gamma(k+r+1)}$ 偶数项

代入幂级数解

$$y_1(x)=\sum_{n=0}^{\infty}a_nx^{n+\rho}\overset{\rho=r}{=}\sum_{n=0}^{\infty}a_nx^{n+r}=\sum_{k=1}^{\infty}\underbrace{a_{2k-1}}_{=0}x^{2k-1+r}+\sum_{k=0}^{\infty}a_{2k}x^{2k+r}$$

递推式从k=1开始，但此处要包含a_0

$$=\sum_{k=0}^{\infty}\frac{(-1)^k}{2^r\cdot 2^{2k}\cdot k!\,\Gamma(k+r+1)}x^{2k+r}=\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\,\Gamma(k+r+1)}\cdot\frac{x^{2k}}{2^{2k}}\cdot\frac{x^r}{2^r}$$

$$=\left(\frac{x}{2}\right)^r\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\,\Gamma(k+r+1)}\left(\frac{x}{2}\right)^{2k}=J_r(x)$$

只需累加偶数项→换为连续序号k

与k无关的移出Σ

r阶Bessel函数

$$J_r(x)=\left(\frac{x}{2}\right)^r\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\,\Gamma(k+r+1)}\left(\frac{x}{2}\right)^{2k}$$

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仅仅是Bessel方程的1个基解?→另一个基解在哪里?

$$x^2y''+xy'+\left(x^2-r^2\right)y=0$$

$$\rho=\begin{cases} r>0 \\ -r<0 \quad \begin{cases} r=N^+ \\ r\neq N^+ \end{cases}, \quad (r\geq 0) \\ r=0 \end{cases}$$

$$y(x)=\sum_{n=0}^{\infty}a_nx^{n+\rho},\ (a_0\neq 0)$$

Bessel方程/Bessel函数

3 阶数 r 分类讨论

2 当 $\rho=-r<0$ 时 $a_0\neq 0$

化简递推式

1. $\rho=\pm r$ ➡ $\rho=-r$
2. $(1+2\rho)a_1=0$ ➡ 不妨仍令 $a_1=0$ ➡ 奇数下标全为0 $a_{2k-1}=\cdots=a_1=0,\ (k\geq 1)$
3. $n(n+2\rho)a_n+a_{n-2}=0,\ (n\geq 2)$ 聚焦→ n 为偶数 $n=2k$ 时递推式 3 的运用

按n为偶数化简

$\rho=-r$

$$n(n+2\rho)a_n+a_{n-2}=0,\ (n\geq 2)$$

$$n(n-2r)a_n+a_{n-2}=0$$

$n=2k$

$$2k(2k-2r)a_{2k}+a_{2k-2}=0,\ (k\geq 1)$$

$$4k(k-r)a_{2k}+a_{2k-2}=0$$

非负整数

若r为正整数→$(k-r)$有可能=0→不能直接除至分母

→需根据 r 是否为正整数分亚类讨论

$$\rho=\begin{cases} r>0 \\ -r<0 \quad \begin{cases} r=N^+ \\ r\neq N^+ \end{cases}, \quad (r\geq 0) \\ r=0 \end{cases}$$

3.2 Bessel函数

$$x^2y''+xy'+\left(x^2-r^2\right)y=0$$

Bessel方程/Bessel函数

3 阶数 r 分类讨论

$$\rho=\begin{cases} r>0 \\ \boxed{-r<0} \begin{cases} r=N^+ \\ r\neq N^+ \end{cases}, \quad (r\geq 0) \\ r=0 \end{cases}$$

$$y(x)=\sum_{n=0}^{\infty}a_n x^{n+\rho},\ \left(a_0\neq 0\right)$$

2 当 $\rho=-r<0$ 时 $a_0\neq 0$

$$\begin{cases} ❶\ \rho=\pm r \\ ❷\ \left(1+2\rho\right)a_1=0 \\ ❸\ n\left(n+2\rho\right)a_n+a_{n-2}=0,\ \left(n\geq 2\right) \end{cases}$$

$$4k\left(k-r\right)a_{2k}+a_{2k-2}=0$$

2.1 当 r 不为正整数时 $k-r\neq 0$

递推式❸ $4k\left(k-r\right)a_{2k}+a_{2k-2}=0$

可除至分母

递推可变型为 $a_{2k}=-\dfrac{1}{4k\left(k-r\right)}a_{2k-2}$ 偶数项递推至a_0 $a_{2k}\to a_{2k-2}\to\cdots\to a_4\to a_2\to a_0$

同理可得

$\rho=-r<0$

$$a_{2k}=\frac{\left(-1\right)^k}{2^{-r}\cdot 2^{2k}\cdot k!\cdot\Gamma\left(k-r+1\right)}$$

对比

相当于r→-r

$\rho=+r>0$

$$a_{2k}=\frac{\left(-1\right)^k}{2^{r}\cdot 2^{2k}\cdot k!\cdot\Gamma\left(k+r+1\right)}$$

得到 r 不为整数时Bessel方程另一个幂级数解

-r 阶Bessel函数

$$J_{-r}\left(x\right)=\left(\frac{x}{2}\right)^{-r}\sum_{k=0}^{\infty}\frac{\left(-1\right)^k}{k!\cdot\Gamma\left(k-r+1\right)}\left(\frac{x}{2}\right)^{2k}$$

3.2 Bessel函数

$$x^2y''+xy'+\left(x^2-r^2\right)y=0$$

Bessel方程/Bessel函数

3 阶数 r 分类讨论

$$\rho=\begin{cases} r>0 \\ -r<0 & \begin{cases} r\neq N^+ \\ \boxed{r=N^+} \end{cases} \\ r=0 \end{cases} \quad (r\geq 0)$$

$$y(x)=\sum_{n=0}^{\infty}a_n x^{n+\rho},\ (a_0\neq 0)$$

2 当 $\rho=-r<0$ 时 $a_0\neq 0$

$$\begin{cases} ❶\ \rho=\pm r \\ ❷\ (1+2\rho)a_1=0 \\ ❸\ n(n+2\rho)a_n+a_{n-2}=0,\ (n\geq 2) \end{cases}$$

$$4k(k-r)a_{2k}+a_{2k-2}=0$$

2.2 当r为正整数时 $r=l$

任意确定正整数

连续变化的 整数k 总有 = l 的可能

当 $k-r=k-l=0$ 时根据递推式

$$4k(k-r)a_{2k}+a_{2k-2}=0,\ (k\geq 1)$$

$$4k\underbrace{(k-l)}_{=0}a_{2l}+a_{2l-2}=0$$

故有 $a_{2l-2}=0$

令 $k=l-1$, 再次使用递推式

$$4\left[(l-1)-1\right]\left[(l-1)-l\right]a_{2l-2}+a_{2l-4}=0$$

只可能 $a_{2l-4}=0$

如此反复运用递推式可得

$$a_{2l-2}=a_{2l-4}=\cdots=a_4=a_2=a_0=0$$

$$a_{2(l-1)}=a_{2(l-2)}=a_{2(l-3)}=\cdots=a_{2\cdot 2}=a_{2\cdot 1}=a_{2\cdot 0}=0$$

与假设$a_0\neq 0$矛盾 问题出在哪里呢?

3.2 Bessel函数

$$x^2y''+xy'+\left(x^2-r^2\right)y=0$$

Bessel方程/Bessel函数

3 阶数 r 分类讨论

$$\rho=\begin{cases} r>0 \\ -r<0 \begin{cases} r\neq N^+ \\ \boxed{r=N^+} \end{cases} & (r\geq 0) \\ r=0 \end{cases}$$

$$y(x)=\sum_{n=0}^{\infty}a_nx^{n+\rho},\ (a_0\neq 0)$$

① $\rho=\pm r$

② $(1+2\rho)a_1=0$

③ $n(n+2\rho)a_n+a_{n-2}=0,\ (n\geq 2)$

2 当 $\rho=-r<0$ 时 $a_0\neq 0$

2.2 当r为正整数时 $r=l$ 任意确定正整数 原幂级数解

$$y(x)=x^{\rho}\sum_{n=0}^{\infty}a_nx^n=x^{-r}\sum_{n=0}^{\infty}a_nx^n=\sum_{k=0}^{\infty}a_{2k}x^{-l+2k},\ (a_0\neq 0)$$

$$=\underbrace{\left[a_0x^{-l}+a_2x^{-l+2}+a_4x^{-l+4}+\cdots+a_{2(l-1)}x^{-l+2(l-1)}\right]}_{a_{2l-2}=a_{2l-4}=\cdots=a_0=0}+\underbrace{a_{2l}x^{-l+2l}+a_{2(l+1)}x^{-l+2(l+1)}+\cdots+a_{2k}x^{-l+2k}+\cdots}_{\text{系数依然满足递推式3}}$$

$$=\sum_{k=l}^{\infty}a_{2k}x^{-l+2k}$$

$$=x^{-l}\sum_{k=l}^{\infty}a_{2k}x^{2k}$$

$$4k(k-l)a_{2l}+a_{2l-2}=0,\ (k>l)$$

说明原幂级数解形式不妥→不应有幂次低于 $2l$ 的项

所有系数只可递推至 a_{2l}

$$a_{2k}\to a_{2k-2}\to a_{2k-4}\to\cdots\to a_{2(l+2)}\to a_{2(l+1)}\to a_{2l}$$

原形式解序号/幂次对应关系不合理→不妨把前 l 项 ($k=0\sim l-1$) 取掉→新假设解为

$$y(x)=x^{\rho}\sum_{k=l}^{\infty}a_{2k}x^{2k}=x^{-l}\sum_{k=l}^{\infty}a_{2k}x^{2k}=\sum_{k=l}^{\infty}a_{2k}x^{-l+2k},\ (a_0\neq 0)$$

从l开始累加

3.2 Bessel函数

$$x^2y''+xy'+\left(x^2-r^2\right)y=0$$

Bessel方程/Bessel函数

3 阶数 r 分类讨论

$$\rho=\begin{cases} r>0 \\ -r<0 \begin{cases} r\neq N^+ \\ \boxed{r=N^+} \end{cases} & (r\geq 0) \\ r=0 \end{cases}$$

$$y(x)=\sum_{n=0}^{\infty}a_nx^{n+\rho},\ (a_0\neq 0)$$

$$\begin{cases} \rho=\pm r \\ (1+2\rho)a_1=0 \\ n(n+2\rho)a_n+a_{n-2}=0,\ (n\geq 2) \end{cases}$$

2 当 $\rho=-r<0$ 时 $a_0\neq 0$

2.2 当r为正整数时 $r=l$

任意确定正整数

新假设解为 $y(x)=x^{-l}\sum_{k=l}^{\infty}a_{2k}x^{2k}$

相当于除去了原级数的前 l 项→新解代入→系数递推式是前述递推式在 $k>l$ 的特殊区段

$$4k(k-l)a_{2k}+a_{2k-2}=0,\ (k>l)$$

$$a_{2k}=-\frac{a_{2k-2}}{4k(k-l)}$$

其系数是 $\rho=-r<0,\ \left(r\neq N^+\right)$ 时系数表达式在 $k>l$ 时的特殊情况

$\rho=-r<0,\ \left(r\neq N^+\right)$

$$a_{2k}=\frac{(-1)^k}{2^{-r}\cdot 2^{2k}\cdot k!\,\Gamma(k-r+1)}$$

$\rho=-r=-l,\ \left(r=N^+,\ k\geq l\right)$

$$a_{2k}=\frac{(-1)^k}{2^{-l}\cdot 2^{2k}\cdot k!\,\Gamma(k-l+1)}$$

代入新幂级数解 $y(x)=\sum_{k=l}^{\infty}\frac{(-1)^k}{2^{-l}\cdot 2^{2k}\cdot k!\,\Gamma(k-l+1)}x^{2k-l}$

$$=\sum_{k=0}^{\infty}\frac{(-1)^k}{2^{-l}\cdot 2^{2k}\cdot k!\,\Gamma(k-l+1)}x^{2k-l}$$

起点可改回0

惊喜 当 $k<l$ 时，$k-l+1$是负整数 $\frac{1}{\Gamma(k-l+1)}=0$

k起点改回0→形式与 $J_r(x)$ 统一 →得另1个非0解

$$J_{-r}(x)=\left(\frac{x}{2}\right)^{-r}\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\,\Gamma(k-r+1)}\left(\frac{x}{2}\right)^{2k}$$

3.2 Bessel函数

Bessel方程/Bessel函数

4 解的线性相关性

$$\rho = \begin{cases} r > 0 \\ -r < 0 \begin{cases} r \neq N^+ \\ r = N^+ \end{cases}, \\ r = 0 \end{cases} \quad (r \geq 0)$$

$$x^2 y'' + xy' + \left(x^2 - r^2\right) y = 0$$

$$y(x) = \sum_{n=0}^{\infty} a_n x^{n+\rho}, \ (a_0 \neq 0)$$

至此, 已得到 $r > 0$ 时Bessel方程的两个解

$$J_r(x) = \left(\frac{x}{2}\right)^r \sum_{k=0}^{\infty} \frac{(-1)^k}{k!\,\Gamma(k+r+1)} \left(\frac{x}{2}\right)^{2k} \qquad J_{-r}(x) = \left(\frac{x}{2}\right)^{-r} \sum_{k=0}^{\infty} \frac{(-1)^k}{k!\,\Gamma(k-r+1)} \left(\frac{x}{2}\right)^{2k}$$

当 $r > 0$ 不为正整数

幂级数部分

$$S_k(0) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!\,\Gamma(k \pm r+1)} \left(\frac{x}{2}\right)^{2k}$$

$$= \underbrace{\frac{1}{\Gamma(\pm r+1)}}_{k=0} + \sum_{k=1}^{\infty} \frac{(-1)^k}{k!\,\Gamma(k \pm r+1)} \underbrace{\left(\frac{0}{2}\right)^{2k}}_{=0}$$

$$= \frac{1}{\Gamma(\pm r+1)} \neq 0$$ 有限值

幂函数部分 $x = 0, \left(\frac{x}{2}\right)^r = 0, \left(\frac{x}{2}\right)^{-r} \to \infty \Rightarrow \frac{J_{-r}(0)}{J_r(0)} \neq \text{Const.}$

说明当 $r > 0$ 不为正整数时, $\{J_{-r}(x), J_r(x)\}$ 线性无关→是Bessel方程的2个基解

3.2 Bessel函数

Bessel方程/Bessel函数

4 解的线性相关性

$$\rho=\begin{cases} r>0 \\ -r<0 & \begin{cases} r\neq N^+ \\ r=N^+ \end{cases}, \quad (r\geq 0) \\ r=0 \end{cases}$$

$$x^2y''+xy'+\left(x^2-r^2\right)y=0$$

$$y(x)=\sum_{n=0}^{\infty}a_n x^{n+\rho},\ \left(a_0\neq 0\right)$$

$$J_r(x)=\left(\frac{x}{2}\right)^r\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\,\Gamma(k+r+1)}\left(\frac{x}{2}\right)^{2k}$$

$$J_{-r}(x)=\left(\frac{x}{2}\right)^{-r}\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\,\Gamma(k-r+1)}\left(\frac{x}{2}\right)^{2k}$$

当 $r>0$ 且为正整数时

$$J_{-n}(x)=\left(\frac{x}{2}\right)^{-n}\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\,\Gamma(k-n+1)}\left(\frac{x}{2}\right)^{2k}$$

$k=n+j$

$$=\left(\frac{x}{2}\right)^{-n}\sum_{j=0}^{\infty}\frac{(-1)^{n+j}}{(n+j)!\,\Gamma(j+1)}\left(\frac{x}{2}\right)^{2n+2j}$$

序号是 j

$$=(-1)^n\left(\frac{x}{2}\right)^{n}\sum_{j=0}^{\infty}\frac{(-1)^j}{j!\,(j+n)\cdot(j+n-1)\cdot\cdots\cdot(j+2)\cdot(j+1)\cdot\Gamma(j+1)}\left(\frac{x}{2}\right)^{2j}$$

$$=(-1)^n\left(\frac{x}{2}\right)^{n}\sum_{j=0}^{\infty}\frac{(-1)^j}{j!\,\Gamma(n+j+1)}\left(\frac{x}{2}\right)^{2j}$$

$\Gamma(\alpha+1)=\alpha\cdot\Gamma(\alpha)$

$$=(-1)^n J_n(x)$$

说明 $J_{-n}(x)$, $J_n(x)$ 线性相关→还需另寻1个基解

当 $r=0$ 时 $J_{-0}(x)=J_0(x)$, 仅有一解

3.2 Bessel函数

Bessel方程/Bessel函数

4 解的线性相关性

$$\rho=\begin{cases} r>0 & \begin{cases} r\neq N^+ \\ \boxed{r=N^+} \end{cases} \\ -r<0 & \\ \boxed{r=0} & \end{cases} \quad (r\geq 0)$$

$$x^2y''+xy'+\left(x^2-r^2\right)y=0$$

$$J_r(x)=\left(\frac{x}{2}\right)^r\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\cdot\Gamma(k+r+1)}\left(\frac{x}{2}\right)^{2k}$$

$$J_{-r}(x)=\left(\frac{x}{2}\right)^{-r}\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\cdot\Gamma(k-r+1)}\left(\frac{x}{2}\right)^{2k}$$

当 $r>0$ 为正整数时 $J_{-n}(x)$, $J_n(x)$ 线性相关

当 $r=0$ 时 $J_{-0}(x)=J_0(x)$，仅有一解

另寻线性无关基解

C.Neumann
1832-1925
German Math.

摄动法 对于任意非负整数 $n\geq 0$，假设阶数 $n<r<n+1$

定义 $N_r(x)=\dfrac{J_r(x)\cos(r\pi)-J_{-r}(x)}{\sin(r\pi)}$ →是Bessel方程解

$$N_n(x)=\lim_{r\to n^+}N_r(x)$$

视为x, r二元函数

$$=\lim_{r\to n^+}\frac{J_r(x)\cos(r\pi)-J_{-r}(x)}{\sin(r\pi)}$$

L'Hospital法则

$$=\lim_{r\to n^+}\frac{\dfrac{\partial\left[J_r(x)\right]}{\partial r}\cos(r\pi)-\pi\sin(r\pi)J_r(x)-\dfrac{\partial\left[J_{-r}(x)\right]}{\partial r}}{\pi\cos(r\pi)}$$

$$=\frac{1}{\pi}\left[\lim_{r\to n^+}\frac{\partial\left[J_r(x)\right]}{\partial r}-(-1)^n\lim_{r\to n^+}\frac{\partial\left[J_{-r}(x)\right]}{\partial r}\right]$$

1. 该极限存在→第二类Bessel函数

$$N_n(x)=\lim_{r\to n^+}\frac{J_r(x)\cos(r\pi)-J_{-r}(x)}{\sin(r\pi)}$$ **1865**

2. $J_n(0)=0,1$ 有限值 $\lim\limits_{x\to 0^+}N_n(x)=-\infty$

$J_n(x)$, $N_n(x)$线性无关→

整数阶Bessel方程基解 $\{J_n(x),\ N_n(x)\}$

3.2 Bessel函数

$$x^2y''+xy'+\left(x^2-r^2\right)y=0$$

Bessel方程/Bessel函数 小结

$$y(x)=\sum_{n=0}^{\infty}a_nx^{n+\rho},\ (a_0\neq 0)$$

*r*阶Bessel方程 $x^2y''+xy'+\left(x^2-r^2\right)y=0$		*r*阶Bessel函数
对任意 $r\geq 0$ 无论是否为整数	$y(x)=C_1J_r(x)+C_2N_r(x)$	$J_r(x)=\left(\frac{x}{2}\right)^r\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\cdot\Gamma(k+r+1)}\left(\frac{x}{2}\right)^{2k}$ $N_n(x)=\lim_{r\to n^+}\frac{J_r(x)\cos(r\pi)-J_{-r}(x)}{\sin(r\pi)}$
对任意 $r>0$ 且不为整数时, 通解为	$y(x)=C_1J_r(x)+C_2J_{-r}(x)$	$J_r(x)=\left(\frac{x}{2}\right)^r\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\cdot\Gamma(k+r+1)}\left(\frac{x}{2}\right)^{2k}$ $J_{-r}(x)=\left(\frac{x}{2}\right)^{-r}\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\cdot\Gamma(k-r+1)}\left(\frac{x}{2}\right)^{2k}$

3.2 Bessel函数

Bessel方程/Bessel函数

$$x^2y''+xy'+\left(x^2-r^2\right)y=0$$

$$J_r(x)=\left(\frac{x}{2}\right)^r\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\,\Gamma(k+r+1)}\left(\frac{x}{2}\right)^{2k}$$

$$J_{-r}(x)=\left(\frac{x}{2}\right)^{-r}\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\,\Gamma(k-r+1)}\left(\frac{x}{2}\right)^{2k}$$

$$N_r(x)=\frac{J_r(x)\cos(r\pi)-J_{-r}(x)}{\sin(r\pi)}$$

例1 写出以下方程的通解

1 $$x^2y''+xy'+\left(x^2-2\right)y=0$$

$$y(x)=C_1J_{\sqrt{2}}(x)+C_2J_{-\sqrt{2}}(x)$$

$$y(x)=C_1J_{\sqrt{2}}(x)+C_2N_{\sqrt{2}}(x)$$

2 $$x^2y''+xy'+\left(4x^2-2\right)y=0$$

解 令 $t=2x$ 原方程中 $y'=y'(x)$

$$y'(x)=y'(t)\frac{dt}{dx}=2y'(t)$$

$$y''(x)=y''(t)\frac{dt}{dx}=4y'(t)$$

代入原方程

$$\frac{t^2}{4}\cdot 4y''(t)+\frac{t}{2}\cdot 2y'(t)+\left(t^2-2\right)y=0$$

$$t^2y''(t)+ty'(t)+\left(t^2-\left(\sqrt{2}\right)^2\right)y=0$$

得 $$y(t)=C_1J_{\sqrt{2}}(t)+C_2J_{-\sqrt{2}}(t)$$

还原变量

$$y(x)=C_1J_{\sqrt{2}}(2x)+C_2J_{-\sqrt{2}}(2x)$$

$$y(x)=C_1J_{\sqrt{2}}(2x)+C_2N_{\sqrt{2}}(2x)$$

3 $$x^2y''+xy'+\left(x^2-16\right)y=0$$

$$y(x)=C_1J_4(x)+C_2N_4(x)$$

4 $$xy''+y'+xy=0$$

$$y(x)=C_1J_0(x)+C_2N_0(x)$$

$$x^2y''+xy'+\left(x^2-r^2\right)y=0$$

Bessel方程/Bessel函数

例2 求ODE的通解 $x^2y''+2xy'+\left[x^2-r\left(r+1\right)\right]y=0$

解 函数代换，令 $y\left(x\right)=x^{-\frac{1}{2}}u\left(x\right)$ 各阶导数 $y'\left(x\right)=-\frac{1}{2}x^{-\frac{3}{2}}u\left(x\right)+x^{-\frac{1}{2}}u'\left(x\right)$

$$y''\left(x\right)=\frac{3}{4}x^{-\frac{5}{2}}u\left(x\right)-\frac{1}{2}x^{-\frac{3}{2}}u'\left(x\right)-\frac{1}{2}x^{-\frac{3}{2}}u'\left(x\right)+x^{-\frac{1}{2}}u''\left(x\right)=\frac{3}{4}x^{-\frac{5}{2}}u\left(x\right)-x^{-\frac{3}{2}}u'\left(x\right)+x^{-\frac{1}{2}}u''\left(x\right)$$

代入原方程

$$x^2\left[\frac{3}{4}x^{-\frac{5}{2}}u\left(x\right)-x^{-\frac{3}{2}}u'\left(x\right)+x^{-\frac{1}{2}}u''\left(x\right)\right]+2x\left[-\frac{1}{2}x^{-\frac{3}{2}}u\left(x\right)+x^{-\frac{1}{2}}u'\left(x\right)\right]+\left[x^2-r\left(r+1\right)\right]x^{-\frac{1}{2}}u\left(x\right)=0$$

$$\left[\frac{3}{4}x^{-\frac{1}{2}}u\left(x\right)-x^{\frac{1}{2}}u'\left(x\right)+x^{\frac{3}{2}}u''\left(x\right)\right]+\left[-x^{-\frac{1}{2}}u\left(x\right)+2x^{\frac{1}{2}}u'\left(x\right)\right]+\left[x^2-r\left(r+1\right)\right]x^{-\frac{1}{2}}u\left(x\right)=0$$

两边同乘以 $x^{\frac{1}{2}}$

$$\left[\frac{3}{4}u\left(x\right)-xu'\left(x\right)+x^2u''\left(x\right)\right]+\left[-u\left(x\right)+2xu'\left(x\right)\right]+\left[x^2-r\left(r+1\right)\right]u\left(x\right)=0$$

$$x^2u''\left(x\right)+xu'\left(x\right)-\frac{1}{4}u\left(x\right)+\left[x^2-r\left(r+1\right)\right]u\left(x\right)=0$$

配平方

$$x^2u''\left(x\right)+xu'\left(x\right)+\left[x^2-r\left(r+1\right)-\frac{1}{4}\right]u\left(x\right)=0$$

原方程的通解可表示为

$$y\left(x\right)=x^{-\frac{1}{2}}u\left(x\right)=x^{-\frac{1}{2}}\left[C_1J_{r+\frac{1}{2}}\left(x\right)+C_2N_{r+\frac{1}{2}}\left(x\right)\right]$$

$$x^2u''\left(x\right)+xu'\left(x\right)+\left[x^2-\left(r+\frac{1}{2}\right)^2\right]u\left(x\right)=0$$

Bessel方程

$$r+\frac{1}{2}\geq 0$$

可能是整数

1. 数学建模和基本原理介绍
2. 分离变量法
3. Bessel函数
    - 3.1 二阶线性常微分方程的幂级数解法
    - 3.2 Bessel函数
        - Gamma函数
        - Bessel方程和Bessel函数
        - Bessel函数的性质
        - Bessel方程的特征值问题
        - 圆域上Laplace算子的特征值问题
    - 3.3 多个自变量分离变量法举例
4. 积分变换法
5. Green函数法
6. 特征线法
7. Legendre多项式

3.2 Bessel函数

$$J_r(x)=\left(\frac{x}{2}\right)^r\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\cdot\Gamma(k+r+1)}\left(\frac{x}{2}\right)^{2k}$$

第一类Bessel函数

$$\sin x=\sum_{k=0}^{\infty}\frac{(-1)^k}{(2k+1)!}x^{2k+1}$$

收敛半径

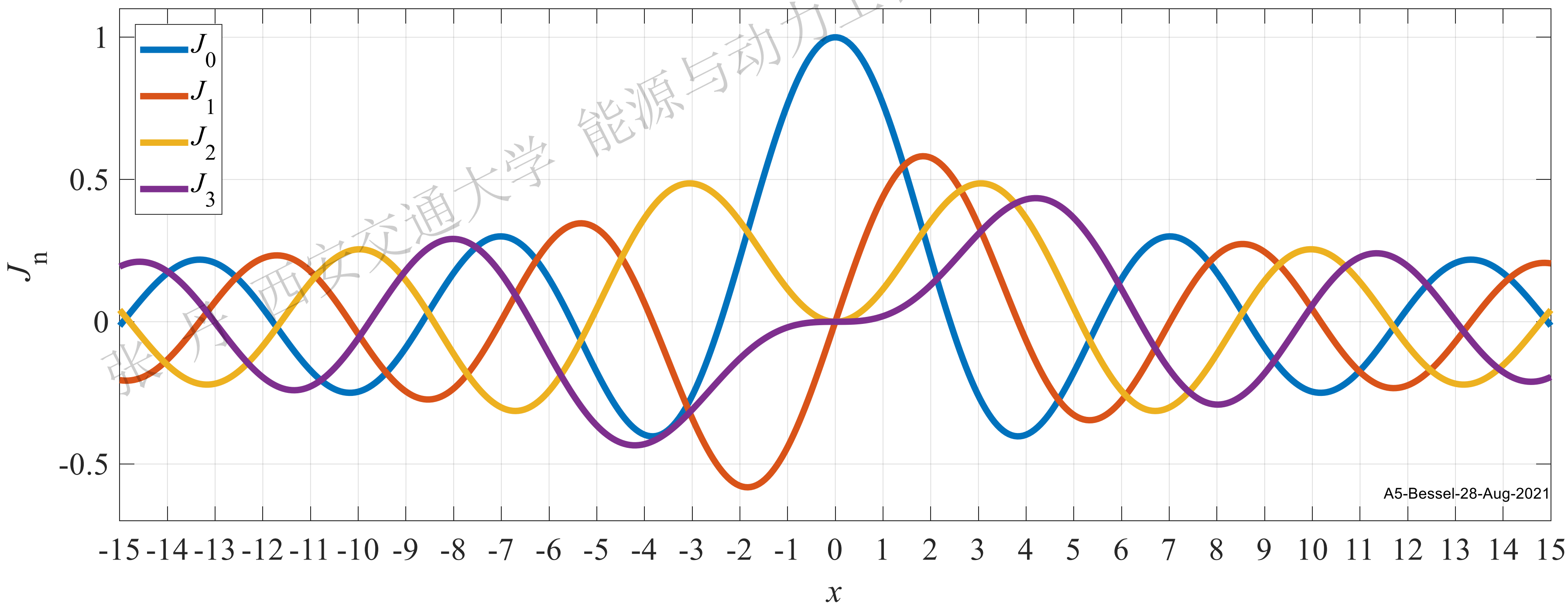
Bessel函数幂级数部分 $S_k(x)=\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\cdot\Gamma(k\pm r+1)}\left(\frac{x}{2}\right)^{2k}$

$$\cos x=\sum_{k=0}^{\infty}\frac{(-1)^k}{(2k)!}x^{2k}$$

D'Alembert 判别法 $\lim\limits_{n\to\infty}\left|\frac{a_n}{a_{n+1}}\right|=\lim\limits_{n\to\infty}\frac{(n+1)!\Gamma(n+2+r)2^{2n+2}}{n!\Gamma(n+1+r)2^{2n}}=\lim\limits_{n\to\infty}4(n+1)(n+1+r)=\infty$

Bessel函数 $J_r(x), J_{-r}(x)$ 收敛半径无穷大→幂级数在实数域 $\boldsymbol{R}$ 上收敛→ $J_r(x)$ 解析

类似 sin(x), cos(x) →后面重点讨论 r 为正整数 n 时→ $J_n(x)$ 在实数域上的性质



3.2 Bessel函数

$$J_n(x)=\left(\frac{x}{2}\right)^n\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\cdot\Gamma(k+n+1)}\left(\frac{x}{2}\right)^{2k}$$

第一类Bessel函数

基本值

❶ $J_n(0)=\begin{cases}1, & (n=0)\\ 0, & (n\ge 1)\end{cases}$

❷ $J_n'(0)=\begin{cases}0, & (n=0,n>1)\\ \frac{1}{2}, & (n=1)\end{cases}$ 后续证明

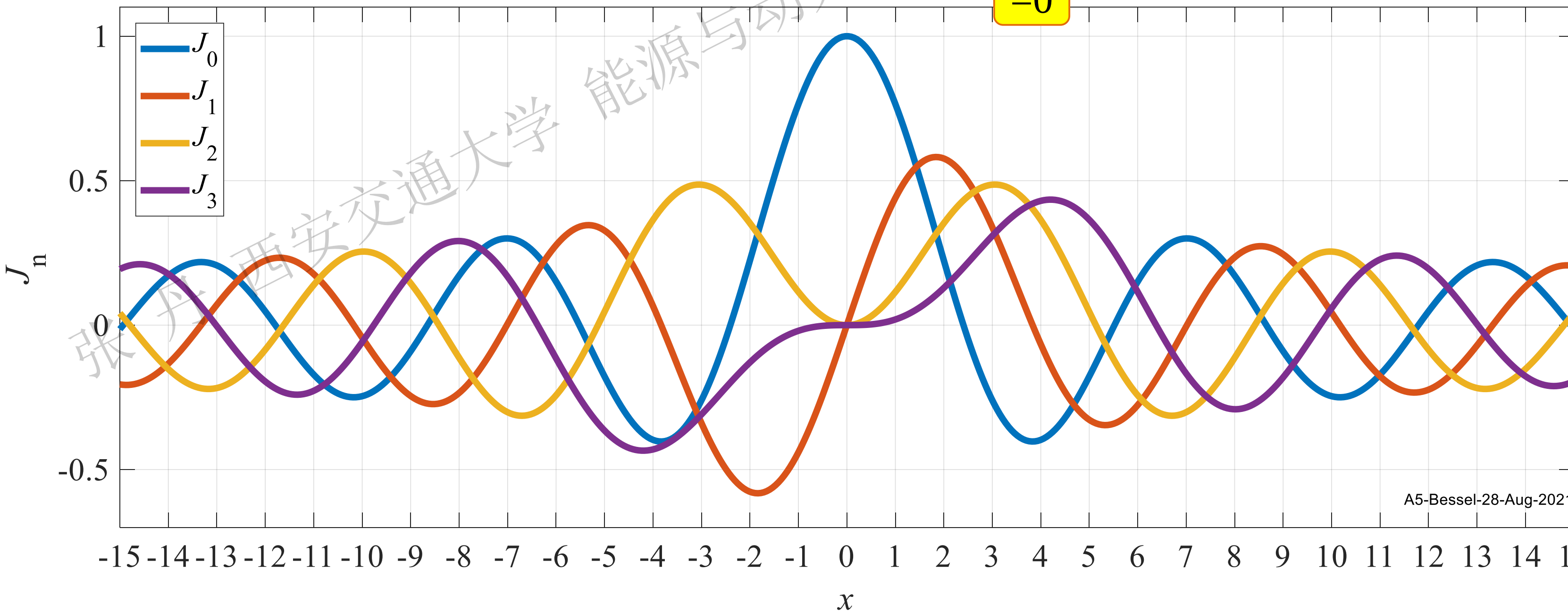
❸ $|J_n(x)|<+\infty$ 有界性

证 ❶

$$J_n(x)=\left(\frac{x}{2}\right)^n\left[\frac{1}{\Gamma(n+1)}+\sum_{k=1}^{\infty}\frac{(-1)^k}{k!\cdot\Gamma(k+n+1)}\left(\frac{x}{2}\right)^{2k}\right]$$

$J_n(0)$ 的不同由幂函数造成

$$J_n(0)=\left(\frac{0}{2}\right)^n\left[\frac{1}{\Gamma(n+1)}+\sum_{k=1}^{\infty}\frac{(-1)^k}{k!\cdot\Gamma(k+n+1)}\underbrace{\left(\frac{0}{2}\right)^{2k}}_{=0}\right]=\begin{cases}1, & (n=0)\\ 0, & (n>0)\end{cases}$$



3.2 Bessel函数

$$J_n(x)=\left(\frac{x}{2}\right)^n\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\,\Gamma(k+n+1)}\left(\frac{x}{2}\right)^{2k}$$

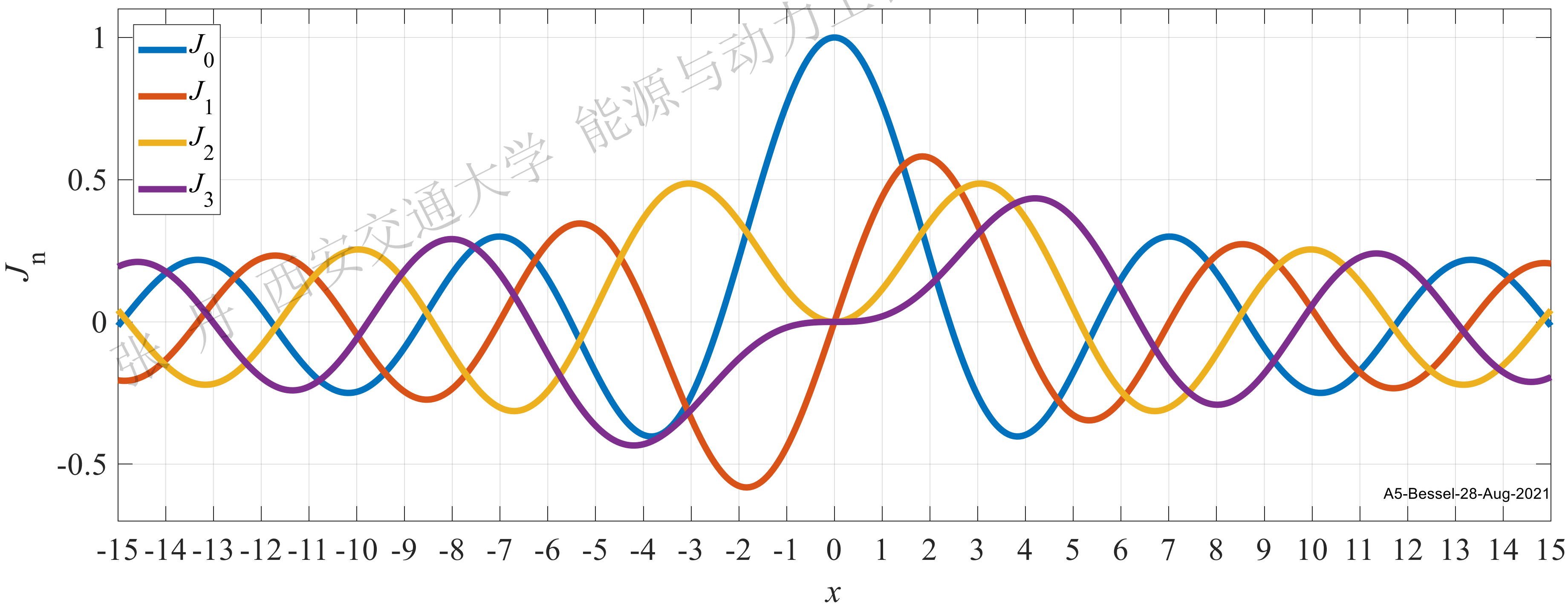
第一类Bessel函数

奇偶性

$$J_n(x)=(-1)^n J_n(-x)\quad\begin{cases} n\text{为奇数时},\ J_n(x)\text{为奇函数}\\ n\text{为偶数时},\ J_n(x)\text{为偶函数}\end{cases}$$

证 由定义式

$$J_n(-x)=\left(\frac{-x}{2}\right)^n\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\,\Gamma(k+n+1)}\left(\frac{-x}{2}\right)^{2k}=(-1)^n\left(\frac{x}{2}\right)^n\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\,\Gamma(k+n+1)}\left(\frac{x}{2}\right)^{2k}=(-1)^n J_n(x)$$



3.2 Bessel函数

$$J_n(x)=\left(\frac{x}{2}\right)^n\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\cdot\Gamma(k+n+1)}\left(\frac{x}{2}\right)^{2k}$$

第一类Bessel函数

递推性

① $\frac{d}{dx}\left[x^nJ_n(x)\right]=x^nJ_{n-1}(x)$

② $\frac{d}{dx}\left[x^{-n}J_n(x)\right]=-x^{-n}J_{n+1}(x)$

③ $J_{n-1}(x)+J_{n+1}(x)=\frac{2n}{x}J_n(x)$

④ $J_{n-1}(x)-J_{n+1}(x)=2J_n'(x)$

证① $\frac{d}{dx}\left[x^nJ_n(x)\right]$

$$=\frac{d}{dx}\left[x^n\left(\frac{x}{2}\right)^n\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\cdot\Gamma(k+n+1)}\left(\frac{x}{2}\right)^{2k}\right]=\frac{d}{dx}\left[2^n\frac{x^{2n}}{2^n\cdot2^n}\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\cdot\Gamma(k+n+1)}\left(\frac{x}{2}\right)^{2k}\right]$$

分子/分母同乘以2^n

$$=2^n\cdot\frac{d}{dx}\left[\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\cdot\Gamma(k+n+1)}\left(\frac{x}{2}\right)^{2k+2n}\right]$$

与幂次合并→方便求导

$$=2^n\sum_{k=0}^{\infty}\frac{(-1)^k\,2(k+n)}{k!\cdot\Gamma(k+n+1)}\frac{1}{2}\left(\frac{x}{2}\right)^{2k+2n-1}$$

$$\frac{\alpha}{\Gamma(\alpha+1)}=\frac{1}{\Gamma(\alpha)}\quad\leftarrow\quad\Gamma(\alpha+1)=\alpha\cdot\Gamma(\alpha)$$

$$=2^n\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\cdot\Gamma(k+n-1+1)}\left(\frac{x}{2}\right)^{2k}\cdot\left(\frac{x}{2}\right)^{n}\cdot\left(\frac{x}{2}\right)^{n-1}$$

$$=2^n\frac{x^n}{2^n}\cdot\left[\left(\frac{x}{2}\right)^{n-1}\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\cdot\Gamma(k+n-1+1)}\left(\frac{x}{2}\right)^{2k}\right]$$

$$=x^nJ_{n-1}(x)$$

同理可证

$$\frac{d}{dx}\left[x^{-n}J_n(x)\right]=-x^{-n}J_{n+1}(x)$$

3.2 Bessel函数

$$J_n(x)=\left(\frac{x}{2}\right)^n\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\cdot\Gamma(k+n+1)}\left(\frac{x}{2}\right)^{2k}$$

第一类Bessel函数

递推性

① $$\frac{d}{dx}\left[x^nJ_n(x)\right]=x^nJ_{n-1}(x)$$

② $$\frac{d}{dx}\left[x^{-n}J_n(x)\right]=-x^{-n}J_{n+1}(x)$$

③ $$J_{n-1}(x)+J_{n+1}(x)=\frac{2n}{x}J_n(x)$$

④ $$J_{n-1}(x)-J_{n+1}(x)=2J_n'(x)$$

证　式①②左端按乘积求导法则展开

$$x^nJ_n'(x)+nx^{n-1}J_n(x)=x^nJ_{n-1}(x)$$

$$x^{-n}J_n'(x)-nx^{-n-1}J_n(x)=-x^{-n}J_{n+1}(x)$$

约去 x^{n-1}, x^{-n-1}

$$\left.\begin{aligned}xJ_n'(x)+nJ_n(x)&=xJ_{n-1}(x)\\ xJ_n'(x)-nJ_n(x)&=-xJ_{n+1}(x)\end{aligned}\right\}\Rightarrow\left\{\begin{aligned}&\text{相减}\quad J_{n-1}(x)+J_{n+1}(x)=\frac{2n}{x}J_n(x)\ ③\\ &\text{相加}\quad J_{n-1}(x)-J_{n+1}(x)=2J_n'(x)\ ④\end{aligned}\right.$$

后续计算→高阶Bessel函数尽量用低阶Bessel表示→变换下标

$$J_{n+2}(x)=\frac{2(n+1)}{x}J_{n+1}(x)-J_n(x)=J_n(x)-2J_{n+1}'(x)$$

$$J_n(x)=\frac{(n+1)}{x}J_{n+1}(x)+J_{n+1}'(x)$$

1. 任意阶Bessel函数最终可用 $J_0(x),J_1(x)$ 表示
2. 递推式对非整数阶Bessel函数亦成立

3.2 Bessel函数

$$J_n(x)=\left(\frac{x}{2}\right)^n\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\,\Gamma(k+n+1)}\left(\frac{x}{2}\right)^{2k}$$

第一类Bessel函数

递推性 **基本值**

利用导数递推式可证明基本值

$$J_n'(0)=\begin{cases}0, & (n=0,n>1)\\ \dfrac{1}{2}, & (n=1)\end{cases}$$

$$J_{n-1}(x)+J_{n+1}(x)=\frac{2n}{x}J_n(x)$$

$$J_{n-1}(x)-J_{n+1}(x)=2J_n'(x)$$

证 ❶ 当 $n=0$ 时

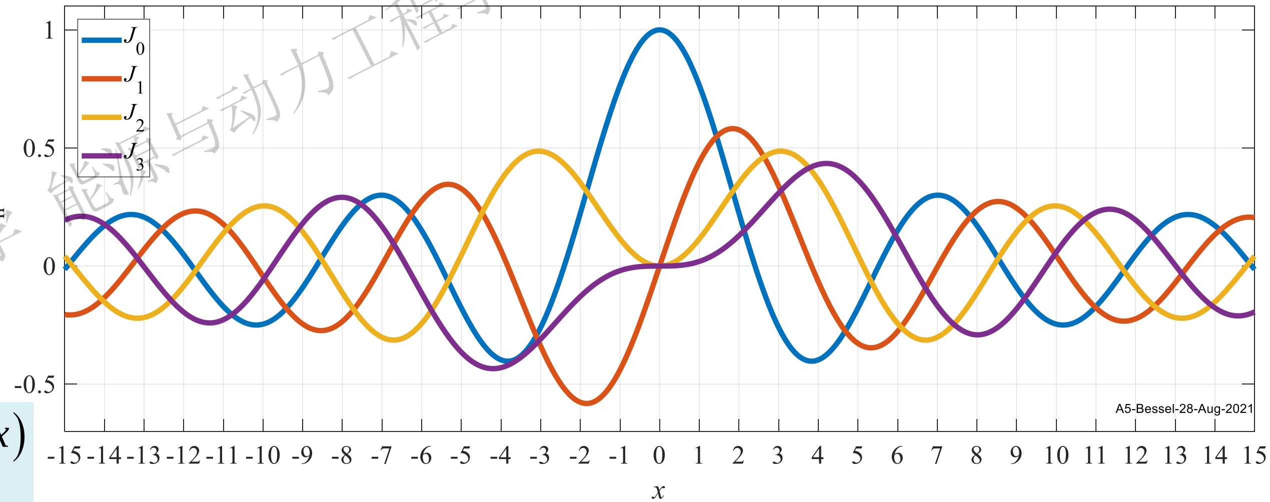
$$J_0'(0)=\frac{d}{dx}\left[1\cdot J_0(0)\right]=\frac{d}{dx}\left[x^{-0}J_0(0)\right]=-\frac{J_1(0)}{x^0}=-\frac{J_1(0)}{1}=0$$

$$\frac{d}{dx}\left[x^{-n}J_n(x)\right]=-x^{-n}J_{n+1}(x)$$

❷ 当 $n=1$ 时

$$J_1'(0)=\frac{1}{2}\left[J_0(0)-J_2(0)\right]=\frac{1}{2}(1-0)=\frac{1}{2}$$

$$J_{n-1}(x)-J_{n+1}(x)=2J_n'(x)$$



❸ 当 $n>1$ 时 $J_n'(0)=\dfrac{1}{2}\left[J_{n-1}(0)-J_{n+1}(0)\right]=\dfrac{1}{2}[0-0]=0$

3.2 Bessel函数

$$J_n(x)=\left(\frac{x}{2}\right)^n\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\,\Gamma(k+n+1)}\left(\frac{x}{2}\right)^{2k}$$

第一类Bessel函数 递推性

Bessel函数积分公式

❶ $$\frac{d}{dx}\left[x^nJ_n(x)\right]=x^nJ_{n-1}(x)$$

❷ $$\frac{d}{dx}\left[x^{-n}J_n(x)\right]=-x^{-n}J_{n+1}(x)$$

❸ $$J_{n-1}(x)+J_{n+1}(x)=\frac{2n}{x}J_n(x)$$

❹ $$J_{n-1}(x)-J_{n+1}(x)=2J_n'(x)$$

变形 ❶ 式→升阶积分公式→积分值用相邻高阶Bessel函数表示

$$\frac{d}{dx}\left[x^nJ_n(x)\right]=x^nJ_{n-1}(x)\ \rightarrow\ \int x^nJ_{n-1}(x)\,dx=x^nJ_n(x)+C$$

变形 ❷ 式→降阶积分公式→积分值用相邻低阶Bessel函数表示

$$\frac{d}{dx}\left[x^{-n}J_n(x)\right]=-x^{-n}J_{n+1}(x)\ \rightarrow\ \int x^{-n}J_{n+1}(x)\,dx=-x^{-n}J_n(x)+C$$

$$\int\frac{J_{n+1}(x)}{x^n}dx=-\frac{J_n(x)}{x^n}+C$$

本质是含有Bessel函数的积分→后续Bessel展开/极坐标PDE的求解发挥重要作用

3.2 Bessel函数

$$J_n(x)=\left(\frac{x}{2}\right)^n\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\cdot\Gamma(k+n+1)}\left(\frac{x}{2}\right)^{2k}$$

第一类Bessel函数

零点 方程 $J_n(x)=0$ 的根

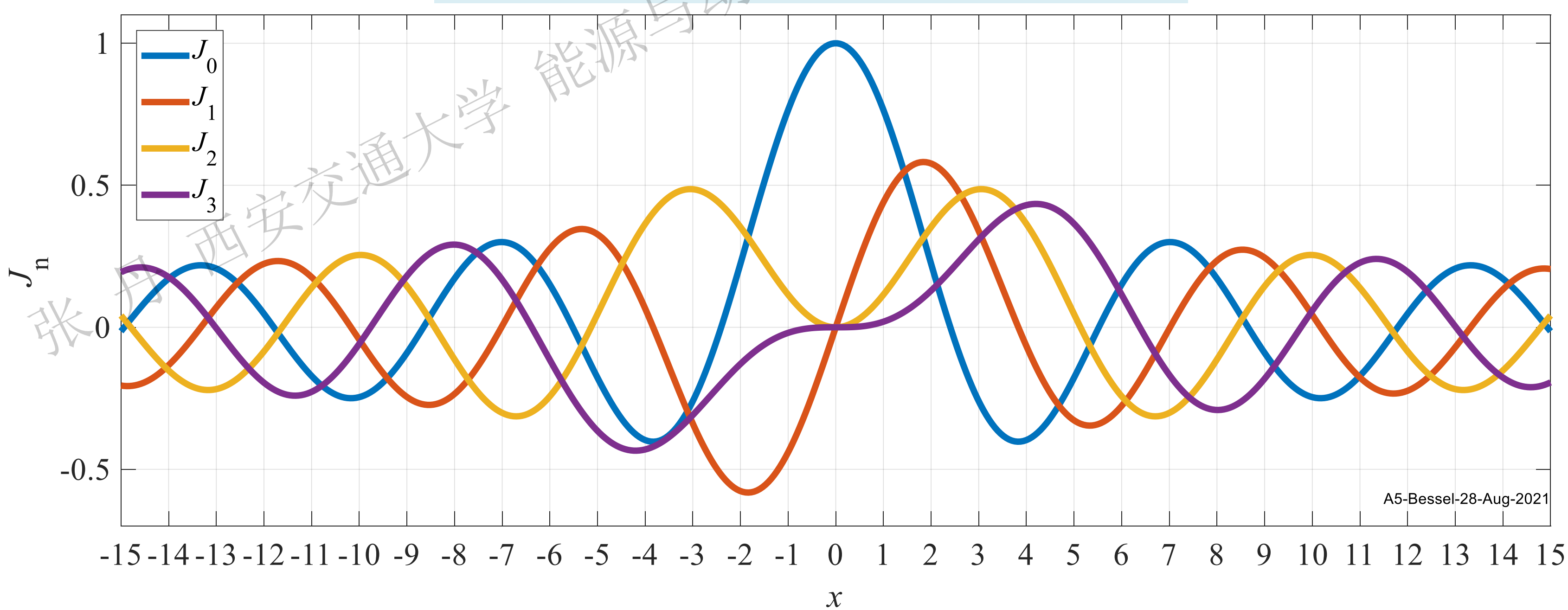
$$\mu_m^{(n)}$$

Bessel函数阶数

原点起正根序号

1. 零点表示符号
2. Bessel函数 $J_n(x)$ 有无数个实零点, 实零点在 x 轴上关于 原点 x=0 对称→关注正根
3. 原点 x=0 不是 $J_0(x)$ 的零点; 但是 $J_n(x)$ $(n > 0)$ 的n重零点. 如 x=0 即是 $J_3(x)$ 的零点也是 $J_5(x)$ 的零点
4. 但除 x=0 外, $J_n(x)$ 的其他零点均为单零点. 如除 x=0外, $J_3(x)$ 的其他零点一定不是 $J_5(x)$ 的零点

$$J_n\left[\mu_m^{(n)}\right]=0,\ J_q\left[\mu_m^{(n)}\right]\neq 0\ \left(m\geq 1,\ n\neq q\right)$$



3.2 Bessel函数

$$J_n(x)=\left(\frac{x}{2}\right)^n\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\cdot\Gamma(k+n+1)}\left(\frac{x}{2}\right)^{2k}$$

第一类Bessel函数

$\mu_m^{(n)}$

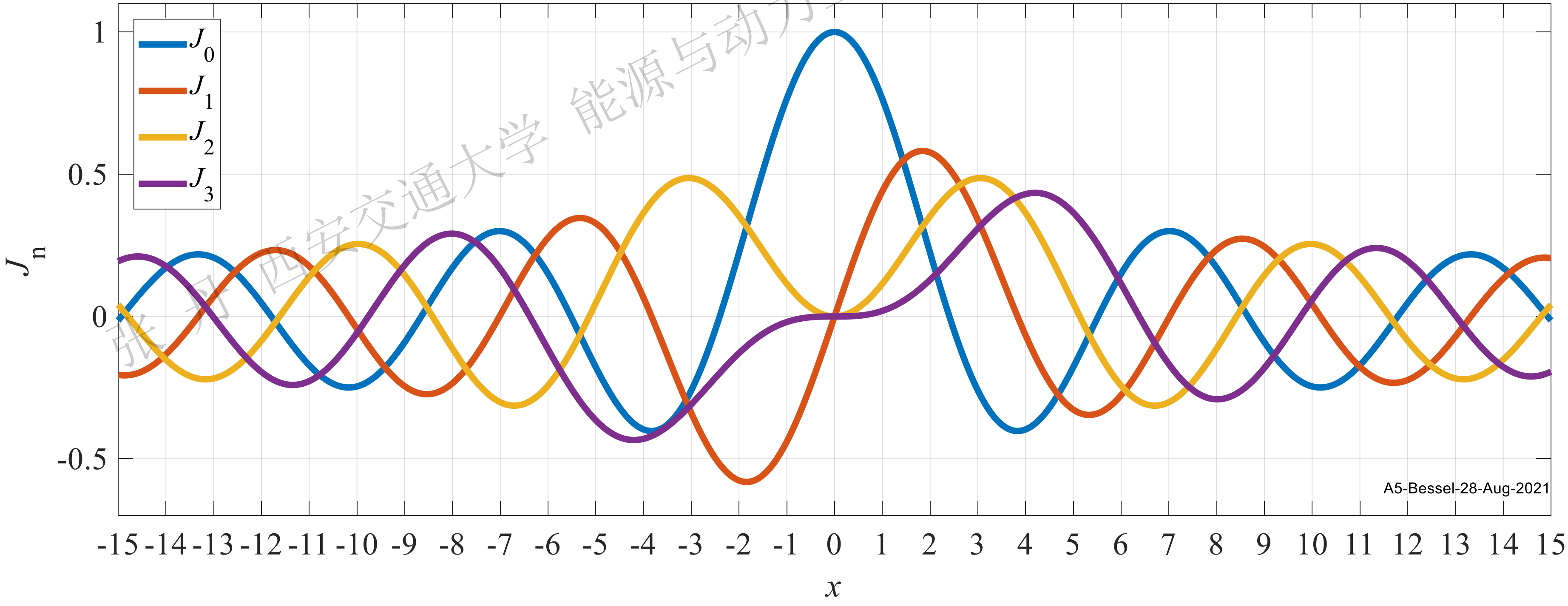
Bessel函数阶数

原点起正根序号

零点 方程 $J_n(x)=0$ 的根

5. 相邻阶数的零点交错分布. 如 2个$J_1(x)$ 的零点之间只有1个 $J_2(x)$ 的零点
6. 随 x 的增大→任意阶Bessel函数的零点间距越来越接近π， $\lim_{m\to\infty}\left(\mu_{m+1}^{(n)}-\mu_m^{(n)}\right)=\pi$
7. 随 x 的增大→任意阶Bessel函数→渐进表达式→趋近于三角函数→减幅振荡/有界

$$\lim_{x\to\infty}J_n(x)=\sqrt{\frac{2}{\pi x}}\cos\left(x-\frac{n\pi}{2}-\frac{\pi}{4}\right)+O\left(x^{-\frac{3}{2}}\right) \Rightarrow |J_n(x)|<+\infty \text{ 有界性}$$



3.2 Bessel函数

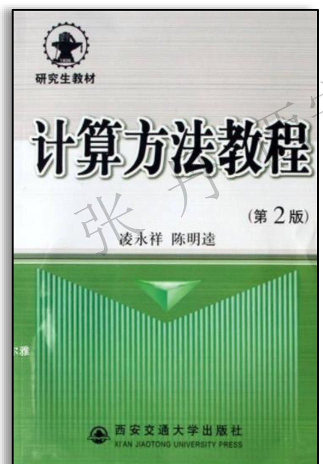
第一类Bessel函数

零点

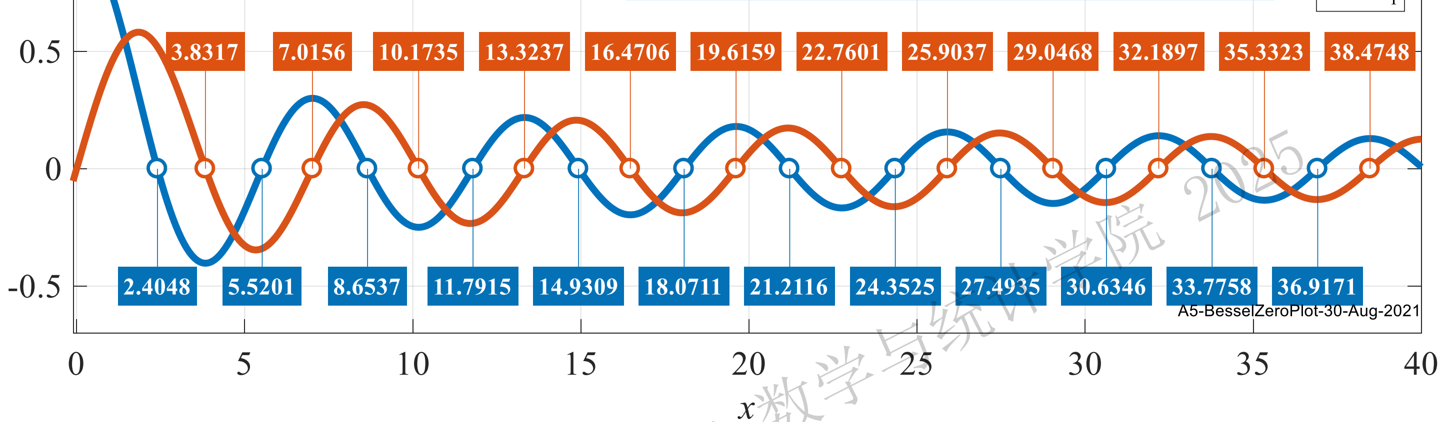
符号/无解析式

$$\mu_m^{(n)}$$

正零点→时空分离常数/Bessel 展开→但Bessel函数正零点无解析式→ 数值计算→高精度近似值



$$J_n(x)=\left(\frac{x}{2}\right)^n\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\cdot\Gamma(k+n+1)}\left(\frac{x}{2}\right)^{2k}$$



各阶Bessel函数的正零点

$\mu_m^{(n)}$ \ $J_n(x)$	0	1	2	3	4	5	......
1	2.404826	3.831706	5.135622	6.380162	7.588342	8.771484	......
2	5.520078	7.015587	8.417244	9.761023	11.06471	12.33860	......
3	8.653728	10.17347	11.61984	13.01520	14.37254	15.70017	......
4	11.79153	13.32369	14.79595	16.22347	17.61597	18.98013	......
5	14.93092	16.47063	17.95982	19.40942	20.82693	22.21780	......
6	18.07106	19.61586	21.11700	22.58273	24.01902	25.43034	......
7	21.21164	22.76008	24.27011	25.74817	27.19909	28.62662	......
8	24.35247	25.90367	27.42057	28.90835	30.37101	31.81172	......
9	27.49348	29.04683	30.56920	32.06485	33.53714	34.98878	......
10	30.63461	32.18968	33.71652	35.21867	36.69900	38.15987	......
11	33.77582	35.33231	36.86286	38.37047	39.85763	41.32638	......
......	......	......	......	......	......	......	......

3.2 Bessel函数

$$J_n(x)=\left(\frac{x}{2}\right)^n\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\cdot\Gamma(k+n+1)}\left(\frac{x}{2}\right)^{2k}$$

第一类Bessel函数 小结

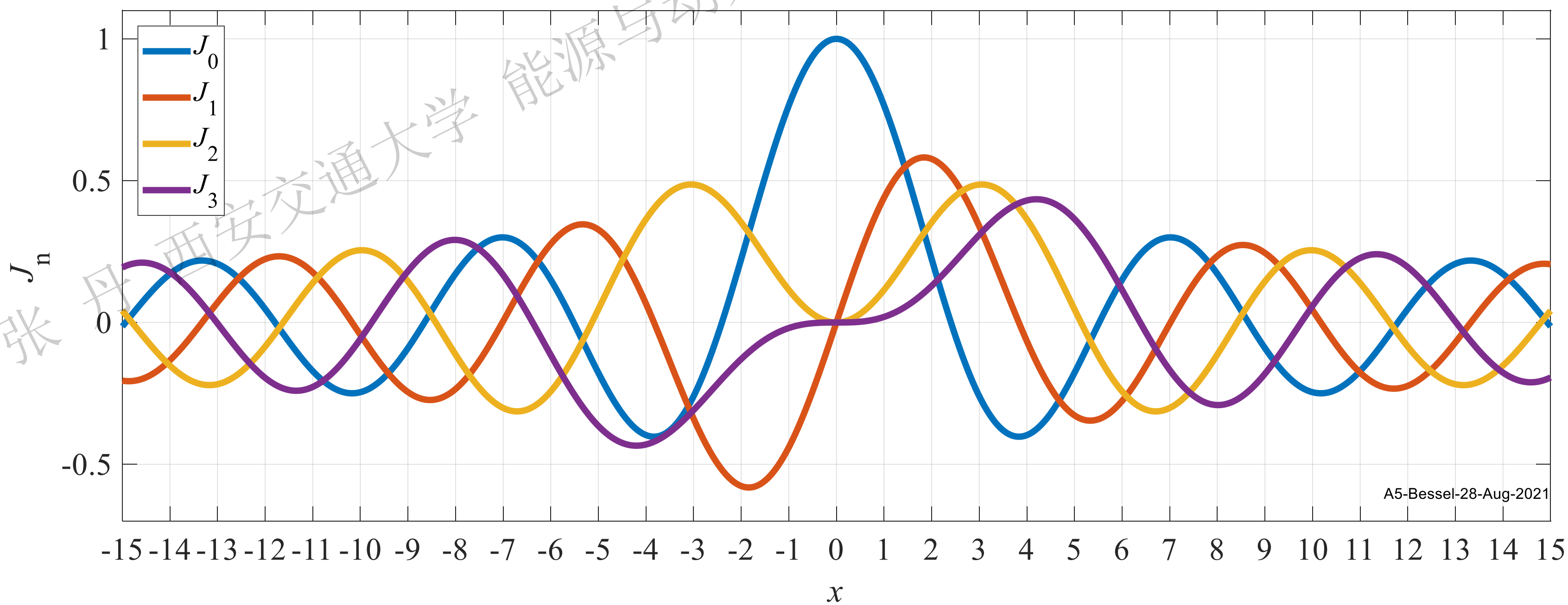
$$J_n(0)=\begin{cases}1, & (n=0)\\ 0, & (n\ge 1)\end{cases}\quad J_n'(0)=\begin{cases}0, & (n=0,n>1)\\ \frac{1}{2}, & (n=1)\end{cases}$$

$$J_n(-x)=(-1)^n J_n(x)\quad \frac{d}{dx}\left[x^nJ_n(x)\right]=x^nJ_{n-1}(x)\quad \frac{d}{dx}\left[x^{-n}J_n(x)\right]=-x^{-n}J_{n+1}(x)$$

$$\int x^nJ_{n-1}(x)dx=x^nJ_n(x)+C,\quad \int\frac{J_{n+1}(x)}{x^n}dx=-\frac{J_n(x)}{x^n}+C$$

$$J_{n-1}(x)+J_{n+1}(x)=\frac{2n}{x}J_n(x)\quad J_{n-1}(x)-J_{n+1}(x)=2J_n'(x)$$

$$\lim_{m\to\infty}\left(\mu_{m+1}^{(n)}-\mu_m^{(n)}\right)=\pi\quad \lim_{x\to\infty}J_n(x)=\sqrt{\frac{2}{\pi x}}\cos\left(x-\frac{n\pi}{2}-\frac{\pi}{4}\right)+O\left(x^{-\frac{3}{2}}\right)$$



3.2 Bessel函数

$$J_n(x)=\left(\frac{x}{2}\right)^n\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\,\Gamma(k+n+1)}\left(\frac{x}{2}\right)^{2k}$$

第二类Bessel函数 常识性的结论

$$J_r(x)=\left(\frac{x}{2}\right)^r\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\,\Gamma(k+r+1)}\left(\frac{x}{2}\right)^{2k}\quad \boxed{N_n(x)=\lim_{r\to n^+}\frac{J_r(x)\cos(r\pi)-J_{-r}(x)}{\sin(r\pi)}}$$

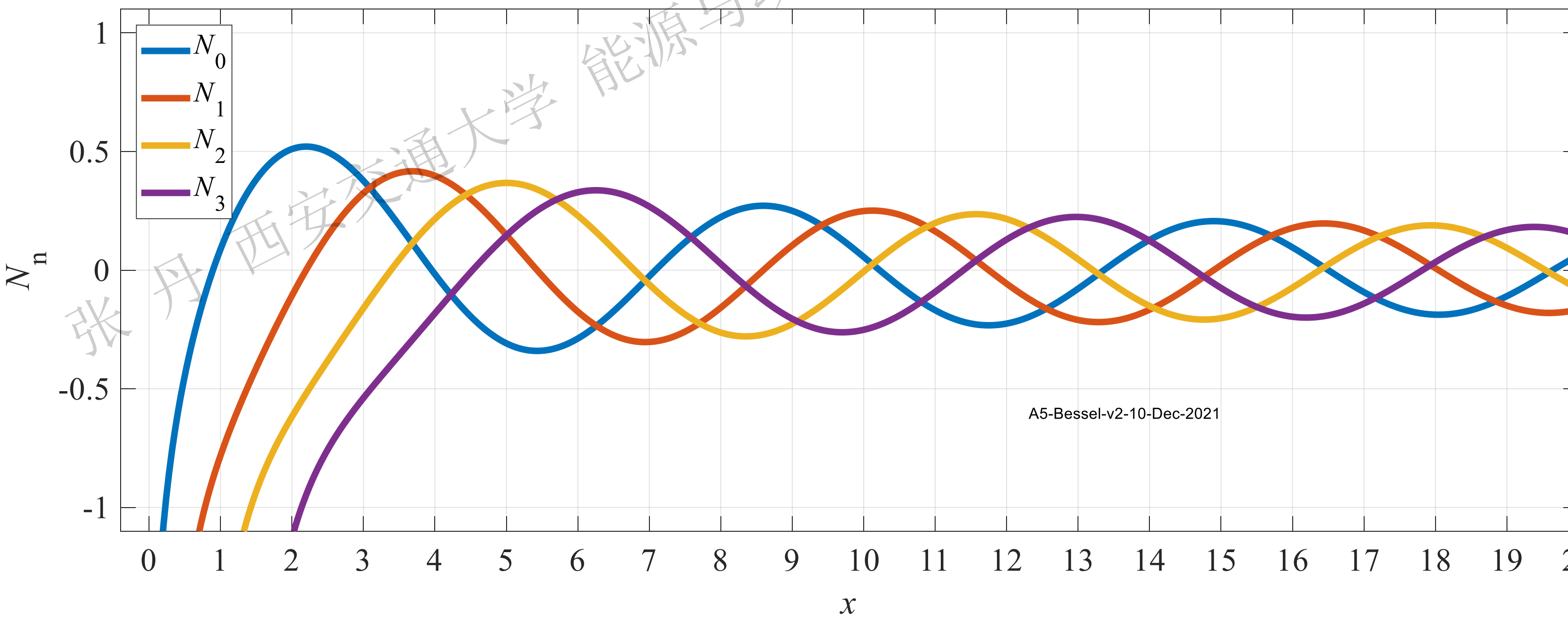
$$\boxed{N_n(0)=-\infty,}\quad N_n(-x)=(-1)^n N_n(x)$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\left[x^n N_n(x)\right]=x^n N_{n-1}(x),\quad \frac{\mathrm{d}}{\mathrm{d}x}\left[x^{-n}N_n(x)\right]=-x^{-n}N_{n+1}(x)$$

$$N_{n-1}(x)+N_{n+1}(x)=\frac{2n}{x}N_n(x),\quad N_{n-1}(x)-N_{n+1}(x)=2N_n'(x)$$

减幅振荡

$$N_n(x)=\sqrt{\frac{2}{x\pi}}\sin\left(x-\frac{1}{4}\pi-\frac{n}{2}\pi\right)+O\left(\frac{1}{x^{3/2}}\right),\quad \boxed{\lim_{x\to\infty}J_n(x)=\lim_{x\to\infty}N_n(x)=0}$$



3.2 Bessel函数

$$J_n(x)=\left(\frac{x}{2}\right)^n\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\,\Gamma(k+n+1)}\left(\frac{x}{2}\right)^{2k}$$

Bessel函数比较

$$J_n(0)=\begin{cases}1, & (n=0)\\ 0, & (n\ge 1)\end{cases}\qquad J_n'(0)=\begin{cases}0, & (n=0,n>1)\\ \dfrac{1}{2}, & (n=1)\end{cases}$$

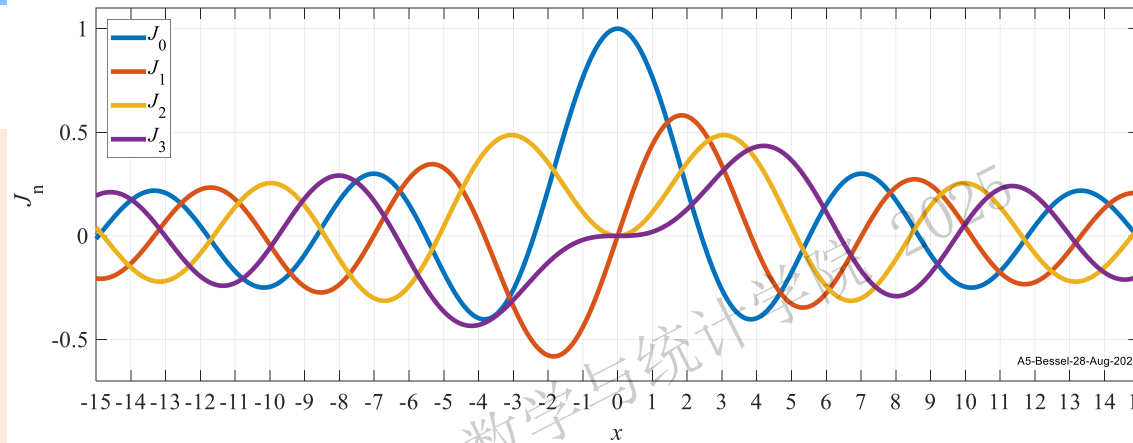
$$J_n(-x)=(-1)^n J_n(x)\qquad \frac{d}{dx}\left[x^n J_n(x)\right]=x^n J_{n-1}(x)$$

$$\frac{d}{dx}\left[x^{-n}J_n(x)\right]=-x^{-n}J_{n+1}(x)$$

$$\int x^n J_{n-1}(x)\,dx=x^n J_n(x)+C,\quad \int\frac{J_{n+1}(x)}{x^n}dx=-\frac{J_n(x)}{x^n}+C$$

$$J_{n-1}(x)+J_{n+1}(x)=\frac{2n}{x}J_n(x)\quad J_{n-1}(x)-J_{n+1}(x)=2J_n'(x)$$

$$\lim_{m\to\infty}\left(\mu_{m+1}^{(n)}-\mu_m^{(n)}\right)=\pi\qquad \lim_{x\to\infty}J_n(x)=\sqrt{\frac{2}{\pi x}}\cos\left(x-\frac{n\pi}{2}-\frac{\pi}{4}\right)+O\left(x^{-\frac{3}{2}}\right)$$



$$J_r(x)=\left(\frac{x}{2}\right)^r\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\,\Gamma(k+r+1)}\left(\frac{x}{2}\right)^{2k},\ N_n(x)=\lim_{r\to n^+}\frac{J_r(x)\cos(r\pi)-J_{-r}(x)}{\sin(r\pi)}$$

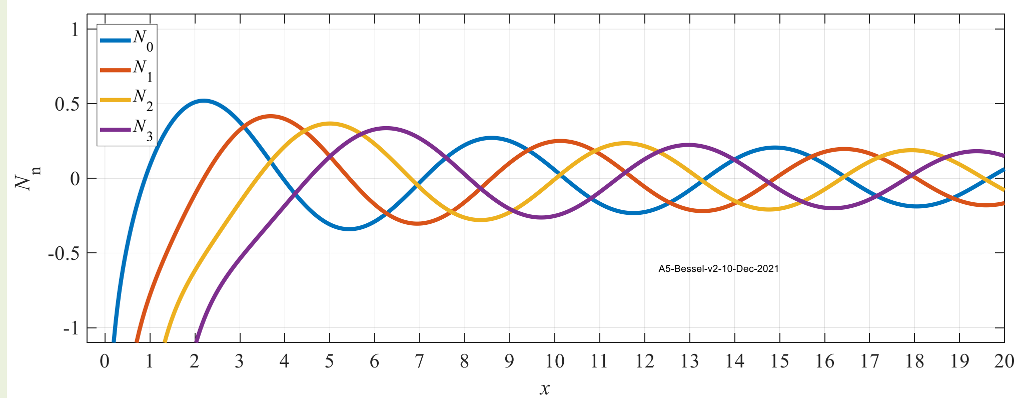
$$N_n(0)=-\infty,\quad N_n(-x)=(-1)^n N_n(x)$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\left[x^n N_n(x)\right]=x^n N_{n-1}(x),\quad \frac{\mathrm{d}}{\mathrm{d}x}\left[x^{-n}N_n(x)\right]=-x^{-n}N_{n+1}(x)$$

$$N_{n-1}(x)+N_{n+1}(x)=\frac{2n}{x}N_n(x),\ N_{n-1}(x)-N_{n+1}(x)=2N_n'(x)$$

$$N_n(x)=\sqrt{\frac{2}{x\pi}}\sin\left(x-\frac{1}{4}\pi-\frac{n}{2}\pi\right)+O\left(\frac{1}{x^{3/2}}\right),$$

$$\lim_{x\to\infty}J_n(x)=\lim_{x\to\infty}N_n(x)=0$$



$$J_n(x)=\left(\frac{x}{2}\right)^n\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\,\Gamma(k+n+1)}\left(\frac{x}{2}\right)^{2k}$$

Bessel函数比较

例

下列说法正确的是

$N_n(x)$ 是有界函数

$N_n(x)$ 是奇函数

$N_n(x)$ 的零点无解析式

$N_n(x)$ 是非振荡衰减函数

对整数阶Bessel函数，下列说法错误的是

$J_n(x)$ 与 $J_{n+1}(x)$ 的零点相间分布

$J_n(x)$ 与 $J_{-n}(x)$ 线性相关

$J_n(x)=0$ 的根全部为单重根

$J_n'(x)=0$ 有无穷多个零点

3.2 Bessel函数

$$J_n(x)=\left(\frac{x}{2}\right)^n\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\cdot\Gamma(k+n+1)}\left(\frac{x}{2}\right)^{2k}$$

Bessel函数比较

例

当 $x\to 0$ 时, 以下表达式不等于0的是

$$J_0'(x)$$

$$\frac{1}{N_1(x)}$$

$$J_1'(x)$$

$$\frac{1}{\Gamma(x)}$$

快速计算手册

边界条件齐次化常用辅助函数 $u(x,t)=v(x,t)+w(x,t)$

边界类型	边界条件	辅助函数
[1,1]	$u(0,t)=g_1(t),\ u(l,t)=g_2(t)$	$w(x,t)=\dfrac{g_2(t)-g_1(t)}{l}x+g_1(t)$
[2,1]	$u_x(0,t)=g_1(t),\ u(l,t)=g_2(t)$	$w(x,t)=g_1(t)(x-l)+g_2(t)$
[1,2]	$u(0,t)=g_1(t),\ u_x(l,t)=g_2(t)$	$w(x,t)=g_2(t)x+g_1(t)$
[2,2]	$u_x(0,t)=g_1(t),\ u_x(l,t)=g_2(t)$	$w(x,t)=\dfrac{g_2(t)-g_1(t)}{2l}x^2+g_1(t)x$

线性常微分方程解法

- 一阶
  - 齐次 $x'+P(t)x=0$ 分离变量法
  - 非齐次 $x'+P(t)x=f(t)$ 常数变易法
- 二阶(高阶)
  - 常系数
    - 齐次
      - 特征方程法 $\lambda^2+p\lambda+q=0$
        - 互异实根 $\{e^{\lambda_1 t},e^{\lambda_2 t}\}$
        - 相同实根 $\{te^{\lambda t},e^{\lambda t}\}$
        - 共轭复根 $\{e^{\alpha t}\cos\beta t,e^{\alpha t}\sin\beta t\}$
      - Liouville公式 基解1→基解2 $x_2(t)=x_1(t)\int\frac{1}{x_1^2(t)}e^{-\int p_1(t)dt}dt$
      - 摄动法
    - 非齐次
      - 通解=通+特
      - 求非齐特解
        - 待定函数法
          - $x''(t)+a_1x'(t)+a_0x(t)=\varphi(t)e^{\mu t}$, $\bar{x}(t)=t^k\Phi(t)e^{\mu t}$
          - $x''(t)+a_1x'(t)+a_0x(t)=\varphi(t)e^{\mu t}\sin\nu t$ $s=\mu\pm\nu i$, $\bar{x}(t)=t^ke^{\mu t}[\Phi_1(t)\cos\nu t+\Phi_2(t)\sin\nu t]$
        - 观察法 $\ddot{x}+x=\sin 2t,\ \bar{x}=C\sin 2t$
  - 变系数
    - Euler方程 $t^n\frac{d^nx}{dt^n}+a_1t^{n-1}\frac{d^{n-1}x}{dt^{n-1}}+\cdots+a_{n-1}t\frac{dx}{dt}+a_nx=F(t),\ t=e^s$ 变量代换法
    - Bernoulli方程 $\frac{dy}{dx}+P(x)\cdot y=Q(x)\cdot y^\alpha,\quad(\alpha\neq0,1),\ u=y^{-\alpha+1}$ 变量代换法
    - Bessel方程 $x^2\frac{d^2y}{dx^2}+x\frac{dy}{dx}+(x^2-r^2)y=0$ 幂级数法
    - Legendre方程 $(1-x^2)\frac{d^2y}{dx^2}-2x\frac{dy}{dx}+n(n+1)y=0$ 幂级数法
- 高阶(可降阶)
  - $y^{(n)}=f(x)$ 无其他导数项, 直接积分法
  - $y''=f(x,y')$ 不显含y ，令 $p=y',y''=\frac{dp}{dx}=f(x,p)$ 变量代换法
  - $y''=f(y,y')$ 不显含 x ，令$p=y',y''=\frac{dp}{dx}=\frac{dp}{dy}\frac{dy}{dx}=p\frac{dp}{dy}$ 变量代换法

算子 $A=-\frac{d^2}{dx^2}$ 不同齐次边界条件下的特征值/特征函数

齐次边界类型	特征值问题	特征值/特征函数
[1,1]	$\begin{cases}X''(x)+\lambda X(x)=0\\X(0)=X(l)=0\end{cases}$	$\lambda_n=\left(\frac{n\pi}{l}\right)^2,\ X_n(x)=\sin\left(\frac{n\pi}{l}x\right),(n=1,2,3\cdots)$
[1,2]	$\begin{cases}X''(x)+\lambda X(x)=0\\X(0)=X'(l)=0\end{cases}$	$\lambda_n=\left[\frac{(2n+1)\pi}{2l}\right]^2,X_n(x)=\sin\left[\frac{(2n+1)\pi}{2l}x\right],(n=0,1,2,3\cdots)$
[2,1]	$\begin{cases}X''(x)+\lambda X(x)=0\\X'(0)=X(l)=0\end{cases}$	$\lambda_n=\left[\frac{(2n+1)\pi}{2l}\right]^2,X_n(x)=\cos\left[\frac{(2n+1)\pi}{2l}x\right],\ (n=0,1,2,3\cdots)$
[2,2]	$\begin{cases}X''(x)+\lambda X(x)=0\\X'(0)=X'(l)=0\end{cases}$	$\lambda_n=\left(\frac{n\pi}{l}\right)^2,\ X_n(x)=\cos\left(\frac{n\pi}{l}x\right),(n=0,1,2,3\cdots)$
周期	$\begin{cases}\Phi''(\theta)+\lambda\Phi(\theta)=0\\\Phi(0)=\Phi(2\pi),\\\Phi'(0)=\Phi'(2\pi)\end{cases}$	$\lambda_0=0,\ \Phi_0(\theta)=1,\ (n=0)$ $\lambda_n=n^2,\ \Phi_n(\theta)=C_1\cos n\theta+C_2\sin n\theta,\ (n=1,2,3\cdots)$

Γ函数常用计算

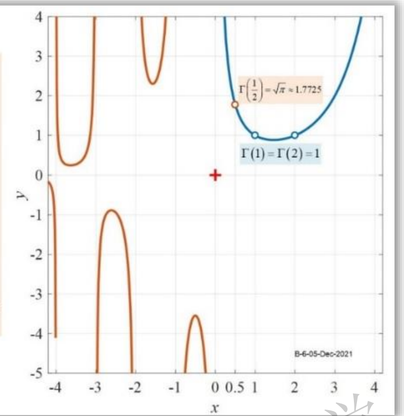
$$\Gamma(\alpha)=\begin{cases}\int_0^\infty x^{\alpha-1}e^{-x}dx, & (\alpha>0)\\ \frac{\Gamma(\alpha+1)}{\alpha}, & (\alpha<0,\alpha\neq Z^-)\\ \infty, & (\alpha=0,Z^-)\end{cases}$$

$\Gamma(\alpha)\neq0$

$\Gamma(\alpha+1)=\alpha\cdot\Gamma(\alpha),\ \frac{1}{\alpha}=\frac{\Gamma(\alpha)}{\Gamma(\alpha+1)}$

$\Gamma(n+1)=n!,\ \Gamma(2)=\Gamma(1)=1$

$\Gamma\left(n+\frac{1}{2}\right)=\frac{(2n-1)!!}{2^n}\sqrt{\pi},\ \Gamma\left(\frac{1}{2}\right)=\sqrt{\pi}$



第一类Bessel函数 小结

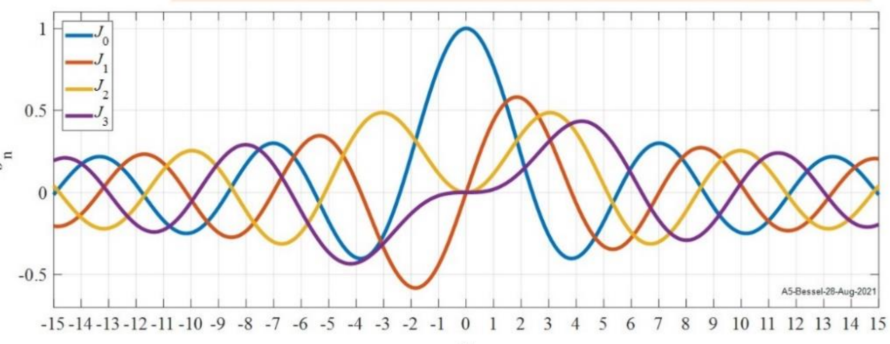
$J_n(0)=\begin{cases}1, & (n=0)\\0, & (n\geq1)\end{cases}\quad J_n'(0)=\begin{cases}0, & (n=0,n>1)\\\frac{1}{2}, & (n=1)\end{cases}$

$J_n(-x)=(-1)^nJ_n(x)\quad \frac{d}{dx}[x^nJ_n(x)]=x^nJ_{n-1}(x)\quad \frac{d}{dx}[x^{-n}J_n(x)]=-x^{-n}J_{n+1}(x)$

$\int x^nJ_{n-1}(x)dx=x^nJ_n(x)+C,\quad \int\frac{J_{n+1}(x)}{x^n}dx=-\frac{J_n(x)}{x^n}+C$

$J_{n-1}(x)+J_{n+1}(x)=\frac{2n}{x}J_n(x)\quad J_{n-1}(x)-J_{n+1}(x)=2J_n'(x)$

$\lim_{m\to\infty}\left(\mu_{m+1}^{(n)}-\mu_m^{(n)}\right)=\pi\quad \lim_{x\to\infty}J_n(x)=\sqrt{\frac{2}{\pi x}}\cos\left(x-\frac{n\pi}{2}-\frac{\pi}{4}\right)+O\left(x^{-\frac{3}{2}}\right)$



r阶Bessel方程 $x^2y''+xy'+(x^2-r^2)y=0$		r阶Bessel函数
对任意 $r>0$ 无论是否为整数	$y(x)=C_1J_r(x)+C_2N_r(x)$	$J_r(x)=\left(\frac{x}{2}\right)^r\sum_{k=0}^\infty\frac{(-1)^k}{k!\Gamma(k+r+1)}\left(\frac{x}{2}\right)^{2k}$ $N_n(x)=\lim_{r\to n^+}\frac{J_r(x)\cos(r\pi)-J_{-r}(x)}{\sin(r\pi)}$
对任意 $r>0$ 且不为整数时, 通解为	$y(x)=C_1J_r(x)+C_2J_{-r}(x)$	$J_r(x)=\left(\frac{x}{2}\right)^r\sum_{k=0}^\infty\frac{(-1)^k}{k!\Gamma(k+r+1)}\left(\frac{x}{2}\right)^{2k}$ $J_{-r}(x)=\left(\frac{x}{2}\right)^{-r}\sum_{k=0}^\infty\frac{(-1)^k}{k!\Gamma(k-r+1)}\left(\frac{x}{2}\right)^{2k}$

3.2 Bessel函数

$$J_n(x)=\left(\frac{x}{2}\right)^n\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\cdot\Gamma(k+n+1)}\left(\frac{x}{2}\right)^{2k}$$

第一类Bessel函数

例1 求不定积分 $\int x^3J_0(x)dx$

解 $\int x^3J_0(x)dx$

升阶积分

$$\int x^{n+1}J_n(x)dx=x^{n+1}J_{n+1}(x)+C$$

$$=\int x^2\left[x\cdot J_0(x)\right]dx$$

$$=\int x^2\cdot d\left[x\cdot J_1(x)\right]$$

$$=x^3\cdot J_1(x)-\int x\cdot J_1(x)\cdot 2x\cdot dx$$

$$=x^3\cdot J_1(x)-2\int x^2\cdot J_1(x)dx$$

$$\int x^{n+1}J_n(x)dx=x^{n+1}J_{n+1}(x)+C$$

$$=x^3\cdot J_1(x)-2x^2\cdot J_2(x)+C$$

求不定积分 $\int xJ_2(x)dx$

解 $\int xJ_2(x)dx$

降阶积分

$$\int\frac{J_n(x)}{x^{n-1}}dx=-\frac{J_{n-1}(x)}{x^{n-1}}+C$$

$$=\int x^2\left[\frac{J_2(x)}{x}\right]dx$$

$$=-\int x^2d\left[\frac{J_1(x)}{x}\right]$$

$$=-\left[xJ_1(x)-\int\frac{J_1(x)}{x}2xdx\right]$$

$$=-\left[xJ_1(x)-2\int\frac{J_1(x)}{x^0}dx\right]$$

$1=x^0$

$$\int\frac{J_n(x)}{x^{n-1}}dx=-\frac{J_{n-1}(x)}{x^{n-1}}+C$$

$$=-\left[xJ_1(x)+2\frac{J_0(x)}{x^0}\right]+C$$

$$=-xJ_1(x)-2J_0(x)+C$$

技巧 $\int x^{\alpha}J_n(x)dx$

1. 积分 x 幂次与 $J_n(x)$ 阶次不一致→调整幂次→升降阶积分
2. 降阶积分中常需要 $1=x^0$

3.2 Bessel函数

$$J_n(x)=\left(\frac{x}{2}\right)^n\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\cdot\Gamma(k+n+1)}\left(\frac{x}{2}\right)^{2k}$$

第一类Bessel函数

$$J_n(0)=\begin{cases}1, & (n=0)\\ 0, & (n\ge 1)\end{cases}\quad J_n'(0)=\begin{cases}0, & (n=0,n>1)\\ \frac{1}{2}, & (n=1)\end{cases}$$

$$J_n(-x)=(-1)^n J_n(x)$$

$$\frac{d}{dx}\left[x^n J_n(x)\right]=x^n J_{n-1}(x)\quad \frac{d}{dx}\left[x^{-n}J_n(x)\right]=-x^{-n}J_{n+1}(x)$$

$$J_{n-1}(x)+J_{n+1}(x)=\frac{2n}{x}J_n(x)\quad J_{n-1}(x)-J_{n+1}(x)=2J_n'(x)$$

$$\boxed{\int x^{n+1}J_n(x)\,dx=x^{n+1}J_{n+1}(x)+C,}\quad \int\frac{J_n(x)}{x^{n-1}}dx=-\frac{J_{n-1}(x)}{x^{n-1}}+C$$

例2 求定积分

1 $\int_0^1 x^2 J_1(2x)\,dx$

解 变量代换 $t=2x,\ x=\frac{t}{2}$

调整积分限

$$\int_0^1 x^2 J_1(2x)\,dx=\int_0^2\left(\frac{t}{2}\right)^2 J_1(t)\,d\left(\frac{t}{2}\right)$$

$$=\frac{1}{8}\int_0^2 t^2 J_1(t)\,dt$$

升阶积分

$$=\frac{1}{8}t^2J_2(t)\Big|_0^2$$

$$=\frac{1}{8}\left[2^2J_2(2)-0^2J_2(0)\right]$$

=0

$$=\frac{1}{2}J_2(2)$$

3.2 Bessel函数

$$J_n(x)=\left(\frac{x}{2}\right)^n\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\cdot\Gamma(k+n+1)}\left(\frac{x}{2}\right)^{2k}$$

第一类Bessel函数

例2 求定积分

2 $\int_0^a x^3 J_0\left(\frac{\mu_m^{(0)}}{a}x\right)dx$

$$J_n(0)=\begin{cases}1, & (n=0)\\ 0, & (n\ge 1)\end{cases}\qquad J_n'(0)=\begin{cases}0, & (n=0,n>1)\\ \frac{1}{2}, & (n=1)\end{cases}$$

$$J_n(-x)=(-1)^n J_n(x)$$

$$\frac{d}{dx}\left[x^n J_n(x)\right]=x^n J_{n-1}(x)\qquad \frac{d}{dx}\left[x^{-n}J_n(x)\right]=-x^{-n}J_{n+1}(x)$$

$$J_{n-1}(x)+J_{n+1}(x)=\frac{2n}{x}J_n(x)\qquad J_{n-1}(x)-J_{n+1}(x)=2J_n'(x)$$

$$\boxed{\int x^{n+1}J_n(x)dx=x^{n+1}J_{n+1}(x)+C,}\qquad \int\frac{J_n(x)}{x^{n-1}}dx=-\frac{J_{n-1}(x)}{x^{n-1}}+C$$

解 变量代换 $t=\frac{\mu_m^{(0)}}{a}x,\ x=\frac{a}{\mu_m^{(0)}}t$

调整积分限

$$\int_0^a x^3 J_0\left(\frac{\mu_m^{(0)}}{a}x\right)dx=\int_0^{\mu_m^{(0)}}\left(\frac{a}{\mu_m^{(0)}}t\right)^3 J_0(t)\,d\left(\frac{a}{\mu_m^{(0)}}t\right)$$

$$=\left(\frac{a}{\mu_m^{(0)}}\right)^4\int_0^{\mu_m^{(0)}}t^3J_0(t)dt=\left(\frac{a}{\mu_m^{(0)}}\right)^4\int_0^{\mu_m^{(0)}}t^2\cdot t\cdot J_0(t)dt$$

升阶积分

$$=\left(\frac{a}{\mu_m^{(0)}}\right)^4\int_0^{\mu_m^{(0)}}t^2\cdot d\left[t\cdot J_1(t)\right]$$

$$=\left(\frac{a}{\mu_m^{(0)}}\right)^4\left[t^3\cdot J_1(t)\Big|_0^{\mu_m^{(0)}}-\int_0^{\mu_m^{(0)}}t\cdot J_1(t)2t\cdot dt\right]$$

$$=\left(\frac{a}{\mu_m^{(0)}}\right)^4\left[t^3\cdot J_1(t)\Big|_0^{\mu_m^{(0)}}-2\int_0^{\mu_m^{(0)}}t^2J_1(t)dt\right]$$

升阶积分

$$=\left(\frac{a}{\mu_m^{(0)}}\right)^4\left[t^3\cdot J_1(t)\Big|_0^{\mu_m^{(0)}}-2t^2J_2(t)\Big|_0^{\mu_m^{(0)}}\right]$$

$$=\left(\frac{a}{\mu_m^{(0)}}\right)^4\left[\left(\mu_m^{(0)}\right)^3\cdot J_1\left(\mu_m^{(0)}\right)-2\left(\mu_m^{(0)}\right)^2J_2\left(\mu_m^{(0)}\right)\right]$$

$$=a^4\frac{\mu_m^{(0)}\cdot J_1\left(\mu_m^{(0)}\right)-2J_2\left(\mu_m^{(0)}\right)}{\left(\mu_m^{(0)}\right)^2}$$

$$J_n(x)=\left(\frac{x}{2}\right)^n\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\,\Gamma(k+n+1)}\left(\frac{x}{2}\right)^{2k}$$

第一类Bessel函数 例3 证明 $J_{1/2}(x)=\sqrt{\dfrac{2}{\pi x}}\sin x$

证 $J_{1/2}(x)=\left(\dfrac{x}{2}\right)^{\frac{1}{2}}\displaystyle\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\,\Gamma\left(k+\frac{1}{2}+1\right)}\left(\frac{x}{2}\right)^{2k}$ 按定义式展开

$$\Gamma\left(n+\frac{1}{2}\right)=\frac{(2n-1)!!}{2^n}\sqrt{\pi}$$

$$=\left(\frac{x}{2}\right)^{\frac{1}{2}}\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\,\dfrac{(2k+1)!!}{2^{k+1}}\sqrt{\pi}\cdot 2^{2k}}\cdot x^{2k}$$

变型分母

$$=\left(\frac{x}{2}\right)^{\frac{1}{2}}\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\,\dfrac{(2k+1)!!}{2^{k}}\cdot\dfrac{\sqrt{\pi}}{2}\cdot 2^k\cdot 2^k}\cdot x^{2k}$$

将 $\dfrac{\sqrt{\pi}}{2}$ 移出Σ

$$=\frac{1}{x}\sqrt{\frac{2x}{\pi}}\sum_{k=0}^{\infty}\frac{(-1)^k}{2^k k!\,(2k+1)!!}x^{2k+1}$$

每项乘以/除以 x

$$=\sqrt{\frac{2}{\pi x}}\sum_{k=0}^{\infty}\frac{(-1)^k}{(2k)!!\,(2k+1)!!}x^{2k+1}$$

半阶乘合并为阶乘

利用级数定义式的经典证明

$$=\sqrt{\frac{2}{\pi x}}\left[\sum_{k=0}^{\infty}\frac{(-1)^k}{(2k+1)!}x^{2k+1}\right]=\sqrt{\frac{2}{\pi x}}\sin x$$

恰为 sin(x) 的Taylor展开

$$\sin x=x-\frac{x^3}{3!}+\frac{x^5}{5!}-\cdots=\sum_{k=0}^{\infty}\frac{(-1)^k}{(2k+1)!}x^{2k+1}$$


N.H.Abel
1802-1829
Norway Math.

$$J_n(x)=\left(\frac{x}{2}\right)^n\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\,\Gamma(k+n+1)}\left(\frac{x}{2}\right)^{2k}$$

第一类Bessel函数

$$J_n(0)=\begin{cases}1, & (n=0)\\ 0, & (n\ge 1)\end{cases}\qquad J_n'(0)=\begin{cases}0, & (n=0,n>1)\\ \frac{1}{2}, & (n=1)\end{cases}$$

$$J_n(-x)=(-1)^n J_n(x)$$

$$\frac{d}{dx}\left[x^n J_n(x)\right]=x^n J_{n-1}(x)\qquad \frac{d}{dx}\left[x^{-n}J_n(x)\right]=-x^{-n}J_{n+1}(x)$$

$$\boxed{J_{n-1}(x)+J_{n+1}(x)=\frac{2n}{x}J_n(x)\qquad J_{n-1}(x)-J_{n+1}(x)=2J_n'(x)}$$

例4 证明 $J_2(x)=J_0''(x)-\dfrac{1}{x}J_0'(x)$

证

$$J_0'(x)=\frac{d}{dx}\left[\frac{J_0(x)}{1}\right]=\frac{d}{dx}\left[\frac{J_0(x)}{x^0}\right]=-\frac{J_1(x)}{x^0}=-J_1(x)$$

即 $-J_0'(x)=J_1(x)$

两侧除以 x $\quad -\dfrac{J_0'(x)}{x}=\dfrac{J_1(x)}{x}=\dfrac{1}{2}\left[J_0(x)+J_2(x)\right]$

两侧求导 $\quad J_0''(x)=-J_1'(x)=-\dfrac{1}{2}\left[J_0(x)-J_2(x)\right]$

消掉 $J_0(x)$

$$J_2(x)=-\frac{1}{2}\left[J_0(x)-J_2(x)\right]+\frac{1}{2}\left[J_0(x)+J_2(x)\right]$$

$$=J_0''(x)-\frac{1}{x}J_0'(x)$$

得证

$$J_n(x)=\left(\frac{x}{2}\right)^n\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\,\Gamma(k+n+1)}\left(\frac{x}{2}\right)^{2k}$$

第一类Bessel函数

$$J_n(0)=\begin{cases}1, & (n=0)\\ 0, & (n\geq 1)\end{cases}\qquad J_n'(0)=\begin{cases}0, & (n=0,n>1)\\ \frac{1}{2}, & (n=1)\end{cases}$$

$$J_n(-x)=(-1)^n J_n(x)$$

$$\frac{d}{dx}\left[x^n J_n(x)\right]=x^n J_{n-1}(x)\qquad \boxed{\frac{d}{dx}\left[x^{-n} J_n(x)\right]=-x^{-n}J_{n+1}(x)}$$

$$J_{n-1}(x)+J_{n+1}(x)=\frac{2n}{x}J_n(x)\qquad J_{n-1}(x)-J_{n+1}(x)=2J_n'(x)$$

$$\int x^{n+1}J_n(x)\,dx=x^{n+1}J_{n+1}(x)+C,\qquad \int\frac{J_n(x)}{x^{n-1}}dx=-\frac{J_{n-1}(x)}{x^{n-1}}+C$$

例5 证明 $\alpha\neq 0$ 时

$$\frac{d}{dx}\left[J_0(\alpha x)\right]=\alpha J_0'(\alpha x)=-\alpha J_1(\alpha x)$$

证

$$J_0'(\alpha x)=\frac{d}{dx}\left[J_0(\alpha x)\right]$$

$$=\frac{d}{dx}\left[\frac{J_0(\alpha x)}{1}\right]$$

$$=\alpha\frac{d}{dx}\left[\frac{J_0(\alpha x)}{(\alpha x)^0}\right]$$

$$=-\alpha\frac{J_1(\alpha x)}{(\alpha x)^0}=-\alpha J_1(\alpha x)$$

$$J_n(x)=\left(\frac{x}{2}\right)^n\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\cdot\Gamma(k+n+1)}\left(\frac{x}{2}\right)^{2k}$$

第一类Bessel函数

$$J_n(0)=\begin{cases}1, & (n=0)\\ 0, & (n\ge 1)\end{cases}\qquad J_n'(0)=\begin{cases}0, & (n=0,n>1)\\ \dfrac{1}{2}, & (n=1)\end{cases}$$

$$J_n(-x)=(-1)^n J_n(x)$$

$$\boxed{\frac{d}{dx}\left[x^n J_n(x)\right]=x^n J_{n-1}(x)}\qquad \frac{d}{dx}\left[x^{-n}J_n(x)\right]=-x^{-n}J_{n+1}(x)$$

$$J_{n-1}(x)+J_{n+1}(x)=\frac{2n}{x}J_n(x)\qquad J_{n-1}(x)-J_{n+1}(x)=2J_n'(x)$$

$$\int x^{n+1}J_n(x)\,dx=x^{n+1}J_{n+1}(x)+C,\quad \int\frac{J_n(x)}{x^{n-1}}dx=-\frac{J_{n-1}(x)}{x^{n-1}}+C$$

例5 证明

$$\frac{d}{dx}\left[xJ_0(x)J_1(x)\right]=x\left[J_0^2(x)-J_1^2(x)\right]$$

证

$$\frac{d}{dx}\left[xJ_0(x)J_1(x)\right]$$

$$=\frac{d}{dx}\left[J_0(x)\cdot xJ_1(x)\right]$$

上例已证明

$$\frac{d}{dx}\left[J_0(\alpha x)\right]=-\alpha J_1(\alpha x)$$

$$=J_0'(x)\left[xJ_1(x)\right]+J_0(x)\left[xJ_1(x)\right]'$$

$$=-J_1(x)\left[xJ_1(x)\right]+J_0(x)\left[xJ_0(x)\right]$$

$$=x\left[-J_1^2(x)+J_0^2(x)\right]$$

几个常用的推论

❶ $$\frac{d}{dx}\left[J_0(\alpha x)\right]=-\alpha J_1(\alpha x)$$

❷ $$\int x\cdot J_0(x)dx=x\cdot J_1(x)+C$$

❸ $$\int x^3\cdot J_0(x)dx$$
$$=\int x^2\cdot x\cdot J_0(x)dx$$
$$=\int x^2\cdot d\left[x\cdot J_1(x)\right]$$
$$=x^3J_1(x)-2\int x^2\cdot J_1(x)dx$$
$$=x^3J_1(x)-2x^2\cdot J_2(x)+C$$
$$=x^3J_1(x)-2x^2\cdot\left[\frac{2}{x}J_1(x)-J_0(x)\right]+C$$
$$=x\left(x^2-4\right)J_1(x)+2x^2J_0(x)+C$$

$$J_{n+1}(x)=\frac{2n}{x}J_n(x)-J_{n-1}(x)$$

后续Bessel展开中反复使用

$$\frac{d}{dx}\left[J_0(\alpha x)\right]=-\alpha J_1(\alpha x)$$
$$\int x\cdot J_0(x)dx=x\cdot J_1(x)+C$$
$$\int x^3\cdot J_0(x)dx=x^3J_1(x)-2x^2\cdot J_2(x)+C$$
$$=x\left(x^2-4\right)J_1(x)+2x^2J_0(x)+C$$

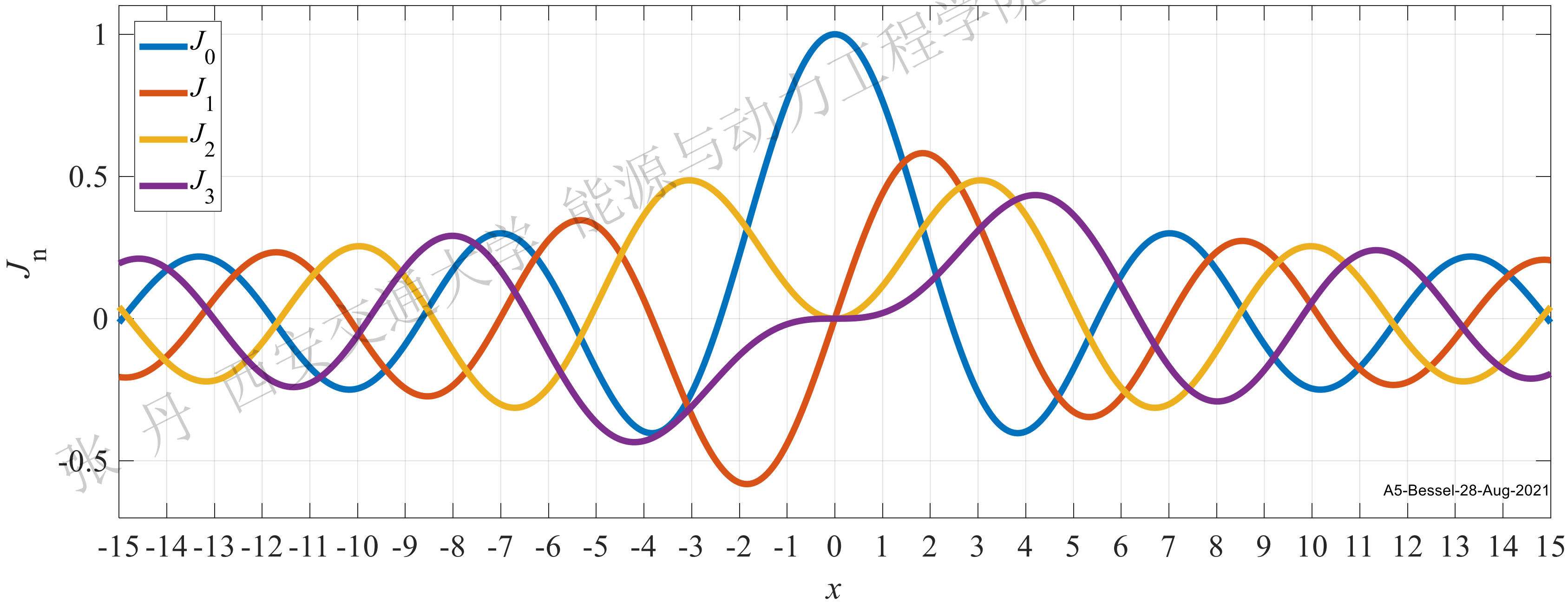
3.2 Bessel函数

$$J_n(x)=\left(\frac{x}{2}\right)^n\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\,\Gamma(k+n+1)}\left(\frac{x}{2}\right)^{2k}$$

第一类Bessel函数

例6 求方程 $J_0'(x)=0$ 的根

解 根据 $J_0'(x)=-J_1(x)=0$ 易知方程的根为 $0,\ \pm\mu_m^{(1)},\ (m=1,2,3\cdots)$



1. 数学建模和基本原理介绍
2. 分离变量法
3. Bessel函数
   - 3.1 二阶线性常微分方程的幂级数解法
   - 3.2 Bessel函数
     - Gamma函数
     - Bessel方程和Bessel函数
     - Bessel函数的性质
     - Bessel方程的特征值问题
     - 圆域上Laplace算子的特征值问题
   - 3.3 多个自变量分离变量法举例
4. 积分变换法
5. Green函数法
6. 特征线法
7. Legendre多项式

极坐标下$-\Delta w(\rho,\theta)=\mu w(\rho,\theta)$展开中几个特征值的关系

周/径向特征值问题

$$\frac{\partial^2 u}{\partial t^2}-a^2\Delta u=0 \qquad u(\rho,\theta,t)=T(t)w(\rho,\theta)$$

$$T''(t)w(\rho,\theta)-a^2T(t)\Delta w(\rho,\theta)=0$$

$$\frac{T''(t)}{a^2T(t)}=\frac{\Delta w(\rho,\theta)}{w(\rho,\theta)}=-\mu$$

时空分离常数

初值问题

$$T''(t)+\mu a^2T(t)=0$$

二维边值问题/圆域-Δ算子特征值问题

$$\begin{cases}\Delta w(\rho,\theta)+\mu w(\rho,\theta)=0,\ (\rho,\theta)\in\Omega\\ w|_{\partial\Omega}=0\end{cases}$$

$$\Delta w=\frac{\partial^2 w}{\partial\rho^2}+\frac{1}{\rho}\frac{\partial w}{\partial\rho}+\frac{1}{\rho^2}\frac{\partial^2 w}{\partial\theta^2}$$

$$w(\rho,\theta)=R(\rho)\Phi(\theta)$$

$$R''(\rho)\Phi(\theta)+\frac{1}{\rho}R'(\rho)\Phi(\theta)+\frac{1}{\rho^2}R(\rho)\Phi''(\theta)=-\mu R(\rho)\Phi(\theta)$$

$$\frac{\Phi''(\theta)}{\Phi(\theta)}=-\frac{\mu R(\rho)+R''(\rho)+\frac{1}{\rho}R'(\rho)}{\frac{1}{\rho^2}R(\rho)}=-\lambda$$

空间分离常数

$\lambda_0=0,\ \Phi_0(\theta)=1,$
$(n=0)$

$\lambda_n=n^2,$
$\Phi_n(\theta)=C_1\cos n\theta+C_2\sin n\theta,$
$(n=1,2,3\cdots)$

周向

$$\begin{cases}\Phi''(\theta)+\lambda\Phi(\theta)=0\\ \Phi(\theta)=\Phi(\theta+2\pi)\end{cases}$$

径向/Bessel方程特征值问题

$$\begin{cases}|R(0)|<+\infty,\ R(\rho_0)=0\\ \rho^2R''(\rho)+\rho R'(\rho)+(\mu\rho^2-\lambda)R(\rho)=0\end{cases}$$

1. 圆域内-Δ算子特征值 μ →周向/径向联合确定→如何关联的呢?
2. 周向: A算子特征值问题/周期性边界→可直接求出特征值/特征函数
3. 径向: ODE能否写成 $AX(x)=\lambda X(x)$ 的形式? →给定边界是否有特征值/特征函数 ?

3.2 Bessel函数

周/径向特征值问题

$$\begin{cases}\Delta w(\rho,\theta)+\mu w(\rho,\theta)=0, & (\rho,\theta)\in\Omega \\ w|_{\partial\Omega}=0\end{cases}$$

分离变量 →

$$-\frac{\mu R(\rho)+R''(\rho)+\frac{1}{\rho}R'(\rho)}{\frac{1}{\rho^2}R(\rho)}=-\lambda=\frac{\Phi''(\theta)}{\Phi(\theta)}$$

二维边值问题/圆域-Δ算子特征值问题

径向

$$-\left[R''(\rho)+\frac{1}{\rho}R'(\rho)-\frac{\lambda}{\rho^2}R(\rho)\right]=\mu R(\rho)$$

定义运算

$$A_\rho=-\left[\frac{\partial^2\square}{\partial\rho^2}+\frac{1}{\rho}\frac{\partial\square}{\partial\rho}-\frac{\lambda}{\rho^2}\square\right] \rightarrow A_\rho\left[R(\rho)\right]=\mu R(\rho)$$

特征值定义

Bessel方程特征值
圆域-Δ的特征值
时空分离常数

周向 → $\lambda=n^2$ 周向特征值

展开成为Bessel方程

$$\begin{cases}\rho^2R''(\rho)+\rho R'(\rho)+\left(\mu\rho^2-n^2\right)R(\rho)=0 \\ \text{圆心自然+外缘1齐 or 圆心自然+外缘2齐}\end{cases}$$

1. 径向ODE 符合特征值定义→解径向ODE =求Bessel方程特征值问题→本质: 寻找给定边界→Bessel方程的非零解
2. 径向/周向→两个维度的特征值→通过Bessel方程零点 $\mu_m^{(n)}$ 关联

3.2 Bessel函数

Bessel方程特征值问题

$$\begin{cases} \rho^2 R''(\rho)+\rho R'(\rho)+\left(\mu\rho^2-n^2\right)R(\rho)=0 \\ \text{圆心自然+外缘1齐 or 圆心自然+外缘2齐} \end{cases}$$

特征值符号变了!!

第一类齐次边界

$(0<\rho<\rho_0)$

$$\begin{cases} \rho^2 R''(\rho)+\rho R'(\rho)+\left(\lambda\rho^2-n^2\right)R(\rho)=0 \\ R(\rho_0)=0,\quad |R(0)|<+\infty \end{cases}$$

Bessel方程特征值改用 λ 表示

特征值/特征函数

$$\lambda_m=\left(\frac{\mu_m^{(n)}}{\rho_0}\right)^2,\ R_m(\rho)=J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right),\quad n\ge 0,\ m\ge 1$$

两组序号

正交性→加权正交

$$\int_0^{\rho_0}\rho R_m(\rho)R_k(\rho)d\rho$$

$$=\int_0^{\rho_0}\rho J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)\cdot J_n\left(\frac{\mu_k^{(n)}}{\rho_0}\rho\right)d\rho$$

$$=\begin{cases} \dfrac{\rho_0^2}{2}\left[J_n'\left(\mu_m^{(n)}\right)\right]^2, & (m=k) \\ 0, & (m\neq k) \end{cases}$$

第二类齐次边界

$(0<\rho<\rho_0)$

$$\begin{cases} \rho^2 R''(\rho)+\rho R'(\rho)+\left(\lambda\rho^2-n^2\right)R(\rho)=0 \\ R_\rho(\rho_0)=0,\ |R(0)|<+\infty \end{cases}$$

特征值/特征函数

$$\begin{cases} n=0, & \begin{cases} \lambda_0=0,\ R_0(\rho)=1, & m=0 \\ \lambda_m=\left(\dfrac{\alpha_m}{\rho_0}\right)^2,\ R_m(\rho)=J_0\left(\dfrac{\alpha_m}{\rho_0}\rho\right), & m\ge 1 \end{cases} \\ n\ge 1, & \lambda_m=\left(\dfrac{\alpha_m}{\rho_0}\right)^2,\ R_m(\rho)=J_n\left(\dfrac{\alpha_m}{\rho_0}\rho\right),\quad m\ge 1 \end{cases}$$

其中 $\alpha_m\,(m\ge 1)$ 为 $J_n'(x)$ 的正零点

正交性→加权正交

$$\int_0^{\rho_0}\rho R_m(\rho)R_k(\rho)d\rho=\begin{cases} \dfrac{\rho_0^2}{2}\left[1-\dfrac{n^2}{\alpha_m^2}\right]J_n^2(\alpha_m), & (m=k) \\ 0, & (m\neq k) \end{cases}$$

Bessel方程特征值问题 第一类边界

$$\begin{cases}\rho^2R''(\rho)+\rho R'(\rho)+\left(\lambda\rho^2-n^2\right)R(\rho)=0, & (0<\rho<\rho_0)\\ R(\rho_0)=0, \quad |R(0)|<+\infty\end{cases}$$

周向外缘齐次　　自然边界条件

$n\ge 0$　周向特征值/Bessel函数阶数

$m\ge 1$　Bessel函数正零点序号/级数项序号

$\mu_m^{(n)}$　n阶Bessel方程第m个正零点

定理1 特征值/特征函数

$$\lambda_m=\left(\frac{\mu_m^{(n)}}{\rho_0}\right)^2,\ R_m(\rho)=J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right),\quad n\ge 0,\ m\ge 1$$

正交性

$$\int_0^{\rho_0}\rho R_m(\rho)R_k(\rho)d\rho=\int_0^{\rho_0}\rho J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)\cdot J_n\left(\frac{\mu_k^{(n)}}{\rho_0}\rho\right)d\rho=\begin{cases}\frac{\rho_0^2}{2}\left[J_n'\left(\mu_m^{(n)}\right)\right]^2, & (m=k)\\ 0, & (m\ne k)\end{cases}$$

权函数加权正交　　平方模

完备性 设 $f(\rho)$ 在 $[0,\rho_0]$ 连续, 且有分段连续的一阶导数,

则 $f(\rho)$ 在该区间上可按特征函数系展开成Fourier-Bessel级数

同阶Bessel函数正交/完备

$$f(\rho)=\sum_{m=1}^{\infty}A_mR_m(\rho)=\sum_{m=1}^{\infty}A_mJ_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)$$

$$A_m=\frac{1}{\frac{1}{2}\left[\rho_0J_n'\left(\mu_m^{(n)}\right)\right]^2}\cdot\int_0^{\rho_0}\rho f(\rho)J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)d\rho$$

3.2 Bessel函数

Bessel方程特征值问题 第一类边界

$$\begin{cases} \rho^2 R''(\rho)+\rho R'(\rho)+\left(\lambda\rho^2-n^2\right)R(\rho)=0, & (0<\rho<\rho_0) \\ R(\rho_0)=0, \quad |R(0)|<+\infty \end{cases}$$

思路 特征值→正交性→平方模

证明 1 特征值 $\lambda>0$

框定特征值的范围

$$\rho^2 R''(\rho)+\rho R'(\rho)+\left(\lambda\rho^2-n^2\right)R(\rho)=0$$

两边同乘以 $\frac{1}{\rho}$

$$\rho R''(\rho)+R'(\rho)+\left(\lambda\rho-\frac{n^2}{\rho}\right)R(\rho)=0$$

求导合并

$$\boxed{\left[\rho R'(\rho)\right]'+\left(\lambda\rho-\frac{n^2}{\rho}\right)R(\rho)=0}$$

重要等式
后续证明起点

两边同乘以 $R(\rho)$

$$R(\rho)\left[\rho R'(\rho)\right]'+\lambda\rho R^2(\rho)-\frac{n^2}{\rho}R^2(\rho)=0$$

$$\int_0^{\rho_0} R(\rho)\left[\rho R'(\rho)\right]' d\rho+\lambda\int_0^{\rho_0}\rho R^2(\rho)-n^2\int_0^{\rho_0}\frac{R^2(\rho)}{\rho}=0$$

两边积分 $[0,\ \rho_0]$

$$\underbrace{R(\rho)\rho R'(\rho)\Big|_0^{\rho_0}}_{=0}-\int_0^{\rho_0}\rho R'(\rho)\cdot R'(\rho)d\rho+\lambda\int_0^{\rho_0}\rho R^2(\rho)-n^2\int_0^{\rho_0}\frac{R^2(\rho)}{\rho}=0$$

$$\lambda=\frac{\int_0^{\rho_0}\left\{\rho\left[R'(\rho)\right]^2+n^2\frac{R^2(\rho)}{\rho}\right\}d\rho}{\int_0^{\rho_0}\rho R^2(\rho)d\rho}\geq 0$$

3.2 Bessel函数

Bessel方程特征值问题 第一类边界

$$\begin{cases}\rho^2R''(\rho)+\rho R'(\rho)+\left(\lambda\rho^2-n^2\right)R(\rho)=0, & (0<\rho<\rho_0)\\ R(\rho_0)=0, \quad |R(0)|<+\infty\end{cases}$$

证明 **1 特征值 $\lambda>0$**

$$\lambda=\frac{\int_0^{\rho_0}\left\{\rho\left[R'(\rho)\right]^2+n^2\frac{R^2(\rho)}{\rho}\right\}d\rho}{\int_0^{\rho_0}\rho R^2(\rho)}\geq 0$$

λ 可否=0 呢?

若 $\lambda=0$，有 $\rho\left[R'(\rho)\right]^2+n^2\dfrac{R^2(\rho)}{\rho}=0$

需同时满足
$$\begin{cases}① \ n=0\\ ② \ R'(\rho)=0 \quad 即 \quad R(\rho)=C\end{cases}$$

由边界条件 $R(\rho_0)=0$ 只可能有 $R(\rho)=C=0$

$\lambda=0$ 不是特征值

故需满足 $\lambda>0$

1. Bessel方程第1类齐次边界下特征值 λ 只能为正数
2. 实际计算时，n可以=0

3.2 Bessel函数

Bessel方程特征值问题 **第一类边界**

$$\begin{cases}\rho^2 R''(\rho)+\rho R'(\rho)+\left(\lambda\rho^2-n^2\right)R(\rho)=0, & (0<\rho<\rho_0)\\ R(\rho_0)=0, \quad |R(0)|<+\infty\end{cases}$$

证明 **2 求出特征值** 变量代换

$$x=\sqrt{\lambda}\rho,\quad \rho=\frac{x}{\sqrt{\lambda}} \qquad y(x)=R(\rho)=R\left(\frac{x}{\sqrt{\lambda}}\right)$$

各阶导数

$$y'(x)=R'(\rho)\frac{d\rho}{dx}=\frac{1}{\sqrt{\lambda}}R'(\rho)$$

$$y''(x)=\frac{1}{\sqrt{\lambda}}R''(\rho)\frac{d\rho}{dx}=\frac{1}{\lambda}R''(\rho)$$

代入原方程

$$\rho^2 R''(\rho)+\rho R'(\rho)+\left(\lambda\rho^2-n^2\right)R(\rho)=0$$

$$\frac{x^2}{\lambda}\lambda y''(x)+\frac{x}{\sqrt{\lambda}}\sqrt{\lambda}y'(x)+\left(\lambda\frac{x^2}{\lambda}-n^2\right)y(x)=0$$

$$x^2y''(x)+xy'(x)+\left(x^2-n^2\right)y(x)=0$$

标准 n 阶Bessel方程

n为整数→通解 $y(x)=C_1J_n(x)+C_2N_n(x)$

还原变量 $R(\rho)=C_1J_n\left(\sqrt{\lambda}\rho\right)+C_2N_n\left(\sqrt{\lambda}\rho\right)$

由自然边界 $|R(0)|<+\infty$ 知 $C_2=0$

由外缘1齐 $R(\rho_0)=0=C_1J_n\left(\sqrt{\lambda}\rho_0\right)$

为取得非零解, $\sqrt{\lambda}\rho_0$ 应取 J_n 的正零点

$$\sqrt{\lambda}\rho_0=\mu_m^{(n)},\quad (m\geq 1)$$

对应无数个零点→无数个特征值/特征函数

$$\lambda_m=\left(\frac{\mu_m^{(n)}}{\rho_0}\right)^2,\ R_m(\rho)=J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right),\quad (m\geq 1)$$

序号为 m

$\lambda\neq 0\to m$ 从1开始

3.2 Bessel函数

Bessel方程特征值问题 第一类边界

$$\begin{cases}\rho^2 R''(\rho)+\rho R'(\rho)+\left(\lambda\rho^2-n^2\right)R(\rho)=0, & (0<\rho<\rho_0)\\ R(\rho_0)=0, & |R(0)|<+\infty\end{cases}$$

$$\lambda_m=\left(\frac{\mu_m^{(n)}}{\rho_0}\right)^2,\ R_m(\rho)=J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right),\ (m\geq 1)$$

证明 3 正交性 特征方程可推出

$$\rho^2 R''(\rho)+\rho R'(\rho)+\left(\lambda\rho^2-n^2\right)R(\rho)=0$$

$$\rho R''(\rho)+R'(\rho)+\left(\lambda\rho-\frac{n^2}{\rho}\right)R(\rho)=0$$

$$\left[\rho R'(\rho)\right]'+\left(\lambda\rho-\frac{n^2}{\rho}\right)R(\rho)=0$$

$$R_\beta(\rho)\cdot ❶-R_\alpha(\rho)\cdot ❷ =$$

$$R_\beta(\rho)\left[\rho R_\alpha'(\rho)\right]'-R_\alpha(\rho)\left[\rho R_\beta'(\rho)\right]'$$

$$+\left(\lambda_\alpha-\lambda_\beta\right)\rho R_\alpha(\rho)R_\beta(\rho)=0$$

特征函数系 $\{R_m(\rho)|m\geq 1\}$ 中任取两个函数

$R_\alpha(\rho),\ R_\beta(\rho),\ (\alpha\neq\beta)$ 均满足上述等式

$$\left[\rho R_\alpha'(\rho)\right]'+\left(\lambda_\alpha\rho-\frac{n^2}{\rho}\right)R_\alpha(\rho)=0 \quad ❶$$

$$\left[\rho R_\beta'(\rho)\right]'+\left(\lambda_\beta\rho-\frac{n^2}{\rho}\right)R_\beta(\rho)=0 \quad ❷$$

3.2 Bessel函数

Bessel方程特征值问题 第一类边界

$$\begin{cases}\rho^2 R''(\rho)+\rho R'(\rho)+\left(\lambda\rho^2-n^2\right)R(\rho)=0, & (0<\rho<\rho_0)\\ R(\rho_0)=0, & |R(0)|<+\infty\end{cases}$$

$$\lambda_m=\left(\frac{\mu_m^{(n)}}{\rho_0}\right)^2,\ R_m(\rho)=J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right),\ (m\geq 1)$$

证明 3 正交性

$$R_\beta(\rho)\left[\rho R'_\alpha(\rho)\right]'-R_\alpha(\rho)\left[\rho R'_\beta(\rho)\right]'+\left(\lambda_\alpha-\lambda_\beta\right)\rho R_\alpha(\rho)R_\beta(\rho)=0$$

移项 $\left(\lambda_\alpha-\lambda_\beta\right)\rho R_\alpha(\rho)R_\beta(\rho)=R_\alpha(\rho)\left[\rho R'_\beta(\rho)\right]'-R_\beta(\rho)\left[\rho R'_\alpha(\rho)\right]'$

两端积分

$$\left(\lambda_\alpha-\lambda_\beta\right)\boxed{\int_0^{\rho_0}\rho R_\alpha(\rho)R_\beta(\rho)d\rho}=\int_0^{\rho_0}R_\alpha(\rho)\left[\rho R'_\beta(\rho)\right]'d\rho-\int_0^{\rho_0}R_\beta(\rho)\left[\rho R'_\alpha(\rho)\right]'d\rho$$

右端积分 $\int_0^{\rho_0}R_\alpha(\rho)d\left[\rho R'_\beta(\rho)\right]-\int_0^{\rho_0}R_\beta(\rho)d\left[\rho R'_\alpha(\rho)\right]$

$$=R_\alpha(\rho)\rho R'_\beta(\rho)\Big|_0^{\rho_0}-\boxed{\int_0^{\rho_0}\rho R'_\beta(\rho)R'_\alpha(\rho)d\rho}-R_\beta(\rho)\rho R'_\alpha(\rho)\Big|_0^{\rho_0}+\boxed{\int_0^{\rho_0}\rho R'_\alpha(\rho)R'_\beta(\rho)d\rho}$$

$$=\rho_0 R_\alpha(\rho_0)R'_\beta(\rho_0)-\rho_0 R_\beta(\rho_0)R'_\alpha(\rho_0)$$

$$=\rho_0 J_n\left(\frac{\mu_\alpha^{(n)}}{\rho_0}\rho_0\right)R'_\beta(\rho_0)-\rho_0 J_n\left(\frac{\mu_\beta^{(n)}}{\rho_0}\rho_0\right)R'_\alpha(\rho_0)=0$$

=0 =0

由 $\lambda_\alpha\neq\lambda_\beta$ 可知 $\int_0^{\rho_0}\rho R_\alpha(\rho)R_\beta(\rho)d\rho=0$，即特征函数 $R_m(\rho)$ 是带权函数 ρ 正交的

3.2 Bessel函数

Bessel方程特征值问题 第一类边界

$$\begin{cases}\rho^2R''(\rho)+\rho R'(\rho)+\left(\lambda\rho^2-n^2\right)R(\rho)=0, & (0<\rho<\rho_0)\\ R(\rho_0)=0, \quad |R(0)|<+\infty\end{cases}$$

$$\lambda_m=\left(\frac{\mu_m^{(n)}}{\rho_0}\right)^2,\ R_m(\rho)=J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right),\ (m\ge1)$$

证明 4 求平方模 借鉴了摄动法思想

记 $\alpha=\sqrt{\lambda_m}=\dfrac{\mu_m^{(n)}}{\rho_0}$, $R_m(\rho)=J_n\left(\dfrac{\mu_m^{(n)}}{\rho_0}\rho\right)=J_n(\alpha\rho)$, $(m\ge1)$ 另取 β，并使 $|\beta-\alpha|=\left|\beta-\dfrac{\mu_m^{(n)}}{\rho_0}\right|\ll1$

$$R_\alpha(\rho)=J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)=J_n(\alpha\rho)$$

是特征函数

是解

$$\begin{cases}\rho^2R_\alpha''(\rho)+\rho R_\alpha'(\rho)+\left(\alpha^2\rho^2-n^2\right)R_\alpha(\rho)=0, & (0<\rho<\rho_0)\\ R_\alpha(\rho_0)=0, \quad |R_\alpha(0)|<+\infty\end{cases}$$

$$\left[\rho R_\alpha'(\rho)\right]'+\left(\alpha^2\rho-\frac{n^2}{\rho}\right)R_\alpha(\rho)=0$$

$$\left[\rho R_\beta'(\rho)\right]'+\left(\beta^2\rho-\frac{n^2}{\rho}\right)R_\beta(\rho)=0$$

方程可改写为

不是特征函数

$$R_\beta(\rho)=J_n(\beta\rho)\neq0$$

是方程解/但不满足外缘1齐

$$\begin{cases}\rho^2R''(\rho)+\rho R'(\rho)+\left(\beta^2\rho^2-n^2\right)R(\rho)=0, & (0<\rho<\rho_0)\\ R(\rho_0)\neq0, \quad |R(0)|<+\infty\end{cases}$$

3.2 Bessel函数

Bessel方程特征值问题 第一类边界

$$\lambda_m=\left(\frac{\mu_m^{(n)}}{\rho_0}\right)^2,\ R_m(\rho)=J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right),\quad (m\geq 1)$$

$$\begin{cases}\rho^2R''(\rho)+\rho R'(\rho)+\left(\lambda\rho^2-n^2\right)R(\rho)=0, & (0<\rho<\rho_0)\\ R(\rho_0)=0, & |R(0)|<+\infty\end{cases}$$

$$R_\alpha(\rho)=J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)=J_n(\alpha\rho),\ R_\beta(\rho)=J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)=J_n(\beta\rho)$$

证明 4 求平方模

$$\left[\rho R'_\alpha(\rho)\right]'+\left(\alpha^2\rho-\frac{n^2}{\rho}\right)R_\alpha(\rho)=0$$

$$\left[\rho R'_\beta(\rho)\right]'+\left(\beta^2\rho-\frac{n^2}{\rho}\right)R_\beta(\rho)=0$$

两端乘以 $R_\beta(\rho)$、$R_\alpha(\rho)$

$$R_\beta(\rho)\left[\rho R'_\alpha(\rho)\right]'+\left(\alpha^2\rho-\frac{n^2}{\rho}\right)R_\alpha(\rho)R_\beta(\rho)=0$$

$$R_\alpha(\rho)\left[\rho R'_\beta(\rho)\right]'+\left(\beta^2\rho-\frac{n^2}{\rho}\right)R_\alpha(\rho)R_\beta(\rho)=0$$

两式相减

$$R_\beta(\rho)\left[\rho R'_\alpha(\rho)\right]'-R_\alpha(\rho)\left[\rho R'_\beta(\rho)\right]'+\left(\alpha^2-\beta^2\right)\rho R_\alpha(\rho)R_\beta(\rho)=0$$

两端在$[0,\ \rho_0]$积分

$$\left(\alpha^2-\beta^2\right)\int_0^{\rho_0}\rho R_\alpha(\rho)R_\beta(\rho)d\rho=\int_0^{\rho_0}R_\alpha(\rho)\left[\rho R'_\beta(\rho)\right]'d\rho-\int_0^{\rho_0}R_\beta(\rho)\left[\rho R'_\alpha(\rho)\right]'d\rho$$

$$=\int_0^{\rho_0}R_\alpha(\rho)d\left[\rho R'_\beta(\rho)\right]-\int_0^{\rho_0}R_\beta(\rho)d\left[\rho R'_\alpha(\rho)\right]$$

$$=\rho R_\alpha(\rho)R'_\beta(\rho)\Big|_0^{\rho_0}-\boxed{\int_0^{\rho_0}\rho R'_\beta(\rho)R'_\alpha(\rho)d\rho}$$

$$-\rho R_\beta(\rho)R'_\alpha(\rho)\Big|_0^{\rho_0}+\boxed{\int_0^{\rho_0}\rho R'_\alpha(\rho)R'_\beta(\rho)d\rho}$$

=0

$$=\rho_0R_\alpha(\rho_0)R'_\beta(\rho_0)-\rho_0R_\beta(\rho_0)R'_\alpha(\rho_0)$$

$$=-\rho_0R'_\alpha(\rho_0)R_\beta(\rho_0)$$

$$\int_0^{\rho_0}\rho R_\alpha(\rho)R_\beta(\rho)d\rho=\frac{-\rho_0R'_\alpha(\rho_0)R_\beta(\rho_0)}{\alpha^2-\beta^2}$$

3.2 Bessel函数

Bessel方程特征值问题　第一类边界

$$\begin{cases} \rho^2 R''(\rho)+\rho R'(\rho)+\left(\lambda\rho^2-n^2\right)R(\rho)=0, & (0<\rho<\rho_0) \\ R(\rho_0)=0, \quad |R(0)|<+\infty \end{cases}$$

$$\lambda_m=\left(\frac{\mu_m^{(n)}}{\rho_0}\right)^2, \; R_m(\rho)=J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right), \; (m\ge 1)$$

$$R_\alpha(\rho)=J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)=J_n(\alpha\rho), \; R_\beta(\rho)=J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)=J_n(\beta\rho)$$

证明　4 求平方模

$\beta\to\alpha$ 的极限即为平方模

$$\int_0^{\rho_0}\rho R_\alpha(\rho)R_\beta(\rho)d\rho=\frac{-\rho_0 R'_\alpha(\rho_0)R_\beta(\rho_0)}{\alpha^2-\beta^2}=\frac{-\rho_0\alpha J'_n(\alpha\rho_0)J_n(\beta\rho_0)}{\alpha^2-\beta^2}$$

$$\int_0^{\rho_0}\rho R_\alpha^2(\rho)d\rho=\lim_{\beta\to\alpha}\int_0^{\rho_0}\rho R_\alpha(\rho)R_\beta(\rho)d\rho$$

β 是变量

$$=\lim_{\beta\to\alpha}\frac{\rho_0\alpha J'_n(\alpha\rho_0)J_n(\beta\rho_0)}{\beta^2-\alpha^2}$$

L'Hospital 法则 对 β 求导

$$=\lim_{\beta\to\alpha}\frac{\rho_0^2\alpha J'_n(\alpha\rho_0)J'_n(\beta\rho_0)}{2\beta}$$

此时$\beta=\alpha$

$$=\frac{\rho_0^2\alpha J'_n(\alpha\rho_0)J'_n(\alpha\rho_0)}{2\alpha}$$

$$=\frac{\rho_0^2}{2}\left[J'_n(\alpha\rho_0)\right]^2$$

$$\alpha=\sqrt{\lambda_m}=\frac{\mu_m^{(n)}}{\rho_0}$$

平方模

$$\int_0^{\rho_0}\rho R_\alpha^2(\rho)d\rho=\frac{\rho_0^2}{2}\left[J'_n\left(\mu_m^{(n)}\right)\right]^2$$

Bessel方程特征值问题

例1 求如下Bessel方程的特征值问题

$$\begin{cases} \rho^2 R''(\rho)+\rho R'(\rho)+\left(\lambda\rho^2-4\right)R(\rho)=0, & (0<\rho<\rho_0) \\ R(\rho_0)=0, \quad |R(0)|<+\infty \end{cases}$$

解 变量代换 $x=\sqrt{\lambda}\rho,\quad \rho=\dfrac{x}{\sqrt{\lambda}}$

各阶导数

$$y(x)=R(\rho)=R\left(\frac{x}{\sqrt{\lambda}}\right)$$

$$y'(x)=R'(\rho)\frac{d\rho}{dx}=\frac{1}{\sqrt{\lambda}}R'(\rho)$$

$$y''(x)=\frac{1}{\sqrt{\lambda}}R''(\rho)\frac{d\rho}{dx}=\frac{1}{\lambda}R''(\rho)$$

代入原方程

$$\rho^2 R''(\rho)+\rho R'(\rho)+\left(\lambda\rho^2-4\right)R(\rho)=0$$

$$\frac{x^2}{\lambda}\lambda y''(x)+\frac{x}{\sqrt{\lambda}}\sqrt{\lambda}y'(x)+\left(\lambda\frac{x^2}{\lambda}-2^2\right)y(x)=0$$

$$x^2y''(x)+xy'(x)+\left(x^2-2^2\right)y(x)=0$$

通解 $y(x)=C_1J_2(x)+C_2N_2(x)$

还原变量 $R(\rho)=C_1J_2\left(\sqrt{\lambda}\rho\right)+C_2N_2\left(\sqrt{\lambda}\rho\right)$

由自然边界 $|R(0)|<+\infty \Rightarrow C_2=0$

由外缘1齐 $R(\rho_0)=C_1J_2\left(\sqrt{\lambda}\rho_0\right)=0$

为取得非零解, $\sqrt{\lambda}\rho_0$ 应取 $J_2(x)$ 的正零点

$$\sqrt{\lambda}\rho_0=\mu_m^{(2)},\quad (m\ge 1)$$

得 特征值/特征函数

$$\lambda_m=\left(\frac{\mu_m^{(2)}}{\rho_0}\right)^2,\ R_m(\rho)=J_2\left(\frac{\mu_m^{(2)}}{\rho_0}\rho\right),\quad (m\ge 1)$$

3.2 Bessel函数

Bessel函数正交/完备

$$\lambda_m=\left(\frac{\mu_m^{(n)}}{\rho_0}\right)^2,\ R_m(\rho)=J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right),\quad (m\geq 1)$$

$$\int_0^{\rho_0}\rho J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)\cdot J_n\left(\frac{\mu_k^{(n)}}{\rho_0}\rho\right)d\rho=\begin{cases}\frac{\rho_0^2}{2}\left[J_n'\left(\mu_m^{(n)}\right)\right]^2, & (m=k)\\ 0, & (m\neq k)\end{cases}$$

阶数n

$$n=0\ \begin{cases}\lambda_m=\left(\frac{\mu_1^{(0)}}{\rho_0}\right)^2,\left(\frac{\mu_2^{(0)}}{\rho_0}\right)^2,\left(\frac{\mu_3^{(0)}}{\rho_0}\right)^2,\cdots,\left(\frac{\mu_m^{(0)}}{\rho_0}\right)^2,\cdots\\ R_m(\rho)=J_0\left(\frac{\mu_1^{(0)}}{\rho_0}\rho\right),J_0\left(\frac{\mu_2^{(0)}}{\rho_0}\rho\right),J_0\left(\frac{\mu_3^{(0)}}{\rho_0}\rho\right),\cdots,J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right),\cdots\end{cases}$$

$$n=1\ \begin{cases}\lambda_m=\left(\frac{\mu_1^{(1)}}{\rho_0}\right)^2,\left(\frac{\mu_2^{(1)}}{\rho_0}\right)^2,\left(\frac{\mu_3^{(1)}}{\rho_0}\right)^2,\cdots,\left(\frac{\mu_m^{(1)}}{\rho_0}\right)^2,\cdots\\ R_m(\rho)=J_1\left(\frac{\mu_1^{(1)}}{\rho_0}\rho\right),J_1\left(\frac{\mu_2^{(1)}}{\rho_0}\rho\right),J_1\left(\frac{\mu_3^{(1)}}{\rho_0}\rho\right),\cdots,J_1\left(\frac{\mu_m^{(1)}}{\rho_0}\rho\right),\cdots\end{cases}$$

$$n=2\ \begin{cases}\lambda_m=\left(\frac{\mu_1^{(2)}}{\rho_0}\right)^2,\left(\frac{\mu_2^{(2)}}{\rho_0}\right)^2,\left(\frac{\mu_3^{(2)}}{\rho_0}\right)^2,\cdots,\left(\frac{\mu_m^{(2)}}{\rho_0}\right)^2,\cdots\\ R_m(\rho)=J_2\left(\frac{\mu_1^{(2)}}{\rho_0}\rho\right),J_2\left(\frac{\mu_2^{(2)}}{\rho_0}\rho\right),J_2\left(\frac{\mu_3^{(2)}}{\rho_0}\rho\right),\cdots,J_2\left(\frac{\mu_m^{(2)}}{\rho_0}\rho\right),\cdots\end{cases}$$

$\cdots$

1. 相同阶数的Bessel函数正交, 阶数不同的Bessel函数不正交
2. 正交是加权正交, 权函数为 ρ

3.2 Bessel函数

Bessel展开

将 $f(\rho)$ 在区间 $[0,\rho_0]$ 上按照Bessel函数系 $\left\{ J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right) \middle| m\geq 1 \right\}$ 展开

$$f(\rho)=\sum_{m=1}^{\infty} A_m J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)=A_1 J_n\left(\frac{\mu_1^{(n)}}{\rho_0}\rho\right)+A_2 J_n\left(\frac{\mu_2^{(n)}}{\rho_0}\rho\right)+A_3 J_n\left(\frac{\mu_3^{(n)}}{\rho_0}\rho\right)+\cdots+A_m J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)+\cdots$$

为求第 m 项展开系数 A_m

两端乘以 $\rho J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)$

$$\rho f(\rho) J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)=\rho A_1 J_n\left(\frac{\mu_1^{(n)}}{\rho_0}\rho\right) J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)+\rho A_2 J_n\left(\frac{\mu_2^{(n)}}{\rho_0}\rho\right) J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)+\cdots$$

$$+\rho A_m J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right) J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)+\cdots$$

两端在 $[0,\ \rho_0]$ 积分

$$\int_0^{\rho_0} \rho f(\rho) J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right) d\rho=A_1\int_0^{\rho_0} \rho J_n\left(\frac{\mu_1^{(n)}}{\rho_0}\rho\right) J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right) d\rho+A_2\int_0^{\rho_0} \rho J_n\left(\frac{\mu_2^{(n)}}{\rho_0}\rho\right) J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right) d\rho+\cdots$$

$$+\boxed{A_m\int_0^{\rho_0} \rho J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right) J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right) d\rho}+\cdots$$

正交性

$$\int_0^{\rho_0} \rho f(\rho) J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right) d\rho=0+0+\cdots+A_m\frac{\rho_0^2}{2}\left[J_n'\left(\mu_m^{(n)}\right)\right]^2+0+\cdots$$

平方模

Bessel展开系数

$$A_m=\frac{1}{\frac{\rho_0^2}{2}\left[J_n'\left(\mu_m^{(n)}\right)\right]^2}\cdot\int_0^{\rho_0} \rho f(\rho) J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right) d\rho$$

3.2 Bessel函数

Bessel展开

例1 将 $f(\rho)=\rho$ 在区间 $[0,\rho_0]$ 上按照1阶Bessel函数系 $\left\{J_1\left(\frac{\mu_m^{(1)}}{\rho_0}\rho\right)\middle|\, m\geq 1\right\}$ 展开

解 欲将原函数展开为

$$f(\rho)=\rho=\sum_{m=1}^{\infty}A_m J_1\left(\frac{\mu_m^{(1)}}{\rho_0}\rho\right)$$

$$A_m=\frac{2}{\left[\rho_0 J_1'\left(\mu_m^{(1)}\right)\right]^2}\int_0^{\rho_0}\rho\cdot\rho\cdot J_1\left(\frac{\mu_m^{(1)}}{\rho_0}\rho\right)d\rho$$

积分上限变化

变量代换 $x=\frac{\mu_m^{(1)}}{\rho_0}\rho,\quad \rho=\frac{\rho_0}{\mu_m^{(1)}}x$

$$A_m=\frac{2}{\left[\rho_0 J_1'\left(\mu_m^{(1)}\right)\right]^2}\int_0^{\mu_m^{(1)}}\left(\frac{\rho_0}{\mu_m^{(1)}}x\right)^2\cdot J_1(x)\frac{\rho_0}{\mu_m^{(1)}}dx$$

$$=\frac{2}{\left[\rho_0 J_1'\left(\mu_m^{(1)}\right)\right]^2}\left(\frac{\rho_0}{\mu_m^{(1)}}\right)^3\int_0^{\mu_m^{(1)}}x^2\cdot J_1(x)dx$$

$$\int x^{n+1}J_n(x)dx=x^{n+1}J_{n+1}(x)+C$$

$$=\frac{2\rho_0}{\left(\mu_m^{(1)}\right)^3\left[J_1'\left(\mu_m^{(1)}\right)\right]^2}x^2\cdot J_2(x)\Big|_0^{\mu_m^{(1)}}$$

$$=\frac{2\rho_0\cdot\left(\mu_m^{(1)}\right)^2 J_2\left(\mu_m^{(1)}\right)}{\left(\mu_m^{(1)}\right)^3\left[J_1'\left(\mu_m^{(1)}\right)\right]^2}$$

$$=\frac{2\rho_0\cdot J_2\left(\mu_m^{(1)}\right)}{\mu_m^{(1)}\left[J_1'\left(\mu_m^{(1)}\right)\right]^2}$$

能否利用其为零点进一步化简?

$$J_{n-1}(x)-J_{n+1}(x)=2J_n'(x)$$

$$J_1'(x)=\frac{1}{2}\left[J_0(x)-J_2(x)\right]$$

$$J_{n-1}(x)=\frac{2n}{x}J_n(x)-J_{n+1}(x)$$

$$=\frac{1}{2}\left[\frac{2}{x}J_1(x)-J_2(x)-J_2(x)\right]$$

$$=\frac{J_1(x)}{x}-J_2(x)$$

$$J_1'\left(\mu_m^{(1)}\right)=\frac{J_1\left(\mu_m^{(1)}\right)}{x}-J_2\left(\mu_m^{(1)}\right)=-J_2\left(\mu_m^{(1)}\right)$$

3.2 Bessel函数

Bessel展开

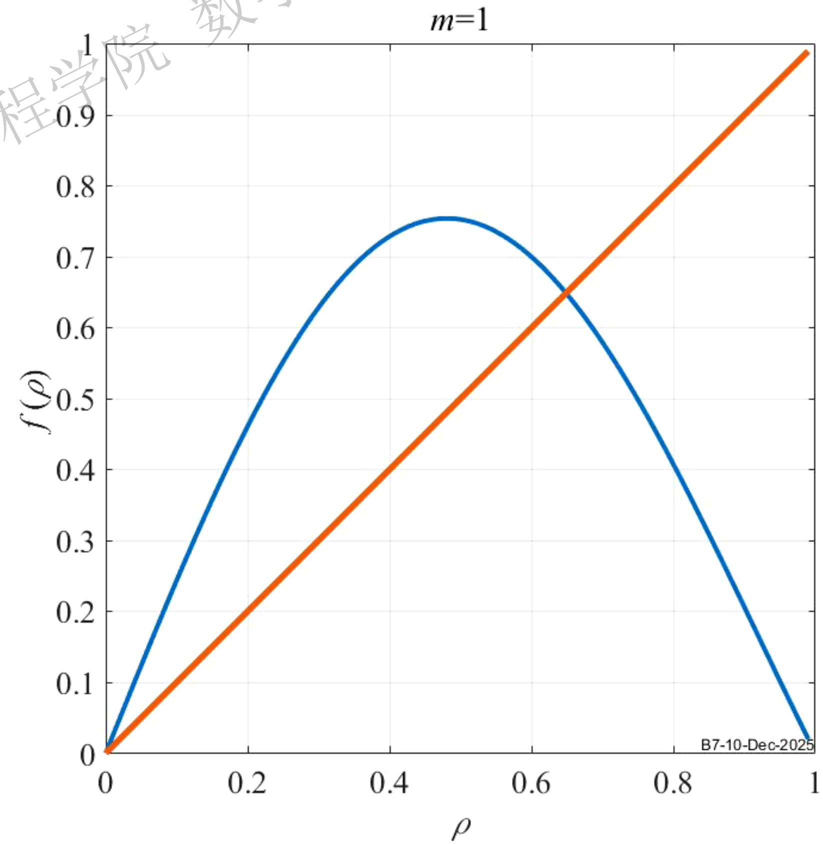
例1 将 $f(\rho)=\rho$ 在区间 $[0,\rho_0]$ 上按照一阶Bessel函数系 $\left\{ J_1\left(\dfrac{\mu_m^{(1)}}{\rho_0}\rho\right) \middle| \ m\geq 1 \right\}$ 展开

$J_1'\left(\mu_m^{(1)}\right)=-J_2\left(\mu_m^{(1)}\right)$ 代入展开系数

$$f(\rho)=\rho=\sum_{m=1}^{\infty}\frac{2\rho_0}{\mu_m^{(1)}J_2\left(\mu_m^{(1)}\right)}\cdot J_1\left(\frac{\mu_m^{(1)}}{\rho_0}\rho\right)$$

$$A_m=\frac{2\rho_0\cdot J_2\left(\mu_m^{(1)}\right)}{\mu_m^{(1)}\left[J_1'\left(\mu_m^{(1)}\right)\right]^2}=\frac{2\rho_0\cdot J_2\left(\mu_m^{(1)}\right)}{\mu_m^{(1)}\left[-J_2\left(\mu_m^{(1)}\right)\right]^2}=\frac{2\rho_0}{\mu_m^{(1)}J_2\left(\mu_m^{(1)}\right)}$$

$\mu_m^{(n)}$ \ $J_n(x)$	0	1	2	3	4	5	······
1	2.404826	3.831706	5.135622	6.380162	7.588342	8.771484	······
2	5.520078	7.015587	8.417244	9.761023	11.06471	12.3386	······
3	8.653728	10.17347	11.61984	13.0152	14.37254	15.70017	······
4	11.79153	13.32369	14.79595	16.22347	17.61597	18.98013	······
5	14.93092	16.47063	17.95982	19.40942	20.82693	22.2178	······
6	18.07106	19.61586	21.117	22.58273	24.01902	25.43034	······
7	21.21164	22.76008	24.27011	25.74817	27.19909	28.62662	······
8	24.35247	25.90367	27.42057	28.90835	30.37101	31.81172	······
9	27.49348	29.04683	30.5692	32.06485	33.53714	34.98878	······
10	30.63461	32.18968	33.71652	35.21867	36.699	38.15987	······
11	33.77582	35.33231	36.86286	38.37047	39.85763	41.32638	······
······	······	······	······	······	······	······	······



3.2 Bessel函数

Bessel展开

例2 将 ❶ $f(\rho)=C$，❷ $f(\rho)=\rho^2$ 在区间 $[0,\rho_0]$ 上按 $\left\{J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)\middle|\, m\geq 1\right\}$ 展开

$$\frac{d}{dx}\left[J_0(\alpha x)\right]=-\alpha J_1(\alpha x)$$
$$\int x\cdot J_0(x)dx=x\cdot J_1(x)+C$$
$$\int x^3\cdot J_0(x)dx=x^3J_1(x)-2x^2\cdot J_2(x)+C$$

解 ❶
$$f(\rho)=C=\sum_{m=1}^{\infty}A_mJ_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)$$

$$A_m=\frac{2}{\left[\rho_0J_0'\left(\mu_m^{(0)}\right)\right]^2}\int_0^{\rho_0}\rho\cdot C\cdot J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)d\rho$$

变量代换 $x=\frac{\mu_m^{(0)}}{\rho_0}\rho,\quad \rho=\frac{\rho_0}{\mu_m^{(0)}}x$

调整积分上限

$$A_m=\frac{2}{\left[\rho_0J_0'\left(\mu_m^{(0)}\right)\right]^2}\int_0^{\mu_m^{(0)}}\frac{\rho_0}{\mu_m^{(0)}}x\cdot C\cdot J_0(x)\frac{\rho_0}{\mu_m^{(0)}}dx$$

$$=\frac{2C}{\left[\rho_0J_0'\left(\mu_m^{(0)}\right)\right]^2}\left(\frac{\rho_0}{\mu_m^{(0)}}\right)^2\int_0^{\mu_m^{(0)}}x\cdot J_0(x)dx$$

升阶积分

$$=\frac{2C}{\left(\mu_m^{(0)}\right)^2\left[J_1\left(\mu_m^{(0)}\right)\right]^2}\mu_m^{(0)}\cdot J_1\left(\mu_m^{(0)}\right)=\frac{2C}{\mu_m^{(0)}\cdot J_1\left(\mu_m^{(0)}\right)}$$

$$f(\rho)=C=\sum_{m=1}^{\infty}\left[\frac{2C}{\mu_m^{(0)}\cdot J_1\left(\mu_m^{(0)}\right)}\right]J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)$$

Bessel展开

例2 将 ❶ $f(\rho)=C$，❷ $f(\rho)=\rho^2$ 在区间 $[0,\rho_0]$ 上按 $\left\{J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)\middle|\, m\ge 1\right\}$ 展开

解 ❷

$$f(\rho)=\rho^2=\sum_{m=1}^{\infty}A_mJ_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)$$

$$A_m=\frac{2}{\left[\rho_0J_0'\left(\mu_m^{(0)}\right)\right]^2}\int_0^{\rho_0}\rho\cdot\rho^2\cdot J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)d\rho$$

变量代换

$$x=\frac{\mu_m^{(0)}}{\rho_0}\rho,\quad \rho=\frac{\rho_0}{\mu_m^{(0)}}x$$

$$\frac{d}{dx}\left[J_0(\alpha x)\right]=-\alpha J_1(\alpha x)$$

$$\int x\cdot J_0(x)\,dx=x\cdot J_1(x)+C$$

$$\int x^3\cdot J_0(x)\,dx=x^3J_1(x)-2x^2\cdot J_2(x)+C$$
$$=x\left(x^2-4\right)J_1(x)+2x^2J_0(x)+C$$

$$A_m=\frac{2}{\left[\rho_0J_0'\left(\mu_m^{(0)}\right)\right]^2}\int_0^{\mu_m^{(0)}}\frac{\rho_0}{\mu_m^{(0)}}x\cdot\left(\frac{\rho_0}{\mu_m^{(0)}}x\right)^2\cdot J_0(x)\frac{\rho_0}{\mu_m^{(0)}}dx$$

$$=\frac{2}{\left[\rho_0J_1\left(\mu_m^{(0)}\right)\right]^2}\left(\frac{\rho_0}{\mu_m^{(0)}}\right)^4\int_0^{\mu_m^{(0)}}x^3\cdot J_0(x)\,dx$$

$$=\frac{2}{\left[\rho_0J_1\left(\mu_m^{(0)}\right)\right]^2}\left(\frac{\rho_0}{\mu_m^{(0)}}\right)^4\left[\left(\mu_m^{(0)}\right)^3J_1\left(\mu_m^{(0)}\right)-2\left(\mu_m^{(0)}\right)^2\cdot J_2\left(\mu_m^{(0)}\right)\right]$$

$$J_{n+1}(x)=\frac{2n}{x}J_n(x)-J_{n-1}(x)$$

$$=\frac{2}{\left[\rho_0J_1\left(\mu_m^{(0)}\right)\right]^2}\left(\frac{\rho_0}{\mu_m^{(0)}}\right)^4\left[\left(\mu_m^{(0)}\right)^3J_1\left(\mu_m^{(0)}\right)-2\left(\mu_m^{(0)}\right)^2\cdot\left(\frac{2}{\mu_m^{(0)}}J_1\left(\mu_m^{(0)}\right)-\underbrace{J_0\left(\mu_m^{(0)}\right)}_{=0}\right)\right]$$

$$=\frac{2\rho_0^2\left[\left(\mu_m^{(0)}\right)^2-4\right]}{\left(\mu_m^{(0)}\right)^3\cdot\left[J_1\left(\mu_m^{(0)}\right)\right]}$$

$$f(\rho)=\rho^2=\sum_{m=1}^{\infty}\frac{2\rho_0^2\left[\left(\mu_m^{(0)}\right)^2-4\right]}{\left(\mu_m^{(0)}\right)^3\cdot\left[J_1\left(\mu_m^{(0)}\right)\right]}J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)$$

习题3, 15

3.2 Bessel函数

Bessel展开

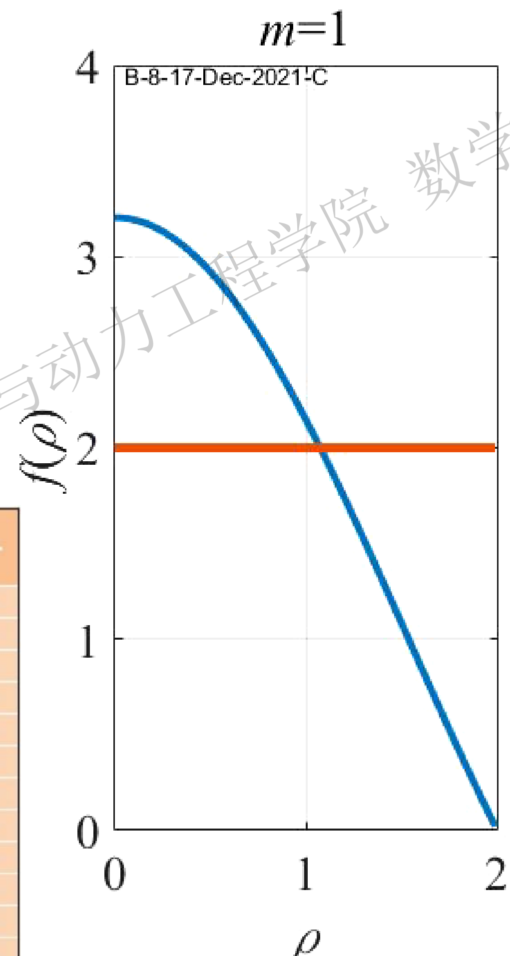
例2 将 ① $f(\rho)=C$，② $f(\rho)=\rho^2$ 在区间 $[0,\rho_0]$ 上按 $\left\{J_0\left(\dfrac{\mu_m^{(0)}}{\rho_0}\rho\right)\middle|\, m\ge 1\right\}$ 展开

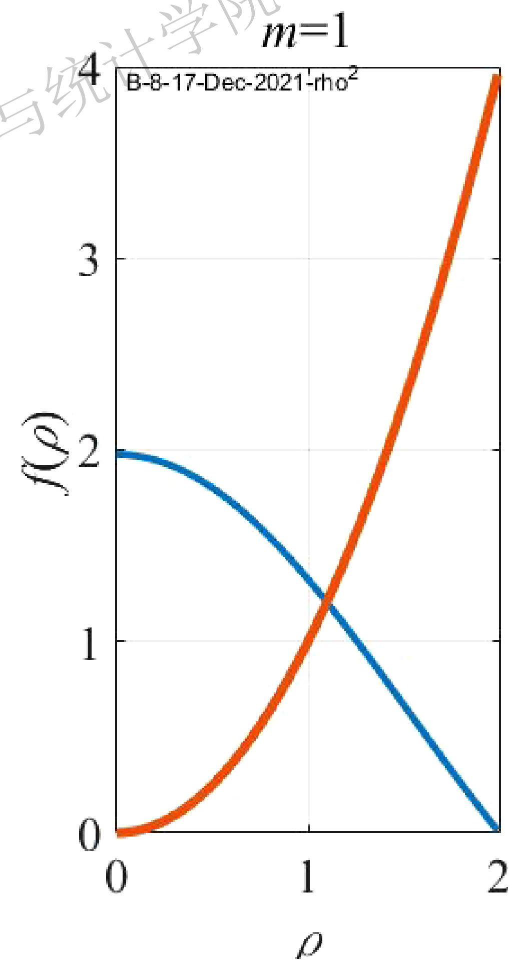
讨论

$$f(\rho)=C=\sum_{m=1}^{\infty}\frac{2C}{\mu_m^{(0)}\cdot J_1\left(\mu_m^{(0)}\right)}J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)$$

$$f(\rho)=\rho^2=\sum_{m=1}^{\infty}\frac{2\rho_0^2\left[\left(\mu_m^{(0)}\right)^2-4\right]}{\left(\mu_m^{(0)}\right)^3\cdot\left[J_1\left(\mu_m^{(0)}\right)\right]}J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)$$

$J_n(x)$ \ $\mu_m^{(n)}$	0	1	2	3	4	5	……
1	2.404826	3.831706	5.135622	6.380162	7.588342	8.771484	……
2	5.520078	7.015587	8.417244	9.761023	11.06471	12.3386	……
3	8.653728	10.17347	11.61984	13.0152	14.37254	15.70017	……
4	11.79153	13.32369	14.79595	16.22347	17.61597	18.98013	……
5	14.93092	16.47063	17.95982	19.40942	20.82693	22.2178	……
6	18.07106	19.61586	21.117	22.58273	24.01902	25.43034	……
7	21.21164	22.76008	24.27011	25.74817	27.19909	28.62662	……
8	24.35247	25.90367	27.42057	28.90835	30.37101	31.81172	……
9	27.49348	29.04683	30.5692	32.06485	33.53714	34.98878	……
10	30.63461	32.18968	33.71652	35.21867	36.699	38.15987	……
11	33.77582	35.33231	36.86286	38.37047	39.85763	41.32638	……
……	……	……	……	……	……	……	……





1. 数学建模和基本原理介绍
2. 分离变量法
3. Bessel函数
    - 3.1 二阶线性常微分方程的幂级数解法
    - 3.2 Bessel函数
        - Gamma函数
        - Bessel方程和Bessel函数
        - Bessel函数的性质
        - Bessel方程的特征值问题
        - 圆域上Laplace算子的特征值问题
    - 3.3 多个自变量分离变量法举例
4. 积分变换法
5. Green函数法
6. 特征线法
7. Legendre多项式

3.2 Bessel函数

-Δ特征值/特征函数与径向/周向特征值/特征函数的关系

圆域上 –Δ算子第1类边界特征值问题

$$\begin{cases}\Delta w(\rho,\theta)+\mu w(\rho,\theta)=0,\ (\rho,\theta)\in\Omega\\ w|_{\partial\Omega}=0\end{cases}$$

$$-\frac{\mu R(\rho)+R''(\rho)+\frac{1}{\rho}R'(\rho)}{\frac{1}{\rho^2}R(\rho)}=\frac{\Phi''(\theta)}{\Phi(\theta)}=-\lambda=-n^2$$

径向 Bessel方程特征值问题

$$\begin{cases}|R(0)|<+\infty,\ R(\rho_0)=0\\ \rho^2R''(\rho)+\rho R'(\rho)+(\mu\rho^2-\lambda)R(\rho)=0\end{cases}$$

周向

$$\begin{cases}\Phi''(\theta)+\lambda\Phi(\theta)=0\\ \Phi(\theta)=\Phi(\theta+2\pi)\end{cases}$$

径向Bessel方程特征值=-Δ特征值

$$\mu=\left(\frac{\mu_m^{(n)}}{\rho_0}\right)^2,\ R_n=J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right),\quad (m\ge1)$$

$$\lambda_0=0,\quad \Phi_0(\theta)=1,\quad (n=0)$$

$$\lambda_n=n^2,\ \Phi_n(\theta)=\{\cos n\theta,\ \sin n\theta\},(n=1,2,3\cdots)$$

圆域上第1类齐次边界下

-Δ特征值/特征函数

$$-\Delta[R(\rho)\Phi(\theta)]$$
$$=-\left[R''(\rho)\Phi(\theta)+\frac{1}{\rho}R'(\rho)\Phi(\theta)+\frac{1}{\rho^2}R(\rho)\Phi''(\theta)\right]$$
$$=-\left[R''(\rho)+\frac{1}{\rho}R'(\rho)-\frac{\lambda}{\rho^2}R(\rho)\right]\cdot\Phi(\theta)$$
$$=\mu R(\rho)\Phi(\theta)$$

n 由周向决定

$$\mu_{n,m}=\left(\frac{\mu_m^{(n)}}{\rho_0}\right)^2,\quad (m\ge1,\ n\ge0)$$

$$W_{n,m}(\rho,\theta)=\left\{J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)\cos n\theta,\ J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)\sin n\theta\right\}$$

$$A_\rho=-\left[\frac{\partial^2\square}{\partial\rho^2}+\frac{1}{\rho}\frac{\partial\square}{\partial\rho}-\frac{\lambda}{\rho^2}\square\right]$$

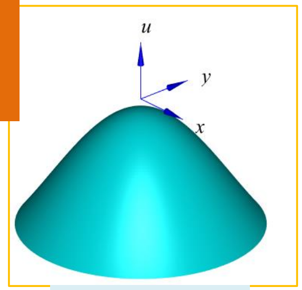
-Δ特征函数=各维特征函数之积

3.2 Bessel函数

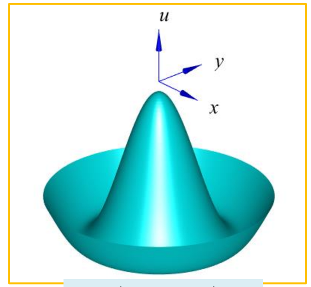
$$\mu_{n,m} = \left(\frac{\mu_m^{(n)}}{\rho_0}\right)^2, W_{n,m}(\rho,\theta) = \left\{ J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)\cos n\theta,\ J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)\sin n\theta \right\}, (m\ge 1,\ n\ge 0)$$

圆域上 $-\Delta$ 算子第1类边界特征函数系

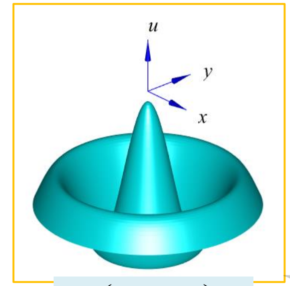
$n=0$



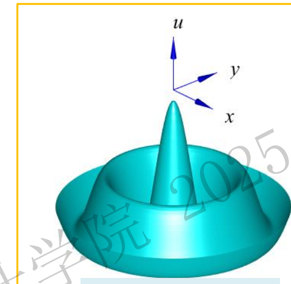
$J_0\left(\frac{\mu_1^{(0)}}{\rho_0}\rho\right)\cdot 1$



$J_0\left(\frac{\mu_2^{(0)}}{\rho_0}\rho\right)\cdot 1$



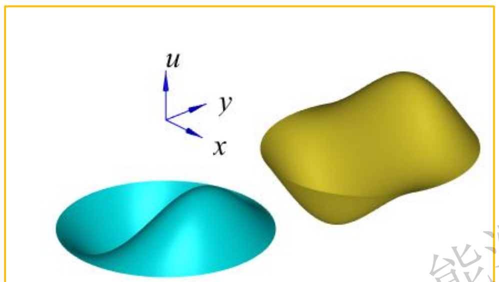
$J_0\left(\frac{\mu_3^{(0)}}{\rho_0}\rho\right)\cdot 1$



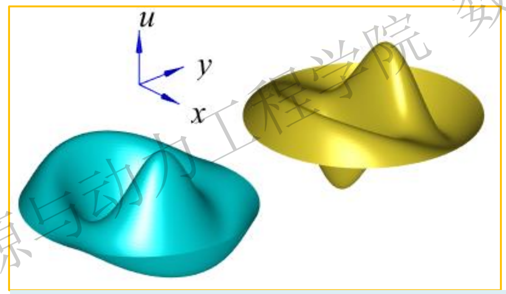
$J_0\left(\frac{\mu_4^{(0)}}{\rho_0}\rho\right)\cdot 1$

…

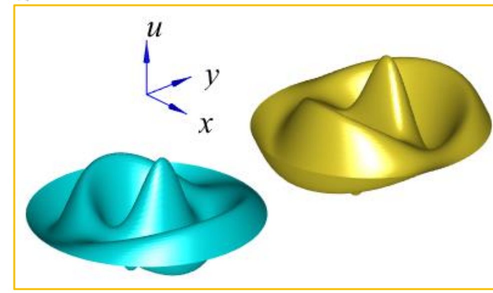
$n=1$



$J_1\left(\frac{\mu_1^{(1)}}{\rho_0}\rho\right)\cos 1\theta,\ J_1\left(\frac{\mu_1^{(1)}}{\rho_0}\rho\right)\sin 1\theta$



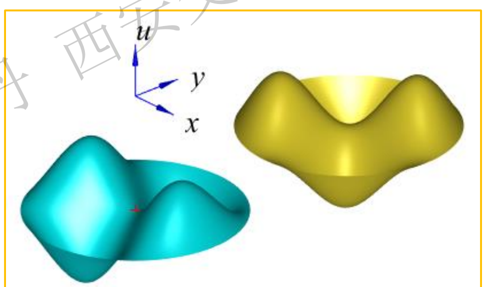
$J_1\left(\frac{\mu_2^{(1)}}{\rho_0}\rho\right)\cos 1\theta,\ J_1\left(\frac{\mu_2^{(1)}}{\rho_0}\rho\right)\sin 1\theta$



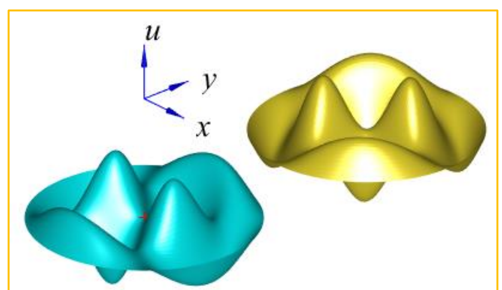
$J_1\left(\frac{\mu_3^{(1)}}{\rho_0}\rho\right)\cos 1\theta,\ J_1\left(\frac{\mu_3^{(1)}}{\rho_0}\rho\right)\sin 1\theta$

…

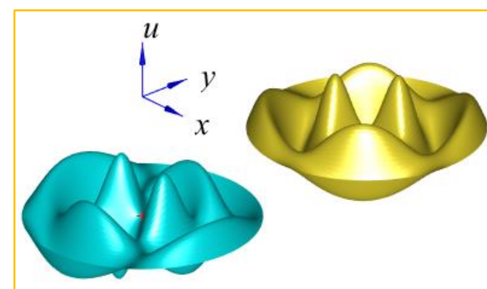
$n=2$



$J_2\left(\frac{\mu_1^{(1)}}{\rho_0}\rho\right)\cos 2\theta,\ J_2\left(\frac{\mu_1^{(1)}}{\rho_0}\rho\right)\sin 2\theta$



$J_2\left(\frac{\mu_2^{(1)}}{\rho_0}\rho\right)\cos 2\theta,\ J_2\left(\frac{\mu_2^{(1)}}{\rho_0}\rho\right)\sin 2\theta$



$J_2\left(\frac{\mu_3^{(1)}}{\rho_0}\rho\right)\cos 2\theta,\ J_2\left(\frac{\mu_3^{(1)}}{\rho_0}\rho\right)\sin 2\theta$

…

3.2 Bessel函数

不同坐标系下-Δ算子 (圆心自然/外缘1齐) 特征值/特征函数

$$\begin{cases} -\Delta w = \mu w, & \Omega \\ w|_{\partial\Omega} = 0 \end{cases}$$

J.Liouville
1809-1882
French Math.

平面直角系

$$\begin{cases} \Delta w(x,y) + \mu w(x,y) = 0, & (x,y)\in\Omega \\ w|_{\partial\Omega} = 0 \end{cases}$$

$$w(x,y) = X(x)Y(y) \qquad \Delta w = \frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2}$$

$$\frac{X''(x)}{X(x)} = -\frac{Y''(y)+\mu Y(y)}{Y(y)} = -\lambda$$

空间分离常数

x方向

$$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l_x) = 0 \end{cases}$$

y方向

$$\begin{cases} Y''(y) + (\mu-\lambda) Y(y) = 0 \\ Y(0) = Y(l_y) = 0 \end{cases}$$

$$\mu_{nx,ny} = \lambda_{nx} + \lambda_{ny} = \left(\frac{n_y\pi}{l_y}\right)^2 + \left(\frac{n_y\pi}{l_y}\right)^2$$

$$W_{nx,ny}(x,y) = \sin\left(\frac{n_x\pi}{l_x}x\right)\sin\left(\frac{n_y\pi}{l_y}y\right)$$

$$(n_y, n_y = 1,2,3\cdots)$$

极坐标系

$$\begin{cases} \Delta w(\rho,\theta) + \mu w(\rho,\theta) = 0, & (\rho,\theta)\in\Omega \\ w|_{\partial\Omega} = 0 \end{cases}$$

$$w(\rho,\theta) = R(\rho)\Phi(\theta) \qquad \Delta w = \frac{\partial^2 w}{\partial \rho^2} + \frac{1}{\rho}\frac{\partial w}{\partial \rho} + \frac{1}{\rho^2}\frac{\partial^2 w}{\partial \theta^2}$$

$$-\frac{\mu R(\rho) + R''(\rho) + \frac{1}{\rho}R'(\rho)}{\frac{1}{\rho^2}R(\rho)} = \frac{\Phi''(\theta)}{\Phi(\theta)} = -\lambda$$

空间分离常数

径向

$$\begin{cases} |R(0)| < +\infty, \quad R(\rho_0) = 0 \\ \rho^2 R''(\rho) + \rho R'(\rho) + (\mu\rho^2 - \lambda) R(\rho) = 0 \end{cases}$$

周向

$$\begin{cases} \Phi''(\theta) + \lambda\Phi(\theta) = 0 \\ \Phi(\theta) = \Phi(\theta + 2\pi) \end{cases}$$

$$\mu_{n,m} = \left(\frac{\mu_m^{(n)}}{\rho_0}\right)^2, \quad (m\ge 1,\ n\ge 0)$$

$$W_{n,m}(\rho,\theta) = \begin{cases} J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right), & (n=0) \\ J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)\cos n\theta,\ J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)\sin n\theta, & (n\ge 1) \end{cases}$$

3.2 Bessel函数

圆域上 -Δ( ) 第1类边界特征值问题

例　下列函数系不是正交函数系的是

$$J_0\left(\frac{\mu_1^{(0)}}{\rho_0}\rho\right),\ J_0\left(\frac{\mu_2^{(0)}}{\rho_0}\rho\right),\ J_0\left(\frac{\mu_3^{(0)}}{\rho_0}\rho\right),\ \cdots,\ J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right),\ \cdots\ m\ge 1$$

$$J_0\left(\frac{\mu_1^{(0)}}{\rho_0}\rho\right),\ J_1\left(\frac{\mu_1^{(1)}}{\rho_0}\rho\right),\ J_2\left(\frac{\mu_1^{(2)}}{\rho_0}\rho\right),\ \cdots J_2\left(\frac{\mu_1^{(m)}}{\rho_0}\rho\right),\ \cdots\ m\ge 1$$

$$J_1\left(\frac{\mu_1^{(1)}}{\rho_0}\rho\right)\cos\theta,\ J_1\left(\frac{\mu_2^{(1)}}{\rho_0}\rho\right)\cos\theta,\ J_1\left(\frac{\mu_3^{(1)}}{\rho_0}\rho\right)\cos\theta,\ \cdots,\ J_1\left(\frac{\mu_m^{(1)}}{\rho_0}\rho\right)\cos\theta,\ \cdots\ m\ge 1$$

$$J_n\left(\frac{\mu_1^{(n)}}{\rho_0}\rho\right)\sin n\theta,\ J_n\left(\frac{\mu_2^{(n)}}{\rho_0}\rho\right)\sin n\theta,\ J_n\left(\frac{\mu_3^{(n)}}{\rho_0}\rho\right)\sin n\theta,\ \cdots,\ J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)\sin n\theta,\ \cdots\ m\ge 1,\ n>0$$

数学物理方程

1. 数学建模和基本原理介绍
2. 分离变量法
3. **Bessel函数**
4. 积分变换法
5. Green函数法
6. 特征线法
7. Legendre多项式

3.1 二阶线性常微分方程的幂级数解法

3.2 Bessel函数

3.3 多个自变量分离变量法举例

- 圆域上的定解问题
- 矩形域上的定解问题

3.3 多个自变量分离变量法举例

例1

$$\begin{cases} u_{tt}-a^2\left(u_{\rho\rho}+\dfrac{1}{\rho}u_\rho+\dfrac{1}{\rho^2}u_{\theta\theta}\right)=0,\ 0\le\rho<\rho_0,\ 0\le\theta<2\pi,\ t>0 \\ |u(0,t)|<+\infty,\ u(\rho_0,\theta,t)=0,\ t\ge0 \\ u(\rho,\theta,0)=\varphi(\rho,\theta),\ u_t(\rho,\theta,0)=\psi(\rho,\theta),\ 0\le\rho\le\rho_0,0\le\theta<2\pi \end{cases}$$

波动/二维/齐次 极坐标	
自然	1齐

解　第1步 时空分离/特征值问题

$$\frac{\partial^2 u}{\partial t^2}-a^2\Delta u=0$$

假设有时空分离的形式解

$$u(\rho,\theta,t)=T(t)w(\rho,\theta)$$

$$\frac{T''(t)}{a^2T(t)}=\frac{\Delta w(\rho,\theta)}{w(\rho,\theta)}=-\mu$$

时空分离常数

初值问题

$$T''(t)+\mu a^2T(t)=0$$

二维边值问题/特征值问题

$$\begin{cases}-\Delta w(\rho,\theta)=\mu w(\rho,\theta),\ (\rho,\theta)\in\Omega \\ |w(0,t)|<+\infty,\ w(\rho_0,t)=0,\ t\ge0\end{cases}$$

圆心自然/外缘1齐

特征值/特征函数

$$\mu_{n,m}=\left(\frac{\mu_m^{(n)}}{\rho_0}\right)^2,\quad (m\ge1,\ n\ge0)$$

两个 μ 意义不同

$$W_{n,m}(\rho,\theta)=\left\{\underbrace{J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)\cdot1}_{n=0},\ \underbrace{J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)\cos n\theta,\ J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)\sin n\theta}_{n\ge1}\right\}$$

3.3 多个自变量分离变量法举例

例1

$$\begin{cases} u_{tt}-a^2\left(u_{\rho\rho}+\frac{1}{\rho}u_\rho+\frac{1}{\rho^2}u_{\theta\theta}\right)=0,\ 0\le\rho<\rho_0,\ 0\le\theta<2\pi,\ t>0 \\ \left|u(0,t)\right|<+\infty,\ u(\rho_0,\theta,t)=0,\ t\ge 0 \\ u(\rho,\theta,0)=\varphi(\rho,\theta),\ u_t(\rho,\theta,0)=\psi(\rho,\theta),\ 0\le\rho\le\rho_0,0\le\theta<2\pi \end{cases}$$

波动/二维/齐次 极坐标	
自然	1齐

第2步 正交分解

$$\mu_{n,m}=\left(\frac{\mu_m^{(n)}}{\rho_0}\right)^2,\quad (m\ge 1,\ n\ge 0)$$

$$W_{n,m}(\rho,\theta)=\left\{J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)\cos n\theta,\ J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)\sin n\theta\right\}$$

原定解问题的形式解可表示为

$$u(\rho,\theta,t)=\sum_{n=0}^{\infty}\sum_{m=1}^{\infty}T_{n,m}(t)W_{n,m}(\rho,\theta)=\sum_{n=0}^{\infty}\sum_{m=1}^{\infty}T_{n,m}(t)R_{n,m}(\rho)\Phi_n(\theta)$$

初始条件按照特征函数系展开

$$\varphi(\rho,\theta)=\sum_{n=0}^{\infty}\sum_{m=1}^{\infty}\varphi_{n,m}W_{n,m}(\rho,\theta)\ \rightarrow\ \varphi_{n,m}=\frac{\int_0^{2\pi}\int_0^{\rho_0}\varphi(\rho,\theta)W_{n,m}(\rho,\theta)\rho d\rho d\theta}{\int_0^{2\pi}\int_0^{\rho_0}\left[W_{n,m}(\rho,\theta)\right]^2\rho d\rho d\theta}$$

$$\psi(\rho,\theta)=\sum_{n=0}^{\infty}\sum_{m=1}^{\infty}\psi_{n,m}W_{n,m}(\rho,\theta)\ \rightarrow\ \psi_{n,m}=\frac{\int_0^{2\pi}\int_0^{\rho_0}\psi(\rho,\theta)W_{n,m}(\rho,\theta)\rho d\rho d\theta}{\int_0^{2\pi}\int_0^{\rho_0}\left[W_{n,m}(\rho,\theta)\right]^2\rho d\rho d\theta}$$

3.3 多个自变量分离变量法举例

例1

$$\begin{cases} u_{tt}-a^2\left(u_{\rho\rho}+\frac{1}{\rho}u_\rho+\frac{1}{\rho^2}u_{\theta\theta}\right)=0,\ 0\le\rho<\rho_0,\ 0\le\theta<2\pi,\ t>0 \\ |u(0,t)|<+\infty,\ u(\rho_0,\theta,t)=0,\ t\ge 0 \\ u(\rho,\theta,0)=\varphi(\rho,\theta),\ u_t(\rho,\theta,0)=\psi(\rho,\theta),\ 0\le\rho\le\rho_0,0\le\theta<2\pi \end{cases}$$

波动/二维/齐次 极坐标	
自然	1齐

第3步 建立初值ODE　将形式解代入原PDE

$$W_{n,m}(\rho,\theta)=R_{n,m}(\rho)\Phi_n(\theta)$$

$$\sum_{n=0}^{\infty}\sum_{m=1}^{\infty}T''_{n,m}(t)W_{n,m}(\rho,\theta)$$

$$=a^2\left(\sum_{n=0}^{\infty}\sum_{m=1}^{\infty}T_{n,m}(t)\left[R''_{n,m}(\rho)+\frac{1}{\rho}R'_{n,m}(\rho)\right]\Phi_n(\theta)+\frac{1}{\rho^2}\sum_{n=0}^{\infty}\sum_{m=1}^{\infty}T_{n,m}(t)R_{n,m}(\rho)\Phi''_n(\theta)\right)$$

$$=a^2\left(\sum_{n=0}^{\infty}\sum_{m=1}^{\infty}T_{n,m}(t)\left[R''_{n,m}(\rho)+\frac{1}{\rho}R'_{n,m}(\rho)\right]\Phi_n(\theta)+\frac{1}{\rho^2}\sum_{n=0}^{\infty}\sum_{m=1}^{\infty}T_{n,m}(t)R_{n,m}(\rho)\left[-\lambda_n\Phi_n(\theta)\right]\right)$$

$$=a^2\sum_{n=0}^{\infty}\sum_{m=1}^{\infty}T_{n,m}(t)\left[R''_{n,m}(\rho)+\frac{1}{\rho}R'_{n,m}(\rho)-\frac{\lambda_n}{\rho^2}R_{n,m}(\rho)\right]\Phi_n(\theta)$$

$$A_\rho=-\left[\frac{\partial^2\square}{\partial\rho^2}+\frac{1}{\rho}\frac{\partial\square}{\partial\rho}-\frac{\lambda}{\rho^2}\square\right]\qquad A_\rho R=\mu R$$

$$=-a^2\sum_{n=0}^{\infty}\sum_{m=1}^{\infty}T_{n,m}(t)\mu_{n,m}R_{n,m}(\rho)\Phi_n(\theta)$$

$$=-a^2\sum_{n=0}^{\infty}\sum_{m=1}^{\infty}T_{n,m}(t)\mu_{n,m}W_{n,m}(\rho,\theta)$$

得初值ODE　$T''_{n,m}(t)+a^2\mu_{n,m}T_{n,m}(t)=0$

3.3 多个自变量分离变量法举例

例1

$$\begin{cases} u_{tt}-a^2\left(u_{\rho\rho}+\dfrac{1}{\rho}u_\rho+\dfrac{1}{\rho^2}u_{\theta\theta}\right)=0,\ 0\le\rho<\rho_0,\ 0\le\theta<2\pi,\ t>0 \\ \left|u(0,t)\right|<+\infty,\ u(\rho_0,\theta,t)=0,\ t\ge0 \\ u(\rho,\theta,0)=\varphi(\rho,\theta),\ u_t(\rho,\theta,0)=\psi(\rho,\theta),\ 0\le\rho\le\rho_0,0\le\theta<2\pi \end{cases}$$

波动/二维/齐次 极坐标	
自然	1齐

第3步 建立初值ODE　根据初始条件

$$u(\rho,\theta,0)=\sum_{n=0}^{\infty}\sum_{m=1}^{\infty}T_{n,m}(0)W_{n,m}(\rho,\theta)=\varphi(\rho,\theta)=\sum_{n=0}^{\infty}\sum_{m=1}^{\infty}\varphi_{n,m}W_{n,m}(\rho,\theta)$$

$$u_t(\rho,\theta,0)=\sum_{n=0}^{\infty}\sum_{m=1}^{\infty}T'_{n,m}(0)W_{n,m}(\rho,\theta)=\psi(\rho,\theta)=\sum_{n=0}^{\infty}\sum_{m=1}^{\infty}\psi_{n,m}W_{n,m}(\rho,\theta)$$

得到初值ODE定解问题

$$\begin{cases} T''_{n,m}(t)+a^2\mu_{n,m}T_{n,m}(t)=0 \\ T_{n,m}(0)=\varphi_{n,m},\ T'_{n,m}(0)=\psi_{n,m} \end{cases}$$

3.3 多个自变量分离变量法举例

例1

$$\begin{cases} u_{tt}-a^2\left(u_{\rho\rho}+\dfrac{1}{\rho}u_\rho+\dfrac{1}{\rho^2}u_{\theta\theta}\right)=0,\ 0\le\rho<\rho_0,\ 0\le\theta<2\pi,\ t>0 \\ |u(0,t)|<+\infty,\ u(\rho_0,\theta,t)=0,\ t\ge 0 \\ u(\rho,\theta,0)=\varphi(\rho,\theta),\ u_t(\rho,\theta,0)=\psi(\rho,\theta),\ 0\le\rho\le\rho_0, 0\le\theta<2\pi \end{cases}$$

波动/二维/齐次 极坐标	
自然	1齐

第4步 解初值ODE

$$\mu_{n,m}=\left(\frac{\mu_m^{(n)}}{\rho_0}\right)^2,\quad (m\ge 1,\ n\ge 0)$$

$$W_{n,m}(\rho,\theta)=\left\{J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)\cos n\theta,\ J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)\sin n\theta\right\}$$

$$\begin{cases} T''_{n,m}(t)+a^2\mu_{n,m}T_{n,m}(t)=0 \\ T_{n,m}(0)=\varphi_{n,m},\ T'_{n,m}(0)=\psi_{n,m} \end{cases}$$

二阶线性齐次ODE, 通解为

$$T_{n,m}(t)=C_{n,m,1}\cos\left(a\sqrt{\mu_{n,m}}\cdot t\right)+C_{n,m,2}\sin\left(a\sqrt{\mu_{n,m}}\cdot t\right)$$

$$T'_{n,m}(t)=-a\sqrt{\mu_{n,m}}\cdot C_{n,m,1}\sin\left(a\sqrt{\mu_{n,m}}\cdot t\right)+a\sqrt{\mu_{n,m}}\cdot C_{n,m,2}\cos\left(a\sqrt{\mu_{n,m}}\cdot t\right)$$

根据初始条件

$$T_{n,m}(0)=C_{n,m,1}=\varphi_{n,m}$$

$$T'_{n,m}(0)=a\sqrt{\mu_{n,m}}\cdot C_{n,m,2}=\psi_{n,m}$$

$$\Rightarrow\quad C_{n,m,1}=\varphi_{n,m},\qquad C_{n,m,2}=\frac{\psi_{n,m}}{a\sqrt{\mu_{n,m}}}$$

初值ODE的解为

$$T_{n,m}(t)=\varphi_{n,m}\cos\left(a\sqrt{\mu_{n,m}}\cdot t\right)+\frac{\psi_{n,m}}{a\sqrt{\mu_{n,m}}}\sin\left(a\sqrt{\mu_{n,m}}\cdot t\right)$$

3.3 多个自变量分离变量法举例

例1

$$\begin{cases} u_{tt}-a^2\left(u_{\rho\rho}+\dfrac{1}{\rho}u_\rho+\dfrac{1}{\rho^2}u_{\theta\theta}\right)=0,\ 0\le\rho<\rho_0,\ 0\le\theta<2\pi,\ t>0 \\ |u(0,t)|<+\infty,\ u(\rho_0,\theta,t)=0,\ t\ge 0 \\ u(\rho,\theta,0)=\varphi(\rho,\theta),\ u_t(\rho,\theta,0)=\psi(\rho,\theta),\ 0\le\rho\le\rho_0, 0\le\theta<2\pi \end{cases}$$

波动/二维/齐次 极坐标	
自然	1齐

第5步 回代　原定解问题的解为

$$\mu_{n,m}=\left(\frac{\mu_m^{(n)}}{\rho_0}\right)^2,\quad (m\ge 1,\ n\ge 0)$$

$$W_{n,m}(\rho,\theta)=\left\{J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)\cos n\theta,\ J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)\sin n\theta\right\}$$

$$T_{n,m}(t)=\varphi_{n,m}\cos\left(a\sqrt{\mu_{n,m}}\cdot t\right)+\frac{\psi_{n,m}}{a\sqrt{\mu_{n,m}}}\sin\left(a\sqrt{\mu_{n,m}}\cdot t\right)$$

$$u(\rho,\theta,t)=\sum_{n=0}^{\infty}\sum_{m=1}^{\infty}T_{n,m}(t)R_{n,m}(\rho)\Phi_n(\theta)$$

$$=\sum_{m=1}^{\infty}\left[\varphi_{0,m}\cos\left(a\sqrt{\mu_{0,m}}\cdot t\right)+\frac{\psi_{0,m}}{a\sqrt{\mu_{n,m}}}\sin\left(a\sqrt{\mu_{0,m}}\cdot t\right)\right]J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\cdot\rho\right)\cdot 1$$

$n=0$

$$+\sum_{n=1}^{\infty}\sum_{m=1}^{\infty}\left[\varphi_{n,m,\cos}\cos\left(a\sqrt{\mu_{n,m}}\cdot t\right)+\frac{\psi_{n,m,\sin}}{a\sqrt{\mu_{n,m}}}\sin\left(a\sqrt{\mu_{n,m}}\cdot t\right)\right]\cdot J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\cdot\rho\right)\cdot\cos n\theta$$

$$+\sum_{n=1}^{\infty}\sum_{m=1}^{\infty}\left[\varphi_{n,m,\cos}\cos\left(a\sqrt{\mu_{n,m}}\cdot t\right)+\frac{\psi_{n,m,\sin}}{a\sqrt{\mu_{n,m}}}\sin\left(a\sqrt{\mu_{n,m}}\cdot t\right)\right]\cdot J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\cdot\rho\right)\cdot\sin n\theta$$

3.3 多个自变量分离变量法举例

波动/二维/齐次 极坐标

自然	1齐

$$\mu_{n,m}=\left(\frac{\mu_m^{(n)}}{\rho_0}\right)^2,\quad (m\ge 1,\ n\ge 0)$$

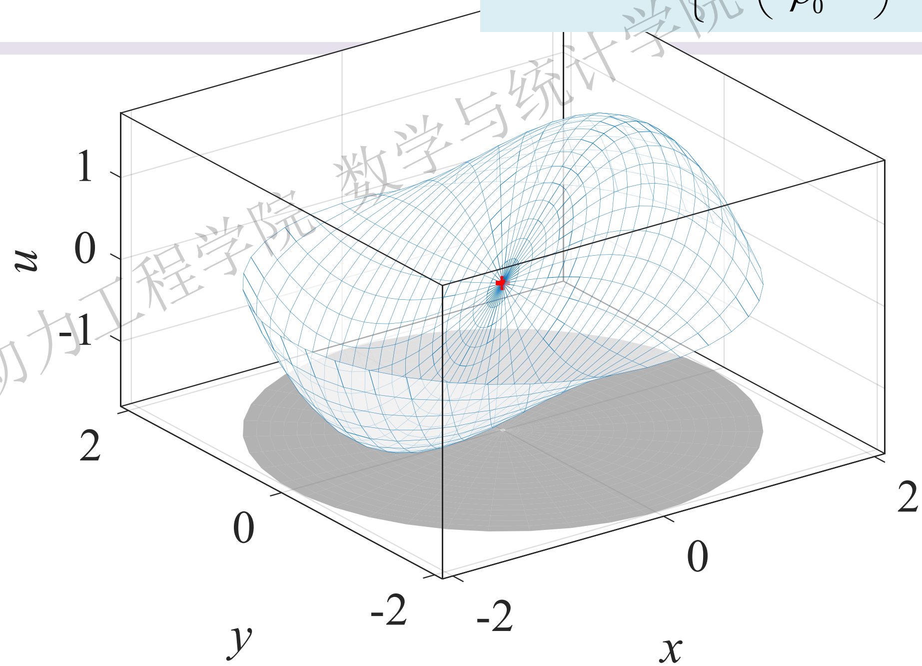
$$W_{n,m}(\rho,\theta)=\begin{cases} J_n\left(\dfrac{\mu_m^{(n)}}{\rho_0}\rho\right)\cos n\theta, \\ J_n\left(\dfrac{\mu_m^{(n)}}{\rho_0}\rho\right)\sin n\theta \end{cases}$$

例1

$$\begin{cases} u_{tt}-a^2\left(u_{\rho\rho}+\dfrac{1}{\rho}u_\rho+\dfrac{1}{\rho^2}u_{\theta\theta}\right)=0,\ 0\le\rho<\rho_0,\ 0\le\theta<2\pi,\ t>0 \\ |u(0,t)|<+\infty,\ u(\rho_0,\theta,t)=0,\ t\ge 0 \\ u(\rho,\theta,0)=\varphi(\rho,\theta),\ u_t(\rho,\theta,0)=\psi(\rho,\theta),\ 0\le\rho\le\rho_0, 0\le\theta<2\pi \end{cases}$$

讨论 假设初始位移/速度分布为

$$\varphi(\rho,\theta)=A\rho\left(1-\frac{\rho^2}{\rho_0^2}\right)\cos\theta \qquad \psi(\rho,\theta)=0$$



易知 $\psi_{n,m}=0$ 下面正交分解初始位移

$$\varphi(\rho,\theta)=\sum_{n=0}^{\infty}\sum_{m=1}^{\infty}\varphi_{n,m}R_{n,m}(\rho)\Phi_n(\theta)$$

$$=\sum_{m=1}^{\infty}\varphi_{0,m}\cdot J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)\cdot 1$$

$$+\sum_{n=1}^{\infty}\sum_{m=1}^{\infty}\varphi_{n,m,\cos}\cdot J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)\cos n\theta$$

$$+\sum_{n=1}^{\infty}\sum_{m=1}^{\infty}\varphi_{n,m,\sin}\cdot J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)\sin n\theta$$

$$\varphi_{n,m}=\frac{\int_0^{2\pi}\int_0^{\rho_0}\varphi(\rho,\theta)R_{n,m}(\rho)\Phi_n(\theta)\rho\, d\rho\, d\theta}{\int_0^{2\pi}\int_0^{\rho_0}\left[R_{n,m}(\rho)\Phi_n(\theta)\right]^2\rho\, d\rho\, d\theta}$$

n→周向特征函数不同→平方模不同→根据n分类讨论

3.3 多个自变量分离变量法举例

波动/二维/齐次
极坐标

自然	1齐

$$\mu_{n,m}=\left(\frac{\mu_m^{(n)}}{\rho_0}\right)^2,\quad (m\ge 1,\ n\ge 0)$$

$$W_{n,m}(\rho,\theta)=\begin{cases}J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)\cos n\theta,\\ J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)\sin n\theta\end{cases}$$

例1

$$\begin{cases}u_{tt}-a^2\left(u_{\rho\rho}+\frac{1}{\rho}u_\rho+\frac{1}{\rho^2}u_{\theta\theta}\right)=0,\ 0\le\rho<\rho_0,\ 0\le\theta<2\pi,\ t>0\\ |u(0,t)|<+\infty,\ u(\rho_0,\theta,t)=0,\ t\ge 0\\ u(\rho,\theta,0)=\varphi(\rho,\theta),\ u_t(\rho,\theta,0)=\psi(\rho,\theta),\ 0\le\rho\le\rho_0,0\le\theta<2\pi\end{cases}$$

讨论

$$\varphi(\rho,\theta)=A\rho\left(1-\frac{\rho^2}{\rho_0^2}\right)\cos\theta\qquad \psi(\rho,\theta)=0$$

❶ 求 $n=0$ 时, 特征函数为 $R_{0,m}=J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)\cdot 1$ 的展开系数

$$\varphi_{0,m}=\frac{\int_0^{2\pi}\int_0^{\rho_0}A\rho\left(1-\frac{\rho^2}{\rho_0^2}\right)\cos\theta\cdot J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)\rho d\rho d\theta}{\int_0^{2\pi}\int_0^{\rho_0}\left[J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)\right]^2\rho d\rho d\theta}=\frac{\underbrace{\int_0^{2\pi}\cos\theta\cdot 1\cdot d\theta}_{=0}\cdot\int_0^{\rho_0}A\rho\left(1-\frac{\rho^2}{\rho_0^2}\right)\cdot J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)\rho d\rho}{\int_0^{2\pi}\int_0^{\rho_0}\left[J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)\right]^2\rho d\rho d\theta}=0$$

3.3 多个自变量分离变量法举例

波动/二维/齐次 极坐标

自然 | 1齐

$$\mu_{n,m}=\left(\frac{\mu_m^{(n)}}{\rho_0}\right)^2,\quad (m\geq 1,\ n\geq 0)$$

$$W_{n,m}(\rho,\theta)=\begin{cases} J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)\cos n\theta, \\ J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)\sin n\theta \end{cases}$$

例1

$$\begin{cases} u_{tt}-a^2\left(u_{\rho\rho}+\frac{1}{\rho}u_\rho+\frac{1}{\rho^2}u_{\theta\theta}\right)=0,\ 0\leq\rho<\rho_0,\ 0\leq\theta<2\pi,\ t>0 \\ |u(0,t)|<+\infty,\ u(\rho_0,\theta,t)=0,\ t\geq 0 \\ u(\rho,\theta,0)=\varphi(\rho,\theta),\ u_t(\rho,\theta,0)=\psi(\rho,\theta),\ 0\leq\rho\leq\rho_0, 0\leq\theta<2\pi \end{cases}$$

讨论

$$\varphi(\rho,\theta)=A\rho\left(1-\frac{\rho^2}{\rho_0^2}\right)\cos\theta \quad \psi(\rho,\theta)=0$$

❷ 求特征函数 $R_{n,m}=J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)\cos n\theta$ 的展开系数

当 $n=1$ 时

$$\varphi_{n=1,m,\cos}=\frac{\int_0^{2\pi}\int_0^{\rho_0}A\rho\left(1-\frac{\rho^2}{\rho_0^2}\right)\cos\theta\cdot J_1\left(\frac{\mu_m^{(1)}}{\rho_0}\rho\right)\cos(1\cdot\theta)\rho d\rho d\theta}{\int_0^{2\pi}\int_0^{\rho_0}\left[J_1\left(\frac{\mu_m^{(1)}}{\rho_0}\rho\right)\right]^2\cos^2(1\cdot\theta)\rho d\rho d\theta}$$

$$=\frac{\int_0^{2\pi}\cos^2\theta d\theta\cdot\int_0^{\rho_0}A\rho\left(1-\frac{\rho^2}{\rho_0^2}\right)\cdot J_1\left(\frac{\mu_m^{(1)}}{\rho_0}\rho\right)\rho d\rho}{\int_0^{2\pi}\cos^2\theta d\theta\cdot\int_0^{\rho_0}\left[J_1\left(\frac{\mu_m^{(1)}}{\rho_0}\rho\right)\right]^2\rho d\rho}$$

分子/分母可约去

$$\int_0^{2\pi}\cos^2\theta d\theta=\frac{1}{2}(\theta+\sin\theta\cos\theta)_0^{2\pi}=\pi\neq 0$$

$$=\frac{\int_0^{\rho_0}A\rho\left(1-\frac{\rho^2}{\rho_0^2}\right)\cdot J_1\left(\frac{\mu_m^{(1)}}{\rho_0}\rho\right)\rho d\rho}{\frac{\rho_0^2}{2}\left[J_1'\left(\mu_m^{(1)}\right)\right]^2}$$

3.3 多个自变量分离变量法举例

波动/二维/齐次 极坐标

自然 | 1齐

例1

$$\begin{cases} u_{tt}-a^2\left(u_{\rho\rho}+\dfrac{1}{\rho}u_\rho+\dfrac{1}{\rho^2}u_{\theta\theta}\right)=0,\ 0\le\rho<\rho_0,\ 0\le\theta<2\pi,\ t>0 \\ |u(0,t)|<+\infty,\ u(\rho_0,\theta,t)=0,\ t\ge 0 \\ u(\rho,\theta,0)=\varphi(\rho,\theta),\ u_t(\rho,\theta,0)=\psi(\rho,\theta),\ 0\le\rho\le\rho_0, 0\le\theta<2\pi \end{cases}$$

$$\mu_{n,m}=\left(\frac{\mu_m^{(n)}}{\rho_0}\right)^2,\quad (m\ge 1,\ n\ge 0)$$

$$W_{n,m}(\rho,\theta)=\begin{cases} J_n\left(\dfrac{\mu_m^{(n)}}{\rho_0}\rho\right)\cos n\theta, \\ J_n\left(\dfrac{\mu_m^{(n)}}{\rho_0}\rho\right)\sin n\theta \end{cases}$$

讨论

$$\varphi(\rho,\theta)=A\rho\left(1-\frac{\rho^2}{\rho_0^2}\right)\cos\theta \qquad \psi(\rho,\theta)=0$$

❷ 求特征函数 $R_{n,m}=J_n\left(\dfrac{\mu_m^{(n)}}{\rho_0}\rho\right)\cos n\theta$ 的展开系数

当 $n=1$ 时

$$\varphi_{n=1,m,\cos}=\frac{A\int_0^{\rho_0}\rho\left(1-\dfrac{\rho^2}{\rho_0^2}\right)\cdot J_1\left(\dfrac{\mu_m^{(1)}}{\rho_0}\rho\right)\rho d\rho}{\dfrac{\rho_0^2}{2}\left[J_1'\left(\mu_m^{(1)}\right)\right]^2}$$

$$\int_0^{\rho_0}\rho\left(1-\frac{\rho^2}{\rho_0^2}\right)\cdot J_1\left(\frac{\mu_m^{(1)}}{\rho_0}\rho\right)\rho d\rho \quad \text{分子积分}$$

$$=\int_0^{\rho_0}\rho^2\cdot J_1\left(\frac{\mu_m^{(1)}}{\rho_0}\rho\right)d\rho-\frac{1}{\rho_0^2}\int_0^{\rho_0}\rho^4\cdot J_1\left(\frac{\mu_m^{(1)}}{\rho_0}\rho\right)d\rho$$

$$=\int_0^{\mu_m^{(1)}}\left(\frac{\rho_0}{\mu_m^{(1)}}x\right)^2\cdot J_1(x)d\left(\frac{\rho_0}{\mu_m^{(1)}}x\right)-\frac{1}{\rho_0^2}\int_0^{\mu_m^{(1)}}\left(\frac{\rho_0}{\mu_m^{(1)}}x\right)^4\cdot J_1(x)d\left(\frac{\rho_0}{\mu_m^{(1)}}x\right)$$

$$=\left(\frac{\rho_0}{\mu_m^{(1)}}\right)^3\left(\mu_m^{(1)}\right)^2\cdot J_2\left(\mu_m^{(1)}\right)-\frac{1}{\rho_0^2}\left(\frac{\rho_0}{\mu_m^{(1)}}\right)^5\boxed{\int_0^{\mu_m^{(1)}}x^4\cdot J_1(x)dx}$$

$$=\frac{2\rho_0^3}{\left(\mu_m^{(1)}\right)^2}J_3\left(\mu_m^{(1)}\right)$$

其中 $\boxed{\int_0^{\mu_m^{(1)}}x^4\cdot J_1(x)dx}$

$$=\int_0^{\mu_m^{(1)}}x^2\cdot d\left[x^2\cdot J_2(x)\right]$$

$$=\left(\mu_m^{(1)}\right)^4\cdot J_2\left(\mu_m^{(1)}\right)-2\int_0^{\mu_m^{(1)}}x^3\cdot J_2(x)dx$$

$$=\left(\mu_m^{(1)}\right)^4\cdot J_2\left(\mu_m^{(1)}\right)-2\left(\mu_m^{(1)}\right)^3\cdot J_3\left(\mu_m^{(1)}\right)$$

例1

$$\begin{cases} u_{tt}-a^2\left(u_{\rho\rho}+\dfrac{1}{\rho}u_\rho+\dfrac{1}{\rho^2}u_{\theta\theta}\right)=0,\ 0\le\rho<\rho_0,\ 0\le\theta<2\pi,\ t>0 \\ \left|u(0,t)\right|<+\infty,\ u(\rho_0,\theta,t)=0,\ t\ge 0 \\ u(\rho,\theta,0)=\varphi(\rho,\theta),\ u_t(\rho,\theta,0)=\psi(\rho,\theta),\ 0\le\rho\le\rho_0,0\le\theta<2\pi \end{cases}$$

$$\mu_{n,m}=\left(\frac{\mu_m^{(n)}}{\rho_0}\right)^2,\quad (m\ge 1,\ n\ge 0)$$

$$W_{n,m}(\rho,\theta)=\begin{Bmatrix} J_n\left(\dfrac{\mu_m^{(n)}}{\rho_0}\rho\right)\cos n\theta, \\ J_n\left(\dfrac{\mu_m^{(n)}}{\rho_0}\rho\right)\sin n\theta \end{Bmatrix}$$

讨论

$$\varphi(\rho,\theta)=A\rho\left(1-\frac{\rho^2}{\rho_0^2}\right)\cos\theta \qquad \psi(\rho,\theta)=0$$

❷ 求特征函数 $R_{n,m}=J_n\left(\dfrac{\mu_m^{(n)}}{\rho_0}\rho\right)\cos n\theta$ 的展开系数

当 $n=1$ 时 求得展开系数为

$$\varphi_{n=1,m,\cos}=\frac{A\int_0^{\rho_0}\rho\left(1-\dfrac{\rho^2}{\rho_0^2}\right)\cdot J_1\left(\dfrac{\mu_m^{(1)}}{\rho_0}\rho\right)\rho d\rho}{\dfrac{\rho_0^2}{2}\left[J_1'\left(\mu_m^{(1)}\right)\right]^2}=\frac{A\dfrac{2\rho_0^3}{\left(\mu_m^{(1)}\right)^2}J_3\left(\mu_m^{(1)}\right)}{\dfrac{\rho_0^2}{2}\left[\dfrac{J_0\left(\mu_m^{(1)}\right)-J_2\left(\mu_m^{(1)}\right)}{2}\right]^2}=\frac{16A\rho_0 J_3\left(\mu_m^{(1)}\right)}{\left(\mu_m^{(1)}\right)^2\left[J_0\left(\mu_m^{(1)}\right)-J_2\left(\mu_m^{(1)}\right)\right]^2}$$

$$J_{n-1}(x)-J_{n+1}(x)=2J_n'(x)$$

3.3 多个自变量分离变量法举例

波动/二维/齐次 极坐标

自然 | 1齐

例1

$$\begin{cases} u_{tt}-a^2\left(u_{\rho\rho}+\dfrac{1}{\rho}u_\rho+\dfrac{1}{\rho^2}u_{\theta\theta}\right)=0,\ 0\le\rho<\rho_0,\ 0\le\theta<2\pi,\ t>0 \\ |u(0,t)|<+\infty,\ u(\rho_0,\theta,t)=0,\ t\ge 0 \\ u(\rho,\theta,0)=\varphi(\rho,\theta),\ u_t(\rho,\theta,0)=\psi(\rho,\theta),\ 0\le\rho\le\rho_0, 0\le\theta<2\pi \end{cases}$$

$$\mu_{n,m}=\left(\frac{\mu_m^{(n)}}{\rho_0}\right)^2,\quad (m\ge 1,\ n\ge 0)$$

$$W_{n,m}(\rho,\theta)=\begin{cases} J_n\left(\dfrac{\mu_m^{(n)}}{\rho_0}\rho\right)\cos n\theta, \\ J_n\left(\dfrac{\mu_m^{(n)}}{\rho_0}\rho\right)\sin n\theta \end{cases}$$

$$\varphi(\rho,\theta)=A\rho\left(1-\frac{\rho^2}{\rho_0^2}\right)\cos\theta \qquad \psi(\rho,\theta)=0$$

讨论

❷ 求特征函数 $R_{n,m}=J_n\left(\dfrac{\mu_m^{(n)}}{\rho_0}\rho\right)\cos n\theta$ 的展开系数

当 $n>1$ 时

$$\varphi_{n,m,\cos}=\frac{\int_0^{2\pi}\int_0^{\rho_0}A\rho\left(1-\dfrac{\rho^2}{\rho_0^2}\right)\cos\theta\cdot J_n\left(\dfrac{\mu_m^{(n)}}{\rho_0}\rho\right)\cos(n\cdot\theta)\rho d\rho d\theta}{\int_0^{2\pi}\int_0^{\rho_0}\left[J_n\left(\dfrac{\mu_m^{(n)}}{\rho_0}\rho\right)\right]^2\cos^2(n\cdot\theta)\rho d\rho d\theta}$$

$$=\frac{\underbrace{\int_0^{2\pi}\cos\theta\cos n\theta d\theta}_{=0}\cdot\int_0^{\rho_0}A\rho\left(1-\dfrac{\rho^2}{\rho_0^2}\right)\cdot J_n\left(\dfrac{\mu_m^{(n)}}{\rho_0}\rho\right)\rho d\rho}{\int_0^{2\pi}\cos^2 n\theta d\theta\cdot\int_0^{\rho_0}\left[J_n\left(\dfrac{\mu_m^{(n)}}{\rho_0}\rho\right)\right]^2\rho d\rho}=0$$

3.3 多个自变量分离变量法举例

波动/二维/齐次 极坐标

自然 | 1齐

$$\mu_{n,m}=\left(\frac{\mu_m^{(n)}}{\rho_0}\right)^2,\quad (m\ge 1,\ n\ge 0)$$

$$W_{n,m}(\rho,\theta)=\begin{cases} J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)\cos n\theta, \\ J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)\sin n\theta \end{cases}$$

例1

$$\begin{cases} u_{tt}-a^2\left(u_{\rho\rho}+\frac{1}{\rho}u_\rho+\frac{1}{\rho^2}u_{\theta\theta}\right)=0,\ 0\le\rho<\rho_0,\ 0\le\theta<2\pi,\ t>0 \\ |u(0,t)|<+\infty,\ u(\rho_0,\theta,t)=0,\ t\ge 0 \\ u(\rho,\theta,0)=\varphi(\rho,\theta),\ u_t(\rho,\theta,0)=\psi(\rho,\theta),\ 0\le\rho\le\rho_0, 0\le\theta<2\pi \end{cases}$$

讨论

$$\varphi(\rho,\theta)=A\rho\left(1-\frac{\rho^2}{\rho_0^2}\right)\cos\theta \qquad \psi(\rho,\theta)=0$$

③ 求特征函数 $R_{n,m}=J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)\sin n\theta$ 的展开系数

$$\varphi_{n,m,\sin}=\frac{\int_0^{2\pi}\int_0^{\rho_0}A\rho\left(1-\frac{\rho^2}{\rho_0^2}\right)\cos\theta\cdot J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)\sin n\theta\cdot\rho d\rho d\theta}{\int_0^{2\pi}\int_0^{\rho_0}\left[J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)\right]^2\sin^2 n\theta\cdot\rho d\rho d\theta}$$

$$=\frac{\underbrace{\int_0^{2\pi}\cos\theta\sin n\theta\cdot d\theta}_{=0}\cdot\int_0^{\rho_0}A\rho\left(1-\frac{\rho^2}{\rho_0^2}\right)\cdot J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)\rho d\rho}{\int_0^{2\pi}\sin^2 n\theta\, d\theta\cdot\int_0^{\rho_0}\left[J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)\right]^2\rho d\rho}=0$$

3.3 多个自变量分离变量法举例

波动/二维/齐次
极坐标

自然	1齐

$$\mu_{n,m}=\left(\frac{\mu_m^{(n)}}{\rho_0}\right)^2,\quad (m\ge 1,\ n\ge 0)$$

$$W_{n,m}(\rho,\theta)=\begin{cases}J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)\cos n\theta,\\ J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)\sin n\theta\end{cases}$$

例1

$$\begin{cases}u_{tt}-a^2\left(u_{\rho\rho}+\frac{1}{\rho}u_\rho+\frac{1}{\rho^2}u_{\theta\theta}\right)=0,\ 0\le\rho<\rho_0,\ 0\le\theta<2\pi,\ t>0\\ |u(0,t)|<+\infty,\ u(\rho_0,\theta,t)=0,\ t\ge 0\\ u(\rho,\theta,0)=\varphi(\rho,\theta),\ u_t(\rho,\theta,0)=\psi(\rho,\theta),\ 0\le\rho\le\rho_0,0\le\theta<2\pi\end{cases}$$

$$\varphi(\rho,\theta)=A\rho\left(1-\frac{\rho^2}{\rho_0^2}\right)\cos\theta\qquad \psi(\rho,\theta)=0$$

讨论

原定解问题的解为

$$u(\rho,\theta,t)=\sum_{n=0}^{\infty}\sum_{m=1}^{\infty}T_{n,m}(t)R_{n,m}(\rho)\Phi_n(\theta)$$

$$=\sum_{m=1}^{\infty}\underbrace{\varphi_{0,m}}_{=0}\cos\left(a\sqrt{\mu_{0,m}}\cdot t\right)\cdot J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\cdot\rho\right)\cdot 1$$

$$+\sum_{n=1}^{\infty}\sum_{m=1}^{\infty}\underbrace{\varphi_{n,m,\cos}}_{\text{仅}n=1\text{时}\neq 0}\cos\left(a\sqrt{\mu_{n,m}}\cdot t\right)\cdot J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\cdot\rho\right)\cdot\cos n\theta$$

$$+\sum_{n=1}^{\infty}\sum_{m=1}^{\infty}\underbrace{\varphi_{n,m,\sin}}_{=0}\cos\left(a\sqrt{\mu_{n,m}}\cdot t\right)\cdot J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\cdot\rho\right)\cdot\sin n\theta$$

$$=\sum_{m=1}^{\infty}\varphi_{n=1,m,\cos}\cos\left(a\sqrt{\mu_{1,m}}\cdot t\right)\cdot J_1\left(\frac{\mu_m^{(1)}}{\rho_0}\cdot\rho\right)\cdot\cos\theta$$

$$=16A\rho_0\cdot\sum_{m=1}^{\infty}\frac{J_3\left(\mu_m^{(1)}\right)}{\left(\mu_m^{(1)}\right)^2\left[J_0\left(\mu_m^{(1)}\right)-J_2\left(\mu_m^{(1)}\right)\right]^2}\cos\left(a\frac{\mu_m^{(1)}}{\rho_0}\cdot t\right)\cdot J_1\left(\frac{\mu_m^{(1)}}{\rho_0}\cdot\rho\right)\cdot\cos\theta$$

3.3 多个自变量分离变量法举例

波动/二维/齐次 极坐标

自然	1齐

例1

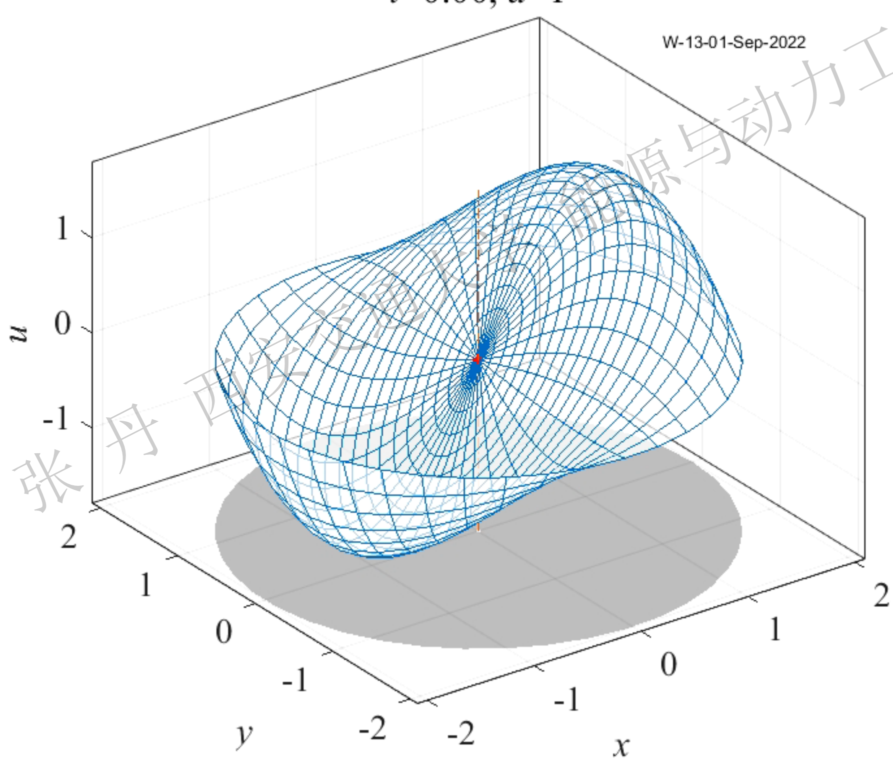
$$\begin{cases} u_{tt}-a^2\left(u_{\rho\rho}+\dfrac{1}{\rho}u_\rho+\dfrac{1}{\rho^2}u_{\theta\theta}\right)=0,\ 0\le\rho<\rho_0,\ 0\le\theta<2\pi,\ t>0 \\ |u(0,t)|<+\infty,\ u(\rho_0,\theta,t)=0,\ t\ge 0 \\ u(\rho,\theta,0)=\varphi(\rho,\theta),\ u_t(\rho,\theta,0)=\psi(\rho,\theta),\ 0\le\rho\le\rho_0, 0\le\theta<2\pi \end{cases}$$

$$\mu_{n,m}=\left(\frac{\mu_m^{(n)}}{\rho_0}\right)^2,\quad (m\ge 1,\ n\ge 0)$$

$$W_{n,m}(\rho,\theta)=\begin{cases} J_n\left(\dfrac{\mu_m^{(n)}}{\rho_0}\rho\right)\cos n\theta, \\ J_n\left(\dfrac{\mu_m^{(n)}}{\rho_0}\rho\right)\sin n\theta \end{cases}$$

讨论

$$u(\rho,\theta,t)=16A\rho_0\cdot\sum_{m=1}^{\infty}\frac{J_3\left(\mu_m^{(1)}\right)}{\left(\mu_m^{(1)}\right)^2\left[J_0\left(\mu_m^{(1)}\right)-J_2\left(\mu_m^{(1)}\right)\right]^2}\cos\left(a\frac{\mu_m^{(1)}}{\rho_0}\cdot t\right)\cdot J_1\left(\frac{\mu_m^{(1)}}{\rho_0}\cdot\rho\right)\cdot\cos\theta$$



$$\varphi(\rho,\theta)=A\rho\left(1-\frac{\rho^2}{\rho_0^2}\right)\cos\theta$$

$$\psi(\rho,\theta)=0$$

$\mu_m^{(n)}$ \ $J_n(x)$	0	1	2	3	4	5	……
1	2.404826	3.831706	5.135622	6.380162	7.588342	8.771484	……
2	5.520078	7.015587	8.417244	9.761023	11.06471	12.3386	……
3	8.653728	10.17347	11.61984	13.0152	14.37254	15.70017	……
4	11.79153	13.32369	14.79595	16.22347	17.61597	18.98013	……
5	14.93092	16.47063	17.95982	19.40942	20.82693	22.2178	……
6	18.07106	19.61586	21.117	22.58273	24.01902	25.43034	……
7	21.21164	22.76008	24.27011	25.74817	27.19909	28.62662	……
8	24.35247	25.90367	27.42057	28.90835	30.37101	31.81172	……
9	27.49348	29.04683	30.5692	32.06485	33.53714	34.98878	……
10	30.63461	32.18968	33.71652	35.21867	36.699	38.15987	……
11	33.77582	35.33231	36.86286	38.37047	39.85763	41.32638	……
……	……	……	……	……	……	……	……

3.3 多个自变量分离变量法举例

圆域 –Δ算子特征值问题的简化

圆域-Δ在第1类边界下特征值问题

$$\begin{cases}\Delta w(\rho,\theta)+\mu w(\rho,\theta)=0, & (\rho,\theta)\in\Omega \\ w|_{\partial\Omega}=0\end{cases}$$

很多物理问题→变量严格轴对称→仅随径向变化→不随周向变化

$$w(\rho,\theta)=R(\rho)\Phi(\theta)$$ 中 $\Phi(\theta)=C$

$$-\frac{\mu R(\rho)+R''(\rho)+\frac{1}{\rho}R'(\rho)}{\frac{1}{\rho^2}R(\rho)}=\frac{\Phi''(\theta)}{\Phi(\theta)}=-\lambda=0$$

即 $\lambda=n^2=0$，径向蜕化为0阶Bessel方程

$$\begin{cases}|R(0)|<+\infty, \; R(\rho_0)=0 \\ \rho^2R''(\rho)+\rho R'(\rho)+(\mu\rho^2-0^2)R(\rho)=0\end{cases}$$

圆域上第1类齐次边界下-Δ特征值/特征函数

$$\mu_{n,m}=\left(\frac{\mu_m^{(n)}}{\rho_0}\right)^2, \quad (m\ge 1, \; n\ge 0)$$

$$W_{n,m}(\rho,\theta)=\left\{J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)\cos n\theta, \; J_n\left(\frac{\mu_m^{(n)}}{\rho_0}\rho\right)\sin n\theta\right\}$$

严格轴对称 ↓ 特征值/特征函数蜕化为

$$\mu_{0,m}=\mu_m=\left(\frac{\mu_m^{(0)}}{\rho_0}\right)^2, \quad (m\ge 1)$$

$$R_{0,m}=R_m=J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)$$

Bessel方程特征值问题形式在 $n=0$ 可简化

$$\rho^2R''(\rho)+\rho R'(\rho)+(\lambda\rho^2-n^2)R(\rho)=0$$

$$-\left[R''(\rho)+\frac{1}{\rho}R'(\rho)-\frac{n^2}{\rho^2}R(\rho)\right]=\lambda R(\rho)$$

$$-\left[R''(\rho)+\frac{1}{\rho}R'(\rho)\right]=\lambda R(\rho)$$

本课程后续计算→以变量不随周向变化为主/以第一类边界为主

3.3 多个自变量分离变量法举例

圆域上的定解问题

例1 圆柱体 $\Omega=\left\{(x,y,z)\middle| x^2+y^2<\rho_0^2\right\}$ 是各相均匀的导热体, 边界温度为0, 初始温度为 $\varphi(x,y,z)$, 且 $\varphi(x,y,z)$ 只与 $\rho=\sqrt{x^2+y^2}$ 有关, 求圆柱体内温度分布随时间的变化 $u(x,y,z,t)$

解 第1步 建立定解问题 该问题在直角系内可表示为

$$\begin{cases} u_t-a^2\Delta u=0,\ (x,y,z)\in\Omega,\ t>0 \\ u=0,\ (x,y,z)\in\partial\Omega,\ \ t\ge 0 \\ u(x,y,z,0)=\varphi\left(\sqrt{x^2+y^2}\right),\ 0\le\sqrt{x^2+y^2}\le\rho_0 \end{cases}$$

$$\Delta u=\frac{\partial^2 u}{\partial\rho^2}+\frac{1}{\rho}\frac{\partial u}{\partial\rho}+\frac{1}{\rho^2}\frac{\partial^2 u}{\partial\theta^2}$$

随高度不变/仅随径向/时间变化

极坐标系更简单

$$\begin{cases} u_t-a^2\left(u_{\rho\rho}+\dfrac{1}{\rho}u_\rho+\dfrac{1}{\rho^2}\dfrac{\partial^2 u}{\partial\theta^2}\right)=0,\ 0\le\rho<\rho_0,\ t>0 \\ |u(0,t)|<+\infty,\ u(\rho_0,t)=0,\ \ t\ge 0 \\ u(\rho,0)=\varphi(\rho),\ 0\le\rho\le\rho_0 \end{cases}$$

=0, 不随周向变化

3.3 多个自变量分离变量法举例

圆域上的定解问题

例1

$$\begin{cases} u_t - a^2\left(u_{\rho\rho} + \dfrac{1}{\rho}u_\rho\right) = 0,\ 0 \le \rho < \rho_0,\ t > 0 \\ |u(0,t)| < +\infty,\ u(\rho_0,t) = 0,\ t \ge 0 \\ u(\rho,0) = \varphi(\rho),\ 0 \le \rho \le \rho_0 \end{cases}$$

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第2步 分离变量

设 $u(\rho,t) = R(\rho)T(t)$, 代入极坐标方程

$$R(\rho)T'(t) = a^2\left(R''(\rho)T(t) + \frac{1}{\rho}R'(\rho)T(t)\right)$$

$$\frac{T'(t)}{a^2T(t)} = \frac{R''(\rho) + \dfrac{1}{\rho}R'(\rho)}{R(\rho)} = -\lambda$$

时间

$$T'(t) + \lambda a^2 T(t) = 0$$

径向 $R''(\rho) + \dfrac{1}{\rho}R'(\rho) + \lambda R(\rho) = 0$

$$\begin{cases} \rho^2 R''(\rho) + \rho R'(\rho) + (\lambda\rho^2 - 0^2)R(\rho) = 0,\ 0 < \rho < \rho_0 \\ |R(0)| < +\infty,\ R(\rho_0) = 0 \end{cases}$$

0阶Bessel方程

3.3 多个自变量分离变量法举例

圆域上的定解问题

例1

$$\begin{cases} u_t - a^2\left(u_{\rho\rho} + \dfrac{1}{\rho}u_\rho\right) = 0,\ 0 \le \rho < \rho_0,\ t > 0 \\ \left|u(0,t)\right| < +\infty,\ u(\rho_0,t) = 0,\ t \ge 0 \\ u(\rho,0) = \varphi(\rho),\ 0 \le \rho \le \rho_0 \end{cases}$$

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第3步 求解特征值问题

径向特征值问题为Bessel方程特征值问题

$$\begin{cases} \rho^2 R''(\rho) + \rho R'(\rho) + \left(\lambda\rho^2 - 0^2\right)R(\rho) = 0,\ 0 < \rho < \rho_0 \\ \left|R(0)\right| < +\infty,\ R(\rho_0) = 0 \end{cases}$$

0阶Bessel方程

圆心自然, 外缘1齐次

特征值/特征函数

$$\lambda_m = \left(\frac{\mu_m^{(0)}}{\rho_0}\right)^2,\ R_m(\rho) = J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right),\ m \ge 1$$

特征值问题解决→即解决了边值问题→下面解决初值问题

3.3 多个自变量分离变量法举例

圆域上的定解问题

例1

$$\begin{cases} u_t - a^2\left(u_{\rho\rho} + \dfrac{1}{\rho}u_\rho\right) = 0,\ 0 \le \rho < \rho_0,\ t > 0 \\ \left|u(0,t)\right| < +\infty,\ u(\rho_0,t) = 0,\ t \ge 0 \\ u(\rho,0) = \varphi(\rho),\ 0 \le \rho \le \rho_0 \end{cases}$$

导热/二维/齐次 极坐标	
自然	1齐

第4步 求解初值问题

$$\lambda_m = \left(\frac{\mu_m^{(0)}}{\rho_0}\right)^2,\ R_m(\rho) = J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right),\ m \ge 1$$

将 λ_m 代入初值问题 $T'(t) + \lambda a^2 T(t) = 0$

$$T(t) = A_m \exp\left[-\left(\frac{\mu_m^{(0)}}{\rho_0}\right)^2 a^2 t\right]$$

ODE任意常数

原定解问题的解可表示为

$$u(\rho,t) = \sum_{m=1}^{\infty} A_m \exp\left[-\left(\frac{\mu_m^{(0)}}{\rho_0}\right)^2 a^2 t\right] \cdot J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)$$

利用初始条件

$$u(\rho,0) = \varphi(\rho) = \sum_{m=1}^{\infty} A_m \cdot J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)$$

相当于把 $\varphi(\rho)$ 按 $J_0(x)$ 展开为Bessel级数

$$A_m = \frac{2}{\left[\rho_0 J_0'\left(\mu_m^{(0)}\right)\right]^2}\int_0^{\rho_0} \rho\varphi(\rho) J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right) d\rho$$

原定解问题的解可表示为

$$u(\rho,t) = \sum_{m=1}^{\infty} A_m \exp\left[-\left(\frac{\mu_m^{(0)}}{\rho_0}\right)^2 a^2 t\right] \cdot J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)$$

3.3 多个自变量分离变量法举例

圆域上的定解问题

例1

$$\begin{cases} u_t - a^2\left(u_{\rho\rho} + \dfrac{1}{\rho}u_\rho\right) = 0,\ 0 \le \rho < \rho_0,\ t > 0 \\ |u(0,t)| < +\infty,\ u(\rho_0,t) = 0,\ t \ge 0 \\ u(\rho,0) = \varphi(\rho),\ 0 \le \rho \le \rho_0 \end{cases}$$

导热/二维/齐次 极坐标	
自然	1齐

$$\lambda_m = \left(\frac{\mu_m^{(0)}}{\rho_0}\right)^2,\ R_m(\rho) = J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right),\ m \ge 1$$

讨论 设初始温度分布为 $\varphi(\rho) = 1 - \dfrac{\rho^2}{\rho_0^2}$

$$A_m = \frac{2}{\left[\rho_0 J_0'\left(\mu_m^{(0)}\right)\right]^2}\int_0^{\rho_0} \rho\varphi(\rho) J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)d\rho = \frac{2}{\left[\rho_0 J_0'\left(\mu_m^{(0)}\right)\right]^2}\int_0^{\rho_0}\left(\rho - \frac{\rho^3}{\rho_0^2}\right)J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)d\rho$$

变量代换

$$x = \frac{\mu_m^{(0)}}{\rho_0}\rho,\quad \rho = \frac{\rho_0}{\mu_m^{(0)}}x$$

$$A_m = \frac{2}{\left[\rho_0 J_0'\left(\mu_m^{(0)}\right)\right]^2}\int_0^{\mu_m^{(0)}}\left[\frac{\rho_0}{\mu_m^{(0)}}x - \frac{1}{\rho_0^2}\left(\frac{\rho_0}{\mu_m^{(0)}}\right)^3 x^3\right]\cdot J_0(x)\cdot\frac{\rho_0}{\mu_m^{(0)}}\cdot dx$$

$$= \frac{2}{\left[\rho_0 J_0'\left(\mu_m^{(0)}\right)\right]^2}\cdot\left(\frac{\rho_0}{\mu_m^{(0)}}\right)^2\cdot\int_0^{\mu_m^{(0)}}\left[x - \frac{1}{\rho_0^2}\left(\frac{\rho_0}{\mu_m^{(0)}}\right)^2 x^3\right]\cdot J_0(x)\cdot dx$$

$$= \frac{2}{\left(\mu_m^{(0)}\right)^2\left[J_0'\left(\mu_m^{(0)}\right)\right]^2}\cdot\left[\int_0^{\mu_m^{(0)}} x\cdot J_0(x)\cdot dx - \frac{1}{\left(\mu_m^{(0)}\right)^2}\int_0^{\mu_m^{(0)}} x^3\cdot J_0(x)\cdot dx\right]$$

3.3 多个自变量分离变量法举例

圆域上的定解问题

例1

$$\begin{cases} u_t - a^2\left(u_{\rho\rho} + \frac{1}{\rho}u_\rho\right) = 0,\ 0 \le \rho < \rho_0,\ t > 0 \\ |u(0,t)| < +\infty,\ u(\rho_0,t) = 0,\ t \ge 0 \\ u(\rho,0) = \varphi(\rho),\ 0 \le \rho \le \rho_0 \end{cases}$$

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自然	1齐

$$\lambda_m = \left(\frac{\mu_m^{(0)}}{\rho_0}\right)^2,\ R_m(\rho) = J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right),\ m \ge 1$$

讨论

$$A_m = \underbrace{\frac{2}{\left(\mu_m^{(0)}\right)^2\left[J_0'\left(\mu_m^{(0)}\right)\right]^2}}_{1} \cdot \left[\underbrace{\int_0^{\mu_m^{(0)}} x \cdot J_0(x)\cdot dx}_{2} - \underbrace{\frac{1}{\left(\mu_m^{(0)}\right)^2}\int_0^{\mu_m^{(0)}} x^3 \cdot J_0(x)\cdot dx}_{3}\right]$$

$$\frac{d}{dx}\left[J_0(\alpha x)\right] = -\alpha J_1(\alpha x)$$

$$\int x \cdot J_0(x)dx = x \cdot J_1(x) + C$$

$$\int x^3 \cdot J_0(x)dx = x^3 J_1(x) - 2x^2 \cdot J_2(x) + C$$

$$= x\left(x^2 - 4\right)J_1(x) + 2x^2 J_0(x) + C$$

1 $$\frac{2}{\left(\mu_m^{(0)}\right)^2\left[J_0'\left(\mu_m^{(0)}\right)\right]^2} = \frac{2}{\left(\mu_m^{(0)}\right)^2\left[J_1\left(\mu_m^{(0)}\right)\right]^2}$$

2 $$\int_0^{\mu_m^{(0)}} x \cdot J_0(x)\cdot dx = x \cdot J_1(x)\Big|_0^{\mu_m^{(0)}} = \mu_m^{(0)} \cdot J_1\left(\mu_m^{(0)}\right)$$

3 $$\frac{1}{\left(\mu_m^{(0)}\right)^2}\int_0^{\mu_m^{(0)}} x^3 \cdot J_0(x)\cdot dx = \frac{1}{\left(\mu_m^{(0)}\right)^2}\left[\left(\mu_m^{(0)}\right)^3 J_1\left(\mu_m^{(0)}\right) - 2\left(\mu_m^{(0)}\right)^2 J_2\left(\mu_m^{(0)}\right)\right]$$

$$= \left(\mu_m^{(0)}\right)J_1\left(\mu_m^{(0)}\right) - 2J_2\left(\mu_m^{(0)}\right)$$

3.3 多个自变量分离变量法举例

圆域上的定解问题

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$$\begin{cases} u_t - a^2\left(u_{\rho\rho} + \dfrac{1}{\rho}u_\rho\right) = 0,\ 0 \le \rho < \rho_0,\ t > 0 \\ |u(0,t)| < +\infty,\ u(\rho_0,t) = 0,\ t \ge 0 \\ u(\rho,0) = \varphi(\rho),\ 0 \le \rho \le \rho_0 \end{cases}$$

导热/二维/齐次 极坐标

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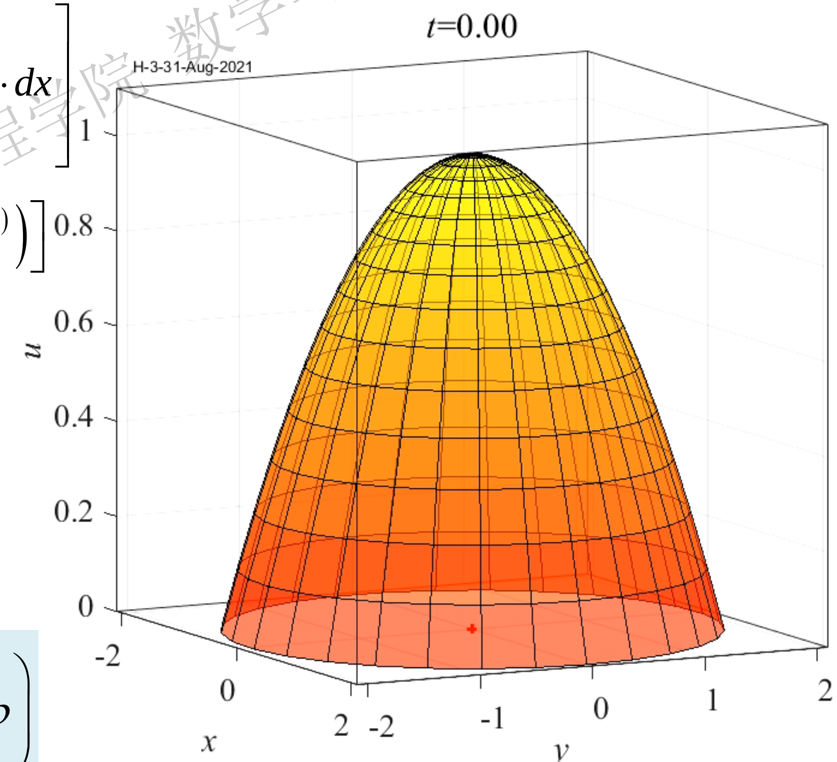
$$A_m = \frac{2}{\left(\mu_m^{(0)}\right)^2\left[J_0'\left(\mu_m^{(0)}\right)\right]^2}\cdot\left[\int_0^{\mu_m^{(0)}} x\cdot J_0(x)\cdot dx - \frac{1}{\left(\mu_m^{(0)}\right)^2}\int_0^{\mu_m^{(0)}} x^3\cdot J_0(x)\cdot dx\right]$$

$$= \frac{2}{\left(\mu_m^{(0)}\right)^2\left[J_1\left(\mu_m^{(0)}\right)\right]^2}\left[\mu_m^{(0)}\cdot J_1\left(\mu_m^{(0)}\right) - \left(\mu_m^{(0)}\right)J_1\left(\mu_m^{(0)}\right) + 2J_2\left(\mu_m^{(0)}\right)\right]$$

$$= \frac{4J_2\left(\mu_m^{(0)}\right)}{\left(\mu_m^{(0)}\right)^2\left[J_1\left(\mu_m^{(0)}\right)\right]^2}$$



原方程的解为

$$u(\rho,t) = \sum_{m=1}^{\infty}\frac{4J_2\left(\mu_m^{(0)}\right)}{\left(\mu_m^{(0)}\right)^2\left[J_1\left(\mu_m^{(0)}\right)\right]^2}\cdot\exp\left[-\left(\frac{\mu_m^{(0)}}{\rho_0}\right)^2 a^2 t\right]\cdot J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)$$

3.3 多个自变量分离变量法举例

圆域上的定解问题

例2

$$\begin{cases} u_t - a^2\left(u_{\rho\rho} + \frac{1}{\rho}u_\rho\right) = A,\ 0 \le \rho < \rho_0,\ t > 0 \\ \left|u(0,t)\right| < +\infty,\ u(\rho_0,t) = 0,\ t \ge 0 \\ u(\rho,0) = \varphi(\rho),\ 0 \le \rho \le \rho_0 \end{cases}$$

导热/二维/非齐 极坐标	
自然	1齐

解法1 方程齐次化

u 仅随径向/时间变化

$$u(\rho,t) = v(\rho,t) + w(\rho,t)$$

欲使 $v(\rho,t)$ 齐次化, 并保持边界

齐次不变, $w(\rho,t)$ 需满足

$$\begin{cases} w_t = a^2\left(w_{\rho\rho} + \frac{1}{\rho}w_\rho\right) + A,\ 0 \le \rho < \rho_0,\ t > 0 \\ w(\rho_0,t) = 0,\quad \left|w(\rho_0,t)\right| < +\infty \end{cases}$$

该方程边界/自由项与时间无关

$$w_t = 0 = a^2\left(w_{\rho\rho} + \frac{1}{\rho}w_\rho\right) + A$$

退化为ODE

$$\rho^2 w_{\rho\rho} + \rho w_\rho = -\frac{A}{a^2}\rho^2$$ 该方程为Euler方程

变量代换 $\rho = e^s,\ s = \ln\rho$ $\quad w_{ss} + w_s = -\frac{A}{a^2}e^{2s}$

对应的齐次方程通解为 $w(s) = C_1 + C_2 e^{-s}$

设特解为 $\bar{w}(s) = C_3 e^{2s}$, 代入原方程

$$4C_3 e^{2s} + 2C_3 e^{2s} = -\frac{A}{a^2}e^{2s} \quad C_3 = -\frac{A}{6a^2} \quad \bar{w}(s) = -\frac{A}{6a^2}e^{2s}$$

非齐次方程通解为 $w(s) = C_1 + C_2 e^{-s} - \frac{A}{6a^2}e^{2s}$

还原变量 $\quad w(\rho) = C_1 + C_2\rho^{-1} - \frac{A}{6a^2}\rho^2$

3.3 多个自变量分离变量法举例

导热/二维/齐次 极坐标

自然	1齐

圆域上的定解问题

例2

$$\begin{cases} u_t - a^2\left(u_{\rho\rho} + \dfrac{1}{\rho}u_\rho\right) = A,\ 0 \le \rho < \rho_0,\ t > 0 \\ \left|u(0,t)\right| < +\infty,\ u(\rho_0,t) = 0,\ t \ge 0 \\ u(\rho,0) = \varphi(\rho),\ 0 \le \rho \le \rho_0 \end{cases}$$

导热/二维/非齐 极坐标

自然	1齐

$$\begin{cases} v_t = a^2\left(v_{\rho\rho} + \dfrac{1}{\rho}v_\rho\right),\ 0 \le \rho < \rho_0,\ t > 0 \\ v(\rho_0,t) = 0,\ \left|v(0,t)\right| < +\infty,\ t \ge 0 \\ v(\rho,0) = \tilde{\varphi}(\rho) = \varphi(\rho) - \dfrac{A}{6a^2}\left(\rho_0^2 - \rho^2\right),\ 0 \le \rho \le \rho_0 \end{cases}$$

改造初始条件

解法1 方程齐次化

通解 $w(\rho) = C_1 + C_2\rho^{-1} - \dfrac{A}{6a^2}\rho^2$

根据边界条件

$$w(\rho_0,t) = 0,\quad \left|w(\rho_0,t)\right| < +\infty$$

确定系数 $C_1 = \dfrac{A}{6a^2},\ C_2 = 0$

得出辅助函数 $w(\rho) = \dfrac{A}{6a^2}\left(\rho_0^2 - \rho^2\right)$

齐次化后的原定解问题如上所示

根据上例,原定解问题的解可表示为

$$u(\rho,t) = \frac{A}{6a^2}\left(\rho_0^2 - \rho^2\right) + \sum_{m=1}^{\infty} A_m \exp\left[-\left(\frac{\mu_m^{(0)}}{\rho_0}\right)^2 a^2 t\right] \cdot J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)$$

$$A_m = \frac{2}{\left[\rho_0 J_0'\left(\mu_m^{(0)}\right)\right]^2}\int_0^{\rho_0} \rho\tilde{\varphi}(\rho) J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right) d\rho$$

1. 方程齐次化可解→还需改造初始条件
2. 自由项/边界条件与时间无关→确定 w_t
3. 得益于→对应齐次PDE定解问题前例已求得

3.3 多个自变量分离变量法举例

圆域上的定解问题

例2

解法2 特征函数法

不含周向导数项

$$\begin{cases} u_t - a^2\left(u_{\rho\rho} + \dfrac{1}{\rho}u_\rho\right) = A,\ 0 \le \rho < \rho_0,\ t > 0 \\ |u(0,t)| < +\infty,\ u(\rho_0,t) = 0,\ t \ge 0 \\ u(\rho,0) = \varphi(\rho),\ 0 \le \rho \le \rho_0 \end{cases}$$

导热/二维/非齐 极坐标	
自然	1齐

解　第1步 分离变量

对比Δu极坐标展开式→不含周向导数项→未知函数严格轴对称

$$\Delta u = \frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho}\frac{\partial u}{\partial \rho} + \frac{1}{\rho^2}\frac{\partial^2 u}{\partial \theta^2}$$

$u(\rho,t) = R(\rho)T(t)$　代入齐次PDE

$$R(\rho)T'(t) = a^2\left[R''(\rho)T(t) + \frac{1}{\rho}R'(\rho)T(t)\right]$$

$$\frac{T'(t)}{a^2T(t)} = \frac{R''(\rho) + \dfrac{1}{\rho}R'(\rho)}{R(\rho)} = -\lambda$$

初值问题

$$T'(t) + \lambda a^2 T(t) = 0$$

径向

$$\begin{cases} \rho^2 R''(\rho) + \rho R'(\rho) + (\lambda\rho^2 - 0^2)R(\rho) = 0,\ 0 < \rho < 1 \\ |R(0)| < +\infty,\ R(\rho_0) = 0 \end{cases}$$

0阶Bessel方程

圆心自然, 外缘1齐

特征值/特征函数

$$\lambda_m = \left(\frac{\mu_m^{(0)}}{\rho_0}\right)^2,\ R_m(\rho) = J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right),\ m \ge 1$$

3.3 多个自变量分离变量法举例

圆域上的定解问题

例2

解法2 特征函数法

$$\begin{cases} u_t - a^2\left(u_{\rho\rho} + \dfrac{1}{\rho}u_\rho\right) = A,\ 0 \le \rho < \rho_0,\ t > 0 \\ \left|u(0,t)\right| < +\infty,\ u(\rho_0,t) = 0,\ t \ge 0 \\ u(\rho,0) = \varphi(\rho),\ 0 \le \rho \le \rho_0 \end{cases}$$

导热/二维/非齐
极坐标

自然	1齐

$$\lambda_m = \left(\frac{\mu_m^{(0)}}{\rho_0}\right)^2,\quad R_m(\rho) = J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right),\quad m \ge 1$$

第2步 正交分解

设原定解问题的解为 $u(\rho,t) = \sum_{m=1}^{\infty} T_m(t) R_m(\rho)$

将自由项/初始条件按特征函数展开

$$A = \sum_{m=1}^{\infty} f_m R_m(\rho)$$

$$f_m = \frac{2}{\left[\rho_0 J_0'\left(\mu_m^{(0)}\right)\right]^2}\int_0^{\rho_0} \rho \cdot A \cdot J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right) d\rho$$

$$\varphi(\rho) = \sum_{m=1}^{\infty} \varphi_m R_m(\rho)$$

$$\varphi_m = \frac{2}{\left[\rho_0 J_0'\left(\mu_m^{(0)}\right)\right]^2}\int_0^{\rho_0} \rho \cdot \varphi(\rho) \cdot J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right) d\rho$$

变量代换 $x = \dfrac{\mu_m^{(0)}}{\rho_0}\rho,\quad \rho = \dfrac{\rho_0}{\mu_m^{(0)}}x$

$$\frac{d}{dx}\left[J_0(\alpha x)\right] = -\alpha J_1(\alpha x)$$

$$\int x \cdot J_0(x)\,dx = x \cdot J_1(x) + C$$

$$\int x^3 \cdot J_0(x)\,dx = x^3 J_1(x) - 2x^2 \cdot J_2(x) + C$$

$$f_m = \frac{2A}{\left[\rho_0 J_0'\left(\mu_m^{(0)}\right)\right]^2}\int_0^{\mu_m^{(0)}} \frac{\rho_0}{\mu_m^{(0)}} x \cdot J_0(x) \frac{\rho_0}{\mu_m^{(0)}} dx$$

$$= \frac{2A}{\left[\rho_0 J_0'\left(\mu_m^{(0)}\right)\right]^2}\left(\frac{\rho_0}{\mu_m^{(0)}}\right)^2 \int_0^{\mu_m^{(0)}} x \cdot J_0(x)\,dx$$

$$= \frac{2A}{\left[\rho_0 J_1\left(\mu_m^{(0)}\right)\right]^2}\left(\frac{\rho_0}{\mu_m^{(0)}}\right)^2 \mu_m^{(0)} \cdot J_1\left(\mu_m^{(0)}\right)$$

$$= \frac{2A}{\mu_m^{(0)} J_1\left(\mu_m^{(0)}\right)}$$

3.3 多个自变量分离变量法举例

圆域上的定解问题

例2

解法2 特征函数法

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导热/二维/非齐 极坐标

自然	1齐

$$\lambda_m = \left(\frac{\mu_m^{(0)}}{\rho_0}\right)^2,\ R_m(\rho) = J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right),\ m \ge 1$$

第3步 建立初值ODE

将展开式代入非齐次PDE

$$u(\rho,t) = \sum_{m=1}^{\infty} T_m(t) R_m(\rho) \qquad \varphi(\rho) = \sum_{m=1}^{\infty} \varphi_m R_m(\rho) \qquad A = \sum_{m=1}^{\infty} f_m R_m(\rho)$$

$$\sum_{m=1}^{\infty} T_m' R_m = a^2\left(\sum_{m=1}^{\infty} T_m R_m'' + \frac{1}{\rho}\sum_{m=1}^{\infty} T_m R_m'\right) + \sum_{m=1}^{\infty} f_m R_m$$

$$\sum_{m=1}^{\infty} T_m' R_m = a^2\sum_{m=1}^{\infty} T_m\left(R_m'' + \frac{1}{\rho}R_m'\right) + \sum_{m=1}^{\infty} f_m R_m$$

$$\sum_{m=1}^{\infty} T_m' R_m = a^2\sum_{m=1}^{\infty} T_m\left(-\lambda_m R_m\right) + \sum_{m=1}^{\infty} f_m R_m$$

Bessel方程特征值问题形式在 $n=0$ 可简化

$$\rho^2 R''(\rho) + \rho R'(\rho) + \left(\lambda\rho^2 - n^2\right)R(\rho) = 0$$

$$-\left[R''(\rho) + \frac{1}{\rho}R'(\rho) - \frac{n^2}{\rho^2}R(\rho)\right] = \lambda R(\rho)$$

$$-\left[R''(\rho) + \frac{1}{\rho}R'(\rho)\right] = \lambda R(\rho)$$

$$\sum_{m=1}^{\infty}\left[T_m' + a^2\lambda_m T_m\right]R_m = \sum_{m=1}^{\infty} f_m R_m$$

$$T_m' + a^2\lambda_m T_m = f_m$$

实现原初始条件

$$\sum_{m=1}^{\infty} T_m(0) R_m(\rho) = \varphi(\rho) = \sum_{m=1}^{\infty} \varphi_m R_m(\rho)$$

$$T_m(0) = \varphi_m$$

初值ODE

$$\begin{cases} T_m' + a^2\lambda_m T_m = f_m \\ T_m(0) = \varphi_m \end{cases}$$

3.3 多个自变量分离变量法举例

圆域上的定解问题

例2

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导热/二维/非齐 极坐标

自然	1齐

$$\lambda_m = \left(\frac{\mu_m^{(0)}}{\rho_0}\right)^2,\ R_m(\rho) = J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right),\ m \ge 1$$

第4步 解初值ODE $\begin{cases} T_m' + a^2\lambda_m T_m = f_m \\ T_m(0) = \varphi_m \end{cases}$

一阶线性非齐次ODE→ 常数变易法

对应的齐次ODE解为 $T_m(t) = C_1 \exp\left(-a^2\lambda_m t\right)$

非齐次ODE通解为 $T_m(t) = C_1(t)\exp\left(-a^2\lambda_m t\right)$

$$T_m'(t) = C_1'(t)\exp\left(-a^2\lambda_m t\right) - a^2\lambda_m C_1(t)\exp\left(-a^2\lambda_m t\right)$$

代入ODE

$$-a^2\lambda_m C_1(t)\exp\left(-a^2\lambda_m t\right) + C_1'(t)\exp\left(-a^2\lambda_m t\right) + a^2\lambda_m C_1(t)\exp\left(-a^2\lambda_m t\right) = f_m$$

$$C_1'(t) = f_m \exp\left(a^2\lambda_m t\right)$$

变易系数 $C_1(t) = \frac{f_m}{a^2\lambda_m}\exp\left(a^2\lambda_m t\right) + C_2$

通解 $T_m(t) = C_2\exp\left(-a^2\lambda_m t\right) + \frac{f_m}{a^2\lambda_m}$

根据初始条件 $T_m(0) = C_2 + \frac{f_m}{a^2\lambda_m} = \varphi_m$

$$C_2 = \varphi_m - \frac{f_m}{a^2\lambda_m}$$

初值ODE的解为

$$T_m(t) = \left(\varphi_m - \frac{f_m}{a^2\lambda_m}\right)\exp\left(-a^2\lambda_m t\right) + \frac{f_m}{a^2\lambda_m}$$

3.3 多个自变量分离变量法举例

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导热/二维/非齐 极坐标

自然 | 1齐

$$\lambda_m = \left(\frac{\mu_m^{(0)}}{\rho_0}\right)^2,\ R_m(\rho) = J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right),\ m \ge 1$$

第5步 回代

$$T_m(t) = \left(\varphi_m - \frac{f_m}{a^2\lambda_m}\right)\exp\left(-a^2\lambda_m t\right) + \frac{f_m}{a^2\lambda_m}$$

原定解问题的解为

$$u(\rho,t) = \sum_{m=1}^{\infty} T_m(t)R_m(\rho)$$

$$= \sum_{m=1}^{\infty}\left[\left(\varphi_m - \frac{f_m}{a^2\lambda_m}\right)\exp\left(-a^2\lambda_m t\right) + \frac{f_m}{a^2\lambda_m}\right]\cdot J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)$$

其中 $\varphi_m = \dfrac{2}{\left[\rho_0 J_0'\left(\mu_m^{(0)}\right)\right]^2}\displaystyle\int_0^{\rho_0} \rho\cdot\varphi(\rho)\cdot J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)d\rho$

$$f_m = \frac{2A}{\mu_m^{(0)} J_1\left(\mu_m^{(0)}\right)}$$

3.3 多个自变量分离变量法举例

圆域上的定解问题

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自然	1齐

$$\lambda_m = \left(\frac{\mu_m^{(0)}}{\rho_0}\right)^2,\ R_m(\rho) = J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right),\ m \ge 1$$

$$\varphi_m = \frac{2}{\left[\rho_0 J_0'\left(\mu_m^{(0)}\right)\right]^2}\int_0^{\rho_0} \rho \cdot \varphi(\rho) \cdot J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right) d\rho = 0$$

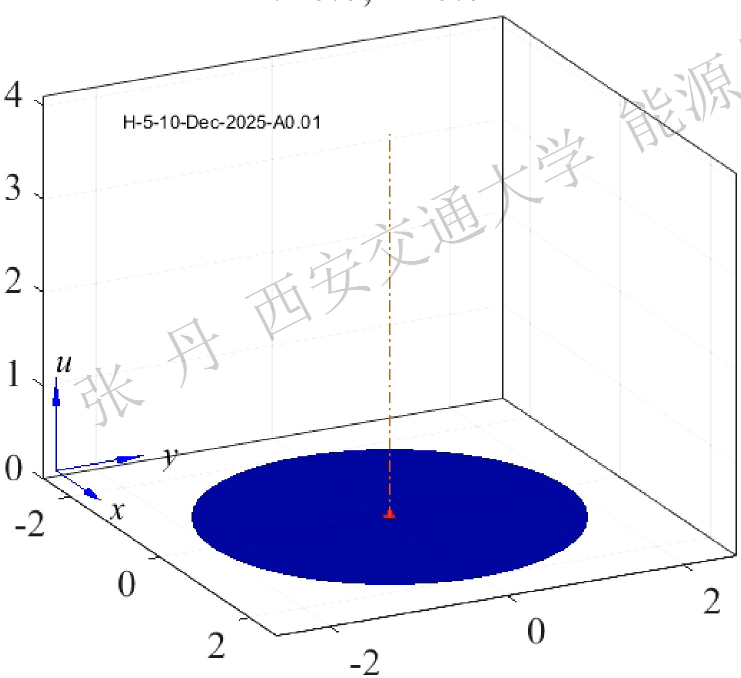
讨论

假设圆域上温度的初始分布为 $\varphi(\rho) = 0$

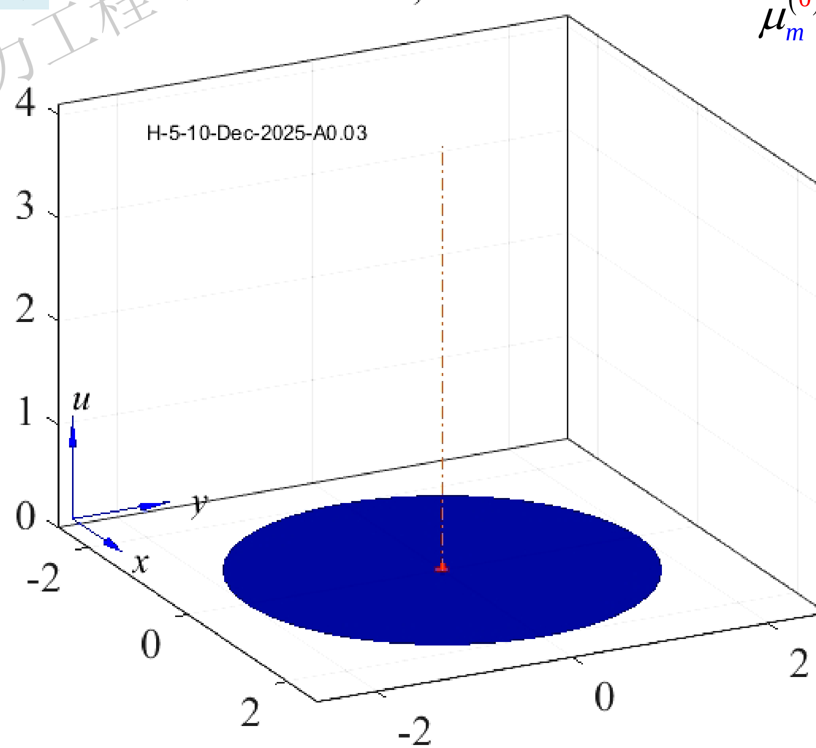
$$u(\rho,t) = \sum_{m=1}^{\infty}\left[\left[1 - \exp\left(-a^2\lambda_m t\right)\right]\frac{f_m}{a^2\lambda_m}\right] \cdot J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)$$

$$f_m = \frac{2A}{\mu_m^{(0)} J_1\left(\mu_m^{(0)}\right)}$$

t=0.0, A=0.01



t=0.0, A=0.03



3.3 多个自变量分离变量法举例

圆域上的定解问题

例3

不含周向导数项

波动/二维/非齐
极坐标

自然	1齐

$$\begin{cases} u_{tt}-a^2\left(u_{\rho\rho}+\dfrac{1}{\rho}u_\rho\right)=A,\ 0\le\rho<\rho_0,\ t>0 \\ \left|u(0,t)\right|<+\infty,\ u(\rho_0,t)=0,\ t\ge 0 \\ u(\rho,0)=\varphi(\rho),\ u_t(\rho,0)=\psi(\rho),\ 0\le\rho\le\rho_0 \end{cases}$$

解 第1步 分离变量

$$\Delta u=\frac{\partial^2 u}{\partial\rho^2}+\frac{1}{\rho}\frac{\partial u}{\partial\rho}+\frac{1}{\rho^2}\frac{\partial^2 u}{\partial\theta^2}$$

PDE中Δu不含周向导数→未知函数

仅随径向/时间变化→可表示为

$u(\rho,t)=R(\rho)T(t)$ 代入齐次PDE

$$R(\rho)T''(t)=a^2\left[R''(\rho)T(t)+\frac{1}{\rho}R'(\rho)T(t)\right]$$

$$\frac{T''(t)}{a^2T(t)}=\frac{R''(\rho)+\dfrac{1}{\rho}R'(\rho)}{R(\rho)}=-\lambda$$

初值问题

$$T''(t)+\lambda a^2T(t)=0$$

径向

$$\begin{cases} \rho^2R''(\rho)+\rho R'(\rho)+\left(\lambda\rho^2-0^2\right)R(\rho)=0,\ 0<\rho<1 \\ \left|R(0)\right|<+\infty,\ R(\rho_0)=0 \end{cases}$$

0阶Bessel方程

圆心自然, 外缘1齐

特征值/特征函数

$$\lambda_m=\left(\frac{\mu_m^{(0)}}{\rho_0}\right)^2,\ R_m(\rho)=J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right),\ m\ge 1$$

3.3 多个自变量分离变量法举例

圆域上的定解问题

例3

$$\begin{cases} u_{tt}-a^2\left(u_{\rho\rho}+\frac{1}{\rho}u_\rho\right)=A,\ 0\le\rho<\rho_0,\ t>0 \\ |u(0,t)|<+\infty,\ u(\rho_0,t)=0,\ t\ge 0 \\ u(\rho,0)=\varphi(\rho),\ u_\rho(\rho,0)=\psi(\rho),\ 0\le\rho\le\rho_0 \end{cases}$$

波动/二维/非齐 极坐标	
自然	1齐

$$\lambda_m=\left(\frac{\mu_m^{(0)}}{\rho_0}\right)^2,\ R_m(\rho)=J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right),\ m\ge 1$$

第2步 正交分解

设原定解问题的解为 $u(\rho,t)=\sum_{m=1}^{\infty}T_m(t)R_m(\rho)$

变量代换 $x=\frac{\mu_m^{(0)}}{\rho_0}\rho,\ \rho=\frac{\rho_0}{\mu_m^{(0)}}x$

将自由项/初始条件按特征函数展开

$$\varphi(\rho)=\sum_{m=1}^{\infty}\varphi_m R_m(\rho)\qquad \psi(\rho)=\sum_{m=1}^{\infty}\psi_m R_m(\rho)$$

$$\varphi_m=\frac{2}{\left[\rho_0 J_0'\left(\mu_m^{(0)}\right)\right]^2}\int_0^{\rho_0}\rho\cdot\varphi(\rho)\cdot J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)d\rho$$

$$\psi_m=\frac{2}{\left[\rho_0 J_0'\left(\mu_m^{(0)}\right)\right]^2}\int_0^{\rho_0}\rho\cdot\psi(\rho)\cdot J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)d\rho$$

$$A=\sum_{m=1}^{\infty}f_m R_m(\rho)$$

$$f_m=\frac{2}{\left[\rho_0 J_0'\left(\mu_m^{(0)}\right)\right]^2}\int_0^{\rho_0}\rho\cdot A\cdot J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)d\rho$$

$$\begin{aligned} f_m&=\frac{2A}{\left[\rho_0 J_0'\left(\mu_m^{(0)}\right)\right]^2}\int_0^{\mu_m^{(0)}}\frac{\rho_0}{\mu_m^{(0)}}x\cdot J_0(x)\frac{\rho_0}{\mu_m^{(0)}}dx \\ &=\frac{2A}{\left[\rho_0 J_0'\left(\mu_m^{(0)}\right)\right]^2}\left(\frac{\rho_0}{\mu_m^{(0)}}\right)^2\int_0^{\mu_m^{(0)}}x\cdot J_0(x)dx \\ &=\frac{2A}{\left[\rho_0 J_1\left(\mu_m^{(0)}\right)\right]^2}\left(\frac{\rho_0}{\mu_m^{(0)}}\right)^2\mu_m^{(0)}\cdot J_1\left(\mu_m^{(0)}\right) \\ &=\frac{2A}{\mu_m^{(0)}J_1\left(\mu_m^{(0)}\right)} \end{aligned}$$

$$\frac{d}{dx}\left[J_0(\alpha x)\right]=-\alpha J_1(\alpha x)$$

$$\int x\cdot J_0(x)dx=x\cdot J_1(x)+C$$

$$\int x^3\cdot J_0(x)dx=x^3J_1(x)-2x^2\cdot J_2(x)+C$$

3.3 多个自变量分离变量法举例

圆域上的定解问题

例3

$$\begin{cases} u_{tt}-a^2\left(u_{\rho\rho}+\dfrac{1}{\rho}u_\rho\right)=A,\ 0\le\rho<\rho_0,\ t>0 \\ |u(0,t)|<+\infty,\ u(\rho_0,t)=0,\ t\ge 0 \\ u(\rho,0)=\varphi(\rho),\ u_t(\rho,0)=\psi(\rho),\ 0\le\rho\le\rho_0 \end{cases}$$

波动/二维/非齐 极坐标

自然	1齐

$$\lambda_m=\left(\frac{\mu_m^{(0)}}{\rho_0}\right)^2,\ R_m(\rho)=J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right),\ m\ge 1$$

第3步 建立初值ODE

$$u(\rho,t)=\sum_{m=1}^{\infty}T_m(t)R_m(\rho)\qquad \varphi(\rho)=\sum_{m=1}^{\infty}\varphi_m R_m(\rho)\qquad \psi(\rho)=\sum_{m=1}^{\infty}\psi_m R_m(\rho)\qquad A=\sum_{m=1}^{\infty}f_m R_m(\rho)$$

将展开式代入非齐次PDE

Bessel方程特征值问题形式在 $n=0$ 可简化

$$\rho^2R''(\rho)+\rho R'(\rho)+(\lambda\rho^2-n^2)R(\rho)=0$$

$$-\left[R''(\rho)+\frac{1}{\rho}R'(\rho)-\frac{n^2}{\rho^2}R(\rho)\right]=\lambda R(\rho)$$

$$-\left[R''(\rho)+\frac{1}{\rho}R'(\rho)\right]=\lambda R(\rho)$$

$$\sum_{m=1}^{\infty}T_m''R_m-a^2\left(\sum_{m=1}^{\infty}T_mR_m''+\frac{1}{\rho}\sum_{m=1}^{\infty}T_mR_m'\right)=\sum_{m=1}^{\infty}f_mR_m$$

$$\sum_{m=1}^{\infty}T_m''R_m-a^2\sum_{m=1}^{\infty}T_m\left(R_m''+\frac{1}{\rho}R_m'\right)=\sum_{m=1}^{\infty}f_mR_m$$

$$\sum_{m=1}^{\infty}T_m''R_m-a^2\sum_{m=1}^{\infty}T_m(-\lambda_mR_m)=\sum_{m=1}^{\infty}f_mR_m$$

$$\sum_{m=1}^{\infty}\left[T_m''+a^2\lambda_mT_m\right]R_m=\sum_{m=1}^{\infty}f_mR_m$$

实现原初始条件

$$\sum_{m=1}^{\infty}T_m(0)R_m(\rho)=\sum_{m=1}^{\infty}\varphi_mR_m(\rho)=\varphi(\rho)$$

$$\sum_{m=1}^{\infty}T_m'(0)R_m(\rho)=\sum_{m=1}^{\infty}\psi_mR_m(\rho)=\psi(\rho)$$

$$T_m(0)=\varphi_m,\ T_m'(0)=\psi_m$$

初值ODE

$$\begin{cases} T''+a^2\lambda_mT_m=f_m \\ T_m(0)=\varphi_m,\ T_m'(0)=\psi_m \end{cases}$$

3.3 多个自变量分离变量法举例

圆域上的定解问题

例3

$$\begin{cases} u_{tt}-a^2\left(u_{\rho\rho}+\dfrac{1}{\rho}u_\rho\right)=A,\ 0\le\rho<\rho_0,\ t>0 \\ |u(0,t)|<+\infty,\ u(\rho_0,t)=0,\ t\ge 0 \\ u(\rho,0)=\varphi(\rho),\ u_\rho(\rho,0)=\psi(\rho),\ 0\le\rho\le\rho_0 \end{cases}$$

波动/二维/非齐 极坐标

自然 | 1齐

$$\lambda_m=\left(\frac{\mu_m^{(0)}}{\rho_0}\right)^2,\ R_m(\rho)=J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right),\ m\ge 1$$

第4步 解初值ODE

$$\begin{cases} T''+a^2\lambda_m T_m=f_m \\ T_m(0)=\varphi_m,\ T_m'(0)=\psi_m \end{cases}$$

线性二阶非齐次ODE

→通解=齐次通解+非齐特解

$$T_m(t)=C_{1m}\cos\left(\sqrt{a^2\lambda_m}\cdot t\right)+C_{2m}\sin\left(\sqrt{a^2\lambda_m}\cdot t\right)+\bar{T}_m$$

原PDE自由项为常数→可设通解亦为常数

$$\bar{T}_m=C$$

各阶导数 $\bar{T}_m'=\bar{T}_m''=0$，代入ODE比较系数

$$0+a^2\lambda_m C=f_m \qquad \bar{T}_m=C=\frac{f_m}{a^2\lambda_m}$$

非齐次ODE通解为

$$T_m(t)=C_{1m}\cos\left(\sqrt{a^2\lambda_m}\cdot t\right)+C_{2m}\sin\left(\sqrt{a^2\lambda_m}\cdot t\right)+\frac{f_m}{a^2\lambda_m}$$

3.3 多个自变量分离变量法举例

圆域上的定解问题

例3

$$\begin{cases} u_{tt} - a^2\left(u_{\rho\rho} + \dfrac{1}{\rho}u_\rho\right) = A,\ 0 \le \rho < \rho_0,\ t > 0 \\ |u(0,t)| < +\infty,\ u(\rho_0,t) = 0,\ t \ge 0 \\ u(\rho,0) = \varphi(\rho),\ u_\rho(\rho,0) = \psi(\rho),\ 0 \le \rho \le \rho_0 \end{cases}$$

波动/二维/非齐 极坐标	
自然	1齐

$$\lambda_m = \left(\frac{\mu_m^{(0)}}{\rho_0}\right)^2,\ R_m(\rho) = J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right),\ m \ge 1$$

第4步 解初值ODE

$$\begin{cases} T'' + a^2\lambda_m T_m = f_m \\ T_m(0) = \varphi_m,\ T_m'(0) = \psi_m \end{cases}$$

$$T_m(t) = C_{1m}\cos\left(\sqrt{a^2\lambda_m}\cdot t\right) + C_{2m}\sin\left(\sqrt{a^2\lambda_m}\cdot t\right) + \frac{f_m}{a^2\lambda_m}$$

$$T_m'(t) = -\sqrt{a^2\lambda_m}C_{1m}\sin\left(\sqrt{a^2\lambda_m}\cdot t\right) + \sqrt{a^2\lambda_m}C_{2m}\cos\left(\sqrt{a^2\lambda_m}\cdot t\right)$$

由ODE初始条件

$$T_m(0) = C_{1m} + \frac{f_m}{a^2\lambda_m} = \varphi_m$$

$$T_m'(0) = \sqrt{a^2\lambda_m}C_{2m} = \psi_m$$

$$\Rightarrow\quad C_{1m} = \varphi_m - \frac{f_m}{a^2\lambda_m}, \qquad C_{2m} = \frac{\psi_m}{\sqrt{a^2\lambda_m}}$$

初值问题的解为

$$T_m(t) = \left(\varphi_m - \frac{f_m}{a^2\lambda_m}\right)\cos\left(\sqrt{a^2\lambda_m}\cdot t\right) + \left(\frac{\psi_m}{\sqrt{a^2\lambda_m}}\right)\sin\left(\sqrt{a^2\lambda_m}\cdot t\right) + \frac{f_m}{a^2\lambda_m}$$

3.3 多个自变量分离变量法举例

圆域上的定解问题

例3

$$\begin{cases} u_{tt}-a^2\left(u_{\rho\rho}+\dfrac{1}{\rho}u_\rho\right)=A,\ 0\le\rho<\rho_0,\ t>0 \\ |u(0,t)|<+\infty,\ u(\rho_0,t)=0,\ t\ge 0 \\ u(\rho,0)=\varphi(\rho),\ u_t(\rho,0)=\psi(\rho),\ 0\le\rho\le\rho_0 \end{cases}$$

波动/二维/非齐
极坐标

自然	1齐

$$\lambda_m=\left(\frac{\mu_m^{(0)}}{\rho_0}\right)^2,\ R_m(\rho)=J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right),\ m\ge 1$$

第5步 回代

$$u(\rho,t)=\sum_{m=1}^{\infty}T_m(t)R_m(\rho)$$

$$=\sum_{m=1}^{\infty}\left[\left(\varphi_m-\frac{f_m}{a^2\lambda_m}\right)\cos\left(\sqrt{a^2\lambda_m}\cdot t\right)+\left(\frac{\psi_m}{\sqrt{a^2\lambda_m}}\right)\sin\left(\sqrt{a^2\lambda_m}\cdot t\right)+\frac{f_m}{a^2\lambda_m}\right]\cdot J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)$$

其中

$$f_m=\frac{2A}{\mu_m^{(0)}J_1\left(\mu_m^{(0)}\right)}$$

$$\varphi_m=\frac{2}{\left[\rho_0 J_0'\left(\mu_m^{(0)}\right)\right]^2}\int_0^{\rho_0}\rho\cdot\varphi(\rho)\cdot J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)d\rho$$

$$\psi_m=\frac{2}{\left[\rho_0 J_0'\left(\mu_m^{(0)}\right)\right]^2}\int_0^{\rho_0}\rho\cdot\psi(\rho)\cdot J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)d\rho$$

3.3 多个自变量分离变量法举例

圆域上的定解问题

例3

$$\begin{cases} u_{tt}-a^2\left(u_{\rho\rho}+\dfrac{1}{\rho}u_\rho\right)=A,\ 0\le\rho<\rho_0,\ t>0 \\ |u(0,t)|<+\infty,\ u(\rho_0,t)=0,\ t\ge 0 \\ u(\rho,0)=\varphi(\rho),\ u_t(\rho,0)=\psi(\rho),\ 0\le\rho\le\rho_0 \end{cases}$$

波动/二维/非齐 极坐标	
自然	1齐

$$\lambda_m=\left(\frac{\mu_m^{(0)}}{\rho_0}\right)^2,\ R_m(\rho)=J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right),\ m\ge 1$$

讨论 设初始条件为 $\varphi(\rho)=0,\ \psi(\rho)=0$

习题3, 20

则

$$\varphi_m=\frac{2}{\left[\rho_0J_0'\left(\mu_m^{(0)}\right)\right]^2}\int_0^{\rho_0}\rho\cdot\varphi(\rho)\cdot J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)d\rho=0$$

$$\psi_m=\frac{2}{\left[\rho_0J_0'\left(\mu_m^{(0)}\right)\right]^2}\int_0^{\rho_0}\rho\cdot\psi(\rho)\cdot J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)d\rho=0$$

$$u(\rho,t)=\sum_{m=1}^{\infty}\left[\left(\varphi_m-\frac{f_m}{a^2\lambda_m}\right)\cos\left(\sqrt{a^2\lambda_m}\cdot t\right)+\left(\frac{\psi_m}{\sqrt{a^2\lambda_m}}\right)\sin\left(\sqrt{a^2\lambda_m}\cdot t\right)+\frac{f_m}{a^2\lambda_m}\right]\cdot J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)$$

原定解问题的解可进一步化简为

$$u(\rho,t)=\sum_{m=1}^{\infty}\frac{f_m}{a^2\lambda_m}\left[1-\cos\left(a\frac{\mu_m^{(0)}}{\rho_0}\cdot t\right)\right]\cdot J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)$$

其中 $f_m=\dfrac{2A}{\mu_m^{(0)}J_1\left(\mu_m^{(0)}\right)}$

3.3 多个自变量分离变量法举例

圆域上的定解问题

例3

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$$\lambda_m=\left(\frac{\mu_m^{(0)}}{\rho_0}\right)^2,\ R_m(\rho)=J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right),\ m\ge 1$$

$\mu_m^{(n)}$ \ $J_n(x)$	0	1	2	3	4	5	......
1	2.404826	3.831706	5.135622	6.380162	7.588342	8.771484	......
2	5.520078	7.015587	8.417244	9.761023	11.06471	12.3386	......
3	8.653728	10.17347	11.61984	13.0152	14.37254	15.70017	......
4	11.79153	13.32369	14.79595	16.22347	17.61597	18.98013	......
5	14.93092	16.47063	17.95982	19.40942	20.82693	22.2178	......
6	18.07106	19.61586	21.117	22.58273	24.01902	25.43034	......
7	21.21164	22.76008	24.27011	25.74817	27.19909	28.62662	......
8	24.35247	25.90367	27.42057	28.90835	30.37101	31.81172	......
9	27.49348	29.04683	30.5692	32.06485	33.53714	34.98878	......
10	30.63461	32.18968	33.71652	35.21867	36.699	38.15987	......
11	33.77582	35.33231	36.86286	38.37047	39.85763	41.32638	......
......	......	......	......	......	......	......	......

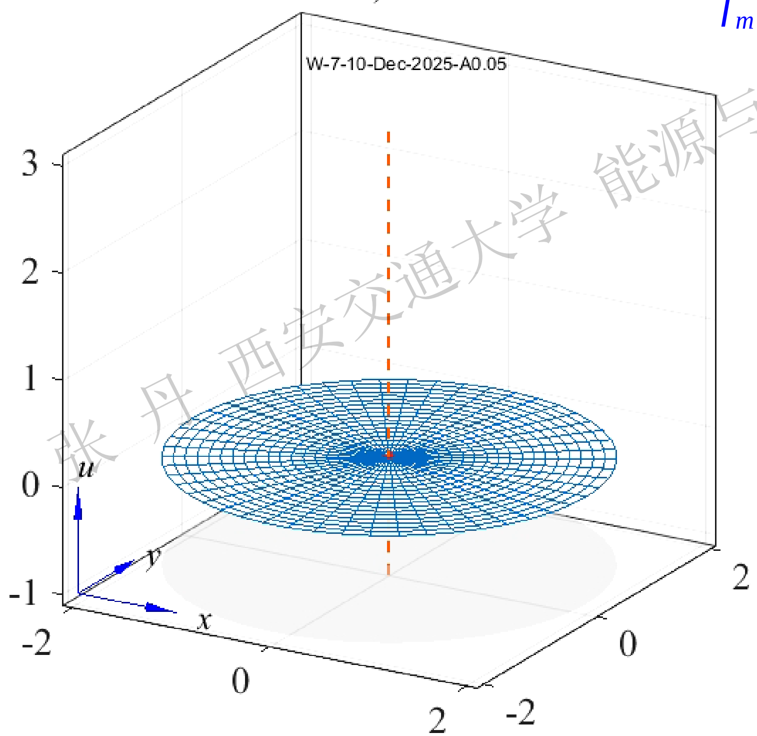
讨论

$$\varphi(\rho)=0,\ \psi(\rho)=0$$

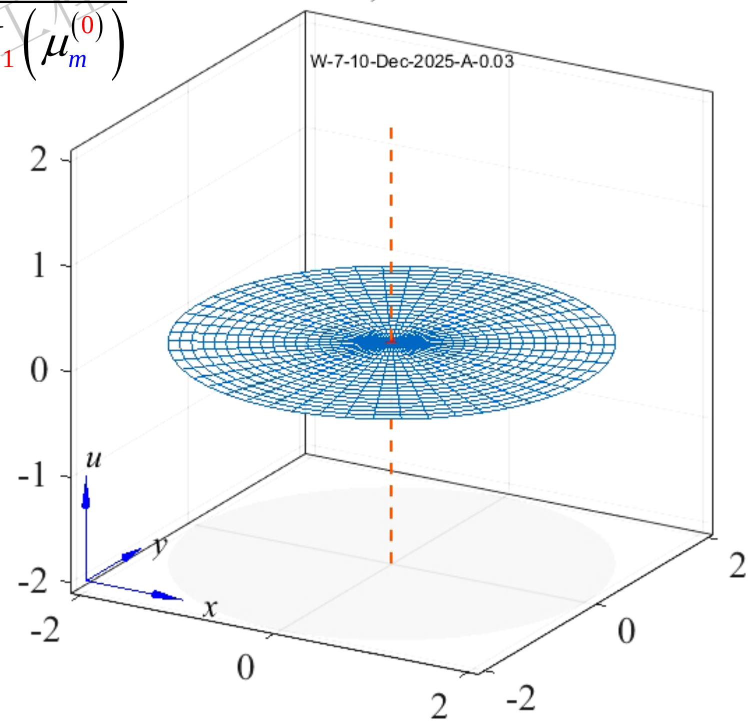
$$u(\rho,t)=\sum_{m=1}^{\infty}\frac{f_m}{a^2\lambda_m}\left[1-\cos\left(a\frac{\mu_m^{(0)}}{\rho_0}\cdot t\right)\right]\cdot J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)$$

$$f_m=\frac{2A}{\mu_m^{(0)}J_1\left(\mu_m^{(0)}\right)}$$

t=0.00, A=0.05



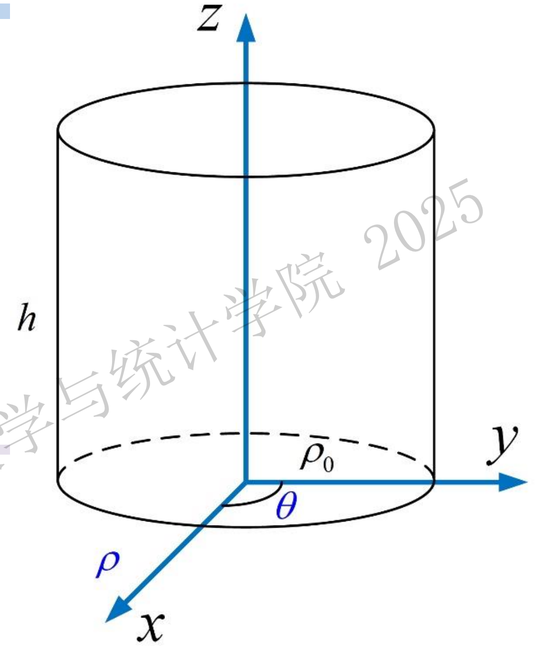
t=0.00, A=-0.03



3.3 多个自变量分离变量法举例

圆域上的定解问题

例4 导体内电位分布满足Laplace方程, 有圆柱体半径为ρ_0, 高为h ，其下底面/侧面电位为0, 上底面电位分布为 x^2+y^2, 求圆柱体内的电位分布



解 第1步 建立定解问题 圆柱体内的电位分布满足 $\Delta u=0$

圆柱系下算子Δ展开式为 $\Delta u=\dfrac{\partial^2 u}{\partial\rho^2}+\dfrac{1}{\rho}\cdot\dfrac{\partial u}{\partial\rho}+\dfrac{1}{\rho^2}\cdot\dfrac{\partial^2 u}{\partial\theta^2}+\boxed{\dfrac{\partial^2 u}{\partial z^2}}$

题设电位不随周向变化→上式中 $\dfrac{\partial^2 u}{\partial\theta^2}=0$

该电位分布可表示为定解问题

$$\begin{cases} u_{\rho\rho}+\dfrac{1}{\rho}u_\rho+u_{zz}=0,\ 0\le\rho<\rho_0,\ 0<z<h \\ u(\rho,0)=0,\ \rho\le\rho_0 & \text{下底面} \\ u(\rho,h)=\rho^2,\ \rho\le\rho_0 & \text{上底面} \\ u(\rho_0,z)=0,\ 0<z<h & \text{侧面} \end{cases}$$

3.3 多个自变量分离变量法举例

圆域上的定解问题

例4 导体内电位分布满足Laplace方程, 有圆柱体半径为 ρ_0，高为 h，其下底面/侧面电位为0, 上底面电位分布为 x^2+y^2, 求圆柱体内的电位分布

$$\begin{cases} u_{\rho\rho}+\dfrac{1}{\rho}u_\rho+u_{zz}=0,\ 0\le\rho<\rho_0,\ 0<z<h \\ u(\rho,0)=0,\ \rho\le\rho_0 \\ u(\rho,h)=\rho^2,\ \rho\le\rho_0 \\ u(\rho_0,z)=0,\ 0\le z\le h \end{cases}$$

Poisson/三维/齐次柱系		
下底 1齐	侧面 1齐	上底 非齐

第2步 分离变量 题设电位仅随径向/高度变化

设 $u(\rho,z)=R(\rho)H(z)$ 代入方程 $R''(\rho)H(z)+\dfrac{1}{\rho}R'(\rho)H(z)+R(\rho)H''(z)=0$

$$\frac{R''(\rho)+\frac{1}{\rho}R'(\rho)}{R(\rho)}=-\frac{H''(z)}{H(z)}=-\lambda$$

轴向

$$H''(z)-\lambda H(z)=0$$

径向

$$\begin{cases} \rho^2R''(\rho)+\rho R'(\rho)+(\lambda\rho^2-0^2)R(\rho)=0 \\ |R(0)|<+\infty,\ R(\rho_0)=0 \end{cases}$$

0阶Bessel方程

圆心自然, 外缘1齐　　特征值/特征函数

$$\lambda_m=\left(\frac{\mu_m^{(0)}}{\rho_0}\right)^2,\ R_m(\rho)=J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right),\ m\ge1$$

3.3 多个自变量分离变量法举例

圆域上的定解问题

例4 导体内电位分布满足Laplace方程, 有圆柱体半径为 ρ_0 , 高为 h ，其下底面/侧面电位为0, 上底面电位分布为 x^2+y^2 , 求圆柱体内的电位分布

$$\begin{cases} u_{\rho\rho}+\dfrac{1}{\rho}u_\rho+u_{zz}=0,\ 0\le\rho<\rho_0,\ 0<z<h \\ u(\rho,0)=0,\ \rho\le\rho_0 \\ u(\rho,h)=\rho^2,\ \rho\le\rho_0 \\ u(\rho_0,z)=0,\ 0\le z\le h \end{cases}$$

Poisson/三维/齐次柱系		
下底1齐	侧面1齐	上底非齐

$$\lambda_m=\left(\frac{\mu_m^{(0)}}{\rho_0}\right)^2,\ R_m(\rho)=J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right),\ m\ge1$$

第3步 解轴向方程

将 λ_m 代入轴向方程 $H''(z)-\lambda_m H(z)=0$

二阶线性齐次ODE **特征方程法**

$$H_m(z)=C_1\exp\left(\sqrt{\lambda_m}z\right)+C_2\exp\left(-\sqrt{\lambda_m}z\right)$$

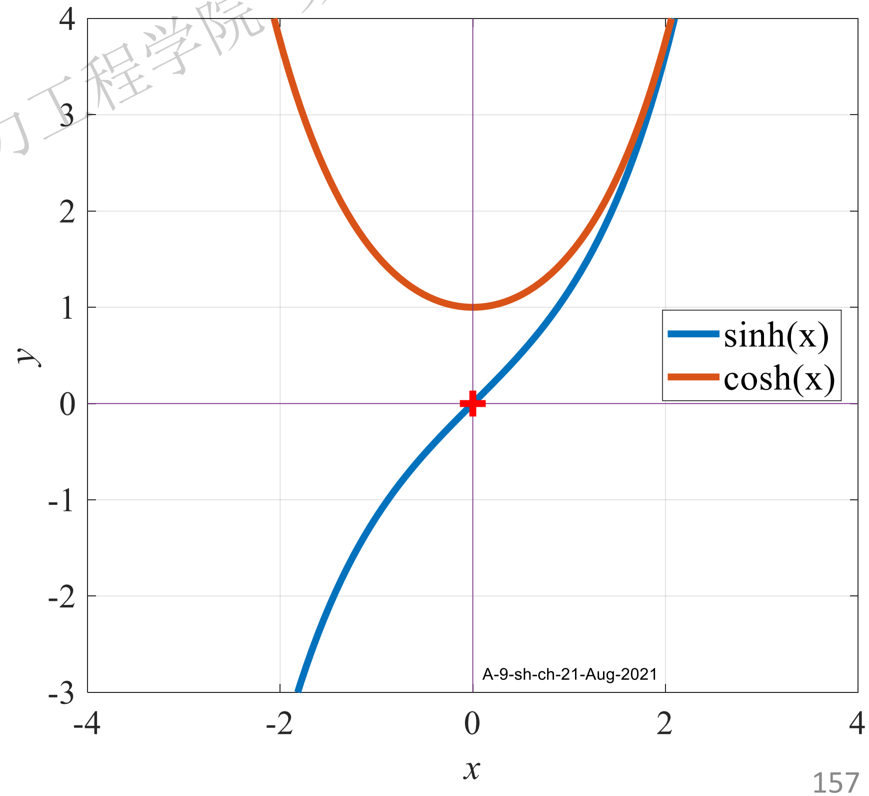
基解重新线性组合→另一种通解形式

$$H_m(z)=C_1\cosh\left(\frac{\mu_m^{(0)}}{\rho_0}z\right)+C_2\sinh\left(\frac{\mu_m^{(0)}}{\rho_0}z\right)$$

双曲正/余弦函数

$$\sinh(x)=\frac{e^x-e^{-x}}{2},\quad \cosh(x)=\frac{e^x+e^{-x}}{2}$$

$$\sinh(0)=0,\quad \cosh(0)=1$$



3.3 多个自变量分离变量法举例

圆域上的定解问题

例4 导体内电位分布满足Laplace方程, 有圆柱体半径为 ρ_0 , 高为 h ，其下底面/侧面电位为0, 上底面电位分布为 x^2+y^2 , 求圆柱体内的电位分布

$$\begin{cases} u_{\rho\rho}+\dfrac{1}{\rho}u_{\rho}+u_{zz}=0,\ 0\le\rho<\rho_0,\ 0<z<h \\ u(\rho,0)=0,\ \rho\le\rho_0 & \text{下底面} \\ u(\rho,h)=\rho^2,\ \rho\le\rho_0 & \text{上底面} \\ u(\rho_0,z)=0,\ 0\le z\le h & \text{侧面} \end{cases}$$

Poisson/三维/齐次柱系		
下底1齐	侧面1齐	上底非齐

$$\lambda_m=\left(\frac{\mu_m^{(0)}}{\rho_0}\right)^2,\ R_m(\rho)=J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right),\ m\ge1$$

第4步 回代 原定解问题的解可表示为

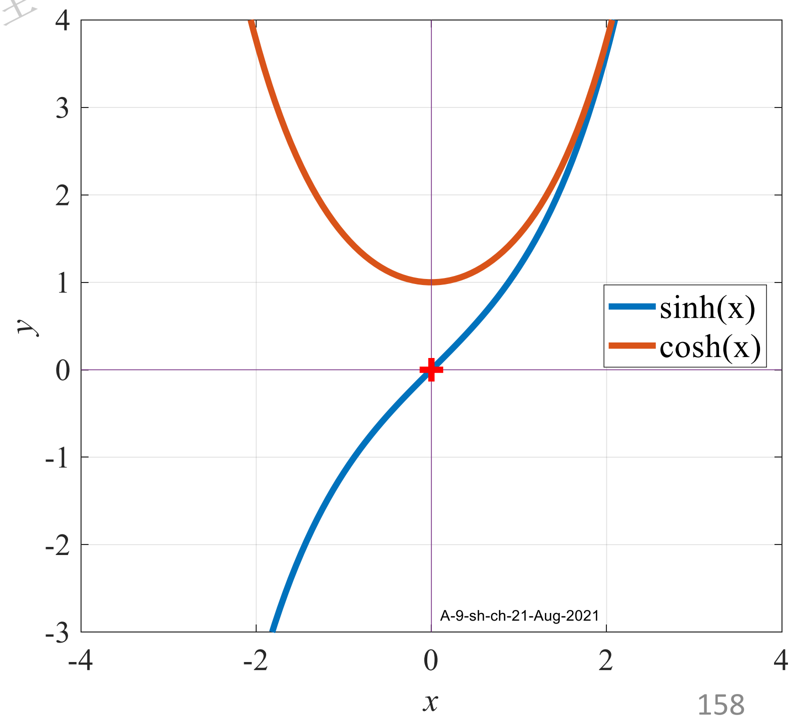
$$u(\rho,z)=\sum_{m=1}^{\infty}H_m(z)_m R(\rho)$$

$$=\sum_{m=1}^{\infty}\left[C_1\cosh\left(\frac{\mu_m^{(0)}}{\rho_0}z\right)+C_2\sinh\left(\frac{\mu_m^{(0)}}{\rho_0}z\right)\right]J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)$$

径向边界→求特征值/特征函数中已经使用→该解自动满足径向边界条件

$$u(\rho_0,z)=0,\ 0\le z\le h$$

下面通过上/下底面边界条件确定系数



圆域上的定解问题

例4 导体内电位分布满足Laplace方程, 有圆柱体半径为 ρ_0, 高为 h ，其下底面/侧面电位为0, 上底面电位分布为 x^2+y^2, 求圆柱体内的电位分布

$$\begin{cases} u_{\rho\rho}+\dfrac{1}{\rho}u_\rho+u_{zz}=0,\ 0\le\rho<\rho_0,\ 0<z<h \\ u(\rho,0)=0,\ \rho\le\rho_0 \quad \text{下底面} \\ u(\rho,h)=\rho^2,\ \rho\le\rho_0 \quad \text{上底面} \\ u(\rho_0,z)=0,\ 0\le z\le h \quad \text{侧面} \end{cases}$$

Poisson/三维/齐次柱系		
下底1齐	侧面1齐	上底非齐

$$\lambda_m=\left(\frac{\mu_m^{(0)}}{\rho_0}\right)^2,\ R_m(\rho)=J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right),\ m\ge1$$

$$u(\rho,z)=\sum_{m=1}^{\infty}\left[C_1\cosh\left(\frac{\mu_m^{(0)}}{\rho_0}z\right)+C_2\sinh\left(\frac{\mu_m^{(0)}}{\rho_0}z\right)\right]J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)$$

由下底面边界

$$u(\rho,0)=\sum_{m=1}^{\infty}\left[C_1\cosh\left(\frac{\mu_m^{(0)}}{\rho_0}z\right)\right]J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)=0 \Rightarrow C_1=0$$

由上底面边界

$$u(\rho,h)=\sum_{m=1}^{\infty}\boxed{C_2\sinh\left(\frac{\mu_m^{(0)}}{\rho_0}h\right)}J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)=\rho^2$$

$$C_2\sinh\left(\frac{\mu_m^{(0)}}{\rho_0}h\right)=\frac{2}{\left[\rho_0J_0'\left(\mu_m^{(0)}\right)\right]^2}\int_0^{\rho_0}\rho\cdot\rho^2J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)d\rho$$

变量代换 $x=\dfrac{\mu_m^{(0)}}{\rho_0}\rho,\quad \rho=\dfrac{\rho_0}{\mu_m^{(0)}}x$

$$C_2\sinh\left(\frac{\mu_m^{(0)}}{\rho_0}h\right)$$

$$=\frac{2}{\left[\rho_0J_0'\left(\mu_m^{(0)}\right)\right]^2}\int_0^{\mu_m^{(0)}}\left(\frac{\rho_0}{\mu_m^{(0)}}x\right)^3J_0(x)\frac{\rho_0}{\mu_m^{(0)}}dx$$

$$=\frac{2\rho_0^2}{\left(\mu_m^{(0)}\right)^4\left[J_0'\left(\mu_m^{(0)}\right)\right]^2}\int_0^{\mu_m^{(0)}}x^3J_0(x)dx$$

❶ ❷

3.3 多个自变量分离变量法举例

圆域上的定解问题

例4 导体内电位分布满足Laplace方程, 有圆柱体半径为ρ_0, 高为 h ，其下底面/侧面电位为0, 上底面电位分布为 x^2+y^2, 求圆柱体内的电位分布

$$\begin{cases} u_{\rho\rho}+\dfrac{1}{\rho}u_\rho+u_{zz}=0,\ 0\le\rho<\rho_0,\ 0<z<h \\ u(\rho,0)=0,\ \rho\le\rho_0 \\ u(\rho,h)=\rho^2,\ \rho\le\rho_0 \\ u(\rho_0,z)=0,\ 0\le z\le h \end{cases}$$

Poisson/三维/齐次柱系		
下底 1齐	侧面 1齐	上底 非齐

$$\lambda_m=\left(\frac{\mu_m^{(0)}}{\rho_0}\right)^2,\ R_m(\rho)=J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right),\ m\ge1$$

$$C_2\sinh\left(\frac{\mu_m^{(0)}}{\rho_0}h\right)=\underbrace{\frac{2\rho_0^2}{\left(\mu_m^{(0)}\right)^4\left[J_0'\left(\mu_m^{(0)}\right)\right]^2}}_{1}\underbrace{\int_0^{\mu_m^{(0)}}x^3J_0(x)\,dx}_{2}$$

$$\frac{d}{dx}\left[J_0(\alpha x)\right]=-\alpha J_1(\alpha x)$$

$$\int x\cdot J_0(x)\,dx=x\cdot J_1(x)+C$$

$$\int x^3\cdot J_0(x)\,dx=x^3J_1(x)-2x^2\cdot J_2(x)+C$$

$$=x\left(x^2-4\right)J_1(x)+2x^2J_0(x)+C$$

1

$$\frac{2\rho_0^2}{\left(\mu_m^{(0)}\right)^4\left[J_0'\left(\mu_m^{(0)}\right)\right]^2}=\frac{2\rho_0^2}{\left(\mu_m^{(0)}\right)^4\left[J_1\left(\mu_m^{(0)}\right)\right]^2}$$

2

$$\int_0^{\mu_m^{(0)}}x^3J_0(x)\,dx=\left(\mu_m^{(0)}\right)^3J_1\left(\mu_m^{(0)}\right)-2\left(\mu_m^{(0)}\right)^2\left[J_2\left(\mu_m^{(0)}\right)\right]$$

$$J_{n+1}(x)=\frac{2n}{x}J_n(x)-J_{n-1}(x)$$

$$=\left(\mu_m^{(0)}\right)^3J_1\left(\mu_m^{(0)}\right)-2\left(\mu_m^{(0)}\right)^2\left[\frac{2}{\mu_m^{(0)}}J_1\left(\mu_m^{(0)}\right)-J_0\left(\mu_m^{(0)}\right)\right]$$

$$=\mu_m^{(0)}\left[\left(\mu_m^{(0)}\right)^2-4\right]J_1\left(\mu_m^{(0)}\right)$$

代入化简

$$C_2=\frac{2\rho_0^2\left[\left(\mu_m^{(0)}\right)^2-4\right]}{\left(\mu_m^{(0)}\right)^3J_1\left(\mu_m^{(0)}\right)\sinh\left(\frac{\mu_m^{(0)}}{\rho_0}h\right)}$$

圆域上的定解问题

例4 导体内电位分布满足Laplace方程, 有圆柱体半径为 ρ_0 , 高为 h , 其下底面/侧面电位为0, 上底面电位分布为 x^2+y^2 , 求圆柱体内的电位分布

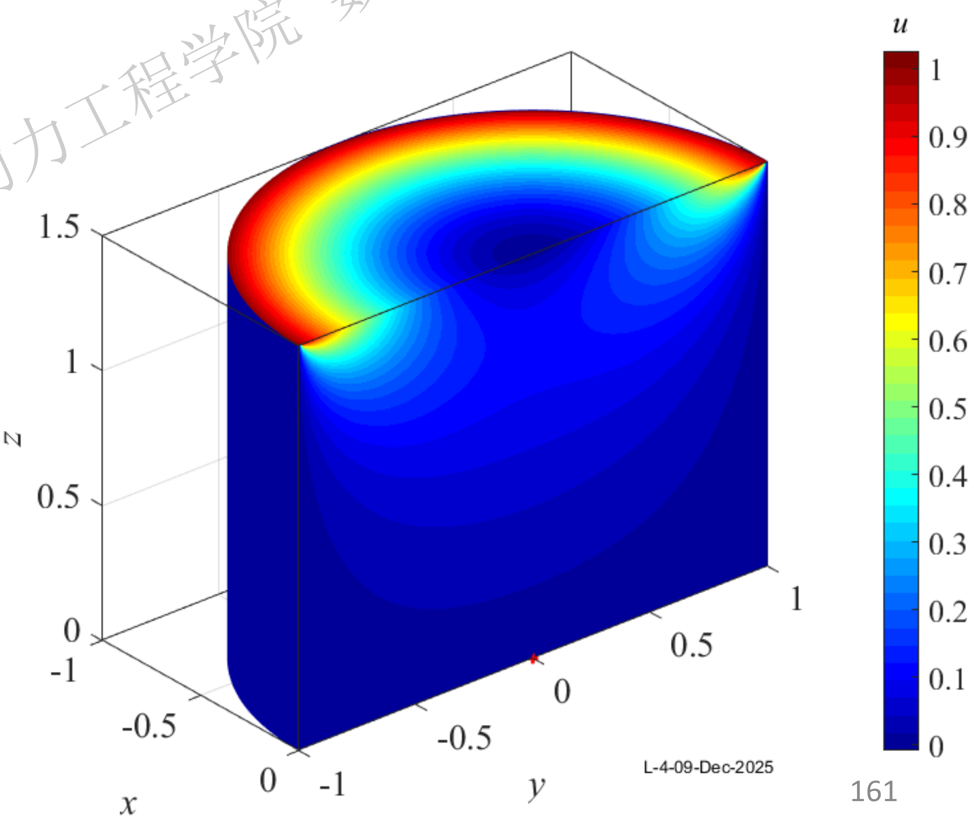
$$
\begin{cases}
u_{\rho\rho}+\dfrac{1}{\rho}u_{\rho}+u_{zz}=0,\ 0\le\rho<\rho_0,\ 0<z<h\\
u(\rho,0)=0,\ \rho\le\rho_0\\
u(\rho,h)=\rho^2,\ \rho\le\rho_0\\
u(\rho_0,z)=0,\ 0\le z\le h
\end{cases}
$$

Poisson/三维/齐次 柱系		
下底 1齐	侧面 1齐	上底 非齐

原定解问题的解

$$
u(\rho,z)=\sum_{m=1}^{\infty}H_m(z)_m R(\rho)
$$

$$
=\sum_{m=1}^{\infty}\frac{2\rho_0^2\left[\left(\mu_m^{(0)}\right)^2-4\right]\sinh\left(\dfrac{\mu_m^{(0)}}{\rho_0}z\right)}{\left(\mu_m^{(0)}\right)^3 J_1\left(\mu_m^{(0)}\right)\sinh\left(\dfrac{\mu_m^{(0)}}{\rho_0}h\right)}J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)
$$



3.3 多个自变量分离变量法举例

圆域上的定解问题

例4 导体内电位分布满足Laplace方程, 有圆柱体半径为ρ_0, 高为 h ，其下底面/侧面电位为0, 上底面电位分布为 x^2+y^2, 求圆柱体内的电位分布

$$\begin{cases} u_{\rho\rho}+\dfrac{1}{\rho}u_\rho+u_{zz}=0,\ 0\le\rho<\rho_0,\ 0<z<h \\ u(\rho,0)=0,\ \rho\le\rho_0 \\ u(\rho,h)=f(\rho),\ \rho\le\rho_0 \\ u(\rho_0,z)=0,\ 0\le z\le h \end{cases}$$

Poisson/三维/齐次 柱系		
下底 1齐	侧面 1齐	上底 非齐

讨论 柱系 变量下底面/侧面为0/上底面为给定分布→变量内部分布与周向无关

变量内部分布服从形式解 $u(\rho,z)=\sum_{m=1}^{\infty}H_m(z)_m R(\rho)=\sum_{m=1}^{\infty}C_2\sinh\left(\dfrac{\mu_m^{(0)}}{\rho_0}z\right)J_0\left(\dfrac{\mu_m^{(0)}}{\rho_0}\rho\right)$

令形式解满足上底面分布 $u(\rho,h)=f(\rho)=\sum_{m=1}^{\infty}C_2\sinh\left(\dfrac{\mu_m^{(0)}}{\rho_0}z\right)J_0\left(\dfrac{\mu_m^{(0)}}{\rho_0}\rho\right)$

系数可通过Bessel展开计算

$$C_2=\frac{2}{\sinh\left(\dfrac{\mu_m^{(0)}}{\rho_0}h\right)\left[\rho_0J_0'\left(\mu_m^{(0)}\right)\right]^2}\int_0^{\rho_0}\rho\cdot f(\rho)\cdot J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)d\rho$$

$$=\frac{2}{\sinh\left(\dfrac{\mu_m^{(0)}}{\rho_0}h\right)\left[\rho_0J_1\left(\mu_m^{(0)}\right)\right]^2}\int_0^{\rho_0}\rho\cdot f(\rho)\cdot J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)d\rho$$

$$\frac{d}{dx}\left[J_0(\alpha x)\right]=-\alpha J_1(\alpha x)$$

$$\int x\cdot J_0(x)dx=x\cdot J_1(x)+C$$

$$\int x^3\cdot J_0(x)dx=x^3J_1(x)-2x^2\cdot J_2(x)+C$$

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圆域上的定解问题

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Poisson/三维/齐次柱系		
下底1齐	侧面1齐	上底非齐

$$u(\rho,z)=\sum_{m=1}^{\infty}C_2\sinh\left(\frac{\mu_m^{(0)}}{\rho_0}z\right)J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)$$

$$C_2=\frac{2}{\sinh\left(\frac{\mu_m^{(0)}}{\rho_0}h\right)\left[\rho_0J_1\left(\mu_m^{(0)}\right)\right]^2}\int_0^{\rho_0}\rho\cdot f(\rho)\cdot J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)d\rho$$

讨论 例如 变量上底面为常数 $f(\rho)=B$

积分部分变量代换 $x=\dfrac{\mu_m^{(0)}}{\rho_0}\rho,\quad \rho=\dfrac{\rho_0}{\mu_m^{(0)}}x$

$$\begin{aligned}&\int_0^{\rho_0}\rho\cdot f(\rho)\cdot J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)d\rho\\ &=\int_0^{\mu_m^{(0)}}\frac{\rho_0}{\mu_m^{(0)}}x\cdot B\cdot J_0(x)\frac{\rho_0}{\mu_m^{(0)}}dx\\ &=B\left(\frac{\rho_0}{\mu_m^{(0)}}\right)^2\int_0^{\mu_m^{(0)}}xJ_0(x)dx\\ &=B\left(\frac{\rho_0}{\mu_m^{(0)}}\right)^2\mu_m^{(0)}J_1\left(\mu_m^{(0)}\right)\end{aligned}$$

$$\begin{aligned}C_2&=\frac{2}{\sinh\left(\frac{\mu_m^{(0)}}{\rho_0}h\right)\left[\rho_0J_1\left(\mu_m^{(0)}\right)\right]^2}\cdot B\cdot\left(\frac{\rho_0}{\mu_m^{(0)}}\right)^2\mu_m^{(0)}J_1\left(\mu_m^{(0)}\right)\\ &=\frac{2B}{\mu_m^{(0)}\sinh\left(\frac{\mu_m^{(0)}}{\rho_0}h\right)J_1\left(\mu_m^{(0)}\right)}\end{aligned}$$

柱系内部的变量分布满足

$$u(\rho,z)=2B\cdot\sum_{m=1}^{\infty}\frac{\sinh\left(\frac{\mu_m^{(0)}}{\rho_0}z\right)J_0\left(\frac{\mu_m^{(0)}}{\rho_0}\rho\right)}{\mu_m^{(0)}\sinh\left(\frac{\mu_m^{(0)}}{\rho_0}h\right)J_1\left(\mu_m^{(0)}\right)}$$

3.3 多个自变量分离变量法举例

习题3, 19

圆域上的定解问题

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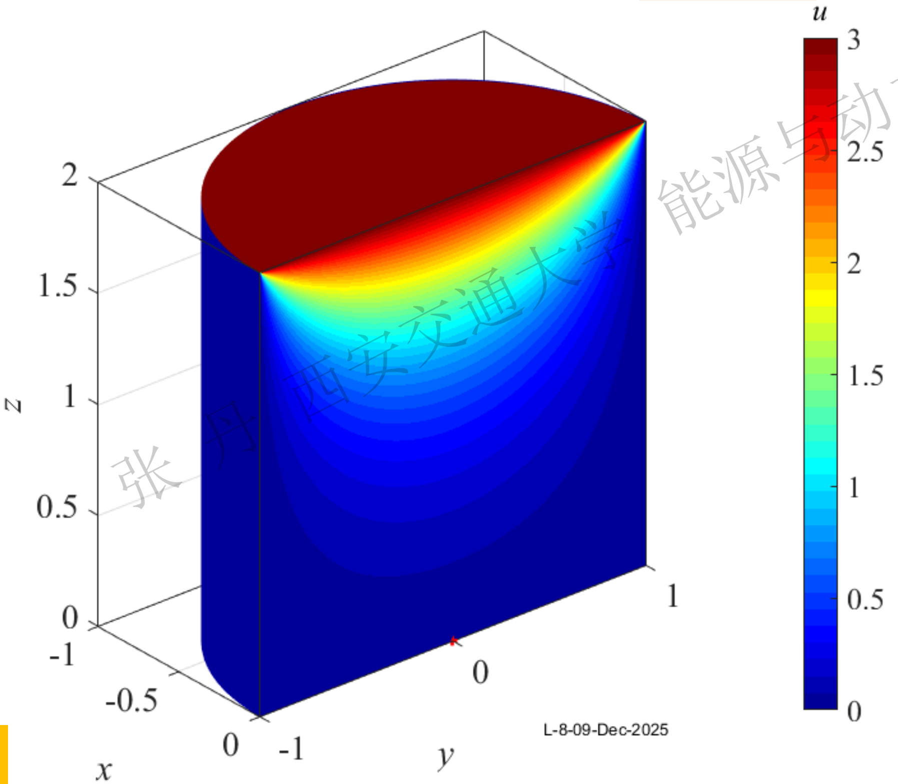
u(\rho,h)=f(\rho),\ \rho\le\rho_0\\
u(\rho_0,z)=0,\ 0\le z\le h
\end{cases}
$$

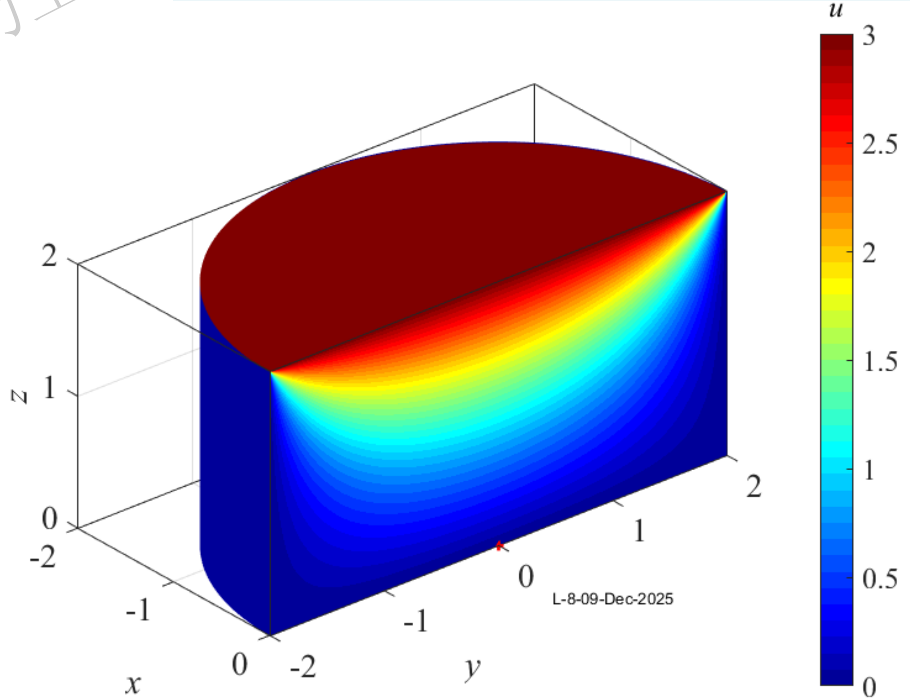
柱系内部变量分布

$$
u(\rho,z)=2B\cdot\sum_{m=1}^{\infty}\frac{\sinh\left(\dfrac{\mu_m^{(0)}}{\rho_0}z\right)J_0\left(\dfrac{\mu_m^{(0)}}{\rho_0}\rho\right)}{\mu_m^{(0)}\sinh\left(\dfrac{\mu_m^{(0)}}{\rho_0}h\right)J_1\left(\mu_m^{(0)}\right)}
$$

可当公式用

讨论 例如 变量上底面为常数 $f(\rho)=B$





本章作业

习题3

❶ 幂级数法求解ODE　　1 (2) (4)

❷ Bessel方程求解　　7 (1) 13 (1)

❸ Bessel函数积分　　14

❹ Bessel级数展开　　15

❺ 极坐标二维导热方程　　例 3.11

❻ 极坐标二维波动方程　　20

❼ 圆柱系Laplace方程　　例 3.13